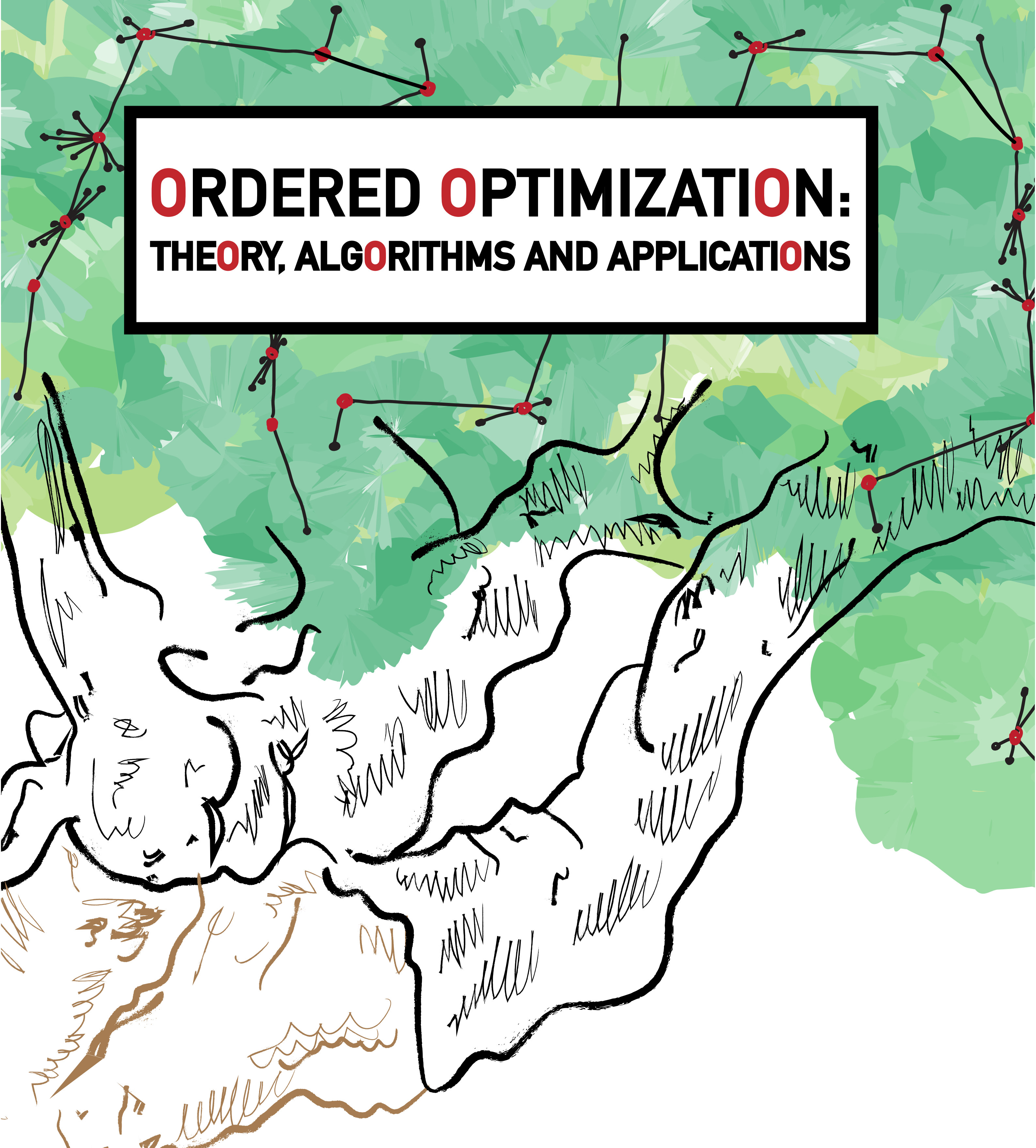
ORDERED OPTIMIZATION:

THEORY, ALGORITHMS AND APPLICATIONS



ALBERTO TORREJÓN VALENZUELA

UNIVERSIDAD
DE SEVILLA

PHD SUPERVISORS:
MIGUEL A. POZO MONTAÑO
JUSTO PUERTO ALBANDOZ

PhD Dissertation

Ordered Optimization:
Theory, Algorithms and Applications

PhD Program in Mathematics
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Alberto Torrejón Valenzuela
Department of Statistics and Operations Research
University of Sevilla
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PhD Supervisors: Prof. Miguel A. Pozo Montaño
Prof. Justo Puerto Albandoz

A mi hermano,
Daniel Torrejón Valenzuela.

Gracias

*"Y el niño crecía y se fortalecía y se llenaba de sabiduría;
y la gracia de Dios era sobre él."*
Lucas 2:40

Caminante, no hay camino, se hace camino al andar. Lo reconozco: aún he caminado poco, he errado mucho, y algún que otro acierto me ha traído hasta aquí. No todo ha sido cuestión de azar. Aunque el árbol que rige la toma de mis decisiones vitales se haya podido bifurcar incontables veces, y cada senda pudiera haberme llevado a otro destino, con esfuerzo, trabajo y constancia, he ido podando ramas que amenazaban con torcerse y fortaleciendo otras que aún a día de hoy siguen creciendo. He caminado poco, sí, mas nunca lo he hecho solo, y eso, eso, es lo que más agradezco.

Gracias, mamá. Gracias, papá. El camino ha tenido muchos oscuros, a veces demasiados. Unas veces podrían haberme vencido, otras eran pequeñas nimiedades que, en su momento, pesaban quintales. Gracias por preparar la tierra, por enseñarme que era yo quien debía cultivarla si algún día quería recoger frutos. Vuestros principios, aquellos que heredasteis de los abuelos, aquellos que nos habéis transmitido a mi hermano y a mí, son sólidos y atemporales, y perdurarán. Porque esto, abuelos, también os pertenece. Todos habéis puesto los cimientos para que este árbol pueda crecer fuerte y sano. El niño sigue creciendo, sigue fortaleciéndose.

Gracias, Justo. Gracias, Miguel. Justo, diste la oportunidad a un niño con muchas ganas pero desorientado. Le has ido prudentemente aconsejando, dando ejemplo, le has ido dando un propósito. Le has abierto puertas y, quizás más importante, le has dado la libertad de decidir si cruzarlas era lo que él realmente quería. Gracias por enseñarme tus teorías y permitirme hacerlas mías también, y disculpa, disculpa a este discípulo, a veces insolente, que no siempre ha sabido ponértelo fácil. Miguel, gracias igualmente por darme margen para hacer, por guiar con tino mis primeros pasos y por ayudarme a construir la base de todo lo que vino después.

Paco, Nacho y yo coincidimos los primeros días en la Facultad de Matemáticas, hace ya diez años de aquello. Gracias, a vosotros y a otros tantos, por acogerme y, sobre todo, por abrirme las puertas de esta ciudad, ahora casa, Sevilla. No obstante, si este manuscrito tiene un alma, Paco, es la tuya. Cada teorema, formulación y demostración lleva tu ayuda constante, tu mirada aguda, tu comprensión sin medida. De nada, Doctor Temprano García. Este camino, a veces abrupto, ha sido mucho más fácil contigo al lado. Y a ti, Moisés, bien sabes que también hay cabida para ti en estas primeras páginas. Nos acogiste a Paco y a mí desde el principio, ofreciendo consejos sinceros que aún hoy seguimos al pie de la letra. Allá donde vayamos, te llevaremos con orgullo, porque incluso tú, que habías perdido el gusto por este mundo, sabías algo que ellos quizás nunca entiendan: que seguiremos haciendo lo de siempre, porque nos gusta, porque nos divierte.

Caminar siempre es más fácil cuando lo haces acompañado, sobre todo si otros antes despejaron el sendero, apartaron la maleza y sembraron abundantes semillas. En mi caso, he tenido la fortuna de contar con uno de los equipos de investigación más sólidos que conozco, no solo en lo profesional, sino también en lo humano. Gracias por vuestra acogida, por crear ese ambiente de confianza donde crecer no era una obligación, sino un privilegio. Al Profesor Antonio M. Rodríguez Chía, al Profesor Víctor Blanco y al Profesor Federico Perea, gracias por tenderme la mano de forma desinteresada, por calmar tempestades, por estar ahí cuando las dudas apretaban.

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que al menos he compartido un café, una charla breve, gracias. El IMUS no es solo un centro de excelencia académica donde el desarrollo profesional es creciente y continuo, es, por encima de todo, un espacio donde el trato humano nunca se queda atrás. Igualmente, gracias a numerosos miembros de la Sociedad de Estadística e Investigación Operativa (y de Ciencia de Datos), de la Comunidad R Hispano, del Grupo de Usuarios de R SevillaR, y de la Red de Localización Española. Haber pertenecido, o pertenecer, a estas instituciones y comunidades ha sido, y sigue siendo, un auténtico honor.

Dicen que París es la ciudad de las luces. No se equivocan. Eso fue lo que encontré durante mi estancia en la ESSEC. Luz en mis compañeros de despacho y profesores con los que compartí tiempo allí. Luz en quienes, desde el Colegio de España en París, me abrieron un universo cultural hasta entonces desconocido. Y si pude reconocer todas esas luces fue porque alguien me brindó la oportunidad. Profesora Ivana Ljubić, gracias. Gracias por compartir tu conocimiento sin reservas, por enseñarme a crecer como investigador y, sobre todo, por darme el coraje para defender mis ideas sin miedo. Gracias por haber guiado con rigor y cercanía muchos de mis pasos, y por encender la luz de tantas ideas. *Merci beaucoup.*

Impartir docencia universitaria era un sueño incluso antes de sentarme por primera vez como alumno en una de sus aulas. Sueño que no acabará aquí. En parte, he tenido la fortuna de hacerlo realidad gracias al Departamento de Estadística e Investigación Operativa de la Universidad de Sevilla. Aquellos a los que una vez llamé profesores, he podido tener el placer posterior de llamarles compañeros. Porque se puede ser un pozo de conocimiento, pero de nada sirve si no sabes transmitirlo, y en esa tarea, he tenido buenos ejemplos. Gracias a todos y cada uno de los profesores que han facilitado mi travesía desde bien temprano. A Don Antonio, por sembrar en mí la semilla de la curiosidad por el saber; a Borja, por mantenerla viva y latente. Al Profesor José Antonio Prado Bassas, por introducirme en el mundo de la divulgación con tanta naturalidad como generosidad. Al Profesor Francisco Saldanha da Gama, por transmitir fuerza, por mostrar en nuestras incontables discusiones que la vida académica es una balanza en la que no se debe dejar atrás el disfrute. *Muito obrigado.* Y a la Profesora Vanesa Guerrero, simplemente gracias, por tu bondad sincera, por estar ahí. *Gracias.*

"¡Tu planeta es muy pequeño!" —dijo el aviador. —"Sí" —respondió el Principito—. "Puedo ver la puesta de sol cuarenta y tres veces en un mismo día". Yo también vengo de un planeta así. Aunque en mi vida académica eché raíces en Sevilla, nací en Los Barrios, un pueblo del Campo de Gibraltar, en Cádiz. Un lugar con la medida justa para sentirlo propio, para abarcarlo con la mirada y el corazón. Con orgullo suelo mostrar en mis charlas un mapa que señala ese punto exacto, porque es allí donde está mi gente: mi familia, mis vecinos, mis amigos y amigas. Mi barullo, mis miserables. Es el lugar donde crecí y al que siempre vuelvo para poder respirar cuando el ruido me desborda. Gracias a todos los que habéis compartido allí puestas de sol conmigo. Seguiré siendo un vecino, un amigo, un barreño orgulloso, allá donde la vida me lleve.

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Seguimos.

"Y todo lo que hagan, háganlo con amor."
1 Corintios 16:14

UNIVERSIDAD DE SEVILLA

PROGRAMA DE DOCTORADO EN MATEMÁTICAS

Tesis doctoral

Resumen

Optimización Ordenada:
Teoría, Algoritmos y Aplicaciones

Alberto Torrejón Valenzuela

Esta disertación presenta un marco general para resolver problemas complejos de optimización, denominado **Optimización Ordenada**. La investigación se basa en la premisa de que problemas de optimización, aparentemente dispares, pueden abordarse de manera unificada al aprovechar su estructura ordenada. Este enfoque introduce la formulación de modelos versátiles y amplía las posibilidades de análisis, sentando así las bases teóricas para unificar y simplificar el cálculo de múltiples medidas de rendimiento e índices de optimización. De esta forma, se ofrece una alternativa ágil y precisa a enfoques tradicionales de optimización.

El potencial de la Optimización Ordenada se hace evidente en su aplicación a diversos ámbitos. En la **localización de recursos**, un área donde el marco ya había sido explorado con anterioridad, se proponen mejoras al estado del arte mediante técnicas de descomposición. Además, se extiende su aplicación al introducir un problema inédito que combina la localización con el **diseño de redes**, donde los recursos se conectan a través de una estructura de árbol interna. El marco también es aplicado al estudio de **soluciones equitativas** y **robustas**. En el contexto de la equidad, se emplea en problemas de **ruteo de vehículos** para distribuir las rutas de manera ecuánime entre los conductores, modelando objetivos que combinan eficiencia operativa con la justicia en el reparto. En cuanto a la robustez, en estadística, el enfoque se aplica para formular estimadores robustos en **regresión lineal**, estudiando enfoques adaptados para modelar una de las familias de estimadores más robustos presentes en la literatura. Esta conexión entre programación matemática y estadística subraya la versatilidad del marco y su capacidad para generar avances metodológicos interdisciplinares.

En conclusión, esta disertación establece un marco de optimización unificado que mejora el abordaje de problemas complejos. Se demuestra que aprovechar la estructura ordenada inherente a estos problemas permite obtener mejoras sustanciales tanto en la eficiencia como en la calidad de las soluciones. Las principales contribuciones de la tesis son la definición de un marco teórico unificado para la optimización con funciones basadas en orden, el desarrollo de modelos y algoritmos que avanzan el estado del arte en campos de la investigación operativa y estadística, específicamente aplicado a la localización de servicios, diseño de redes, ruteo de vehículos y regresión lineal, así como la apertura de nuevas líneas de investigación que conectan la equidad y la robustez con la optimización. Estos resultados subrayan la relevancia de los enfoques ordenados como herramientas transversales en la optimización matemática.

UNIVERSITY OF SEVILLE
PHD PROGRAM IN MATHEMATICS

Phd Dissertation

Abstract

Ordered Optimization:
Theory, Algorithms and Applications

Alberto Torrejón Valenzuela

This dissertation presents a general framework for addressing complex optimization problems, namely **Ordered Optimization**. The research builds on the premise that seemingly disparate optimization problems can be approached in a unified manner by exploiting their ordered structure. This perspective allows the formulation of versatile models and broadens the scope of analysis, thereby laying the theoretical foundations for unifying and simplifying the computation of multiple performance measures and optimization indices. As a result, the framework provides an agile and accurate alternative to traditional optimization approaches.

The potential of Ordered Optimization becomes evident through its application in several contexts. In **facility location**, an area where the framework had previously been explored, improvements over the state-of-the-art are proposed through decomposition techniques. Its applicability is further extended by introducing a novel problem that combines facility location with **network design**, in which resources are internally connected through a tree structure. The framework is also applied to the study of **fair** and **robust solutions**. Concerning fairness, it is introduced in **vehicle routing** problems to distribute routes equitably among drivers, modeling objectives that integrate operational efficiency with fairness in workload allocation. With respect to robustness, the framework is applied in statistics to formulate robust estimators in **linear regression**, exploring tailored approaches to model one of the most robust families of estimators in the literature. This connection between mathematical programming and statistics highlights the versatility of the framework and its capacity to foster interdisciplinary methodological advances.

In conclusion, this dissertation establishes a unified optimization framework that enhances the formulation and resolution of complex problems. It shows that leveraging the inherent ordered structure of these problems yields substantial improvements in both efficiency and solution quality. The main contributions of the thesis are the formulation of a unified theoretical framework for optimization with order-based functions, the development of models and algorithms that advance the state-of-the-art across several fields of operations research and statistics, particularly in facility location, network design, vehicle routing, and linear regression, and the identification of promising research lines linking fairness and robustness with optimization. These findings underscore the relevance of ordered approaches as cross-cutting tools in mathematical optimization.

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"Take all that you can of this Book upon reason, and balance the rest on faith, and you will live and die a happier man."

Abraham Lincoln

1 Preface

It is evident that *decision-making* constitutes, perhaps not the whole of the human being, but certainly the central axis of their development and maturity. We need to make decisions to move forward in life. Maturing involves making increasingly complex choices deeply intertwined with ourselves, our fellow human beings, our experiences, and a deep understanding of our environment, while accepting their probable consequences. The scaffolding of our scientific, philosophical, and religious knowledge, even our most primal instincts, are connections that may help us develop the confidence to ensure a previous result. However, even if we choose objectively, without the proper guidelines, the best solution is not always guaranteed. Sometimes, our judgment leads us to heavens or hells yet unforeseen. This perhaps eccentric explanation of decision-making and its consequences can be illustrated by the paradox of *Jean Buridan's Donkey*, which was put forward by his detractors as a mockery of his excessive rationalism:

> *A farmer used to feed his donkey every day with two sacks of hay, one larger than the other. The donkey would always choose the larger one. One day, the farmer gave the donkey two identical sacks, and the donkey, accustomed to knowing which sack to pick, could not make a clear decision and ended up collapsing.*

This paradox shows that if there is no reason guiding the decision, no preferences over others, there will be no action. It highlights that the maxim of always choosing the best option based on weak knowledge becomes problematic when faced with two options equally "better", or "worse".

Strategic reasoning, which can be understood as the step prior to decision-making, has always been our way of preparing to face problems. The search for knowledge that allows us to improve our outcomes has always been present. The aphorism found in the pronaos of the Temple of Apollo at Delphi, *"know thyself"*, is the reflection that underpins the structure of our choices. First know yourself, then act in consequence. Another example is the ancient Chinese treatise *The Art of War*, by Sun Tzu (*Master Tzu*), dating back to approximately the 5-th century B.C., which compiles the main military strategies of the time and is considered one of the first publications in history on strategic study. In the modern era, strategic reasoning has been extensively studied from a mathematical lens perspective, giving rise to disciplines that attempt to axiomatize the decision-making process. Key fields such as *Operations Research*, which uses analytical methods to find optimal solutions for complex problems, and *Game Theory*, which models interactions between rational agents, provide a rigorous framework for planning. Furthermore, *Statistics* and *Probability Theory* equip us with the tools to manage data, face uncertainty and evaluate risks, while *Utility Theory* helps to quantify preferences and model individual choices. These disciplines do not seek to replace human intuition but rather to complement it, offering a quantitative framework to structure thought, analyze complexity, and ultimately refine the art of making the best possible decision.

Regarding the human capacity to study and propose options, one must recall the concept of *free will*, which has been a cornerstone of debate for millennia. From a philosophical perspective, ancient Greek philosophers like Plato and Aristotle explored the nature of choice and responsibility, often linking it to virtue and moral character. They considered whether human actions were predetermined by fate or if individuals had the capacity for genuine choice. This idea also runs transversely through the Christian thought, as stated in *Deuteronomy* 30:19-20, *"with the ability to choose and make decisions"*; and reaffirmed in *Revelation* 3:20, *"man must make his decisions"*. Later, during the Enlightenment, thinkers took a more structured approach, arguing that a truly free will is one that acts according to a self-imposed moral law, or the Categorical Imperative (Kant, 1785). For most of them, freedom wasn't about simply doing whatever you want, but about the ability to reason and choose actions based on universal moral principles, independent of personal desires or consequences. Nowadays, the debate has been further complicated by the rise of scientific determinism. With advances in neuroscience, psychology and artificial intelligence, some argue that our choices are the result of biological processes, genetic predispositions, or environmental factors, leaving little room for a truly free will. However, this perspective has its own critics, who point to the subjective experience of making choices and the moral responsibility we instinctively attribute to human actions. Whether viewed through a religious, philosophical, or scientific lens, the concept of free will remains central to decision-making.

But, if it is clear that decision-making is intrinsic to human nature, why can the process be surprisingly difficult? The journey from strategic reasoning to a final choice is a complex interplay of factors, and several elements can hinder our initiative. One significant obstacle is what might be termed the *"curse of combinatorics"*, akin to the *"curse of dimensionality"*, and also known as *"choice overload"*. When faced with numerous valid options, the decision-making process can become paralyzing, often leading to exhaustion and less satisfactory outcomes. This phenomenon is well-illustrated by the game of chess. As pioneering information theorist, Claude Shannon proposed in his seminal work (Shannon, 1950), that the game of chess can be modeled as a decision tree. Figure 1.1, which shows a hand-drawn analysis by former world champion M.M. Botvinnik (see Botvinnik, 1982), exemplifies this. It displays an exhaustive analysis of a specific chess position, with a complex tree of variations branching out from the initial board setup. This tree consists of 145 nodes representing possible moves, depicting the immense number of choices the human brain must evaluate. While the combinatorics of this near-endgame position may be manageable for a computer, the complexity of earlier stages in the game is often nearly impossible to fully comprehend, not only for the human mind but even for existing computational systems.

Moreover, it is widely recognized that human decision-making faces significant challenges under conditions of intense pressure and uncertainty. In such high-stakes environments, the process is often clouded by emotional stress. Notable examples of this phenomenon occur in imperfect-information games, such as poker, where critical elements like an opponent's hand remain unknown, or in televised game shows, where contestants confront immense pressure in front of a large audience. It is precisely in these scenarios that strategic reasoning proves indispensable. By providing a systematic framework, it clarifies the decision-making process and promotes shifting from intuition-driven choices to more rigorous and structured analysis (see, e.g., the game-theoretic approaches in Schmid et al. (2023) for imperfect games and Perea & Puerto (2007) for the famous *Who Wants to Be a Millionaire?* TV show).

Another significant adversary to clear decision-making can be the well-known *conventional wisdom*. True insight often emerges when you question something people deeply believe in with

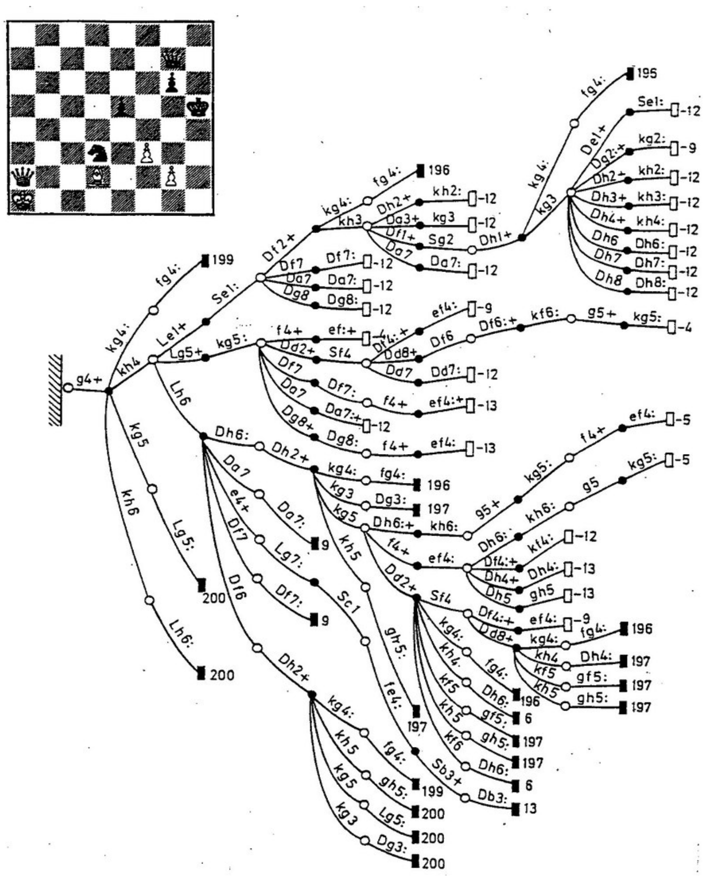


Figure 1.1: Decision tree of a chess position (Botvinnik, 1982).

almost blind faith. The key is to successfully overturn the prevailing narrative. As commented in Levitt & Dubner (2014), the economist John Kenneth Galbraith, who coined the term (Galbraith, 1998), never intended it as a compliment. *"We associate truth with convenience"*, he wrote, *"with what most closely accords with self-interest and personal well-being or promises best to avoid awkward effort or unwelcome dislocation of life"*. While conventional wisdom is not always wrong, identifying where it might be built on sloppy or self-interested thinking is a powerful starting point for asking important questions. Furthermore, beyond the allure of conventional wisdom, several other adverse rationales, in the form of *cognitive biases*, can distort our decision-making. These mental shortcuts often lead to systematic errors in judgment. A brief representation of examples, since there exist hundreds of these biases recognized in the literature, includes the following:

- *Conformity Bias* or *Bandwagon Effect* (Leibenstein, 1950), which is the tendency to do or believe things because many other people do or believe the same. This pressure to conform can stifle innovation and lead individuals to make choices that might not align with themselves.
- *Confirmation Bias* (Wason, 1960), which is the tendency to search for, interpret, favor, and recall information that confirms or supports one's preexisting beliefs or hypotheses. It causes us to selectively notice evidence that proves us right while ignoring anything that proves us wrong.
- *Gambler's Fallacy* (Tversky & Kahneman, 1971), which is the mistaken belief that if a random event has occurred more (or less) frequently than expected in the past, it is less (or more) likely to occur in the future. In other words, it is the idea that the probabilities of an independent event change based on previous outcomes, e.g., *"Tails, Tails, Tails... then the next one is for sure Heads!"*

- *Overconfidence Bias* (Tversky & Kahneman, 1974), which is the tendency for a person's subjective confidence in their judgments to be reliably greater than their objective accuracy. A related phenomenon is the *Dunning-Kruger effect* (Kruger & Dunning, 1999), which is a cognitive bias whereby people with low ability at a task overestimate their ability, failing to recognize their own incompetence.

These adversarial conditions, among many others, demand more than just intuition, they call for a structured approach to navigate complexity. Therefore, there is a clear need to build a framework that guides decision-making in a manner that is both efficient, making proper use of available resources such as time, information, or computational power, and effective, meaning it is accurate in achieving the intended objective.

This ambition aligns with the modest contributions of this dissertation, which aims to deduce a unified and generic framework for decision-making. Rather than presenting a narrow or domain-specific method, this work adopts a panoramic perspective. The goal is not simply to catalogue the vast body of existing work but to weave these disparate threads together into a cohesive tapestry. This synthesis meticulously connects well established theories while introducing novel concepts to bridge the conceptual gaps that have remained unexplored. The resulting framework is therefore a holistic construct, capable of encompassing and structuring a multitude of decision-making paradigms. To fully grasp the scope of this research, it is essential to unravel the two pillars upon which it is built: **optimization** and **order**.

Optimization is the engine of rational choice. It is the formal method of finding the best possible outcome from a set of alternatives by measuring them against specific goals, whether that means, e.g., maximizing profit, minimizing travel time, or balancing competing interests. In the words of George Dantzing, one of the fathers of Operations Research, *"real optimization is the most revolutionary contribution of modern research to decision-making processes."* However, as desirable as it may be, optimizing a decision-making process is generally not straightforward and can lead to misleading outcomes if it is not designed carefully. A classic illustration of this is the well-known *Braess's Paradox* (see Braess, 1968), which demonstrates how an apparently beneficial addition to a system can unexpectedly reduce its overall efficiency.

This paradox is depicted in Figure 1.2, where there are two road systems, one with an additional lane. The red arcs represent high-capacity highways where travel time does not depend on the number of vehicles on the road. In this case, it takes 45 minutes to travel from A to Y, similarly from X to B. The orange arcs represent low-capacity roads that depend on V, the number of vehicles traveling on the road. In this case, it takes $V/100$ minutes to travel from A to X and the same from Y to B. Consider 4000 vehicles traveling from point A to point B in these networks. In Figure 1.2 (a), drivers have two routes with the same travel time, and the traffic naturally splits, resulting in a stable equilibrium where the average travel time for every vehicle is 65 minutes. Paradoxically, when a new high-speed road is added connecting the intermediate points of the network X and Y as shown in Figure 1.2 (b), the system's performance degrades. From an individual driver's perspective, this new shortcut always appears to be part of a faster route. Acting on this local, rational incentive, every driver chooses the new path. However, this creates massive congestion on the segments leading to and from the shortcut. As a result, the new travel time for every single vehicle increases to 81 minutes. Therefore, it seems that adding one more lane does not always solve traffic problems. The paradox highlights a critical principle: local optimization by individual agents does not guarantee global optimization for the whole system.

The search for an optimal solution within the space of all possibilities is, as a rule, a formidable task. As the number of variables and constraints in a problem grows, the number of potential solutions can explode combinatorially, making an exhaustive search computationally intractable. The *curse of combinatorics* arise again, now in practice. Also, this landscape is typically populated with numerous *local optima*, solutions that are superior to their immediate neighbors but inferior

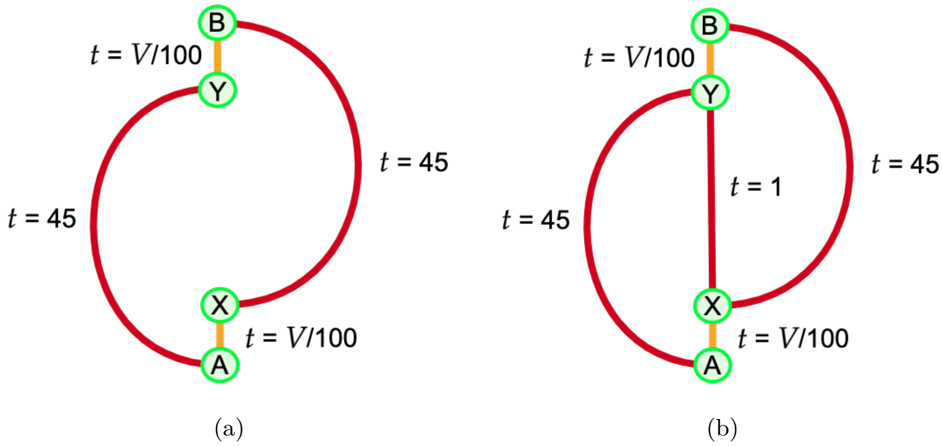


Figure 1.2: Illustation of the *Braess's Paradox*.

to the one true *global optimum*. Distinguishing the highest peak in the entire mountain range is another challenge that optimization faces. To navigate this challenge, optimization techniques are broadly categorized into two families. On the one hand, exact optimization comprises methods that are mathematically guaranteed to find the global optimum. These techniques conduct a rigorous and exhaustive search of the solution space, and while they assure ultimate success, their computational cost is often substantial. On the other hand, heuristic optimization encompasses a range of techniques designed to find a sufficiently good solution within a reasonable timeframe. These methods sacrifice the guarantee of global optimality in favor of speed and efficiency, making them a practical choice when a near-optimal solution is acceptable. This establishes a clear trade-off between the quality of the solution and the computational resources required to find it.

The following dissertation positions itself firmly within the domain of exact methods, while does not discard the heuristic approaches when exact methods fail to yield a feasible solution. Its primary focus is to explore and advance exact optimization techniques that enable us to solve contemporary problems at scale and, critically, to formally define the search for the global optimum in new problem domains. To undertake this endeavor, a powerful and systematic framework for modeling and solving complex optimization problems is required. The tool chosen for this purpose is *mathematical optimization*, also known as *mathematical programming*, a versatile and robust paradigm that provides the foundation for this work. The principles and components of mathematical programming will be explained in greater detail in the next chapter on background concepts.

On the other hand, *order* is the antidote to chaos in decision-making. It represents the crucial act of imposing structure upon an unstructured problem. This can manifest as the ranking of alternatives, prioritization, the sequencing of actions, or the very architecture of the decision model itself. In many real-life decisions, it is not necessary to know the precise value of every alternative, rather, it is enough to compare them. Which location for a facility is better? Which route is better? Which client should be prioritized? Which solution is fairer, more robust, more reasonable? Order transforms an intimidating maze of possibilities into a navigable map, making complex problems tractable by aggregating the available information. Furthermore, establishing an ordered structure within optimization problems yields significant advantages. First, an ordered framework allows for the generalization of numerous criteria found in the literature, enabling a more unified and consistent approach to evaluation. By applying order, one can move beyond simple scalar objectives like minimizing cost or maximizing profit and begin to incorporate more nuanced, qualitative dimensions into the models. Second, and just as importantly, order provides a clear mechanism for assessing the distribution of outcomes among different agents. Through a ranked structure, it becomes possible to precisely identify which agent is most disadvantaged or, conversely, who benefits the most from a given solution, as well as to detect isolated agents or outliers. This capacity is indispensable for addressing questions of fairness, equity, and balance in complex multi-agent systems, and for bringing robustness to the decision-making process

This dissertation is built upon the results of the following works including peer-reviewed published articles (Ljubić et al., 2024; Pozo et al., 2024; Puerto & Torrejon, 2025) and preprints:

[AT1] Blanco,V., Pozo, M.A., Puerto, J., and Torrejon, A. "Linear, nested, and quadratic ordered measures: Computation and incorporation into optimization problems." *arXiv*, 2025. `https://arxiv.org/abs/2503.18097`

[AT2] Ljubić, I., Pozo, M.A., Puerto, J., and Torrejon, A. "Benders decomposition for the discrete ordered median problem." *European Journal of Operational Research*, 2024. Elsevier. `https://doi.org/10.1016/j.ejor.2024.04.030`

[AT3] Pozo, M.A., Puerto, J., and Torrejon, A. "The Ordered Median Tree Location Problem." *Computers & Operations Research*, 169, 106746, 2024. Elsevier. `https://doi.org/10.1016/j.cor.2024.106746`

[AT4] Puerto, J., and Torrejon, A. "A fresh view on Least Quantile of Squares Regression based on new optimization approaches." *Expert Systems with Applications*, 127705, 2025. Elsevier. `https://doi.org/10.1016/j.eswa.2025.127705`

Therefore, the contributions of this dissertation can be summarized as follows. At its core, this work proposes a unified framework for addressing a broad class of objectives in optimization problems, referred to as *Ordered Optimization*. The work is structured to guide the reader from first principles to practical implementation, organized across three fundamental parts: *Theory*, which lays the mathematical groundwork, *Algorithms*, which translate theory into executable procedures through mathematical models, and *Applications*, which demonstrate the framework's utility in solving real-world problems. The specific contributions are detailed across the subsequent chapters of the dissertation, which is organized as follows. Chapter 2 establishes the preliminary concepts, foundational theory, and notation that are used throughout the dissertation. Chapter 3, building on the results of [**AT1**], develops a novel framework that leverages an ordered structure to unify and simplify the computation of diverse performance measures and indices. Chapters 4 and 5, based on the works [**AT2**] and [**AT3**], respectively, detail significant advances in the application of ordered optimization to facility location problems, presenting contributions that advance the state-of-the-art and extend the existing literature. Chapter 6, presents the novel application of the ordered framework to vehicle routing problems, with a specific focus on modeling and generating equitable and balanced solutions. Chapter 7, based on the results [**AT4**], demonstrates the framework's utility in the context of linear regression, showcasing advances in the computation of more robust estimators. Finally, Chapter 8 provides concluding remarks and promising lines for future research.

"Le meilleur des mondes possibles"
Gottfried Leibniz - *Essais de Théodicée* (1710)

2
Background

This chapter aims to introduce the fundamental mathematical tools and theoretical foundations used throughout this dissertation. Standard definitions, key theorems, and essential properties of mathematical optimization are presented, with particular emphasis on linear programming, mixed-integer linear programming, and bilevel optimization. The material covered here provides the necessary background for understanding the methodologies and algorithms developed in subsequent chapters.

2.1 Mathematical optimization

In his work *Essais de Théodicée sur la bonté de Dieu, la liberté de l'homme et l'origine du mal* (*Essays on Theodicy on the Goodness of God, the Freedom of Man and the Origin of Evil*), from 1710, the German mathematician Gottfried Leibniz coined the phrase *"le meilleur des mondes possibles"* (*"the best of all possible worlds"*), referring to the world that has been imposed upon us since our birth. A central concern for Leibniz was reconciling God and human freedom with the determinism inherent in his theory. His solution portrays God as an *optimizer* of all conceivable possibilities: since He is good and omnipotent, God selected this world from among the set of all possible worlds, and thus it must be the best. *Mathematical optimization*, also known as *mathematical programming*, is essentially the same, the search for the best of all possible worlds.

Mathematical optimization is a branch of mathematics concerned with selecting the most favorable element from a feasible set, according to a criterion expressed by an objective function. Formally, an optimization problem seeks to maximize or minimize a real-valued function by choosing input values from a feasible set and evaluating the resulting outcomes. An optimization problem can be defined as follows. Given a function $f : X \to \mathbb{R}$, defined over a set $X \subseteq \mathbb{R}^n$, find a x_0 in X such that $f(x_0) \leq f(x)$ for all $x \in X$ (*minimization*) or such that $f(x_0) \geq f(x)$ for all $x \in X$ (*maximization*).

However, mathematical optimization is a more versatile tool as it allows for constraints that limit the space of solutions. This is known as constrained optimization and forms the basis of the models that will follow in subsequent chapters. Accordingly, the general definition of optimization problem that will be used throughout this dissertation is the following:

Definition 2.1.1 A general optimization problem can be formulated as:

$$\underset{x}{\text{opt}} \quad f(x) \tag{2.1a}$$

$$\text{s.t.} \quad g_i(x) \leq 0, \qquad i \in \{1, \ldots, \ell\}, \tag{2.1b}$$

$$h_j(x) = 0, \qquad j \in \{1, \ldots, p\}, \tag{2.1c}$$

$$x \in X \tag{2.1d}$$

where opt states the sense of the optimization process, minimization or maximization, $f : \mathbb{R}^n \to \mathbb{R}$ is the objective function, $g_i : \mathbb{R}^n \to \mathbb{R}$ are inequality constraint functions, for all $i \in \{1, \ldots, \ell\}$, $h_j : \mathbb{R}^n \to \mathbb{R}$ are equality constraint functions, for all $j \in \{1, \ldots, p\}$, $X \subseteq \mathbb{R}^n$ is the domain constraint set, and $x \in \mathbb{R}^n$ is the vector of decision variables.

Hereafter, unless explicitly stated, it is assumed the minimization sense in problems. The *feasible region* (or feasible set) of the problem (2.1.1) is the set of all points that satisfy its constraints, defined as $\mathcal{F} = \{x \in X \mid g_i(x) \leq 0, i \in \{1, \ldots, \ell\}; h_j(x) = 0, j \in \{1, \ldots, p\}\}$. A point $x^* \in \mathcal{F}$ is called a *global optimal solution* if it yields the best possible value for the objective function over the feasible region. For a minimization problem, this means $f(x^*) \leq f(x)$ for all $x \in \mathcal{F}$. In contrast, a point x^* is a *local optimal solution* if it is optimal within a neighborhood of that point. Formally, there exists an $\epsilon > 0$ such that $f(x^*) \leq f(x)$ for all $x \in \mathcal{F} \cap B(x^*, \epsilon)$, where $B(x^*, \epsilon)$ is the ϵ-ball centered at x^*.

Although the definition of optimization problem in (2.1.1) includes an immense variety of problems, there are specific properties and methods to certify optimality for some of these problems, such as Linear Programming or Convex Programming. This is the case of the well-known Karush-Kuhn-Tucker (KKT) conditions (see Karush, 1939; Kuhn & Tucker, 1951).

Theorem 2.1.1 — KKT conditions. Consider the optimization problem (2.1.1) where the functions $f, g_i, h_j : \mathbb{R}^n \to \mathbb{R}$ are continuously differentiable. Let x^* be a local minimum for this problem, and assume that a constraint qualification (such as LICQ, see Bazaraa et al., 2006) holds at x^*. Then, there exist Lagrange multiplier vectors $\mu^* \in \mathbb{R}^\ell$ and $\lambda^* \in \mathbb{R}^p$ such that the following four conditions hold:

1. **Stationarity:**

$$\nabla f(x^*) + \sum_{i=1}^{\ell} \mu_i^* \nabla g_i(x^*) + \sum_{j=1}^{p} \lambda_j^* \nabla h_j(x^*) = 0$$

2. **Primal Feasibility:**

$$g_i(x^*) \leq 0, \qquad i \in \{1, \ldots, \ell\},$$
$$h_j(x^*) = 0, \qquad j \in \{1, \ldots, p\},$$

3. **Dual Feasibility:**

$$\mu_i^* \geq 0, \qquad i \in \{1, \ldots, \ell\},$$

4. **Complementary Slackness:**

$$\mu_i^* g_i(x^*) = 0, \qquad i \in \{1, \ldots, \ell\}.$$

KKT conditions are not generally sufficient for optimality on their own. They are first-order necessary conditions, meaning that for any local minimum that satisfies certain regularity conditions, it must fulfill the KKT criteria. However, these conditions can also be met at points that are not minima, such as local maxima or saddle points, especially in non-convex problems. The KKT conditions become sufficient for global minimum if the optimization problem is convex.

Theorem 2.1.2 — KKT sufficiency. Consider the optimization problem (2.1.1) where the functions $f, g_i, h_j : \mathbb{R}^n \to \mathbb{R}$ are continuously differentiable. Suppose the problem is a convex optimization problem. Then, if a point $x^* \in \mathbb{R}^n$ and multipliers $\mu^* \in \mathbb{R}^\ell, \lambda^* \in \mathbb{R}^p$ exist such that (x^*, λ^*, μ^*) satisfy the KKT conditions, then x^* is a global minimum of the optimization problem.

2.2 Taxonomy of optimization problems

The mathematical optimization problems considered throughout this dissertation are the following:

1. **Linear Programming (LP):** f, g_i and h_j are all linear functions.
2. **Convex Programming:** f and g_i, and h_j are convex.
3. **Mixed-Integer Linear Programming (MILP):** Some variables are continuous, others are integer.
4. **Non-linear Programming (NLP):** At least one of f, g_i or h_j is non-linear.
5. **Quadratic Programming (QP):** The objective function f is a quadratic from.

Out of the scope of this dissertation are *Conic Programming*, *Second Order Cone Programming* and *Semidefinite Programming* (see Nesterov & Nemirovsky, 1992; Alizadeh, 1995; Ben-Tal & Nemirovski, 2001). The interested reader is referred to Puerto (2024) for a gentle review of these other optimization paradigms.

In depth, a Linear Programming (LP) problem, first introduced by Dantzig (1951), is an optimization problem that can be expressed as:

$$\begin{aligned} (P) : \min_{x} \quad & c^\top x, && (2.2a)\\ \text{s.t.} \quad & Ax = b, && (2.2b)\\ & x \geq 0, && (2.2c) \end{aligned}$$

where $c \in \mathbb{R}^n$ and $b \in \mathbb{R}^p$ are given vectors, $A \in \mathbb{R}^{p\times n}$ is a given matrix and $(A, b) \in \mathbb{R}^{p\times(n+1)}$ is a full rank matrix. For the ease of presentation, inequality constraints are omitted.

The problem (2.2) can equivalently be expressed as finding a vector x within the polyhedron defined by the equality constraints $Ax = b$ that minimizes the linear functional $c^\top x$. Due to the linearity of both the objective function and the constraints, any optimal solution to (2.2), whenever it exists, must be attained at an extreme point (vertex) of the feasible region $\{x \in \mathbb{R}^n : Ax = b\}$. In this context, Dantzig (1951) introduced the celebrated *Simplex algorithm*, a method for solving linear programming problems. The algorithm proceeds by moving from one extreme point of the feasible polyhedron to an adjacent one, strictly improving the value of the objective function at each iteration, until a minimum is reached. Each extreme point of the feasible set corresponds uniquely to a subset of exactly p variables, known as the *basic variables*. The coordinates of an extreme point can thus be obtained by solving the reduced system $Ax = b$ restricted to its basic variables, with all non-basic variables set to zero. Consequently, solving an LP problem amounts to determining the optimal basis, i.e., the subset of basic variables associated with the optimal extreme point.

It should be noted that an LP problem does not necessarily admit a finite optimal solution. Specifically, the problem is *infeasible* if the feasible set $\{x \in \mathbb{R}^n : Ax = b\}$ is empty, and it is *unbounded* if the objective function can be decreased indefinitely within the feasible region.

Furthermore, to every *primal* LP problem of the form (2.2), there exists an associated *dual* LP problem, defined as follows:

$$
\begin{aligned}
\text{(D)}: \max_{u} \quad & b^\top u && \text{(2.3a)}\\
\text{s.t.} \quad & A^\top u \leq c, && \text{(2.3b)}\\
& u \textit{ free} \text{ (unrestricted in sign).} && \text{(2.3c)}
\end{aligned}
$$

According to George Dantzig, the principle of duality in linear optimization was first conjectured by von Neumann (1928), following his own introduction of the linear programming problem. The equivalence and properties of both primal and dual problems are key to the solving algorithms of this family of problems. The importance of duality in LP is summarized in the following Theorem (Wolfe, 1961):

Theorem 2.2.1 Given a primal LP problem in the form of (2.2), and its dual (2.3), they verify:

1. **Weak Duality**: For any feasible solution x of (2.2), and any feasible solution u of (2.3), then:
$$c^t x \geq b^t u.$$
2. **Strong Duality**: The optimal values of (2.2) and (2.3) are the same, i.e., if $\bar{x}$ and $\bar{u}$ are the optimal solution of (2.2) and (2.3), respectively, then:
$$c^t \bar{x} = b^t \bar{u}.$$
3. Exactly one of the following three cases occurs:
   (a) Both (2.2) and (2.3) are infeasible.
   (b) One of (2.2) and (2.3) is unbounded, and the other is infeasible.
   (c) Both (2.2) and (2.3) have optimal solutions, and their optimal values are identical.

On the other hand, Mixed-Integer Linear Programming (MILP) problems constitute a family of more computationally challenging problems belonging to the NP-hard class (see Cook, 1971; Karp, 1972). An MILP problem is an extension of an LP problem where, in addition to continuous variables, some decision variables are restricted to take on only integer values. The general formulation of an MILP problem is:

$$
\begin{aligned}
\min_{x,y} \quad & c^\top x + d^\top y, && \text{(2.4a)}\\
\text{s.t.} \quad & Ax + By = b, && \text{(2.4b)}\\
& x \geq 0, && \text{(2.4c)}\\
& y \in \mathbb{Z}_+^m, && \text{(2.4d)}
\end{aligned}
$$

where $x \in \mathbb{R}^n$ is the vector of continuous decision variables, $y \in \mathbb{Z}_+^m$ is the vector of non-negative integer decision variables, and $d \in \mathbb{R}^m$ and $B \in \mathbb{R}^{p\times m}$ are the input matrix and independent vector corresponding to integer part of the problem.

Note that MILP problems contain as a particular case Integer Programming (IP), where all variables are integer. In special cases, such as when the entire constraint matrix of a pure IP is *totally unimodular*, solving the LP relaxation, i.e., allowing integer variables to be continuous, is guaranteed to yield an integral solution, if one exists (see Hoffman & Kruskal, 2009). However, for a general MILP problem, this does not hold.

Also, the above theorem and duality equivalences between (2.2) and (2.3) are only valid when the variables of (2.2) are defined over a continuous domain. However, even with MILP problems, it is possible to take advantage of the above theorem and the duality properties, so that the performance of the existing algorithms is improved. Methods based on the above duality theorem, such as Lagrangian Relaxation (Everett, 1963) and Benders Decomposition (Benders, 1962), which will be reviewed in the next section, work well and are highly useful for solving MILP problems.

MILP serves as the foundational framework for modeling a significant subset of the optimization problems addressed in this dissertation, in particular, *combinatorial optimization problems*. Combinatorial optimization is a subfield of mathematical optimization concerned with identifying the optimal solution from a finite, discrete set of feasible solutions. Such problems typically involve the selection, arrangement, or partitioning of discrete structures, such as graphs, sets, or sequences, to minimize or maximize an objective function while satisfying a given set of constraints.

The final class of problems considered in this work is Quadratic Programming (QP). Quadratic Programming generalizes LP by allowing the objective function to be a quadratic function of the decision variables, while maintaining linear constraints. Quadratic programming is thus a particular case of Non-Linear Programming (NLP). The earliest known formal introduction of QP is attributed to Karush (1939), who established the foundational optimality conditions, while the first algorithmic approach to solve pure QP problems with linear constraints was proposed by Frank & Wolfe (1956). The general form of a QP problem is given by:

$$\min_{x} \quad x^\top Q x + c^\top x, \tag{2.5a}$$
$$\text{s.t.} \quad Ax = b, \tag{2.5b}$$
$$x \geq 0, \tag{2.5c}$$

where $Q \in \mathbb{R}^{n\times n}$ defines the quadratic form. The problem is convex if and only if Q is positive semi-definite.

Although a broad range of problem classes have been introduced, many other combinations and generalizations exist. For instance, NLP problems may be either convex or non-convex. Similarly, there are Mixed-Integer Quadratic Programming (MIQP) and Mixed-Integer Non-Linear Programming (MINLP) problems. Figure 2.1 depicts the complexity spectrum of these problems, ranging from those solvable in polynomial time to those that are NP-complete or NP-hard.

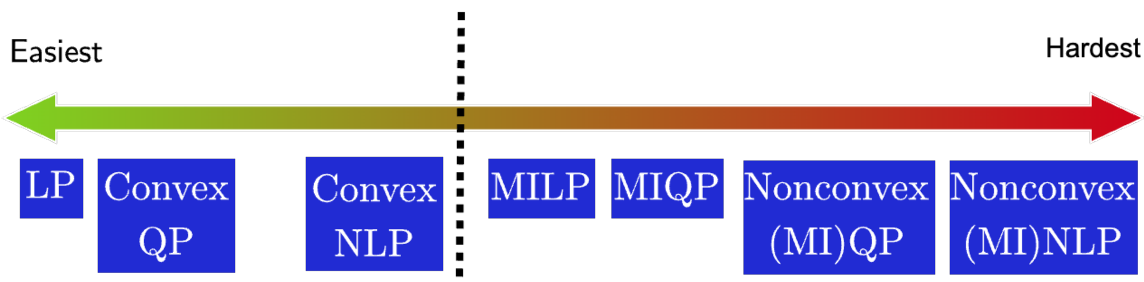


Figure 2.1: Problem types and degree of difficulty.

2.3 Solution methods

As previously discussed, deriving analytical optimality conditions (such as KKT conditions or closed-form solutions) for most of the problems in this dissertation is not feasible, since they primarily belong to the class of MILPs. The combinatorial nature of these problems, where discrete decisions introduce non-convexities and NP-hard complexity, calls for specialized algorithmic strategies capable of obtaining solutions while providing optimality guarantees or bounds on the solution quality. In this section, the principal algorithmic frameworks and decomposition techniques employed in this work to address these challenges are presented.

2.3.1 Branch-and-Bound algorithm

Exact methods, such as Branch-and-Bound (B&B) and its enhancement Branch-and-Cut (B&C), form the backbone of modern mixed-integer optimization. These techniques leverage mathematical programming to systematically explore the solution space of the MILP while certifying optimality through the generation of dual bounds. The B&B method, in particular, is the foundational framework for nearly all commercial MILP solvers, including state-of-the-art software such as Gurobi or CPLEX.

At its core, B&B (Land & Doig, 1960) is a *divide-and-conquer* algorithm. It strategically partitions the original, complex problem into a series of smaller, more manageable, subproblems. These subproblems, which are typically LP relaxations, are solved sequentially to progressively narrow the search space until an optimal integer solution is found and verified. Consider the following general formulation for an MILP:

$$J^* = \min_{x,y} \left\{ c^T x + d^T y \mid (x,y) \in X \right\}, \tag{2.6}$$

where X represents the set of feasible solutions, defined as $X = \left\{(x,y) \in \mathbb{R}^n_+ \times \mathbb{Z}^p_+ : Ax + By \geq b\right\}$. The divide and conquer strategy relies then on decomposing the feasible region X. This principle can be formalized as follows:

Proposition 2.3.1 Let $X = X_1 \cup X_2 \cup \ldots \cup X_K$ be a decomposition of the feasible region X into smaller sets X_k, and let $J^k = \min \left\{ c^T x + d^T y : (x,y) \in X_k \right\}$ for $k = \{1, \ldots, K\}$. Then, the optimal solution to the original problem is given by $J^* = \min_k J^k$.

The B&B algorithm operationalizes this proposition by creating a search tree, where each node represents a subproblem corresponding to a subset of the feasible region. The process involves two main operations: *branching* and *pruning*.

In the branching step, the algorithm begins by solving the initial LP relaxation of the MILP, where the integrality constraints on the y variables are relaxed, allowing them to take on real values.

$$\begin{aligned} J_{\mathrm{R}} = \min_{(x,y)} \; & c^T x + d^T y \\ \text{s.t. } & (x,y) \in P \end{aligned} \tag{2.7}$$

where $P = \left\{(x,y) \in \mathbb{R}^n_+ \times \mathbb{R}^p_+ : Ax + By \geq b\right\}$. Let the solution to this relaxation be $(x^{\mathrm{R}}, y^{\mathrm{R}})$. If all integer variables y^{R} happen to take integer values, i.e., $y^{\mathrm{R}} \in \mathbb{Z}^p_+$, then this solution is also the optimal solution to the original MILP, and the process terminates. However, it is common for one or more integer variables to have a fractional value. When a variable y_j is fractional in the solution, i.e., $y_j^{\mathrm{R}} \notin \mathbb{Z}$, the algorithm branches. Branching eliminates this fractional solution by partitioning the problem into two new, disjoint subproblems. Since any feasible integer solution must satisfy either $y_j \leq \left\lfloor y_j^{\mathrm{R}} \right\rfloor$ or $y_j \geq \left\lceil y_j^{\mathrm{R}} \right\rceil$, two new LP subproblems with updated feasible regions can be created:

$$P^1 := P \cap \left\{ y : y_j \leq \left\lfloor y_j^R \right\rfloor \right\} \tag{2.8}$$

$$P^2 := P \cap \left\{ y : y_j \geq \left\lceil y_j^R \right\rceil \right\} \tag{2.9}$$

This process generates two new nodes in the search tree, descending from the current node. The algorithm then continues by selecting a new node to explore.

The efficiency of B&B comes from its ability to intelligently prune branches of the search tree, avoiding the exhaustive enumeration of all possible integer solutions. This is achieved by maintaining global upper and lower bounds on the optimal objective value, J^*. For a minimization problem:

- **Lower Bound (LB):** The objective value J_{R} provides a valid lower bound on J^*. The global lower bound, J_{LB}, is the minimum of the objective values of all active (unexplored) nodes in the tree.
- **Upper Bound (UB):** Any feasible integer solution $(\bar{x}, \bar{y})$ provides a valid upper bound, $\bar{J} = c^T\bar{x} + d^T\bar{y}$, on J^*. The global upper bound is the value of the best integer solution found so far throughout the search.

As the algorithm progresses, one has $J_{LB} \leq J^* \leq \bar{J}$. A node (and the entire subtree below it) can be pruned based on three criteria:

1. **Pruning by optimality:** The solution to the node's LP relaxation is integer-feasible. This node requires no further branching. Its objective value provides a new potential upper bound. If this value is better than the current global UB, the UB is updated.
2. **Pruning by bound:** The objective value J^i of the LP relaxation at node i is greater than or equal to the current global upper bound $\bar{J}$ (i.e., $J^i \geq \bar{J}$). No better integer solution can be found by exploring this branch further.
3. **Pruning by infeasibility:** The LP relaxation at the current node is infeasible. This branch cannot lead to a feasible solution.

If a node cannot be pruned, it becomes a candidate for further branching. The choice of which node to solve next is determined by a *node selection rule*, such as depth-first search, breadth-first search, or a best-bound search that selects the active node with the most promising (lowest) objective value. The algorithm terminates when there are no more active nodes to explore. At this point, the optimal solution is the incumbent integer solution that provided the final upper bound. A toy example can be found next.

■ **Example 2.1 — Subset sum problem.** Given a set of integers $S = \{2, 3, 5, 7, 9\}$, find all subsets $T \subseteq S$ such that the sum of the items in T is 12.

The problem is solved using a Branch-and-Bound approach, according to the following steps:

1. **Branching**: At each level, decide to include/exclude an element.
2. **Bounding**: Prune branches where:
   - Partial sum > 12.
   - Partial sum + remaining elements < 12.

The corresponding **Branch-and-Bound** search tree is listed below:

```
START [SUM: 0]
|-- [INCLUDE 2] (SUM: 2)
|   |-- [INCLUDE 3] (SUM: 5)
|   |   |-- [INCLUDE 5] (SUM: 10) -> BOUND: 10 + 7 > 12
|   |   |-- [EXCLUDE 5]
|   |       |-- [INCLUDE 7] (SUM: 12) -> SOLUTION {2, 3, 7}
|   |       |-- [EXCLUDE 7] -> [INCLUDE 9] (SUM: 14) -> NO
|   |-- [EXCLUDE 3]
|       |-- [INCLUDE 5] (SUM: 7)
|       |   |-- [INCLUDE 7] (SUM: 14) -> NO
        |   |-- [EXCLUDE 7] -> BOUND: 7 + 9 > 12
|       |-- [EXCLUDE 5]
|           |-- [INCLUDE 7] (SUM: 9) -> BOUND: 9 + 9 > 12
|           |-- [EXCLUDE 7]
|               |-- [INCLUDE 9] (SUM: 11) -> NO
|               |-- [EXCLUDE 9] (SUM: 2)  -> NO
|-- [EXCLUDE 2]
    |-- [INCLUDE 3] (SUM: 3)
    |   |-- [INCLUDE 5] (SUM: 8) -> BOUND: 8 + 7 > 12
    |   |-- [EXCLUDE 5]
    |       |-- [INCLUDE 7] (SUM: 10) -> BOUND: 10 + 9 > 12
    |       |-- [EXCLUDE 7]
    |           |-- [INCLUDE 9] (SUM: 12) -> SOLUTION {3, 9}
    |           |-- [EXCLUDE 9] (SUM: 3)  -> NO
    |-- [EXCLUDE 3]
        |-- [INCLUDE 5] (SUM: 5)
        |   |-- [INCLUDE 7] (SUM: 12) -> SOLUTION {5, 7}
        |   |-- [EXCLUDE 7]
        |       |-- [INCLUDE 9] (SUM: 14) -> NO
        |       |-- [EXCLUDE 9] (SUM: 5)  -> NO
        -- [EXCLUDE 5]
           |-- [INCLUDE 7] (SUM: 7) -> BOUND: 7 + 9 > 12
           |-- [EXCLUDE 7]
               |-- [INCLUDE 9] (SUM: 9) -> NO
               |-- [EXCLUDE 9] (SUM: 0)  -> NO
```

Therefore, the solutions found $T_1 = \{2, 3, 7\}$, $T_2 = \{3, 9\}$ and $T_2 = \{5, 7\}$.

■

A common termination condition in practice is the *optimality gap*, which measures the relative difference between the best upper bound ($\bar{J}$) and the best lower bound (J_{LB}).

$$\mathtt{GAP} = 100\% \times \frac{|\bar{J} - J_{\mathrm{LB}}|}{|\bar{J}|} \tag{2.10}$$

The algorithm is often stopped when this gap falls below a user-defined tolerance (e.g., 0.01%), certifying that the current best integer solution is within a certain percentage of the true optimal value, or when a time stopping condition is imposed. The complete B&B process is summarized in Algorithm 2.3.1. The efficiency of the algorithm is critically dependent on several implementation details, including node selection rules, branching variable selection strategies (e.g., strong branching), and preprocessing techniques that tighten the formulation.

To further enhance the performance of B&B, modern solvers integrate it with the *Cutting Plane method*. Cutting planes are valid inequalities that cut off parts of the LP relaxation's feasible region containing fractional solutions, without removing any feasible integer solutions. Notable families of cuts include Chvátal-Gomory cuts and covering inequalities. The synergy of these two methods gives rise to the *Branch-and-Cut (B&C)* algorithm. In B&C, cutting planes are added at various nodes of the B&B tree to tighten the LP relaxations. This results in better lower bounds, which in turn allows for more effective pruning of the tree. This hybrid approach is generally the most efficient algorithm for solving large-scale MILP problems.

Algorithm 2.3.1 Branch-and-Bound.

Initialize: list of active nodes $\mathcal{L} = \{P_0\}$, upper bound $\bar{J} = \infty$, best solution $(\bar{x}, \bar{y}) = \text{null}$.
while $\mathcal{L} \neq \emptyset$ **do**
  Select a subproblem P_k from $\mathcal{L}$ (according to the node selection rule).
  Solve the LP relaxation of P_k to get the solution (x^k, y^k) and objective value J^k.
  if subproblem P_k is infeasible **or** $J^k \geq \bar{J}$ **then**
    Prune node k (by infeasibility or by bound).
  else if the solution y^k is integer ($y^k \in \mathbb{Z}^p_+$) **then**
    An integer solution has been found. Update the upper bound $\bar{J} \leftarrow J^k$.
    Store the best solution found so far $(\bar{x}, \bar{y}) \leftarrow (x^k, y^k)$.
    Prune node k (by optimality).
  else
    Branch: Choose a y^k_j with a fractional value according to branching rule.
    Create two new subproblems by adding the following constraints:
      $P_{\text{new}1} \leftarrow P_k \cap \{(x, y) \mid y_j \leq \lfloor y^k_j \rfloor\}$
      $P_{\text{new}2} \leftarrow P_k \cap \{(x, y) \mid y_j \geq \lceil y^k_j \rceil\}$
    Update the list of nodes: $\mathcal{L} \leftarrow (\mathcal{L} \setminus \{P_k\}) \cup \{P_{\text{new}1}, P_{\text{new}2}\}$.
  end if
end while
Return the best solution found $(\bar{x}, \bar{y})$ and its objective value $\bar{J}$.

2.3.2 Benders decomposition

Many real-world optimization problems are of such a large scale and complexity that solving them with general-purpose algorithms as monolithic models is computationally intractable. Fortunately, these complex problems often possess special structures that can be exploited to break them down into smaller, more manageable pieces. Decomposition methods are a class of advanced algorithms designed to accomplish this task. They usually decompose a large problem into a master problem and one or more subproblems, which are solved iteratively to find the optimal solution of the original problem.

Several powerful decomposition techniques have been developed, including *Dantzig-Wolfe decomposition*, *column generation* method, and the *Branch-and-Price* algorithm, which integrates column generation into a Branch-and-Bound framework. This section focuses on another fundamental technique particularly well-suited for problems with a separable structure and complicating variables: *Benders decomposition*.

First proposed by Jacques F. Benders in 1962 (Benders, 1962), Benders decomposition is an exact algorithm designed for problems where fixing a subset of variables simplifies the remaining problem significantly. These *complicating variables* are typically integer or strategic decision variables, while the remaining variables are often continuous and part of an easier-to-solve subproblem (e.g., an LP).

The classical Benders decomposition algorithm divides the original mixed-integer problem into two distinct parts that are solved iteratively: a *master problem* and a *subproblem*.

- **The Master Problem (MP)** works with the complicating variables only, along with an additional scalar variable, η, which approximates the cost or consequence of the subproblem. The MP's role is to propose a promising solution for the complicating variables.
- **The Subproblem (SP)** takes the solution of the complicating variables from the MP as fixed parameters. Its role is to solve the remaining, simpler optimization problem and evaluate the consequences of the MP's decision.

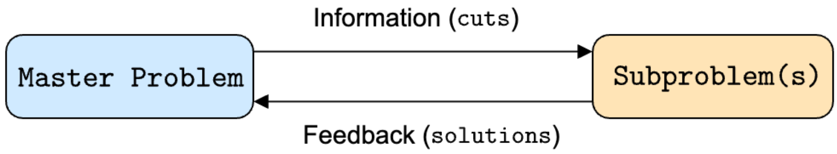


Figure 2.2: Classic Benders decomposition iterative scheme.

The two problems communicate with each other in an iterative loop as depicted in Figure 2.2. The MP passes a candidate solution to the SP. The SP, in turn, generates and sends back a special constraint to the MP, known as a *Benders cut*. At each iteration k, the MP is solved to obtain a candidate solution for the complicating variables, which is denoted by s^k. This solution is passed to the SP. When solving the dual of the SP, two mutually exclusive outcomes can occur:

1. **The SP is feasible.** This means the decision s^k is viable. The optimal solution of the SP, combined with s^k, provides a complete, feasible solution to the original problem and thus a valid upper bound (UB) on the optimal objective. From the dual solution of the SP (an extreme point), an *optimality cut* is generated. This cut refines the MP's approximation of the subproblem cost, η, making the MP's formulation more accurate for the next iteration.
2. **The SP is infeasible.** This implies that the decision s^k proposed by the MP cannot lead to a feasible solution for the overall problem. The dual of the SP will be unbounded, and from its extreme ray, a *feasibility cut* is generated. This cut is added to the MP, rendering the previous solution s^k (and other similarly poor solutions) infeasible for the next iteration.

The objective value of the MP at each iteration provides a valid lower bound (LB) on the true optimal value. The algorithm terminates when the upper bound (from the best feasible SP found) and the lower bound (from the latest MP solution) converge to within a desired tolerance.

This iterative process can be formalized to yield a final reformulation equivalent to the original problem (2.11), namely *Benders reformulation*, as follows. Consider a general optimization problem with the following structure:

$$
\begin{aligned}
\min_{x,y} \quad & c^\top x + d^\top y, && \text{(2.11a)}\\
\text{s.t.} \quad & Ax + By \geq b, && \text{(2.11b)}\\
& x \in \mathbb{R}^n_+, && \text{(2.11c)}\\
& y \in Y. && \text{(2.11d)}
\end{aligned}
$$

Here, the variables are split into two sets: $x \in \mathbb{R}^n_+$ are the continuous variables, and $y \in Y$ are the complicating variables. The set Y often represents discrete or integer choices making the problem difficult to solve directly. The key insight of Benders decomposition is to rephrase the problem by choosing a value for variables y first, and then solving for the optimal x. If variables y are fixed to a specific value $\bar{y} \in Y$, the problem simplifies to finding:

$$\min_{\bar{y}\in Y}\left\{d^T\bar{y}+\min_{x\geq 0}\{c^Tx \mid Ax\geq b-B\bar{y}\}\right\}. \tag{2.12}$$

The inner problem is defined as the *Benders subproblem*, whose optimal value is a function of $\bar{y}$:

$$\Phi(\bar{y})=\min_{x\geq 0}\{c^Tx \mid Ax\geq b-B\bar{y}\}. \tag{2.13}$$

The original problem is now equivalent to the problem $\min_{\bar{y}\in Y}\{d^T\bar{y}+\Phi(\bar{y})\}$. The function $\Phi(\bar{y})$ is difficult to evaluate directly. However, one can use LP duality to find an alternative representation. Assuming the SP is feasible and bounded for a given $\bar{y}$, its dual can be formulated as:

$$\Phi(\bar{y})=\max_{u\geq 0}\{(b-B\bar{y})^Tu \mid A^Tu\leq c\}. \tag{2.14}$$

A crucial observation is that the feasible region of the dual, $\mathcal{F}=\{u\in\mathbb{R}^m \mid A^Tu\leq c, u\geq 0\}$, does *not* depend on $\bar{y}$. The optimal value of this dual LP will always be achieved at one of the extreme points of the polyhedron $\mathcal{F}$. Let the set of extreme points be $\{u^p\}_{p=1}^P$, then the value function can be rewritten as:

$$\Phi(\bar{y})=\max_{p=\{1,\ldots,P\}}\{(b-B\bar{y})^Tu^p\}. \tag{2.15}$$

By substituting this dual representation back into the MP and introducing a new continuous variable η to represent $\Phi(y)$, one arrives to the *Benders reformulation*:

$$\begin{aligned}
\min_{y,\eta}\quad & d^Ty+\eta, && (2.16a)\\
\text{s.t.}\quad & \eta\geq (b-By)^Tu^p, \quad p\in\{1,\ldots,P\}, && (2.16b)\\
& y\in Y, \eta\in\mathbb{R}. && (2.16c)
\end{aligned}$$

The constraints $\eta\geq (b-By)^Tu^p$ are the *optimality cuts*. They ensure that η is a valid upper bound on the subproblem's optimal value. This formulation assumes the SP is always feasible. If for some choice of $\bar{y}$ the SP is infeasible, its dual is unbounded. This unboundedness can be characterized by an extreme ray of the dual feasible region, $\mathcal{F}$. Let $\{u^r\}_{r=1}^R$ be the set of extreme rays of $\mathcal{F}$. For the dual to be bounded (and thus the primal to be feasible), one must ensure that $(b-B\bar{y})^Tu^r\leq 0$ for all extreme rays. This gives rise to *feasibility cuts*. Combining both types of cuts gives the complete Benders reformulation presented in the next result.

Theorem 2.3.2 — Benders reformulation. The problem defined in (2.11) can be equivalently reformulated as the following problem:

$$\begin{aligned}
\min_{y,\eta}\quad & d^Ty+\eta, && && (2.17a)\\
\text{s.t.}\quad & \eta\geq (b-By)^Tu^p, && u^p\in U_p, \quad \text{(Optimality Cuts)} && (2.17b)\\
& 0\geq (b-By)^Tu^r, && u^r\in U_r, \quad \text{(Feasibility Cuts)} && (2.17c)\\
& y\in Y, \eta\in\mathbb{R}, && && (2.17d)
\end{aligned}$$

where U_p is the set of extreme points and U_r is the set of extreme rays of the dual feasible region $\mathcal{F}$.

Next, an illustrative example is provided.

Example 2.2 — Benders decomposition example. Consider the following optimization problem:

$$\begin{aligned} \min \quad & 3x + y_1 + 2y_2, \\ \text{s.t.} \quad & y_1 \geq 4 - 2x, \\ & y_2 \geq 6 - 3x, \\ & y_1 + y_2 \geq 8 - x, \\ & x \in \{0,1,2,3\}, y_1, y_2 \geq 0. \end{aligned}$$

Here, x is the integer (or *complicating*) variable. The problem is decomposed into a master problem (MP) that handles the integer variable x and a subproblem (SP) that handles the continuous variables y for a fixed value of x. The initial MP is formulated by replacing the continuous part of the objective function $(y_1 + 2y_2)$ with a single variable θ:

$$\min_{x \in \{0,1,2,3\}, \theta} \quad 3x + \theta \quad \text{s.t. } \theta \geq \Phi(x).$$

Here, θ represents a lower bound on the optimal value of the SP, denoted as $\Phi(x)$. For a fixed integer value $x = \bar{x}$, the SP becomes:

$$\begin{aligned} Q(\bar{x}) = \min_{y \geq 0} \; & y_1 + 2y_2, \\ \text{s.t.} \quad & y_1 \geq 4 - 2\bar{x}, \\ & y_2 \geq 6 - 3\bar{x}, \\ & y_1 + y_2 \geq 8 - \bar{x}. \end{aligned}$$

The dual of this SP provides the necessary information for the Benders cuts. Introducing dual multipliers $u_1, u_2, u_3 \geq 0$ for the three constraints, the dual subproblem is:

$$\begin{aligned} \Phi(\bar{x}) = \max_{u \geq 0} \; & (4 - 2\bar{x})u_1 + (6 - 3\bar{x})u_2 + (8 - \bar{x})u_3, \\ \text{s.t.} \quad & u_1 + u_3 \leq 1, \\ & u_2 + u_3 \leq 2, \\ & u_1, u_2, u_3 \geq 0. \end{aligned}$$

Note that the feasible region of the dual, $\mathcal{F} = \{u \geq 0 : \; u_1 + u_3 \leq 1, \; u_2 + u_3 \leq 2\}$, is a fixed polyhedron that does not depend on x. By strong duality, since the SP is feasible and bounded, one has $Q(\bar{x}) = \Phi(\bar{x})$. This means only optimality cuts are needed in this case, although feasibility cuts could be easily derived and included. Since the dual's objective function is linear, its maximum value will always be found at one of the extreme points (vertices) of its feasible region $\mathcal{F}$. Each vertex u^p of $\mathcal{F}$ defines a linear function of x:

$$\Phi_p(x) = (4u_1^p + 6u_2^p + 8u_3^p) \; - \; (2u_1^p + 3u_2^p + u_3^p)\, x.$$

The SP's objective function, $\Phi(x)$, is the upper convex hull of these lines: $\Phi(x) = \max_p \Phi_p(x)$. The resulting Benders cuts that are added to the MP are of the form $\theta \geq \Phi_p(x)$.

The relevant extreme points of the dual's feasible region $\mathcal{F}$ are found to be:

$$\begin{aligned} u^{(1)} &= (0,0,0), & u^{(2)} &= (0,0,1), \\ u^{(3)} &= (0,1,1), & u^{(4)} &= (0,2,0), \\ u^{(5)} &= (1,0,0), & u^{(6)} &= (1,2,0). \end{aligned}$$

From each vertex, a cut of the form $\theta \geq \text{constant}_p + (\text{slope}_p) \cdot x$ is generated:

$$\begin{aligned} &\text{cut 1: } \theta \geq 0 + 0x, && \text{cut 4: } \theta \geq 12 - 6x,\\ &\text{cut 2: } \theta \geq 8 - x, && \text{cut 5: } \theta \geq 4 - 2x,\\ &\text{cut 3: } \theta \geq 14 - 4x, && \text{cut 6: } \theta \geq 16 - 8x. \end{aligned}$$

These cuts are iteratively added to the master problem until a solution that satisfies all conditions is found. The function $\Phi(x) = \max_p \Phi_p(x)$ is the explicit representation of the subproblem's value, which is now incorporated into the master problem via these linear constraints. By analytically evaluating (or solving the LP of the subproblem) for x, one obtains:

x	0	1	2	3
$Q(x)$	16	10	6	5
$z(x) = 3x + Q(x)$	16	13	12	14

Therefore, the optimal solution is $x^\star = 2$ with a cost of $z^\star = 12$. Figure 2.3 shows the lines generated by all vertices (optimality cuts), the convex hull $\Phi(x) = \max_p \Phi_p(x)$ (thick line), the points $(x, Q(x))$ for every x, and the marked optimal solution. ■

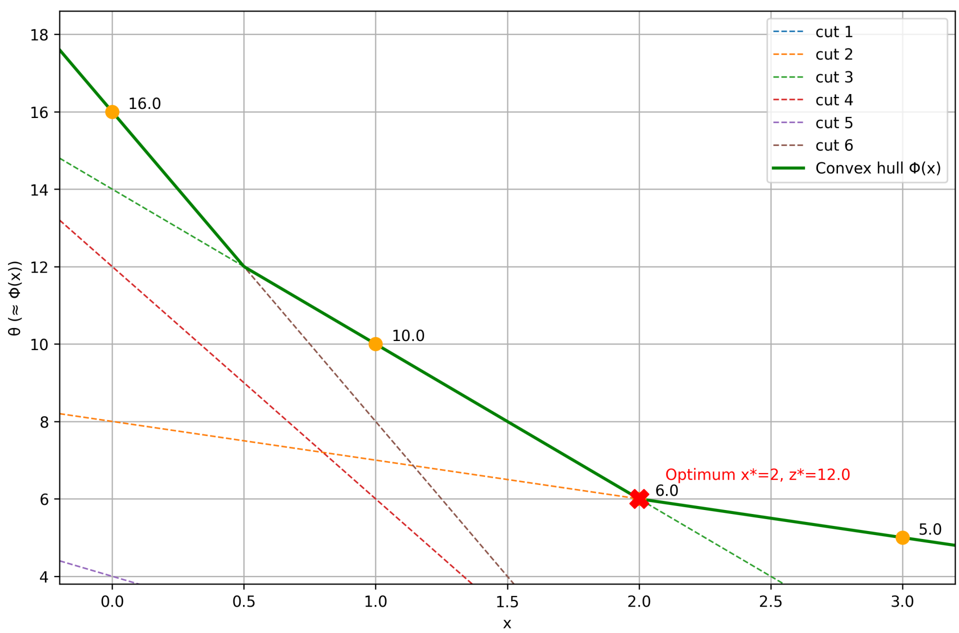


Figure 2.3: Benders decomposition example graphic.

In many applications, such as stochastic programming and facility location, the problem exhibits a special *block-separable structure*. This means the constraint matrix and cost vector are partitioned according to independent subproblems or scenarios, $k \in \{1, \ldots, K\}$. Then, the overall constraint matrix takes the following block-angular form

$$\begin{pmatrix} B & A_1 & 0 & \cdots & 0\\ B & 0 & A_2 & \cdots & 0\\ \vdots & \vdots & \vdots & \ddots & \vdots\\ B & 0 & 0 & \cdots & A_K \end{pmatrix}, \tag{2.18}$$

and the corresponding cost vector is given by

$$\begin{pmatrix} d & c_1 & c_2 & \cdots & c_K \end{pmatrix}. \tag{2.19}$$

If there is a block structure on A, once y is fixed, the SP decouples into K independent smaller problems:

$$\Phi(y) = \sum_{k=1}^{K} \Phi_k(y) \quad \text{where} \quad \Phi_k(y) = \min_{x_k \geq 0} \{c_k^T x_k \mid A_k x_k \geq b_k - B_k y\}. \tag{2.20}$$

This structure allows for two main types of reformulations: *multi-cut reformulation* or *single-cut reformulation*. For the first, a separate variable η_k is introduced for each SP's value, leading to a master problem where cuts are added for each subproblem independently.

$$\begin{aligned}
&\min_{y,\eta} \ d^T y + \sum_{k=1}^{K} \eta_k, && && (2.21a)\\
&\text{s.t. } \ \eta_k \geq (b_k - B_k y)^T u^{p_k}, && u^{p_k} \in U_{p,k}, k \in \{1, \dots, K\}, && (2.21b)\\
&\qquad 0 \geq (b_k - B_k y)^T u^{r_k}, && u^{r_k} \in U_{r,k}, k \in \{1, \dots, K\}, && (2.21c)\\
&\qquad y \in Y, \eta_k \in \mathbb{R}. && && (2.21d)
\end{aligned}$$

Here, $U_{p,k}$ and $U_{r,k}$ are the extreme points and rays of the dual feasible region for subproblem k. For the second, a single variable η is used to represent the sum of the subproblem values. This results in fewer variables but denser cuts.

$$\begin{aligned}
&\min_{y,\eta} \ d^T y + \eta, && && (2.22a)\\
&\text{s.t. } \ \eta \geq \sum_{k=1}^{K} (b_k - B_k y)^T u^{p_k}, && (u^{p_1}, \dots, u^{p_K}) \in U_{p,1} \times \cdots \times U_{p,K}, && (2.22b)\\
&\qquad 0 \geq (b_k - B_k y)^T u^{r_k}, && u^{r_k} \in U_{r,k}, k \in \{1, \dots, K\}, && (2.22c)\\
&\qquad y \in Y, \eta \in \mathbb{R}. && && (2.22d)
\end{aligned}$$

The feasibility cuts remain the same, but the optimality cuts are formed by aggregating the results from each subproblem. The choice between multi-cut and single-cut depends on the specific problem structure and can impact the algorithm's performance.

While the classical algorithm described above provides a standard way to solve the Benders master problem, it is not the only strategy. The core challenge of the Benders reformulation is that it often contains an exponential number of optimality and feasibility cuts making it impractical. In order to be effective, this needs an iterative approach where cuts are added on-the-fly as they are needed. In cases where the dual subproblem's structure is well-understood, it is sometimes possible to analytically derive the expression for the entire family of Benders cuts. Even if the set of all cuts is infinitely large, one can treat them as a "pool" of valid inequalities. This insight allows to re-frame the method entirely, moving from an iterative scheme to a pure *cutting plane algorithm*. Instead of solving a subproblem to generate a single cut, a *separation algorithm* is used. A separation algorithm takes a solution from the current relaxed master problem and attempts to efficiently find a valid Benders cut from the entire family of cuts that this solution violates.

An overall algorithm proceeds as follows:

1. Start with the MP containing no Benders cuts.
2. Within a B&B tree, solve the LP relaxation of the current node's problem.
3. Call the Benders separation algorithm. If violated cuts are found, add them to the LP relaxation and re-solve. Repeat until no more violated cuts can be found.
4. If the resulting solution has fractional values for the complicating variables, branch on one of these variables and repeat the process for the child nodes.

This approach, known as the *Branch-and-Benders-Cut* framework, mitigates the cost of repeatedly solving a full LP subproblem at each iteration by replacing it with a typically more efficient specialized separation algorithm. It is especially effective when the subproblem structure is consistent across all scenarios, enabling the development of a single, highly optimized separation routine. This procedure is often referred to as the modern form of the Benders decomposition.

Over the years, several improvements have been proposed in order to enhance the performance of the Benders decomposition algorithm, (see Geoffrion, 1972; Magnanti & Wong, 1981; Fischetti et al., 2010; Fortz & Poss, 2009; Naoum-Sawaya & Elhedhli, 2013; Conforti & Wolsey, 2019; Brandenberg & Stursberg, 2021; Ramírez-Pico et al., 2023). The reader could find a more detailed description about improvements in Benders decomposition in Rahmaniani et al. (2017). Nevertheless, despite its power, both classical and modern Benders decomposition frameworks assume the subproblem is a Linear Program. However, the core idea can be generalized in what is known as *Logic-Based Benders Decomposition* (see Hooker, 2023). In this extension, the subproblem can be any type of optimization problem (e.g., a constraint programming problem or a non-linear problem). The "cuts" are no longer restricted to linear inequalities derived from dual solutions but can be more general logical propositions or constraints that serve the same purpose: to prune infeasible or suboptimal solutions from the master problem's search space.

2.4 Bilevel optimization

Classic, or single-level, optimization models are invaluable tools for situations where a single entity makes all decisions. However, many real-world applications involve hierarchical decision-making, where multiple actors with potentially conflicting objectives interact. A decision-maker often must act in anticipation of the reaction of another. Since traditional optimization models do not capture these dynamics, the field of *bilevel optimization* emerged to fill this gap.

The origins of bilevel optimization can be traced to the economic concept of Stackelberg competition (von Stackelberg, 1952). It was formally introduced to the field of mathematical optimization by Bracken and McGill, who framed it as a *"mathematical program with optimization problems in the constraints"* (Bracken & McGill, 1973). This description perfectly captures the essence of bilevel programming: a nested structure where one optimization problem is embedded within the constraints of another. This structure makes bilevel optimization a powerful tool for modeling hierarchical processes, but it also renders the problems intrinsically difficult to solve. The challenges were recognized early by Candler & Norton (1977), and formal complexity analysis later confirmed that even linear bilevel problems are strongly NP-hard (see Jeroslow, 1985; Hansen et al., 1992). Despite this inherent difficulty, researchers continue to develop more complex bilevel models to address real-world applications, incorporating challenges such as mixed-integer variables, non-linearities, and uncertainty.

A bilevel optimization problem models a sequential game between two players: a *leader* and a *follower*. The leader acts first, anticipating the follower's optimal response. After observing the leader's decision, the follower solves their own optimization problem to make a decision that is best for them. This hierarchical relationship can be formally defined as follows.

Definition 2.4.1 — Bilevel optimization problem. For an upper-level (leader's) objective function $F(x, y)$ and a lower-level (follower's) objective function $f(x, y)$, a bilevel optimization problem is given by:

$$\min_{x\in X, y\in Y} \quad F(x, y), \tag{2.23a}$$

$$\text{s.t.} \quad G(x, y) \geq 0, \tag{2.23b}$$

$$y \in S(x), \tag{2.23c}$$

where $S(x)$ is the set of optimal solutions to the follower's problem, which is parameterized by the leader's decision x:

$$S(x) = \arg\min_{y' \in Y} \quad \{f(x, y') \mid g(x, y') \geq 0\}. \tag{2.24}$$

Here, $x \in X \subseteq \mathbb{R}^{n_x}$ are the leader's variables and $y \in Y \subseteq \mathbb{R}^{n_y}$ are the follower's variables. The sets X and Y typically define variable domains, including any integrality constraints. For instance, $Y = \mathbb{Z}^{n_y}$ makes the lower-level problem an integer program. The functions $F, f : \mathbb{R}^{n_x} \times \mathbb{R}^{n_y} \to \mathbb{R}$ are the objective functions, and $G : \mathbb{R}^{n_x} \times \mathbb{R}^{n_y} \to \mathbb{R}^m$ and $g : \mathbb{R}^{n_x} \times \mathbb{R}^{n_y} \to \mathbb{R}^\ell$ define the constraint sets. For ease of presentation, the definition is here restricted to inequality constraints only.

A critical issue arises if the follower's optimal solution is not unique, i.e., $|S(x)| > 1$. In this case, the leader's objective value is ambiguous. To resolve this, one typically adopts an *optimistic* or *pessimistic* stance. In the optimistic approach, it is assumed the follower will choose a solution from $S(x)$ that is most favorable to the leader. In the pessimistic approach, the leader anticipates the follower will choose the worst possible solution for her.

2.4.1 Single-level reformulation strategies

The most common strategy for solving bilevel problems is to reformulate them as equivalent single-level optimization problems, which can then be tackled with standard solvers. Below the two primary approaches that will be used in subsequent chapters are discussed.

One way to create a single-level problem is by using the follower's optimal value function. Let $\varphi(x)$ be the optimal value of the follower's problem for a given leader decision x:

$$\varphi(x) := \min_{y' \in Y} \{f(x, y') \mid g(x, y') \geq 0\}. \tag{2.25}$$

The bilevel problem can then be rewritten as the following single-level problem:

$$\begin{aligned}
\min_{x \in X, y \in Y} \quad & F(x, y), && (2.26a)\\
\text{s.t.} \quad & G(x, y) \geq 0, && (2.26b)\\
& g(x, y) \geq 0, && (2.26c)\\
& f(x, y) \leq \varphi(x). && (2.26d)
\end{aligned}$$

This is known as the *value-function reformulation*. While it appears to be a standard single-level problem, the constraint $f(x, y) \leq \varphi(x)$ poses significant challenges. The value function $\varphi(x)$ is generally non-smooth, non-convex, and not available in a closed form, making the problem very difficult to solve directly. Nonetheless, this reformulation is the foundation for many algorithms in mixed-integer bilevel optimization, which often work by relaxing the difficult constraint and iteratively adding approximations of it (see Fischetti et al., 2017).

An alternative and widely used approach is to replace the follower's problem with its optimality conditions. This is only possible when the follower's problem has a compact and sufficient optimality certificate. For the remainder of this section, it is assumed that the follower's problem (2.24) is a convex optimization problem for any fixed x (i.e., f is convex in y and g is concave in y), is continuously differentiable, and satisfies a constraint qualification.

Under these assumptions, the KKT conditions are both necessary and sufficient for optimality. By replacing the lower-level problem with its KKT conditions, the following single-level reformulation is obtained:

Theorem 2.4.1 — KKT reformulation. The optimistic bilevel problem is equivalent to:

$$\min_{x,y,\lambda} \quad F(x,y), \tag{2.27a}$$
$$\text{s.t.} \quad G(x,y) \geq 0, \tag{2.27b}$$
$$g(x,y) \geq 0, \tag{2.27c}$$
$$\nabla_y f(x,y) - \sum_{i=1}^{\ell} \lambda_i \nabla_y g_i(x,y) = 0, \tag{2.27d}$$
$$\lambda_i g_i(x,y) = 0, \qquad i \in \{1,\dots,\ell\}, \tag{2.27e}$$
$$\lambda_i \geq 0, \qquad i \in \{1,\dots,\ell\}, \tag{2.27f}$$
$$x \in X, y \in Y. \tag{2.27g}$$

where λ are the dual multipliers associated with the lower-level constraints.

This KKT reformulation transforms the bilevel problem into a single-level problem with continuous and discrete variables. However, it introduces the non-convex *complementarity constraints* (2.27e), which take the form $\lambda_i g_i(x,y) = 0$. These constraints are the primary source of difficulty in solving the KKT reformulation. In practice, they are often handled using techniques from mixed-integer programming, such as introducing binary variables and "big-M" constants (Fortuny-Amat & McCarl, 1981) or using special ordered sets (SOS1). Both approaches have their own complexities, particularly the challenge of selecting valid and tight big-M values (see Pineda & Morales, 2019; Kleinert et al., 2020). Another approach, which often yields better computational performance, is to replace the KKT conditions with a strong duality reformulation. Instead of the stationarity and complementarity constraints, one enforces primal and dual feasibility for the lower level and equates their objective functions. For a comprehensive survey of modern and classical exact methods and examples for bilevel optimization, the interested reader is referred to Kleinert et al. (2021) and Beck & Schmidt (2021).

As an illustrative exercise, the following reviews these single-level reformulation techniques for a family of bilevel problems that will arise in the subsequent chapters, specifically those with a lower-level linear problem. For clarity of presentation, assume that both the upper-level and lower-level problems are linear. A general LP-LP bilevel problem can be written as:

$$\min_{x,y} \quad c_x^T x + c_y^T y, \tag{2.28a}$$
$$\text{s.t.} \quad Ax + By \geq a, \tag{2.28b}$$
$$y \in \arg\min_{\bar{y}} \{d^T \bar{y} \mid Cx + D\bar{y} \geq b\}. \tag{2.28c}$$

The lower-level problem in (2.28) can be seen as the x-parameterized linear problem:

$$\min_y d^\top y, \quad \text{s.t.} \quad Dy \geq b - Cx, \tag{2.29}$$

whose dual problem is $\max_{\lambda \geq 0}\{(b - Cx)^T \lambda \mid D^T \lambda = d\}$, where $\lambda \in \mathbb{R}^\ell$ are the dual variables.

Assuming it is feasible and bounded, one can replace it with its KKT conditions. This yields a single-level problem with linear constraints and the non-convex complementarity conditions as follows:

$$\min_{x,y,\lambda} \quad c_x^T x + c_y^T y, \tag{2.30a}$$
$$\text{s.t.} \quad Ax + By \geq a, \tag{2.30b}$$

$$
\begin{aligned}
& Cx + Dy \geq b, && && (2.30c)\\
& D^T \lambda = d, \quad \lambda \geq 0, && && (2.30d)\\
& \lambda_i (C_i x + D_i y - b_i) = 0, && i \in \{1, \ldots, \ell\}. && (2.30e)
\end{aligned}
$$

The non-convexity of this formulation arises exclusively from the bilinear terms in the complementarity constraints (2.30e). These constraints can be linearized by introducing auxiliary binary variables. The disjunction $\lambda_i = 0$ or $(C_i x + D_i y - b_i) = 0$ can be modeled using a binary variable z_i and a sufficiently large constant M. Therefore, the KKT reformulation (2.30) is equivalent to the following MILP:

$$
\begin{aligned}
\min_{x,y,\lambda,z} \quad & c_x^T x + c_y^T y, && && (2.31a)\\
\text{s.t.} \quad & Ax + By \geq a, && && (2.31b)\\
& Cx + Dy \geq b, && && (2.31c)\\
& D^T \lambda = d, \quad \lambda \geq 0, && && (2.31d)\\
& \lambda_i \leq M z_i, && i \in \{1, \ldots, \ell\}, && (2.31e)\\
& b_i - C_i x - D_i y \leq M(1 - z_i), && i \in \{1, \ldots, \ell\}, && (2.31f)\\
& z_i \in \{0, 1\}, && i \in \{1, \ldots, \ell\}. && (2.31g)
\end{aligned}
$$

If $z_i = 1$, the first inequality is relaxed while the second forces the primal constraint to be active. If $z_i = 0$, the dual variable λ_i is forced to zero while the primal constraint is relaxed. This reformulation, while linear, requires careful selection of the big-M values to ensure correctness without introducing numerical instability.

Next, the strong duality reformulation is introduced. To this end, equating the primal and dual lower-level objectives gives a single non-linear constraint. This constraint, together with both upper and lower levels feasibility, yields the following single-level program which is equivalent to the LP-LP bilevel problem (2.28):

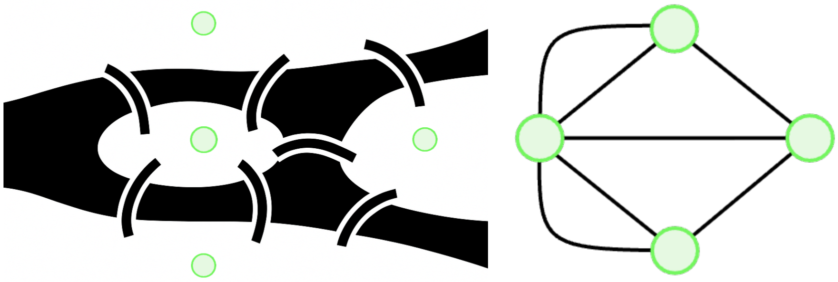
$$
\begin{aligned}
\min_{x,y,\lambda} \quad & c_x^T x + c_y^T y, && (2.32a)\\
\text{s.t.} \quad & Ax + By \geq a, && (2.32b)\\
& Cx + Dy \geq b, && (2.32c)\\
& D^T \lambda = d, \quad \lambda \geq 0, && (2.32d)\\
& d^T y \leq (b - Cx)^T \lambda. && (2.32e)
\end{aligned}
$$

While still non-convex because of (2.32e), this structure can be more amenable to global optimization solvers and often performs better in practice than the KKT reformulation.

2.5 Graphs and mathematical optimization

> *Is it possible to take a walk through the city of Königsberg, crossing each of its seven bridges exactly once?*

This playful yet deep question, posed in the 18th century, marked the beginning of *Graph Theory*. The city of Königsberg (now Kaliningrad, Russia) was set on the Pregel River, which divided it into four landmasses connected by seven bridges. The puzzle of finding a path that crossed each bridge just once captivated its residents, but a solution remained elusive. In 1736, the mathematician Leonhard Euler approached the problem not by exhaustively searching for a route, but by creating an abstraction. He recognized that the physical layout and distances were irrelevant, the crucial element was the pattern of connections. He simplified the problem by representing each landmass as a point (*vertex*) and each bridge as a line (*edge*) connecting two points. Euler proved that such a walk was impossible with a simplified but powerful network of points and lines. Figure 2.4 shows the configuration of (a) the bridges of Königsberg and (b) the representation stated by Euler.



(a) (b)

Figure 2.4: Bridges of Königsberg and Euler's representation.

This act of abstracting a real-world problem into a network of vertices and edges is the core idea of graph representation. Its power lies in distilling complex systems, such as transportation routes, communication networks, or social connections, into a structured format that can be rigorously analyzed. A striking example of the utility and extent of graph-based modeling nowadays is depicted in Figure 2.5, obtained from Gómez-Giménez (2022). This figure depicts historical labor-related mobility flows in Spain between 1986 and 2016, representing the movement patterns of workers traveling from their places of residence to their workplaces.

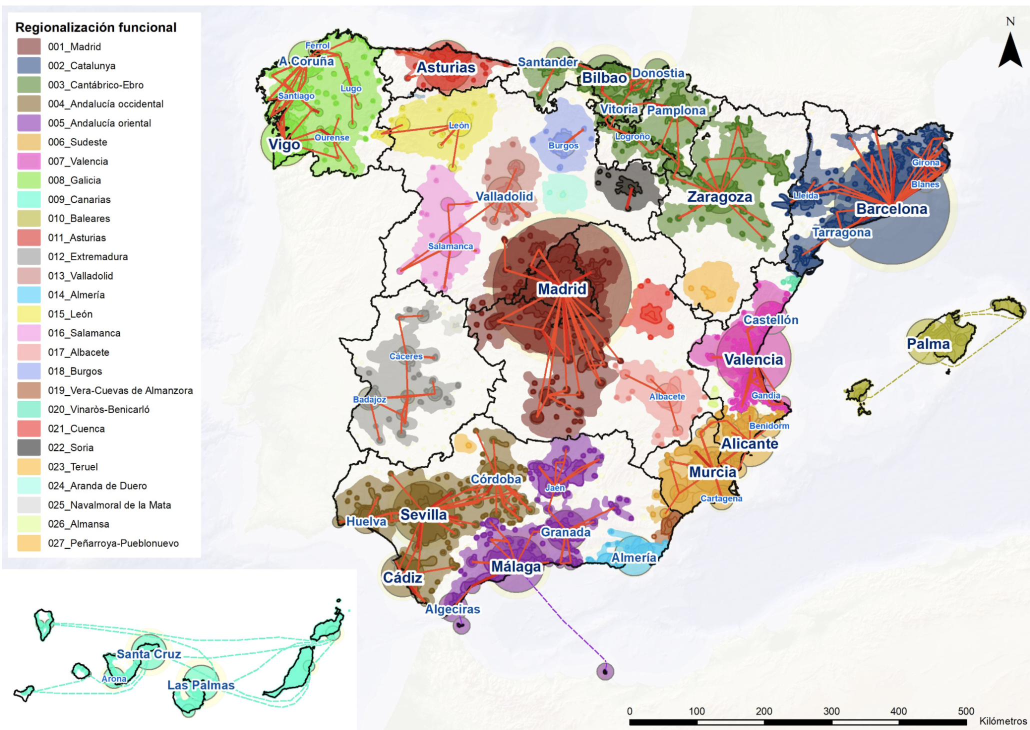


Figure 2.5: Labor-related mobility flows in Spain from 1986 to 2016 (Gómez-Giménez, 2022).

Formally, a graph models relationships between objects. The definitions can vary slightly depending on whether you need to represent loops (edge connecting a vertex to itself) or multiple edges between the same two vertices. A general graph definition is given next.

Definition 2.5.1 A *general graph* G is an ordered triple $G = (V, E, \psi)$, where:

- V is a non-empty finite set of *vertices*.
- E is a finite set of *edges*.
- $\psi : E \to \mathcal{P}_2(V)$ is an *incidence function* that maps each edge to an unordered pair of vertices from V, where $\mathcal{P}_2(V)$ denotes the set of subsets of V with one or two elements.

The vertices $u, v \in V$ in the pair $\psi(e) = \{u, v\}$ are called the *endpoints* of the edge e. The edge e is said to be *incident* on u and v. Two vertices are *adjacent* if they are the endpoints of a common edge. An edge whose endpoints are the same vertex, i.e., $\psi(e) = \{v, v\}$, is called a *loop*. The *order* of a graph is its number of vertices, $|V|$, and its *size* is its number of edges, $|E|$. The *degree* of a vertex v, denoted $\deg(v)$, is the number of edges incident to it.

Connectivity is another fundamental concept in graphs. A *walk* in a graph is a sequence of vertices $v_0, v_1, \ldots, v_k$ such that $\{v_{i-1}, v_i\} \in E$ for all $i \in \{1, \ldots, k\}$. A *path* is a walk in which all vertices are distinct. A graph is *connected* if there is a path between any two of its vertices. If not, it is *disconnected* and consists of several *connected components*, which are usually represented as $G_1 = (V_1, E_1), ..., G_2 = (V_M, E_M)$.

In many real-world models, relationships between objects are directional. For example, in a city with hills, it is not the same going up or down the hills. A *directed graph*, or *digraph*, captures this by giving edges an orientation. A definition of directed graph is provided next.

Definition 2.5.2 — Directed graph. A *directed graph*, also known as *digraph*, is an ordered pair $D = (V, A)$, where:

- V is the set of vertices.
- $A \subseteq \{(u, v) \mid u, v \in V, u \neq v\}$ is a set of *arcs* (or directed edges), where each arc is an ordered pair of distinct vertices.

Figure 2.6 depicts the structural differences between the above defined graphs. For an arc (u, v), u is the *tail* and v is the *head*. The arc is said to be an *outgoing arc* for u and an *incoming arc* for v. Given a digraph D, one can associate with it an *underlying* undirected graph G defined on the same vertex set: for each arc of D, G contains an undirected edge with the same endpoints. Conversely, starting from any undirected graph G one can obtain a digraph, known as *orientation* graph of G, by assigning a direction to each edge. A digraph is said to be *weakly connected* if its underlying undirected graph is connected. It is *strongly connected* if, for every ordered pair of vertices (u, v), there exists a directed path from u to v.

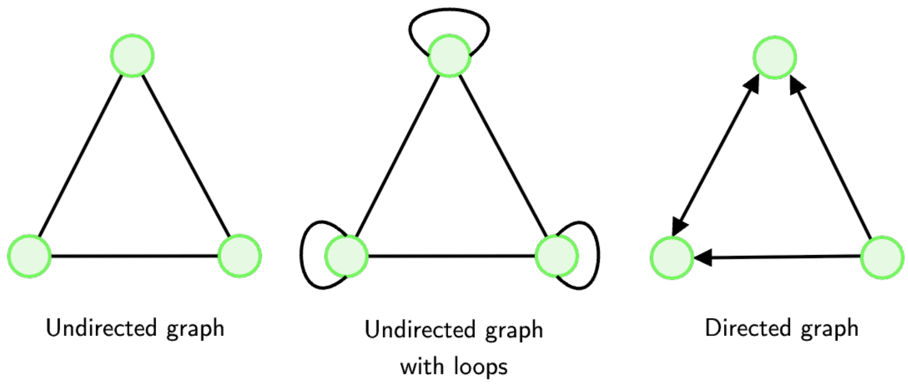


Figure 2.6: Examples of undirected and directed graphs.

Deep in notation, given a subset of nodes $S \subset V$, denote by $E(S)$ and $A(S)$ the subsets of edges of E and arcs of A connecting nodes in S, i.e., $E(S) = \{(u, v) \in E \mid u, v \in S\}$ and $A(S) = \{(u, v) \in A \mid u, v \in S\}$. The *cut-set* $\delta(S)$ associated with $S \subset V$ contains all edges with one node in S and the other node outside S, i.e. $\delta(S) = \{(u, v) \in E \mid (u \in S, v \in V \setminus S) \text{ or } (v \in S, u \in V \setminus S)\}$. Equivalently, in the digraph, the *outgoing cut-set* of S is defined as $\delta^+(S) = \{(u, v) \in A \mid u \in S, v \in V \setminus S\}$ and the *ingoing cut-set* of S is defined as $\delta^-(S) = \{(u, v) \in A \mid v \in S, u \in V \setminus S\}$.

Graphs can be extended by assigning a numerical value, or *weight*, to each edge. A *weighted graph*, often called a *network*, is a triple $G = (V, E, \omega)$, where (V, E) is a graph and $\omega : E \to \mathbb{R}$ is a function that assigns a weight $\omega(e)$ to each edge $e \in E$. These weights can represent costs, distances, capacities, or any other relevant metric.

Among the vast family of graphs, one class frequently referenced throughout this dissertation stands out for its simplicity and fundamental importance: the *tree graph*. They constitute the simplest family of connected graphs. Formally, a *tree*, namely $T = (V_T, E_T)$ is a connected graph that contains no cycles. This simple definition gives rise to a structure that is both rigid and efficient. Let $\mathcal{T}_G$ represent the family of all trees for a given graph G. Trees exhibit several equivalent and defining properties, as stated in the following result.

Proposition 2.5.1 — Tree characterization. For a graph $T = (V_T, E_T)$, the following statements are equivalent:

1. T is a tree (i.e., it is connected and acyclic).
2. There is a *unique path* between any two vertices in T.
3. T is *minimally connected*, i.e., if any single edge is removed it becomes disconnected.
4. T is connected and has exactly $|V_T| - 1$ edges.
5. T is acyclic and has exactly $|V_T| - 1$ edges.

A collection of one or more disjoint trees is called a *forest*. Vertices in a tree with a degree of 1 are known as *leaves*. In many applications, particularly in computer science, one vertex is designated as the *root* node, creating a *rooted tree*. This establishes a natural hierarchy of parent-child relationships, forming the basis for data structures like binary search trees and organizational charts. Because of their unique properties, trees are foundational for modeling hierarchical data, representing decision processes, and designing efficient network backbones.

The theoretical framework of graph theory is fundamentally connected to mathematical optimization, particularly within the domain of combinatorial optimization. Numerous classical optimization problems admit natural formulations on graphs, wherein the objective is to identify a subset of vertices or edges satisfying prescribed constraints while minimizing or maximizing a well-defined objective function. This interplay is especially pronounced in combinatorial optimization, which is concerned with determining an optimal solution from a discrete, often combinatorially large, feasible set. Graph theory provides the formal vocabulary necessary for the rigorous formulation of such problems. Within this paradigm, the decision variables correspond to combinatorial structures such as subsets of vertices, paths, trees, or cycles, whose selection is governed by both problem constraints and the optimization criterion.

Numerous canonical problems in combinatorial optimization are inherently graph-theoretic in nature. The *Shortest Path Problem*, for instance, entails identifying a path of minimal aggregate weight between two designated vertices, serving as a cornerstone in routing, navigation, and network analysis (see Ford & Fulkerson, 1962). The *Minimum Spanning Tree* (MST) problem seeks a subset of edges interconnecting all vertices of a network with the minimum possible total weight (see Edmonds et al., 1967). At the more computationally challenging end of the spectrum, the *Traveling Salesperson Problem* (TSP), a paradigmatic NP-hard problem, requires the determination of the shortest Hamiltonian cycle visiting each vertex exactly once (see Dantzig et al., 1954a). Given their relevance to this dissertation, some of these problems will be examined in detail in the following section.

Among many others, these examples highlight the deep synergies between the two fields. Graph theory offers the perfect language to formulate problems, while optimization theory provides the powerful algorithms to solve them, turning abstract puzzles into solutions for real-world challenges. Contrary to the question posed at the outset of the section, graph theory and mathematical optimization are not disparate shores but firmly connected by a conceptual bridge.

2.6 Examples of optimization problems

In the final section of this chapter on preliminary concepts, several well-known optimization problems that will reappear throughout the following chapters are reviewed. These problems are foundational to two major branches of mathematics, Operations Research and Statistics, highlighting the broad relevance of mathematical optimization across disciplines.

2.6.1 p-Median Problem

> *Which p facilities should be opened to minimize the overall distance from clients to their nearest facility?*

The *p-Median Problem* (pMED), is a classic challenge in operations research and computational geometry that belongs to the broader family of Location Problems. General background on Location Theory, definitions, applications, and solution methods can be found in, e.g., Drezner & Hamacher (2002); Melo et al. (2009); Laporte et al. (2020); Nickel & Puerto (2005); Saldanha-da-Gama (2022); Dönmez et al. (2021); Saldanha-da-Gama & Wang (2024). The objective is to determine p optimal locations, known as *medians*, from a set of potential sites, so that the sum of the distances from each demand point to its closest median is minimized.

Formally, the pMED can be expressed as an integer linear programming problem. Let I be the set of demand points, J the set of candidate locations, and c_{ij} the cost (or distance) between demand point i and location j. Let x_{ij} be binary variables equal to 1 if demand point i is assigned to median j, and 0 otherwise, and let $y_j \in 0, 1$ equal 1 if a median is located at site j, and 0 otherwise. The formulation is:

$$
\begin{aligned}
\min \quad & \sum_{i \in I} \sum_{j \in J} c_{ij} x_{ij}, && && (2.33a)\\
\text{s.t.} \quad & \sum_{j \in J} x_{ij} = 1, && i \in I, && (2.33b)\\
& \sum_{j \in J} y_j = p, && && (2.33c)\\
& x_{ij} \leq y_j, && i \in I, j \in J, && (2.33d)\\
& x_{ij} \in \{0, 1\}, && i \in I, j \in J, && (2.33e)\\
& y_j \in \{0, 1\}, && j \in J. && (2.33f)
\end{aligned}
$$

The objective function (2.33a) minimizes the total assignment cost. Constraint (2.33b) ensures that each demand point is assigned to exactly one facility. Constraint (2.33c) enforces that exactly p facilities are opened. Constraint (2.33d) links assignment and facility-opening decisions: a demand point can only be assigned to a location if a facility is actually opened there.

The pMED is NP-hard (Kariv & Hakimi, 1979). Its origins trace back to the Weber problem (Weber, 1929), but its formal definition stems from the seminal works of Hakimi (1964, 1965) on graphs, and ReVelle & Swain (1970), who proposed an integer programming formulation inspired by Balinski (1965). Recent state-of-the-art exact methods include the Benders decomposition approach of Duran-Mateluna et al. (2023). For an in-depth treatment of polyhedral properties, exact and heuristic methods, and applications, the reader is referred to Laporte et al. (2020). That text also presents related problems such as the p-center and covering problems. Two noteworthy applications of the pMED are:

1. *Clustering.* One of the earliest applications of the pMED is k-median clustering, which partitions data into k clusters by minimizing the sum of distances between each data point and the median of its assigned cluster. Compared to other clustering methods, it is notably robust to outliers and well-suited for discrete or categorical data. The first contribution using mathematical optimization techniques is due to Vinod (1969), with further developments by Mulvey & Crowder (1979); Rao (1971); Klastorin (1985); Hansen & Jaumard (1997); Brusco & Köhn (2008); Benati & García (2014); Puerto et al. (2020).

2. *Obnoxious facility location,* also known as *undesirable* or *controversial* facility location, is a variation of the classical p-median where facilities generate negative externalities (e.g., noise, pollution, danger). The goal is to maximize the distance from demand points rather than minimize it, which is relevant when locating waste treatment plants, chemical or nuclear plants, airports, noisy factories, prisons, or hazardous material storage. Early works include Drezner & Wesolowsky (1983, 1985), with recent bilevel approaches in Labbé et al. (2019). Surveys and key references include Church & Garfinkel (1978); Erkut & Neuman (1989); Tamir (1991); Church & Drezner (2022).

2.6.2 Knapsack Problem

How many elements can be introduced in a knapsack with a given capacity?

The *Knapsack Problem* (KnaP) is another of the most famous problems in combinatorial optimization. It serves as a simple yet powerful model for resource allocation problems where one must choose from a set of non-divisible objects under a fixed budget or resource constraint. It has been studied for over a century, with early works dating back to Mathews (1896).

The most common variant is the *0/1* KnaP, where each item is indivisible, i.e, it must be taken entirely or not at all. Let n be the number of items. For each item $i \in \{1, \dots, n\}$, a value v_i and a weight w_i are given. The knapsack has a maximum weight capacity of W. The decision variable x_i is binary, which is 1 if item i is selected, and 0 otherwise. Therefore, a valid formulation is the following:

$$
\begin{aligned}
\max \quad & \sum_{i=1}^{n} v_i x_i, && && (2.34a)\\
\text{s.t.} \quad & \sum_{i=1}^{n} w_i x_i \leq W, && && (2.34b)\\
& x_i \in \{0, 1\}, && i \in \{1, \dots, n\}, && (2.34c)
\end{aligned}
$$

where the objective function (2.34a) maximizes the total value of the selected items, while constraint (2.34b) enforces the knapsack's weight capacity, and constraint (2.34c) ensures that each item is either fully included or excluded.

Like the pMED, the 0/1 KnaP is NP-hard (Karp, 1972). A related variant, the *fractional* KnaP, allows taking fractions of items. This version is much easier and can be solved optimally in polynomial time with a simple greedy algorithm:

1. Calculate the value-to-weight ratio, v_i/w_i, for each item.
2. Sort the items in descending order of this ratio.
3. Add items to the knapsack in this order until the knapsack is full. If the next item does not fit entirely, take a fraction of it to fill the remaining capacity.

Next, an illustrative example is provided.

Example 2.3 — KnaP example. Consider a knapsack with capacity $W = 50$ and three items:

- **Item 1**: Value=60, Weight=10 (Ratio: 6.0)
- **Item 2**: Value=100, Weight=20 (Ratio: 5.0)
- **Item 3**: Value=120, Weight=30 (Ratio: 4.0)

For the 0/1 KnaP version, the optimal choice is to take **Item 2** and **Item 3**, for a total value of $100 + 120 = 220$ and a total weight of $20 + 30 = 50$. For the fractional KnaP solution, using the greedy algorithm:

1. Take all of **Item 1** (Value=60, Weight=10). Remaining capacity: 40.
2. Take all of **Item 2** (Value=100, Weight=20). Remaining capacity: 20.
3. Take $20/30 = 2/3$ of **Item 3** (Value= $(2/3) \times 120 = 80$, Weight=20).

The total value is $60 + 100 + 80 = 240$. ■

The literature on knapsack problems is extensive, encompassing variants such as multiple knapsack problems and quadratic formulations, see Cacchiani et al. (2022b). A solid understanding of the knapsack problem and its extensions is essential for addressing more complex optimization models and designing efficient algorithms applicable to real-world scenarios. For comprehensive overviews, the reader is referred to the surveys by Martello & Toth (1990) and Kellerer et al. (2004), as well as the most recent review by Cacchiani et al. (2022a).

2.6.3 Traveling Salesman Problem

How can 1.33 billion stars be visited exactly once and return to the first star?

The *Traveling Salesman Problem* (TSP) is one of the most studied problems in Operations Research, particularly in Transportation Science. Given a set of cities and the pairwise distances between them, the objective is to determine the shortest possible tour that visits each city exactly once and returns to the starting city. Formally, let $N = \{1, \ldots, n\}$ denote the set of cities, and let d_{ij} be the distance from city i to city j. A tour is a permutation π of N such that the total length

$$\sum_{i=1}^{n} d_{\pi(i),\pi(i+1)}$$

is minimized, where $\pi(n+1) = \pi(1)$ ensures a return to the starting city.

Several MILP models exist to solve the TSP. One of the earliest is the *Dantzig–Fulkerson–Johnson formulation* (DFJ), introduced in Dantzig et al. (1954a). In this model, a binary variable x_{ij} is 1 if the path from city i to j is in the tour, and 0 otherwise. The formulation is the following:

$$\min \quad \sum_{i \in N} \sum_{\substack{j \in N \\ j \neq i}} d_{ij} x_{ij}, \tag{2.35a}$$

$$\text{s.t.} \quad \sum_{\substack{j \in N \\ j \neq k}} x_{jk} = 1, \qquad k \in N, \tag{2.35b}$$

$$\sum_{\substack{i \in N \\ i \neq k}} x_{ki} = 1, \qquad k \in N, \tag{2.35c}$$

$$\sum_{i \in S} \sum_{\substack{j \in S \\ j \neq i}} x_{ij} \leq |S| - 1, \qquad S \subset N,\ 2 \leq |S| \leq n-1, \tag{2.35d}$$

$$x_{ij} \in \{0,1\}, \qquad i, j \in N,\ i \neq j. \tag{2.35e}$$

In this formulation, the *degree constraints* (2.35b)–(2.35c) ensure that each city is entered and exited exactly once. The core of the model lies in the *subtour elimination constraints* (SEC) (2.35d). These prevent the formation of multiple small tours by stating that for any proper subset of cities S, the number of selected edges within that subset cannot exceed $|S| - 1$. A subtour within S would require $|S|$ edges, thus violating the constraint. While the DFJ formulation is very strong, its primary drawback is the exponential number of SEC, which are typically added dynamically using branch-and-cut algorithms.

To avoid an exponential number of constraints, the *Miller–Tucker–Zemlin formulation* (MTZ), introduced in Miller et al. (1960), considers auxiliary continuous variables $u_i \geq 0$, $i \in N$, to represent the position of each city in the tour. The formulation is then the following:

$$\min \quad \sum_{i \in N} \sum_{\substack{j \in N \\ j \neq i}} d_{ij} x_{ij}, \tag{2.36a}$$

$$\text{s.t.} \quad \sum_{\substack{j \in N \\ j \neq i}} x_{ij} = 1, \qquad i \in N, \tag{2.36b}$$

$$\sum_{\substack{i \in N \\ i \neq j}} x_{ij} = 1, \qquad j \in N, \tag{2.36c}$$

$$u_1 = 0, \tag{2.36d}$$

$$u_j \geq u_i + 1 - n(1 - x_{ij}) \qquad 1 < i \neq j \in N, \tag{2.36e}$$

$$x_{ij} \in \{0,1\}, \qquad i, j \in N,\ i \neq j, \tag{2.36f}$$

$$u_i \in \{0, \ldots, n-1\}, \qquad i \in N. \tag{2.36g}$$

Here, subtours are eliminated by the constraints (2.36e). If a path exists from city i to j ($x_{ij} = 1$), the constraint simplifies to $u_i + 1 \leq u_j$, enforcing a sequential order. A subtour would create a logical contradiction (e.g., $u_i < u_j < \cdots < u_i$), making it impossible. The MTZ model is polynomially sized and thus easier to implement, though its linear relaxation is weaker than that of the DFJ model.

An alternative intuitive approach uses *flow variables*, $f_{ij} \geq 0$, to model the salesman's journey as a commodity flow, with city 1 as the source, as follows:

$$\min \quad \sum_{i \in N} \sum_{\substack{j \in N \\ j \neq i}} d_{ij} x_{ij}, \tag{2.37a}$$

$$\text{s.t.} \quad \sum_{\substack{j \in N \\ j \neq k}} x_{jk} = 1, \qquad k \in N, \tag{2.37b}$$

$$\sum_{\substack{i \in N \\ i \neq k}} x_{ki} = 1, \qquad k \in N, \tag{2.37c}$$

$$\sum_{\substack{j \in N \\ j \neq i}} f_{ij} - \sum_{\substack{j \in N \\ j \neq i}} f_{ji} = 1, \qquad i \in N \setminus \{1\}, \tag{2.37d}$$

$$f_{ij} \leq (n-1)\, x_{ij}, \qquad i, j \in N,\ i \neq j, \tag{2.37e}$$

$$x_{ij} \in \{0, 1\}, \qquad i, j \in N,\ i \neq j, \tag{2.37f}$$
$$f_{ij} \geq 0, \qquad i, j \in N,\ i \neq j. \tag{2.37g}$$

This model ensures that each non-source city "consumes" one unit of flow via the *balance constraints* (2.37d). A subtour disconnected from the source would have no flow, making it impossible to satisfy the balance constraints for its member cities, thus eliminating subtours.

The TSP is NP-hard (Karp, 1972), meaning exact solutions become computationally intractable as the number of cities n grows. The literature on the TSP is vast, covering exact algorithms, cutting-plane methods, and sophisticated heuristics (see Lawler, 1985; Gutin & Punnen, 2006; Applegate et al., 2011, among others). For the state-of-the-art on exact solutions, the *Concorde TSP Solver* (Applegate et al., 2011) outstands, implementing advanced branch-and-cut algorithms capable of solving benchmark instances with thousands of cities to proven optimality. The solver's source code and documentation are publicly available, making it a popular tool for both academic research and practical applications requiring high-quality exact solutions to the TSP.

The TSP stands as one of the most prominent examples demonstrating the power of mathematical optimization, and linear programming in particular. The strength of modern TSP algorithms is spectacularly showcased in the Gaia TSP Challenge (Applegate et al., 2025), where researchers computed a provable optimal tour of 119,614 stars, the largest solved instance at the time of writing this dissertation, as well as a likely optimal tour through approximately 1.33 billion stars mapped by the Gaia space telescope, depicted in Figure 2.7. This accomplishment highlights remarkable progress on a classical and notoriously difficult computational problem. Yet, as the authors themselves note: *"There's always a bigger TSP."*

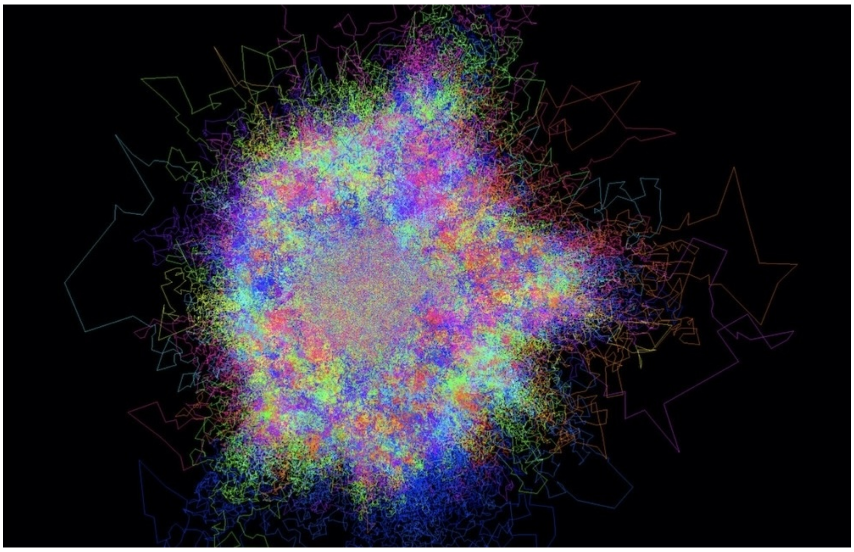


Figure 2.7: 1.33 billion stars nearly optimal TSP tour (Applegate et al., 2025).

2.6.4 Minimum Spanning Tree Problem

How can we connect all the cities investing the least amount possible in roads?

The *Minimum Spanning Tree Problem* (MST) is a key problem in graph theory, with also great relevance in combinatorial optimization. It seeks to find the cheapest way to connect a set of vertices together. Formally, given an undirected weighted graph $G = (V, E)$ with a cost (or weight) c_e associated with each edge $e \in E$, the goal is to find a subset of edges that connects all vertices into a single tree with the minimum possible total cost. A *spanning tree* is a tree that includes all vertices of the original graph. There are well-known optimality conditions that characterize when a spanning tree is a MST, which form the basis for efficient solution methods. Some of the most relevant are stated below.

Proposition 2.6.1 — Cut optimality condition. A spanning tree $T = (V, E_T)$ is a MST if and only if

$$\forall e \in E_T, \quad c_e = \min\{c_f \mid f \in \delta(S_{e,T})\},$$

where $S_{e,T} \subset V$ is the set of vertices in one of the two connected components of the graph $(V, E_T \setminus \{e\})$, and $\delta(S_{e,T})$ is the *fundamental cut* with respect to E_T and e, defined as the set of edges $\{(u, v) \in E \mid u \in S_{e,E_T} \text{ and } v \in V \setminus S_{e,T}\}$.

Proposition 2.6.2 — Path optimality condition. A spanning tree $T = (V, E_T)$ is a MST if and only if

$$\forall f = (u, v) \in E \setminus E_T, \quad c_f \geq \max\{c_e \mid e \in P_T(u, v)\}$$

where $E \setminus E_T$ is the set of edges in graph G that are not in the spanning tree T, and $P_T(u, v)$ is the *fundamental path*, which is the unique simple path connecting vertices u and v in T.

Proposition 2.6.3 — Cycle optimality condition. A spanning tree $T = (V, E_T)$ is a MST if and only if

$$c_f \leq c_e, \forall e \in E \setminus E_T, f \in \mathcal{C}(e, T) : f \neq e.$$

where $\mathcal{C}(e, S)$ is the set of cycles of G that include edge e and edges of the set $S \subseteq E \setminus \{e\}$.

Unlike NP-hard problems, the MST can be solved in polynomial time using greedy algorithms, which select the locally optimal choice at each step. Two classical algorithms, Kruskal (1956) and Prim (1957), are particularly celebrated for their elegance and efficiency in solving the MST. In the following, a gentle description of both algorithms and an illustrative example are provided.

Kruskal's algorithm builds the MST by growing a forest of trees that eventually merge into one. The idea is to always add the next cheapest edge that doesn't create a cycle. The algorithm performs the following steps:

1. Initialize a forest with each vertex as an individual tree.
2. Sort all edges in non-decreasing order of weight.
3. For each edge in sorted order:
   - If its endpoints belong to different trees, add the edge and merge the trees.
   - Otherwise, discard the edge (to avoid cycles).

In contrast, *Prim's algorithm* grows a single tree from an arbitrary starting vertex. It works by continuously adding the cheapest edge that connects a vertex inside the growing tree to a vertex outside of it. The algorithm performs the following steps:

1. Start with a tree containing a single, arbitrarily chosen vertex.
2. Select the minimum-weight edge connecting the tree to any vertex not yet included, and add both the edge and the new vertex to the tree.
3. Repeat until all vertices are included in the tree.

■ **Example 2.4 — MST example.** Consider the graph in Figure 2.8 with 5 vertices and weighted edges $((A,B),2), ((A,C),3), ((B,C),1), ((B,D),4), ((C,D),5), ((C,E),6), ((D,E),7)$.

Using Kruskal's Algorithm one has:

1. **Sort edges by cost**: (B,C,1), (A,B,2), (A,C,3), (B,D,4), (C,D,5), (C,E,6), (D,E,7).
2. **Add edges**:
   - Add **(B,C)** with cost 1. Trees: {A}, {B,C}, {D}, {E}.
   - Add **(A,B)** with cost 2. Trees: {A,B,C}, {D}, {E}.
   - Add (A,C) with cost 3? **No**, it forms a cycle A-B-C-A.
   - Add **(B,D)** with cost 4. Trees: {A,B,C,D}, {E}.
   - Add (C,D) with cost 5? **No**, it forms a cycle B-C-D-B.
   - Add **(C,E)** with cost 6. All vertices are connected. Stop.

The MST includes edges {(B,C), (A,B), (B,D), (C,E)}. Total cost $= 1+2+4+6 = 13$.

On the other hand, using Prim's Algorithm (starting at A) one has:

1. **Start at A**. Tree: {A}.
2. **Step 1**: Cheapest edge from A is **(A,B)** with cost 2. Tree: {A,B}.
3. **Step 2**: Cheapest edge from {A,B} is **(B,C)** with cost 1. Tree: {A,B,C}.
4. **Step 3**: Cheapest edge from {A,B,C} is **(B,D)** with cost 4. Tree: {A,B,C,D}.
5. **Step 4**: Cheapest edge from {A,B,C,D} is **(C,E)** with cost 6. Tree: {A,B,C,D,E}.

The MST includes edges {(A,B), (B,C), (B,D), (C,E)}. Total cost $= 2+1+4+6 = 13$. Both algorithms yield the same optimal result, which is highlighted in blue in Figure 2.8. ■

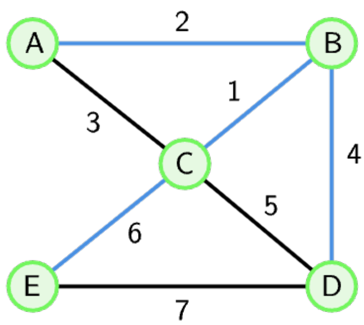


Figure 2.8: Graph example for the MST example and solution (blue).

While greedy algorithms are the practical methods to solve MST, the problem can also be modeled using mathematical programming formulations (see Magnanti & Wolsey, 1995). These formulations are of significant theoretical importance, offering deep insights into the structure of the problem and serving as essential building blocks within more complex optimization models, such as network design or vehicle routing.

At the core of these formulations is a binary decision variable z_e for each edge e in the graph's edge set E. The variable is defined as:

$$z_e = \begin{cases} 1 & \text{if edge (or arc) } e \text{ is selected for the MST} \\ 0 & \text{otherwise} \end{cases} \tag{2.38}$$

The goal is to build valid spanning tree polytope descriptions and select a set of edges with the minimum total cost. Two classic formulations based on fundamental tree properties are the *subtour elimination* and *cutset* formulations.

The subtour elimination formulation approach (Edmonds et al., 1967) is based on the property that a spanning tree on a graph with $n = |V|$ vertices must contain exactly $n-1$ edges and be acyclic. To this end, the SEC used in the DFJ formulation of the TSP are considered. This leads to the following formulation:

$$\min \quad \sum_{e \in E} c_e z_e, \tag{2.39a}$$

$$\text{s.t.} \quad \sum_{e \in E} z_e = n - 1, \tag{2.39b}$$

$$\sum_{e \in E(S)} z_e \leq |S| - 1, \qquad S \subset V, S \neq \emptyset, \tag{2.39c}$$

$$z_e \in \{0, 1\}, \qquad e \in E. \tag{2.39d}$$

Here, the objective function (2.39a) minimizes the cost of the selected edges, the first constraint (2.39b) enforces that exactly $n-1$ edges are chosen, and the second set of constraints (2.39c) are the already introduced SEC. For any proper, non-empty subset of vertices $S \subset V$, these constraints prevent the edges within that subset from forming a cycle. While this formulation is perfectly valid, its major drawback is the exponential number of SEC (on the order of 2^n), which makes it impractical to explicitly list for all but the smallest graphs. In practice, SEC are incorporated into branch-and-bound procedures and added dynamically as needed.

A slight modification of the previous formulation offers an alternative perspective, defining a tree as a minimally connected graph. A graph is connected if, for every possible "cut" that partitions the vertices into two disjoint sets, S and $V \setminus S$, there is at least one edge connecting the two sets. This leads to the cut-set formulation:

$$\min \quad \sum_{e \in E} c_e z_e, \tag{2.40a}$$

$$\text{s.t.} \quad \sum_{e \in E} z_e = n - 1, \tag{2.40b}$$

$$\sum_{e \in \delta(S)} z_e \geq 1, \qquad S \subset V, S \neq \emptyset \tag{2.40c}$$

$$z_e \in \{0, 1\}, \qquad e \in E. \tag{2.40d}$$

Constraints (2.40c) ensures that every possible cut is crossed by at least one selected edge, thereby guaranteeing the resulting graph is connected. Similar to the subtour model, the cutset formulation also suffers from an exponential number of constraints. Although both formulations correctly define the MST polytope, the linear relaxation of the subtour formulation is known to be tighter (stronger) than that of the cutset formulation.

Beyond the two fundamental undirected formulations (subtour elimination and cut-set), several other models exist. Many of these are derived from the more general problem of finding an optimal arborescence (a directed tree rooted at a specific node) in a directed graph. Consequently, it's common and efficient to use arc set notation (A) to describe these MST formulations. These alternative models, such as the *Miller-Tucker-Zemlin* (MTZ), *flow*, and *Kipp Martin* formulations, which are often more compact than the classic subtour model.

Depending on the case, different variables must be considered. The MTZ formulation for the MST uses variables $y_{ij} \in \{0, 1\}$, $(i, j) \in A$, which take value 1 if and only if arc (i, j) belongs to the modeled arborescence, and continuous variables $l_i \geq 0$, $i \in V$, denoting the position that node i occupies in the arborescence with respect to the root node. In the *flow* based formulation introduced by Gavish (1983), continuous flow variables $f_{ij} \geq 0$, $(i, j) \in A$, are defined on the arcs of the directed network indicating the amount of flow through arc (i, j). The formulation proposed by Martin (1991) models as many arborescences as network nodes using three index decision variables $q_{kij} \in \{0,1\}$, $(i,j) \in A$, which indicate whether or not arc (i,j) belongs to the arborescence rooted at k. Table 2.1 summarizes the key properties of these models, including the notation used, their main constraints, whether the formulation is rooted, the number of variables and constraints, and whether the integrality property holds, that is, whether the LP relaxation is guaranteed to yield an integer solution.

Formulation	Notation	Main constraints	root	#vars	#const	int
Subtour Edmonds et al. (1967)	$\mathcal{T}^{sub}$	$\sum_{(i,j)\in E(S)} z_{ij} \leq \lvert S\rvert - 1,\ S \neq \emptyset, S \subset V$		$O(\lvert E\rvert)$	$Exp(n)$	Yes
MTZ Miller et al. (1960)	$\mathcal{T}^{mtz}$	$l_j \geq l_i + 1 - n(1-y_{ij}),\ (i,j) \in A$	r	$O(\lvert E\rvert)$	$O(\lvert E\rvert)$	No
Flow Gavish (1983)	$\mathcal{T}^{flow}$	$\sum_{(i,j)\in\delta^+(i)} f_{ij} - \sum_{(j,i)\in\delta^-(i)} f_{ji} = \begin{cases} n-1, & i = r \\ -1, & i \in V\setminus\{r\} \end{cases}$	r	$O(\lvert E\rvert)$	$O(\lvert E\rvert)$	No
Kipp Martin Martin (1991)	$\mathcal{T}^{km}$	$\sum_{(i,j)\in\delta^+(i)} q_{kij} \leq \begin{cases} 1, & k \in V, i \in V : i \neq k \\ 0, & k \in V, i = k \end{cases}$	$\forall k$	$O(n\lvert E\rvert)$	$O(n\lvert E\rvert)$	Yes

Table 2.1: Main properties of the spanning tree polytope formulations considered.

The Kipp Martin formulation is particularly relevant because it has the integrality property while maintaining a polynomial number of constraints. This characteristic allows the application of duality theory, making it the foundation for some of the solution methods presented in Chapter 5. The formulation models an undirected spanning tree by ensuring that for every possible root node k, the selected edges can form a valid arborescence rooted at k. It can be stated as follows:

$$
\begin{aligned}
\min\quad & \sum_{(u,v)\in E} c_{uv} z_{uv}, && && \text{(2.41a)}\\
\text{s.t.}\quad & \sum_{(u,v)\in E} z_{uv} = |V| - 1, && && \text{(2.41b)}\\
& \sum_{v:(k,v)\in A} q_{kkv} = 0, && k \in V, && \text{(2.41c)}\\
& \sum_{u:(u,v)\in A} q_{kuv} = 1, && k, v \in V, v \neq k, && \text{(2.41d)}\\
& q_{kuv} + q_{kvu} = z_{uv}, && k \in V, (u,v) \in E, && \text{(2.41e)}\\
& z_{uv} \geq 0, && (u,v) \in E, && \text{(2.41f)}\\
& q_{kuv} \geq 0, && k \in V, (u,v) \in A. && \text{(2.41g)}
\end{aligned}
$$

The logic is as follows. Constraints (2.41c)-(2.41d) define the arborescence structure for each potential root k. Constraints (2.41c) state that the root node k can have no outgoing arcs in its own arborescence. Constraints (2.41d) ensure that every other node v has exactly one incoming arc. Together, these two constraints guarantee that for any given k, the arcs selected by the q variables form a valid arborescence rooted at k, thereby preventing cycles. Constraints (2.41e) link the undirected edge selection (z_{uv}) to the directed arc selections (q_{kuv}). It enforces that if an edge (u,v) is chosen ($z_{uv} = 1$), then for every potential root k, exactly one of its corresponding directed arcs, either (u,v) or (v,u), must be chosen for the k-arborescence. By forcing the selected edges to form a valid arborescence from every possible root node, the model implicitly eliminates all possible undirected cycles, resulting in a single, connected spanning tree.

The elegance and efficiency of the algorithms for finding the MST have led to their application in diverse fields. The canonical application of the MST is in the design of least-cost infrastructure networks. In this context, the vertices of the graph represent geographical locations or service points, and the edges represent potential connections between them. This is the case, e.g., for telecommunications and computer networks. When deploying physical infrastructure such as fiber optic cables or local area networks (LANs), the MST can be used for connecting all nodes (e.g., households, servers, or network switches) minimizing material and installation costs. Furthermore, the Spanning Tree Protocol (Perlman, 2000) used in Ethernet networks is a dynamic application of this principle. See Ahuja et al. (1994) or Ramanathan & Steenstrup (1996) for further details.

Beyond physical networks, the MST has another interesting applications, such as similarly to the pMED, clustering analysis. In this application, the observations of the dataset are treated as vertices and the weight of an edge between any two vertices is defined by a metric of dissimilarity or distance between them. A small edge weight implies high similarity, while a large weight indicates low similarity. The resulting MST forms a structural "skeleton" of the dataset, connecting all points based on their local proximity. This structure is fundamental to *single-linkage hierarchical clustering*. Clusters manifest as dense subgraphs connected by edges of low weight. By identifying and removing the longest edges in the MST, one can systematically partition the dataset into clusters. For instance, removing the $k-1$ longest edges yields k distinct clusters. The interested reader is referred to the seminal work of Gower & Ross (1969), or the most recent work of Labbé et al. (2023).

2.6.5 Sample Quantile Problem

> *Which value divides a data vector so that a specified proportion of the observations lie below it and the rest lie above it?*

The last example in this chapter involves shifting from purely combinatorial problems to the realm of statistics, where mathematical optimization provides a powerful tool. Many fundamental problems in statistics, such as, classification, regression, variable selection, and parameter estimation, can be naturally reformulated as optimization tasks. This perspective has gained increasing prominence in both robust statistics and machine learning, where the design of estimators and predictive models often reduces to solving structured mathematical programs.

Optimization plays a central role in robust statistics. A canonical example is *Quantile Regression* (Koenker & Bassett Jr, 1978), which estimates conditional quantiles by minimizing an asymmetric loss function. Advances in optimization, including MILP, have helped in finding solutions to complex statistical challenges, such as ensuring *non-crossing quantiles* (Bondell et al., 2010) or computing robust estimators like the *Least Quantile of Squares* (Rousseeuw, 1984; Bertsimas & Mazumder, 2014). Moreover, compact optimization formulations naturally accommodate side constraints allowing, e.g., for regularization techniques like *LASSO* (Tibshirani, 1996) to be embedded directly into the estimation process. Machine learning models also benefit from an optimization-based perspective. Classic models such as *support vector machines* (Cortes & Vapnik, 1995), *decision trees* (Bertsimas & Dunn, 2017), and *clustering* algorithms (e.g., for applications of the pMED and MST problems in clustering tasks, see Sections 2.6.1 and 2.6.4, respectively), have inspired both continuous and discrete optimization formulations. Furthermore, these formulations allow us to incorporate fairness, interpretability, and statistical guarantees, among other desirable properties, into model design.

Conversely, optimization problems frequently depend on statistical measures (e.g., means, medians, percentiles) as objectives or constraints, particularly in data-driven decision-making contexts. Statistical measures are used to summarize and describe the characteristics of a dataset. They are typically classified into three main categories:

- **Measures of tendency or location:** They indicate where the data are centered relative to a typical value in the dataset. Common examples include centrality measures, such as the mean, median, and mode, or non-central, like quantiles, minimum or maximum.

- **Measures of dispersion or scale:** These quantify the spread or variability of the data points around a reference value. They tell how tightly grouped or spread out the observations are. Key examples are the variance, standard deviation, or the interquartile range.
- **Measures of shape:** These describe the distribution, or shape, of the data. The two primary shape measures are *skewness*, which indicates the asymmetry of the distribution, and *kurtosis*, which describes the "tailedness" or the propensity of the distribution to produce outliers.

Let $\mathbf{x} \in \mathbb{R}^n$ be a given data vector, and let $N = \{1, \ldots, n\}$ be the index set for its components. Many simple location measures can be computed by solving straightforward LP formulations. For instance, the mean, maximum, and minimum of $\mathbf{x}$ can be found using:

$$\text{mean}(\mathbf{x}) = \min \gamma \quad (2.42a)$$
$$\text{s.t.} \sum_{i \in N} x_i \leq n\gamma, \quad (2.42b)$$
$$\gamma \in \mathbb{R}. \quad (2.42c)$$

$$\max(\mathbf{x}) = \min \gamma \quad (2.43a)$$
$$\text{s.t. } x_i \leq \gamma, i \in N, \quad (2.43b)$$
$$\gamma \in \mathbb{R}. \quad (2.43c)$$

$$\min(\mathbf{x}) = \max \gamma \quad (2.44a)$$
$$\text{s.t. } x_i \geq \gamma, i \in N, \quad (2.44b)$$
$$\gamma \in \mathbb{R}. \quad (2.44c)$$

Although these metrics are straightforward to compute directly, casting them as optimization problems is helpful in decision-making problems, e.g., under uncertainty, including stochastic and robust optimization, when addressing objectives such as minimizing the average or maximizing the worst-case scenario over a set of outcomes. Moreover, this perspective creates a conceptual bridge to more sophisticated estimators, such as the quantile, which is reviewed next.

For $0 < \tau < 1$, consider a X real-valued random variable with cumulative distribution function $F_X(x) = P[X \leq x]$. The τ-th quantile of X, namely at times τ-quantile, is given by

$$q_X(\tau) = F_X^{-1}(\tau) = \inf \{x : F_X(x) \geq \tau\}. \quad (2.45)$$

Consider the *pinball* or *check* loss function defined as

$$\rho_\tau(u) = u\left(\tau - \mathbb{I}_{(u<0)}\right) = \begin{cases} (\tau - 1)u, & \text{if } u < 0, \\ \tau u, & \text{if } u \geq 0, \end{cases} \quad (2.46)$$

where $\mathbb{I}(A)$ is an indicator function which takes value 1 if A is true, and 0 otherwise. It is known that the τ-quantile can be found by minimizing the expected loss of $X - u$ with respect to u, i.e.,

$$q_X(\tau) = \arg\min_u \mathbb{E}\left[\rho_\tau(X - u)\right] \quad (2.47)$$
$$= \arg\min_u \int_{-\infty}^{+\infty} \rho_\tau(x - u) dF_X(x) \quad (2.48)$$
$$= \arg\min_u \left\{(\tau - 1)\int_{-\infty}^{u} (x - u) dF_X(x) + \tau \int_{u}^{\infty} (x - u) dF_X(x)\right\}. \quad (2.49)$$

In this case, the stationary condition obtained by equating the derivative to zero, is

$$0 = (\tau - 1)\int_{-\infty}^{q} dF_X(u) + \tau \int_{q}^{\infty} dF_X(u) = (\tau - 1)F_X(q) + \tau(1 - F_X(q)), \quad (2.50)$$

This expression reduces to $F_X(q) = \tau$, from which one can conclude that $q_\tau = F_X^{-1}(\tau)$ is the τ-quantile of F_X. If the solution q_τ is not unique, then one has to take the smallest such solution to obtain the τ-quantile of the random variable X.

Therefore, given a sample of size n, $\mathbf{x} = (x_i)_{i \in N} \in \mathbb{R}^n$, the τ*-th sample quantile* can be obtained by solving the minimization problem

$$\widehat{q}_\tau = \arg\min_{\gamma \in \mathbb{R}} \sum_{i=1}^{n} \rho_\tau(x_i - \gamma) = \arg\min_{\gamma \in \mathbb{R}} \mathcal{L}_\tau(\mathbf{x}, \gamma), \tag{2.51}$$

where $\mathcal{L}_\tau$ is known as the sample τ-quantile function for $\mathbf{x}$,

$$\mathcal{L}_\tau(\mathbf{x}, \gamma) = (\tau - 1) \sum_{x_i < \gamma} (x_i - \gamma) + \tau \sum_{x_i \geq \gamma} (x_i - \gamma). \tag{2.52}$$

In the work Koenker & Bassett Jr (1978), authors stated that the τ-th sample quantile, $\widehat{q}_\tau$, can be defined as the solution to an LP problem. To this end, they introduced a continuous variable $\gamma \in \mathbb{R}$ which represents the quantile estimate, and two sets of non-negative variables, u_i and v_i for each observation $i \in N$, to represent the positive and negative deviations of x_i from γ:

- $u_i = \max(0, x_i - \gamma)$, that represents the positive deviation.
- $v_i = \max(0, \gamma - x_i)$, that represents the negative deviation.

Any deviation $x_i - \gamma$ can be split into its positive and negative parts such that $x_i - \gamma = u_i - v_i$, where $u_i, v_i \geq 0$, and crucially, $u_i v_i = 0$, since a deviation cannot be simultaneously positive and negative. The objective function of the LP will automatically enforce this last condition. The objective is to minimize the weighted sum of these deviations. This leads to the following LP formulation for the τ-th sample quantile:

$$\begin{aligned}
\min_{\gamma \in \mathbb{R}} \ & \tau \sum_{i=1}^{n} u_i + (1 - \tau) \sum_{i=1}^{n} v_i, & & \text{(2.53a)}\\
\text{s.t.} \ & x_i - \gamma \leq u_i, & i \in N, \quad & \text{(2.53b)}\\
& \gamma - x_i \leq v_i, & i \in N, \quad & \text{(2.53c)}\\
& u_i, v_i \geq 0, & i \in N, \quad & \text{(2.53d)}
\end{aligned}$$

or equivalently,

$$\begin{aligned}
\min_{\gamma \in \mathbb{R}} \ & \tau \sum_{i=1}^{n} u_i + (1 - \tau) \sum_{i=1}^{n} v_i, & & \text{(2.54a)}\\
\text{s.t.} \ & u_i - v_i + \gamma = x_i, & i \in N, \quad & \text{(2.54b)}\\
& u_i, v_i \geq 0, & i \in N. \quad & \text{(2.54c)}
\end{aligned}$$

Next, to illustrate the application of the linear program presented in (2.53), a numerical example which computes the sample median of a small dataset is provided.

■ **Example 2.5 — Sample quantile example.** Consider a sample of $n = 5$ observations:

$$x = \{1, 2, 5, 8, 10\}.$$

The objective is to compute the sample median, which corresponds to the $\tau = 0.5$ quantile. By inspection of the ordered data, the median is 5. With $\tau = 0.5$ and our five data points, the LP is formulated as follows. The objective function is:

$$\min \quad 0.5 \sum_{i=1}^{5} u_i + (1 - 0.5) \sum_{i=1}^{5} v_i \quad \equiv \quad \min \quad 0.5 \sum_{i=1}^{5} (u_i + v_i)$$

For each data point x_i, two main constraints are obtained:

$$\begin{aligned} 1-\gamma &\leq u_1, & \gamma - 1 &\leq v_1, \\ 2-\gamma &\leq u_2, & \gamma - 2 &\leq v_2, \\ 5-\gamma &\leq u_3, & \gamma - 5 &\leq v_3, \\ 8-\gamma &\leq u_4, & \gamma - 8 &\leq v_4, \\ 10-\gamma &\leq u_5, & \gamma - 10 &\leq v_5, \end{aligned}$$

and the non-negativity constraints: $u_i, v_i \geq 0$, $i = 1, \ldots, 5$.

The optimal solution to this LP yields the quantile estimate $\gamma^* = 5$. This result can be understood by interpreting the objective function. At the optimum, the auxiliary variables are driven to their lower bounds, such that $u_i = \max(0, x_i - \gamma)$ and $v_i = \max(0, \gamma - x_i)$. These variables represent the positive and negative deviations of each observation from γ, respectively. The values of the auxiliary variables at the optimal solution $\gamma^* = 5$ are given below:

x_i	u_i(positive deviation)	v_i(negative deviation)
1	0	4
2	0	3
5	0	0
8	3	0
10	5	0
Sum:	8	7

The minimum value of the objective function is therefore:

$$0.5 \times (\sum_{i=1}^{5} u_i) + 0.5 \times (\sum_{i=1}^{5} v_i) = 0.5 \times (8) + 0.5 \times (7) = 7.5$$

Any other choice for γ would result in a larger value for the objective function. For example, if $\gamma = 4$ is chosen, the objective value would be 8.0. If $\gamma = 6$ is chosen, the value would also be 8.0. Thus, the LP correctly identifies the sample median as $\gamma = 5$. ■

Transforming the estimation of a sample quantile into a linear program provides a compelling illustration of how an optimization perspective can enrich statistical methodology. What begins as a seemingly simple descriptive statistic, identifying the point below which a given proportion of observations fall, can be reinterpreted within the rigorous structure of mathematical programming. This reformulation not only offers a deeper theoretical understanding of quantiles as solutions to well-defined optimization problems, but also enables the development of more versatile and powerful statistical tools. By embedding quantile estimation into the optimization framework, one gains the flexibility to extend classical concepts to more complex settings, such as high-dimensional data, regression models, or robust decision-making under uncertainty. In this sense, the transition from a basic statistical measure to an optimization-based formulation exemplifies the broader paradigm of leveraging mathematical programming to unify, generalize, and strengthen methods of statistical inference.

Más que nada en el mundo, Juan Salvador Gaviota amaba volar.
Este modo de pensar, descubrió, no es la manera con que uno se hace popular entre los demás pájaros. Hasta sus padres se desilusionaron al ver a Juan pasarse días enteros, solo, haciendo cientos de planeos a baja altura, experimentando...
-¿Por qué, Juan, por qué? - preguntaba su madre.

Richard Bach - *Juan Salvador Gaviota* (1970)

3
Ordered measures

This chapter introduces a unified mathematical optimization framework to define and compute a wide range of measures central to Statistics, Data Science and Operations Research, among other fields. The primary goal is to embed these metrics directly within general optimization models, facilitating their efficient computation and enabling their use in complex decision-making problems. The framework is designed mainly to compute three families of measures: *linear ordered measures*, or L-statistics, *nested ordered measures*, including variance, skewness, and kurtosis, and *quadratic ordered measures*, for which a novel and general operator is defined and studied in detail. The effectiveness of the proposed formulations is validated through extensive computational experiments, which demonstrate significant improvements in efficiency and accuracy over standard implementations in numerical software packages. The resulting framework provides the basic methodology that will be extended in subsequent chapters.

3.1 Introduction

The choice of appropriate measures to summarize a set of values from a given variable is a persistent challenge across diverse fields, including Statistics, Data Science, Operations Research, or Game Theory. Descriptive measures (at times called *statistics*) are scalars that allow one to summarize and describe the essential characteristics of the variable. The choice of measure depends on the particular aspect of the data to be emphasized. Among the large number of proposals, the most popular are: the *measures of tendency*, that can be central (as the mean, median, mode) or non-central (as the quantiles, maximum, minimum), the *measures of dispersion* (as the variance, standard deviation, range), and the *measures of shape* (as the skewness or kurtosis). Furthermore, these families are interconnected. For example, measures of central tendency often serve as the basis for defining other indices, e.g., the variance measures dispersion relative to the mean, while skewness quantifies asymmetry around the mean. Beyond these classical categories, additional families have been proposed, including *robust measures* (e.g., regret, k-sums) and *fairness measures* (e.g., equity or envy-freeness criteria).

A general framework for defining and studying the behavior and properties of a remarkably large number of these measures is to express them as a weighted linear combination of ordered values. Hereafter, a *linear ordered measure*, sometimes simplified to *ordered measure*, is defined as any metric that can be expressed in the form (3.1), given in the next definition.

Definition 3.1.1 — Linear ordered measure. Let $\mathbf{x} \in \mathbb{R}^n$ be a sample and $\lambda \in \mathbb{R}^n$ a vector of weights, a *linear ordered measure* for the sample $\mathbf{x}$ is given by:

$$\ell(\mathbf{x}, \lambda) := \sum_{k=1}^{n} \lambda_k x_{(k)}, \tag{3.1}$$

where $x_{(i)} \in \{x_1, \ldots, x_n\}$ with $x_{(1)} \leq x_{(2)} \leq \ldots \leq x_{(n)}$.

In statistical contexts, expression (3.1) is known as L-statistic or L-estimator. L-statistics were first outlined by Daniell (1920), but practically considered as a useful framework in Robust Statistics by Huber (1964b), since some of them have been proven to be statistically robust under the presence of outliers (whereas others, as the minimum, maximum, mean, and mid-range are known to be sensitive to extreme values). For a complete definition of L-statistics, concerning properties and applications, the interested reader is referred to Huber (1964b) or Serfling (1980). L-estimators achieve robustness by means of the weights associated to the ordered values of a dataset. Their resistance to outliers is mathematically quantified by the *breakdown point*, which is the smallest fraction of contaminated data that can cause the estimator to take on an arbitrarily large or small value. This concept, originally introduced by (Hampel, 1968, 1971) and later formalized for samples by Donoho (1982); Donoho & Huber (1983), is briefly reviewed here due to its relevance for L-estimators, which will be studied later in Chapter 7.

Definition 3.1.2 — Finite-sample breakdown point. The *finite-sample breakdown point* (ϵ_n^*) of a statistic T for a dataset $X = \{x_1, \ldots, x_n\}$ is the minimum proportion of data points that must be replaced by arbitrary large values to make the statistic *"break"* (i.e., move to infinity or negative infinity). Let X' be any dataset obtained by replacing m of the original n data points with arbitrarily values $y_1, \ldots, y_m$. The maximum bias caused by this contamination is:

$$\text{bias}(m; T, X) = \sup_{X'} |T(X') - T(X)| \tag{3.2}$$

The statistic *breaks* if this bias can be made infinite. Then, the *finite-sample breakdown point* is defined as:

$$\epsilon_n^*(\ell, X) = \frac{m^*}{n} \tag{3.3}$$

where m^* is the smallest number of contaminations (m) for which $\text{bias}(m; \ell, X)$ is infinite.

The robustness of an L-statistic $\ell(\mathbf{x}, \lambda)$ depends entirely on its weight structure, specifically the weights assigned to the smallest and largest order statistics ($x_{(1)}$ and $x_{(n)}$). For instance, the sample mean is an L-statistic where every weight $\lambda_i = 1/n$. To break the mean, one only needs to corrupt one data point, e.g., replace x_1 with an arbitrary value y. Then, the new mean is $\bar{X}' = \frac{y + \sum_{i=2}^{n} x_i}{n}$. As $y \to \infty$, it is clear that $\bar{X}' \to \infty$. Since only one data point ($m = 1$) is needed, the breakdown point is $\epsilon_n^* = \frac{1}{n}$. As $n \to \infty$, this approaches 0, which implies no robustness.

The median is another example of L-statistic that places all weight on the central order statistic. For simplicity, let $n = 2k + 1$, so the median is $\ell = x_{(k+1)}$. To make the median arbitrarily large, one must corrupt data points and move them to the right of the current median. If k points are corrupted by replacing them with very large values, these new values will occupy the positions from $x_{(k+2)}$ to $x_{(2k+1)}$. The new median will be the original $x_{(k+1)}$, which is unchanged. However, if one more point ($m = k + 1$) is corrupted, one can force the median itself to become one of the arbitrarily large values. Therefore, $k + 1$ points are needed to break the median, i.e.,

$$\epsilon_n^* = \frac{k+1}{n} = \frac{k+1}{2k+1} \tag{3.4}$$

As $n \to \infty$ then $\epsilon_n^* \to \frac{1}{2}$, which is the highest possible, making the median extremely robust.

Another robust statistic is the α-*trimmed mean*, obtained by discarding the smallest and largest $100\alpha\%$ observations of the sample. This measure can also be expressed by allowing different amounts of trimming at each tail, leading to the more general (r, s)-*trimmed mean*. For simplicity, let $k = \lfloor n\alpha \rfloor$ be the number of observations trimmed from each end, then for the the k-trimmed mean weights are $\lambda_i = 0$ for $i \leq k$ and $i > n-k$, and $\lambda_i = \frac{1}{n-2k}$ for the central observations. The statistic is:

$$\ell(\mathbf{x}, \lambda) = \frac{1}{n-2k} \sum_{i=k+1}^{n-k} x_{(i)} \tag{3.5}$$

To break the trimmed mean, one must corrupt enough data points to affect the observations that receive non-zero weights. By replacing k data points with large values, they become the new $x_{(n-k+1)}, \ldots, x_{(n)}$ which are trimmed away, so the statistic remains unaffected. But if one replaces $k+1$ points, at least one contaminated point will fall within the central part of the ordered data, receiving a non-zero weight. By making this contaminated value arbitrarily large, the trimmed mean can be made arbitrarily large. The number of contaminations needed to break the statistic is $m = k+1$. Therefore, one obtains that:

$$\epsilon_n^* = \frac{k+1}{n} = \frac{\lfloor n\alpha \rfloor + 1}{n} \tag{3.6}$$

As $n \to \infty$, the breakdown point approaches α. This shows a direct trade-off: increasing the trimming proportion α increases the robustness of the statistic. For example, a 25% trimmed mean ($\alpha = 0.25$) has a breakdown point of approximately 25%. Therefore, the key is assigning zero weight to extreme order statistics. By "trimming" the influence of the smallest and largest values, L-statistics can achieve a high breakdown point, providing reliable estimates even when a significant portion of the data is contaminated.

As it may be apparent at this point, the linear ordered structure presented in (3.1) offers great modeling flexibility. By carefully selecting the λ-vector of weights, this single framework can be used to model a vast array of statistical operators, extending from simple location estimators to more complex measures of dispersion. Each unique choice of the λ-weights results in a different operator, demonstrating the power and adaptability of this approach. Tendency measures are the most representative family of such metrics that can be written by means of an ordered approach. Table 3.1 illustrates several of the most popular tendency measures and their specific choices of these λ-weights.

Beyond tendency, the framework naturally extends to measures of dispersion, which quantify the spread, variability, or diversity of data points. Conceptually, these measures are zero when all data points are identical and their value increases as the data become more scattered. Dispersion is usually assessed using as basis certain tendency measures and very often are sustained on measuring the deviation from this reference value. For example, the variance and the mean absolute deviation (MeanAD) quantifies dispersion from the mean. Robust measures of dispersion, additionally, aim at being resistant to the influence of outliers. Common robust dispersion measures include the interquartile range (IQR) and the median absolute deviation (MedianAD). This ability to quantify deviation is particularly powerful for modeling abstract concepts like fairness or equity, where one might measure the deviations from an ideal central point (Hoover, 1941) or among all pairs of values (Gini, 1921). For a comprehensive overview of fairness and equity measures, the interested reader is referred to Marsh & Schilling (1994); Sen & Foster (1997); Young (1995); Bertsimas et al. (2011), or to Chapter 6 of this dissertation.

Statistic	Expression	λ
Arithmetic mean	$\frac{1}{n}\sum_{i=1}^{n} x_i$	$\frac{1}{n}(1, \ldots, 1)$
Weighted mean	$\frac{1}{\sum_{i=1}^{n} w_i}\sum_{i=1}^{n} w_i x_i, \quad w_i \geq 0, \sum_{i=1}^{n} w_i = 1$	$(\frac{w_1}{\sum_{i=1}^{n} w_i}, \ldots, \frac{w_n}{\sum_{i=1}^{n} w_i})$
Maximum	$\max x_i$	$(0, \ldots, 0, 1)$
Minimum	$\min x_i$	$(1, 0, \ldots, 0, 0)$
Median	$\text{median}(x) = \begin{cases} x_{(m+1)}, & n = 2m+1 \\ \frac{x_{(m)}+x_{(m+1)}}{2}, & n = 2m \end{cases}$	$(0, \ldots, 0, \underbrace{1}_{(m+1)-th}, 0, \ldots, 0)$ $(0, \ldots, 0, \underbrace{\frac{1}{2}}_{m-th}, \underbrace{\frac{1}{2}}_{(m+1)-th}, 0, \ldots, 0)$
q-th quantile (or τ-quantile)	$x_{(q)}, \, q = \lfloor \tau n \rfloor$	$(0, \ldots, 0, \underbrace{1}_{q-th}, 0, \ldots, 0)$
Midrange	$\frac{1}{2}(\max x_i + \min x_i)$	$\frac{1}{2}(1, 0, \ldots, 0, 1)$
Tukey's trimean	$\frac{1}{4}(x_{(\lfloor n/4 \rfloor)} + 2x_{(\lfloor n/2 \rfloor)} + x_{(\lfloor 3n/4 \rfloor)})$	$\frac{1}{4}(0, \ldots, 0, \underbrace{1}_{\lfloor n/4 \rfloor - th}, 0, \ldots, 0, \underbrace{2}_{\lfloor n/2 \rfloor - th}, 0, \ldots, 0, \underbrace{1}_{\lfloor 3n/4 \rfloor - th}, 0, \ldots, 0)$
General midrange (midhinge, etc.)	$\frac{1}{2}(x_{(r)} + x_{(s)}), \, (r \leq s)$	$\frac{1}{2}(0, \ldots, 0, \underbrace{1}_{r-th}, 0, \ldots, 0, \underbrace{1}_{s-th} \ldots, 0)$
(r, s)-trimmed mean	$\frac{1}{n-r-s}\sum_{i=r+1}^{n-s} x_{(i)}$	$\frac{1}{n-r-s}(\underbrace{0, \ldots, 0}_{r}, \underbrace{1, \ldots, 1}_{n-r-s}, \underbrace{0, \ldots, 0}_{s})$
(r, s)-mid mean	$\frac{1}{r}\sum_{i=1}^{r} x_{(i)} + \frac{1}{s}\sum_{i=n-s}^{n} x_{(i)}$	$(\underbrace{\frac{1}{r}, \ldots, \frac{1}{r}}_{r}, \underbrace{0, \ldots, 0}_{n-r-s}, \underbrace{\frac{1}{s}, \ldots, \frac{1}{s}}_{s})$
(r, s)-winsorized mean	$\frac{1}{n}\left(r x_{(r+1)} + \sum_{i=r+1}^{n-s} x_{(i)} + s x_{(n-s)}\right)$	$\frac{1}{n}(0, \ldots, 0, \underbrace{r+1}_{(r+1)-th}, 1, \ldots, 1, \underbrace{s+1}_{(n-s)-th}, 0, \ldots, 0)$

Table 3.1: Location or tendency measures and λ-vectors.

Many prominent dispersion indices can be formulated as linear ordered measures by choosing the appropriate λ-vector. Examples include the IQR, Gini's mean differences, and other efficient scale estimators as proposed in Welsh & Morrison (1990). Table 3.2 lists some of the most popular dispersion measures, along with their corresponding λ weights when they can be cast as linear ordered measures. While the expressions for most λ-vectors in the table are straightforward, detailed derivations for the Gini coefficient can be found in Downton (1966) or Mesa et al. (2003), and for the mean absolute deviation from the q-th quantile ($\mathsf{MeanAD}_{(q)}$) in Harmati et al. (2024).

While linear ordered measures are defined by elegant, explicit expressions, their practical computation is inherently a two-step procedure: first sort the data, then compute the measure. This reliance on order statistics introduces a significant computational prerequisite for which, particularly when dealing with large-scale datasets, efficiency is of utmost importance. Obtaining the order statistics from a raw sample is functionally equivalent to solving one of the most fundamental problems in Computer Science: the *Sorting Problem*. Sorting involves rearranging a sequence of elements into a specified order, in our case non-decreasing. The problem can be formally defined as follows:

Statistic	Expression	λ
Range	$\max x_i - \min x_i$	$(-1, 0, ..., 0, 1)$
General interquantile range (e.g., interquartile or interdecile ranges)	$x_{(r)} - x_{(s)}, (r < s)$	$(0, ..., 0, \underbrace{-1}_{r-th}, 0, ..., 0, \underbrace{1}_{s-th} ..., 0)$
Variance (as squared deviation from the mean)	$\frac{1}{n}\sum_{i=1}^{n}\left(x_i - \frac{1}{n}\sum_{j=1}^{n} x_j\right)^2$	
Mean squared deviation from a location measure $m(x)$ (e.g., from the q-th quantile)	$\frac{1}{n}\sum_{i=1}^{n}\left(x_i - m(x)\right)^2$	
Gini differences or mean absolute difference	$\frac{1}{n^2}\sum_{i,j=1}^{n} \lvert x_i - x_j\rvert$	$\frac{2}{n^2}((2i - n - 1))_{i \in \{1,\dots,n\}}$
Mean absolute deviation from mean	$\frac{1}{n}\sum_{i=1}^{n}\left\lvert x_i - \frac{1}{n}\sum_{j=1}^{n} x_j\right\rvert$	
Mean absolute deviation from a location measure $m(x)$	$\frac{1}{n}\sum_{i=1}^{n}\left\lvert x_i - m(x)\right\rvert$	e.g., from the q-th quantile $m(x) = x_{(q)}$ ($\mathsf{MeanAD}_{(q)}$): $\frac{1}{n}(-1, ..., -1, \underbrace{2q - n - 1}_{q-th}, 1, ..., 1)$
Median absolute deviation from a location measure $m(x)$ (e.g., from the mean)	$median(\left\lvert x_i - m(x)\right\rvert)$	

Table 3.2: Dispersion or scale measures and λ-vectors.

Definition 3.1.3 — Sorting Problem. Given a sequence of n elements $a_1, a_2, \dots, a_n$, find a permutation $\pi : \{1, 2, \dots, n\} \to \{1, 2, \dots, n\}$ such that:

$$a_{\pi(1)} \leq a_{\pi(2)} \leq \cdots \leq a_{\pi(n)}.$$

The essence of sorting is that it establishes a relative structure allowing for meaningful comparisons (who is the shortest, who is the tallest, who is most harmed, who benefits the most). While the problem is simple to state, the design and analysis of efficient sorting algorithms reveal deep connections between computational complexity, algorithm correctness, and practical performance trade-offs. A rich body of literature exists on algorithms designed to solve the sorting problem. These algorithms can be broadly categorized by their design strategy and computational complexity. Elementary algorithms like *Bubble Sort*, *Insertion Sort*, and *Selection Sort* are intuitive but exhibit quadratic $\mathcal{O}\left(n^2\right)$ time complexity in the worst case. Advanced techniques, such as *Merge Sort* and *Quick Sort*, leverage divide-and-conquer to achieve $\mathcal{O}(n \log n)$ average-case performance. Merge Sort guarantees optimal worst-case behavior, while Quick Sort, despite its $\mathcal{O}\left(n^2\right)$ worst-case scenario, is often preferred in practice. Moreover, when data satisfies additional constraints (e.g., integers within a bounded range), non-comparison sorts like *Counting Sort*, *Radix Sort*, and *Bucket Sort* can achieve linear $\mathcal{O}(n)$ time, breaking the $\mathcal{O}(n \log n)$ barrier. These algorithms exploit structural properties of the input, demonstrating how domain-specific knowledge can lead to dramatic efficiency gains. A summary of the complexity of these algorithms is presented in Table 3.3. Further details can be found, e.g., in the book by Cormen et al. (2022).

`Algorithm`	`Best case`	`Avg case`	`Worst case`
Bubble Sort	$\mathcal{O}(n)$	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$
Insertion Sort	$\mathcal{O}(n)$	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$
Selection Sort	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$
Merge Sort	$\mathcal{O}(n \log n)$	$\mathcal{O}(n \log n)$	$\mathcal{O}(n \log n)$
Quick Sort	$\mathcal{O}(n \log n)$	$\mathcal{O}(n \log n)$	$\mathcal{O}(n^2)$
Counting Sort	$\mathcal{O}(n+k)$	$\mathcal{O}(n+k)$	$\mathcal{O}(n+k)$
Radix Sort	$\mathcal{O}(nk)$	$\mathcal{O}(nk)$	$\mathcal{O}(nk)$
Bucket Sort	$\mathcal{O}(n+k)$	$\mathcal{O}(n+k)$	$\mathcal{O}(n^2)$

Table 3.3: Comparison of sorting algorithms complexity.

While the aforementioned algorithms provide efficient procedural solutions for sorting, their iterative nature presents a fundamental challenge in decision-making modeling, particularly in the field of mathematical optimization. Optimization models require a mathematical formulation of a problem, as a system of equations and inequalities. Standard sorting algorithms, being defined by sequential operations, cannot be directly embedded into such static frameworks. This is a critical barrier. Although these algorithms are fast for a one-time computation, they are unsuitable when the ordering of values, or variables, must be determined within an optimization process. For example, if an objective function or a set of constraints depends on the sorted order of decision variables, a pre-processing sorting step is not possible, as the values of these variables are not known a priori. Therefore, there is a clear need in optimization for the development of mathematical formulations that integrate the logic of both sorting and statistical computation into a single, unified model. Such approach would bypass the need for external algorithms and enable the direct use of order-dependent optimization contexts. This challenge, and its relevance in real-world problems, will be a central theme in the final section discussion of this chapter, and a guiding thread for the issues addressed in the following chapters.

The remainder of the chapter is organized as follows. Section 3.2 presents the LP formulations to compute the ordered measures. Section 3.3 extends these concepts to measures with a nested structure, such as variance and higher-order moments. Section 3.4 introduces our novel quadratic ordered operator, which generalizes existing quadratic measures. Section 3.5 presents an empirical comparison of the proposed solution methods against benchmark techniques. Finally, Section 3.6 concludes with a discussion of the practical implications of this framework and its integration into sophisticated optimization-based decision-making presenting some examples.

3.2 Linear ordered measures

In this section, LP models to compute the linear ordered measures introduced in (3.1) are presented. Some of these models build on existing approaches to order-related problems in the literature, while others constitute original contributions of this dissertation. To this end, let $\mathbf{x} \in \mathbb{R}^n_+$ be a given vector and $N = \{1, \dots, n\}$ the index set for the components of this vector. For simplicity, the vector is assumed to be non-negative, consistent with most problem settings in the literature which involves positive costs, and with the assumptions adopted in the subsequent chapters.

The objective of this section is to compute $\ell(\mathbf{x}, \lambda)$ for a given set of weights $\lambda \in \mathbb{R}^n$. The first approach, presented in the following proposition, builds upon the work of Boland et al. (2006) and relies on an assignment-based ILP formulation.

Proposition 3.2.1 — ℓ_1 model. Let $\mathbf{x} \in \mathbb{R}^n_+$ be a given vector and $\lambda \in \mathbb{R}^n$ define the linear ordered measure in (3.1), then the following formulation provides a valid model for its computation:

$$
\begin{aligned}
\ell_1(\mathbf{x}, \lambda) = \min \ & \sum_{i,k=1}^{n} \lambda_k z_{ik} x_i, && && \text{(3.7a)}\\
\text{s.t.} \ & \sum_{i=1}^{n} z_{ik} = 1, && k \in N, && \text{(3.7b)}\\
& \sum_{k=1}^{n} z_{ik} = 1, && i \in N, && \text{(3.7c)}\\
& \sum_{i=1}^{n} z_{ik} x_i \le \sum_{i=1}^{n} z_{i,k+1} x_i, && k \in N, k < n, && \text{(3.7d)}\\
& z_{ik} \in \{0, 1\}, && i, k \in N. && \text{(3.7e)}
\end{aligned}
$$

Proof. The proof follows by identifying the z-variables with the assignment variables that determine the order in the sorting sequence for the values in the sample $\mathbf{x}$:

$$
z_{ik} = \begin{cases} 1, & \text{if the } i\text{-th element is in the } k\text{-th position of the ordered vector,} \\ 0, & \text{otherwise,} \end{cases} \qquad i, k \in N.
$$

These variables are properly defined through constraints (3.7b) and (3.7c) which ensure that each component of $\mathbf{x}$ is allocated to exactly one position and each position is assigned to exactly one component of $\mathbf{x}$, and (3.7d) which enforce that the components after the sorting are in non-decreasing order. With these variables $x_{(k)} = \sum_{i \in N} z_{ik} x_i$ indicates the k-th component in the sorted vector. Therefore, the objective function is exactly the expression (3.1). ■

Note that this model is an integer linear program with $\mathcal{O}(n^2)$ variables, which may result in computational difficulties when applied to large datasets.

The second model for computing $\ell(\mathbf{x}, \lambda)$ is derived from the formulation in Marín et al. (2020) to the *Discrete Ordered Median Location Problem* (problem to be examined in Chapter 4). This approach relies on the k-sum operator, which is defined as the sum of the largest $n - k + 1$ elements of $\mathbf{x}$, namely:

$$
S_k(x) = \sum_{r=k}^{n} x_{(r)}, \tag{3.8}
$$

Using this operator, the expression of the ordered measure can be rewritten as:

$$
\ell(\mathbf{x}, \lambda) = \sum_{k=1}^{n} \lambda_k x_{(k)} = \sum_{k=1}^{n} \Delta_k S_k(x), \tag{3.9}
$$

where $\lambda_0 := 0$ and $\Delta_k := \lambda_k - \lambda_{k-1}$, for all $k \in \{1, ..., n\}$. For further purposes, let $\Delta^- = \{k \in \{1, ..., n\} | \Delta_k < 0\}$, and $\Delta^+ = \{k \in \{1, ..., n\} | \Delta_k > 0\}$. Observe that computing (3.8) is equivalent to solve the following linear problem or its dual (see, e.g., Ogryczak & Tamir, 2003; Kalcsics et al., 2002; Puerto et al., 2017):

$$
\begin{aligned}
\max \ & \sum_{i=1}^{n} d_i x_i, && && \text{(3.10a)}\\
\text{s.t.} \ & \sum_{i=1}^{n} d_i = n - k + 1, && && \text{(3.10b)}\\
& 0 \le d_i \le 1, && i \in N, && \text{(3.10c)}
\end{aligned}
\qquad
\begin{aligned}
\min \ & (n - k + 1)t + \sum_{i=1}^{n} v_i, && && \text{(3.11a)}\\
\text{s.t.} \ & t + v_i \ge x_i, && i \in N, && \text{(3.11b)}\\
& t \in \mathbb{R}, v_i \ge 0, && i \in N, && \text{(3.11c)}
\end{aligned}
$$

where if $\mathbf{x} \in \mathbb{R}^n_+$, which is our case, variable t can be considered non-negative, i.e., $t \ge 0$.

Proposition 3.2.2 — ℓ_2 model. Let $\mathbf{x} \in \mathbb{R}^n_+$ be a given vector and $\lambda \in \mathbb{R}^n$ define a linear ordered measure, then the following formulation provides a valid model for its computation:

$$
\begin{aligned}
\ell_2(\mathbf{x}, \lambda) = \min \; & \sum_{k \in \Delta^+} \Delta_k((n-k+1)t_k + \sum_{i=1}^n v_{ik}) + \sum_{k \in \Delta^-} \Delta_k \sum_{i=1}^n d_{ik} x_i, && && (3.12a)\\
\text{s.t. } \; & t_k + v_{ik} \geq x_i, && i \in N, k \in \Delta^+, && (3.12b)\\
& \sum_{i=1}^n d_{ik} = n-k+1, && k \in \Delta^-, && (3.12c)\\
& t_k \in \mathbb{R}, v_{ik} \geq 0, && i \in N, k \in \Delta^+, && (3.12d)\\
& 0 \leq d_{ik} \leq 1, && i \in N, k \in \Delta^-. && (3.12e)
\end{aligned}
$$

Proof. The proof proceeds by partitioning the right-hand-side of equivalence in (6.3) into the two subsets Δ^+ and Δ^-. The proposed model is then derived by combining two problem formulations: the dual problem (3.11) is applied if $k \in \Delta^+$, and the primal problem (3.10) is applied when $k \in \Delta^-$. ■

Observe that, although the model still uses $\mathcal{O}(n^2)$ variables and $\mathcal{O}(n^2)$ constraints, it is a linear optimization problem that avoids using binary variables, which provides a great computational advantage with respect to model ℓ_1.

Finally, an alternative model to the ℓ_2 can be build by leveraging the connection between the *k-sums operator* (3.8) and the *sample τ-quantile function* (2.52) introduced in Chapter 2. This relationship, formalized in Theorem 3.2.3, establishes that the k-sums operator is lower-bounded by the τ-quantile loss function.

Theorem 3.2.3 Let $y = (y_1, \dots, y_n) \in \mathbb{R}^n_+$, $\tau \in [0, 1]$, and $q \in \mathbb{Z}_+$. For $\gamma = y_{(q)}$, the q-th order statistic of y, and under the condition that $q \geq \tau n + 1$, the following inequality holds:

$$\mathcal{L}_\tau(y, \gamma) + (1-\tau) \sum_{i=1}^n y_i \geq S_q(y), \tag{3.13}$$

where $\mathcal{L}_\tau$ is the quantile loss function and $S_q(y)$ is the sum of the $(n-q+1)$ largest components of y. The equality holds when $q = \tau n + 1$.

Proof. First, it is straightforward that

$$\mathcal{L}_\tau(y, \gamma) = (1-\tau) \sum_{y_i \leq \gamma} (\gamma - y_i) + \tau \sum_{y_i > \gamma} (y_i - \gamma),$$

As a consequence, it follows that

$$
\begin{aligned}
\mathcal{L}_\tau(y, \gamma) + (1-\tau) \sum_{i=1}^n y_i &= (1-\tau) \sum_{y_i \leq \gamma} \gamma - (1-\tau) \sum_{y_i \leq \gamma} y_i + \tau \sum_{y_i > \gamma} y_i - \tau \sum_{y_i > \gamma} \gamma + (1-\tau) \sum_{i=1}^n y_i \\
&= (1-\tau) \sum_{y_i \leq \gamma} \gamma + \sum_{y_i > \gamma} y_i - \tau \sum_{y_i > \gamma} \gamma \\
&= (1-\tau) \sum_{y_i \leq \gamma} \gamma + \sum_{y_i > \gamma} (y_i - \gamma) + (1-\tau) \sum_{y_i > \gamma} \gamma \\
&= (1-\tau) \sum_{i=1}^n \gamma + \sum_{y_i > \gamma} (y_i - \gamma) \\
&= (1-\tau) \sum_{i=1}^n \gamma + \sum_{y_i \geq \gamma} (y_i - \gamma)
\end{aligned}
$$

$$= (1-\tau)\gamma n + \sum_{y_i \geq \gamma} y_i - \sum_{y_i \geq \gamma} \gamma.$$

Let $\gamma = y_{(q)}$ be the q-th value in the ordered sequence of y. Therefore,

$$\sum_{y_i \geq \gamma} 1 = n - q + 1, \text{ and}$$
$$\sum_{y_i \geq \gamma} y_i = \sum_{\ell=q}^{n} y_{(\ell)} = S_q(y).$$

Consequently,

$$L_\tau(y,\gamma) + (1-\tau)\sum_{i=1}^{n} y_i = (1-\tau)\gamma n + S_q(y) - \gamma(n-q+1) = \gamma(q-\tau n-1) + S_q(y) \geq S_q(y),$$

which holds since $q \geq \tau n + 1$. Also, the equality holds when $q = \tau n + 1$, as claimed. ■

As a result of Theorem 3.2.3, another single-level formulation following a rationale similar to that in Proposition 3.2.2 can be introduced.

Proposition 3.2.4 — ℓ_3 model. Let $\mathbf{x} \in \mathbb{R}^n_+$ be a given vector and $\lambda \in \mathbb{R}^n$ define a linear ordered measure, then the following formulation provides a valid model for its computation:

$$\begin{aligned}
\ell_3(\mathbf{x},\lambda) = \min \; & \sum_{k\in\Delta^+}\sum_{i\in N} \Delta_k\Big(\frac{k}{n}u_{ik} + (1-\frac{k}{n})v_{ik}\Big) + \sum_{k\in\Delta^-}\sum_{i\in N}\Delta_k\alpha_{ik}x_i \\
& + \sum_{k\in N}\sum_{i\in N}\Delta_k(1-\frac{k}{n})x_i, && && (3.14a)\\
\text{s.t. } & u_{ik} - v_{ik} + \gamma_k = x_i, && i\in N, k\in\Delta^+, && (3.14b)\\
& \sum_{i\in N}\alpha_{ik} = 0, && k\in\Delta^-, && (3.14c)\\
& \gamma_k\in\mathbb{R}, u_{ik}, v_{ik} \geq 0, && i\in N, k\in\Delta^+, && (3.14d)\\
& -\frac{k}{n} \leq \alpha_{ik} \leq 1-\frac{k}{n}, && i\in N, k\in\Delta^-. && (3.14e)
\end{aligned}$$

Proof. Recall that the sample τ-quantile problem can be solved as the following LP:

$$\begin{aligned}
\min \; & \tau\sum_{i=1}^{n} u_i + (1-\tau)\sum_{i=1}^{n} v_i, && && (3.15a)\\
\text{s.t. } & u_i - v_i + \gamma = x_i, && i\in N, && (3.15b)\\
& u_i, v_i \geq 0, && i\in N, && (3.15c)\\
& \gamma\in\mathbb{R}, && && (3.15d)
\end{aligned}$$

which dual is the following problem:

$$\begin{aligned}
\max \; & \sum_{i=1}^{n}\alpha_i x_i, && && (3.16a)\\
\text{s.t. } & \sum_{i=1}^{n}\alpha_i = 0, && && (3.16b)\\
& \tau - 1 \leq \alpha_i \leq \tau, && i\in N. && (3.16c)
\end{aligned}$$

Next, by Theorem 3.2.3, the following equation holds:

$$\sum_{k=1}^{n} \lambda_k x_{(k)} = \sum_{k=1}^{n} \Delta_k S_k(x) = \sum_{k\in\Delta^+} \Delta_k \left(\min_{\gamma_k} \mathcal{L}_\tau(x,\gamma_k) + (1-\tau)\sum_{i=1}^{n} x_i \right)$$
$$+ \sum_{k\in\Delta^-} \Delta_k \left(\min_{\gamma_k} \mathcal{L}_\tau(x,\gamma_k) + (1-\tau)\sum_{i=1}^{n} x_i \right),$$

where $\tau = k/n$.

One can reformulate the expression $\ell(\mathbf{x}, \boldsymbol{\lambda})$ into the objective function (3.14a) by applying a strategic, mixed primal-dual substitution. The choice of formulation depends on the sign of the coefficient Δ_k. For terms where $\Delta_k > 0$ (i.e., $k \in \Delta^+$), the quantile loss is replaced with its primal formulation (3.15). Since these are all minimization problems, they can be readily combined into one larger minimization. For terms where $\Delta_k < 0$ (i.e., $k \in \Delta^-$), the dual formulation (3.16) is used. The maximization of the negative coefficient (Δ_k) in the dual objective is equivalent to set a minimization objective. This technique aligns all components into a consistent minimization structure. By unifying these primal and dual substitutions, $\ell(\mathbf{x}, \boldsymbol{\lambda})$ is transformed into a single, comprehensive optimization problem, yielding the desired objective function. ■

Note that models ℓ_2 and ℓ_3, given by (3.12) and (3.14), respectively, can both be decomposed as two independent linear optimization problems that can be solved independently, and then to obtain the value of $\ell(\mathbf{x}, \lambda)$ as the sum of the objectives values of both problems, namely:

$$\ell_i(\mathbf{x}, \lambda) = \ell_i^+(\mathbf{x}, \lambda) - \ell_i^-(\mathbf{x}, \lambda) + c_i, \tag{3.17}$$

for $i \in \{2, 3\}$, and where ℓ_i^+ is given by the part of the objective function and the constraints concerning the Δ^+ set, ℓ_i^- is given by the part of the objective function and the constraints concerning the Δ^- set, and c_i is a constant term that must be added, i.e., $c_2 = 0$ and $c_3 = \sum_{k\in\Delta_k^+}\sum_{i\in N} \Delta_k(1-\frac{k}{n})x_i - \sum_{k\in\Delta_k^-}\sum_{i\in N}\Delta_k x_i$. These linear problems have $\mathcal{O}(n|\Delta^+|)$ and $\mathcal{O}(n|\Delta^-|)$ variables, respectively.

3.3 Nested ordered measures

This section addresses a class of optimization problems involving the minimization of measures that contain a nested linear ordered measure. Given a vector $\mathbf{x} \in \mathbb{R}^n_+$ and a set of weights $\lambda \in \mathbb{R}^n$ which defines a linear ordered measure $\ell(\mathbf{x}, \lambda) = \sum_{k=1}^{n} \lambda_k x_{(k)}$, our objective is to compute the value of a function $\mathfrak{b}(\mathbf{x}, \rho)$ for a given loss function ρ. This is formally expressed as:

$$\mathfrak{b}(\mathbf{x}, \rho) := \min \rho\left(\mathbf{x}, \ell(\mathbf{x}, \lambda)\right). \tag{3.18}$$

Expression $\mathfrak{b}$ is termed as *nested ordered measure*, while ℓ is linear ordered measure nested in $\mathfrak{b}$.

The nested structure of this problem, where the loss function ρ depends on the outcome of the linear ordered measure ℓ, lends itself to a bilevel optimization framework where the lower-level problem computes the value of ℓ. This value is then passed to the upper-level problem via a non-negative variable $\theta \geq 0$. The formulation is as follows:

$$\mathfrak{b}(\mathbf{x}, \rho) = \min\ \rho(\mathbf{x}, \theta), \tag{3.19a}$$
$$\text{s.t. } \theta = \ell(\mathbf{x}, \lambda). \tag{3.19b}$$

To this moment, a natural question could arise: *is this bilevel structure strictly necessary?* The primary justification lies in the role of the loss function ρ, which is allowed to be arbitrarily complex. For instance, the upper-level can be a LP if ρ is linear, a QP for a least-squares objective, or a general NLP for more complex forms. In some cases, the bilevel structure may not be strictly necessary, as ℓ can be represented directly using the LP models already introduced in the previous section. However, the bilevel approach is essential for creating a general and robust framework. This is the case when dealing with highly non-convex measures such as skewness or kurtosis coefficients, where this explicit separation becomes essential to ensure the correct representation of ℓ. Since our ultimate goal is to accommodate a wide variety of loss functions, the bilevel formulation provides the most general and appropriate framework.

Furthermore, the necessity of the bilevel framework becomes undeniable in contexts where $\mathbf{x}$ is not a fixed input vector but a set of decision variables within a larger optimization problem that *must* be optimal. Consider the application that will be introduced in Chapter 6, where the elements of $\mathbf{x}$ represent optimal vehicle route lengths. The objective may be to minimize a complex fairness function of these lengths, where θ will represent the average route length. The value of θ is therefore endogenous to the problem, it is determined by the optimal routing decisions for $\mathbf{x}$. A single-level formulation would fail to capture this hierarchical dependency. Although the specific approach developed in Chapter 6 differs, the bilevel structure in (3.19) faithfully reflects the nature of the problem: the upper-level determines the values for $\mathbf{x}$, while the lower-level computes the ordered measure θ based on these decisions.

In summary, the bilevel formulation is not an unnecessary complication but a mathematically precise method for modeling this class of problems. It provides the generality to handle a wide variety of loss functions and accurately captures the nested dependency between the primary decision variables and the ordered measure they define.

Note that constraint (3.19b) can be represented through any of the optimization models described in the previous sections. Since our aim is to reformulate the problem as a single-level program via strong duality, the scope is restricted to the two linear optimization representations associated with ℓ_1 and ℓ_2. In particular, using the k-sum based model provided in Proposition 3.2.2, the bilevel problem (3.19) can be equivalently reformulated as a single-level optimization model, as stated in the following result.

Proposition 3.3.1 — $\mathfrak{b}_1$ model. Given $\mathbf{x} \in \mathbb{R}^n_+$, a loss function ρ, and a weight vector $\lambda \in \mathbb{R}^n$ defining the linear ordered measure $\ell(\mathbf{x}, \lambda)$, the nested ordered measure $\mathfrak{b}(\mathbf{x}, \rho)$ can be computed through the following formulation:

$$
\begin{aligned}
\mathfrak{b}_1(\mathbf{x}, \rho) = \min \;& \rho(\mathbf{x}, \theta), && && (3.20a)\\
\text{s.t. } & \theta = \sum_{k \in \Delta^+} |\Delta_k| \big((n-k+1)t_k + \sum_{i=1}^n v_{ik}\big) - \sum_{k \in \Delta^-} |\Delta_k| \sum_{i=1}^n d_{ik} x_i, && && (3.20b)\\
& \theta = -\sum_{k \in \Delta^-} |\Delta_k| \big((n-k+1)t_k + \sum_{i=1}^n v_{ik}\big) + \sum_{k \in \Delta^+} |\Delta_k| \sum_{i=1}^n d_{ik} x_i, && && (3.20c)\\
& t_k + v_{ik} \geq x_i, && i, k \in N, && (3.20d)\\
& \sum_{i=1}^n d_{ik} = n - k + 1, && k \in N, && (3.20e)\\
& t_k \in \mathbb{R}, v_{ik} \geq 0, && i, k \in N, && (3.20f)\\
& 0 \leq d_{ik} \leq 1, && i, k \in N, && (3.20g)\\
& \theta \geq 0. && && (3.20h)
\end{aligned}
$$

Proof. Note that the dual problem of the k-sum problem (3.12a)-(3.12e) is:

$$
\max \sum_{k \in \Delta^+} \sum_{i=1}^n d^*_{ik} x_i + \sum_{k \in \Delta^-} \big((n-k+1)t^*_k + \sum_{i=1}^n v^*_{ik}\big), \qquad (3.21a)
$$

$$\text{s.t.} \sum_{i=1}^{n} d^*_{ik} = (n-k+1)\Delta_k, \qquad k \in \Delta^+, \tag{3.21b}$$

$$0 \le d^*_{ik} \le \Delta_k, \qquad i,k \in \Delta^+, \tag{3.21c}$$

$$t^*_k + v^*_{ik} \le x_i \Delta_k, \qquad i,k \in \Delta^-, \tag{3.21d}$$

$$t^*_k \in \mathbb{R}, v^*_{ik} \ge 0, \qquad i,k \in \Delta^-. \tag{3.21e}$$

Then, by scaling the decision variables and rewriting:

$$d_{ik} = \frac{d^*_{ik}}{|\Delta_k|}, i \in N, k \in \Delta^+, t_k = -\frac{t^*_k}{|\Delta_k|}, v_{ik} = -\frac{v^*_{ik}}{|\Delta_k|}, i \in N, k \in \Delta^-, \tag{3.22}$$

it follows that problem (3.21a)–(3.21e) is equivalent to:

$$\max \sum_{k \in \Delta^+} |\Delta_k| \sum_{i=1}^{n} d_{ik} x_i - \sum_{k \in \Delta^-} |\Delta_k| ((n-k+1)t_k + \sum_{i=1}^{n} v_{ik}), \tag{3.23a}$$

$$\text{s.t.} \sum_{i=1}^{n} d_{ik} = n-k+1, \qquad k \in \Delta^+, \tag{3.23b}$$

$$0 \le d_{ik} \le 1, \qquad i,k \in \Delta^+, \tag{3.23c}$$

$$t_k + v_{ik} \ge x_i, \qquad i,k \in \Delta^-, \tag{3.23d}$$

$$t_k \in \mathbb{R}, v_{ik} \ge 0, \qquad i,k \in \Delta^-. \tag{3.23e}$$

Therefore, by considering the lower-level primal constraints (3.12b)-(3.12e) and dual constraints (3.23b)-(3.23e), and both objective functions equivalence, one obtains to (3.20a)-(3.20h) ■

On the other hand, using the τ-quantile based model provided in Proposition 3.2.4, the bilevel problem (3.19) can be equivalently reformulated as described in the following result.

Proposition 3.3.2 — $\mathfrak{b}_2$ model. Given $\mathbf{x} \in \mathbb{R}^n_+$, a loss function ρ, and a weight vector $\lambda \in \mathbb{R}^n$ defining the linear ordered measure $\ell(\mathbf{x}, \lambda)$, the nested ordered measure $\mathfrak{b}(\mathbf{x}, \rho)$ can be computed through the following formulation:

$$\mathfrak{b}_2(\mathbf{x}, \rho) = \min \ \rho(\mathbf{x}, \theta), \tag{3.24a}$$

$$\text{s.t.} \ \theta = \sum_{k \in \Delta^+} \sum_{i \in N} |\Delta_k| \Big(\frac{k}{n} u_{ik} + (1 - \frac{k}{n}) v_{ik} \Big) + \sum_{k \in \Delta^-} \sum_{i \in N} |\Delta_k| \alpha_{ik} x_i + \sum_{k \in N} \sum_{i \in N} \Delta_k (1 - \frac{k}{n}) x_i, \tag{3.24b}$$

$$\theta = - \sum_{k \in \Delta^-} \sum_{i \in N} |\Delta_k| \Big(\frac{k}{n} u_{ik} + (1 - \frac{k}{n}) v_{ik} \Big) - \sum_{k \in \Delta^+} \sum_{i \in N} |\Delta_k| \alpha_{ik} x_i + \sum_{k \in N} \sum_{i \in N} \Delta_k (1 - \frac{k}{n}) x_i, \tag{3.24c}$$

$$u_{ik} - v_{ik} + \gamma_k = x_i, \qquad i,k \in N, \tag{3.24d}$$

$$\sum_{i \in N} \alpha_{ik} = 0, \qquad k \in N, \tag{3.24e}$$

$$\gamma_k \in \mathbb{R}, u_{ik}, v_{ik} \ge 0, \qquad i,k \in N, \tag{3.24f}$$

$$-\frac{k}{n} \le \alpha_{ik} \le 1 - \frac{k}{n}, \qquad i,k \in N, \tag{3.24g}$$

$$\theta \ge 0. \tag{3.24h}$$

Note that the two proposed single-level optimization models to compute $\mathfrak{b}(\mathbf{x},\rho)$ consist of linear constraints, and the non-linearity arises when ρ is non-linear. This is true even for the most popular loss functions, the absolute or quadratic deviations, or shape measures such as the skewness or kurtosis coefficients, as well as some of the already introduced functions. In the following, an illustrative example detailing how to proceed in the case where the loss function is the absolute deviation is provided.

■ **Example 3.1** Consider the loss function $\rho(\mathbf{x},\theta) = \sum_{i\in N} |\mathbf{x}_i - \theta|$ which induces the MeanAD with respect to any linear ordered measure $\ell(\mathbf{x},\lambda) = \sum_{k=1}^{n} \lambda_k x_{(k)}$:

$$\mathsf{MeanAD}_\lambda(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} |\mathbf{x}_i - \ell(\mathbf{x},\lambda)|\,, \tag{3.25}$$

By introducing non-negative variables $r_i^+, r_i^- \geq 0$, for each $i \in N$, to linearize the absolute values, each term in the above sum can be expressed as follows:

$$|x_i - \sum_{k=1}^{n} \lambda_k x_{(k)}| = r_i^+ + r_i^- \tag{3.26a}$$

$$x_i - \sum_{k=1}^{n} \lambda_k x_{(k)} = r_i^+ - r_i^-, \tag{3.26b}$$

$$r_i^+, r_i^- \geq 0, r_i^+ r_i^- = 0, i \in N. \tag{3.26c}$$

Thus, it follows that:

$$\begin{aligned}
\mathsf{MeanAD}_\lambda(\mathbf{x}) = \min & \sum_{i=1}^{n} (r_i^+ + r_i^-) && && (3.27a)\\
\text{s.t. } & r_i^+ - r_i^- = x_i - \theta, && i \in N, && (3.27b)\\
& r_i^+, r_i^- \geq 0, && i \in N, && (3.27c)\\
& (r_i^+, r_i^-) : \mathsf{SOS1}, && i \in N, && (3.27d)\\
& \theta = \sum_{k=1}^{n} \lambda_k x_{(k)}. && && (3.27e)
\end{aligned}$$

where, SOS1 stands for the Special Ordered Sets of Type 1, introduced by Beale & Tomlin (1970), which are a special set of variables for which at most one of them can take a non-zero value, while all other remain zero. ■

Finally, note that in cases where the linear ordered measure ℓ admits a simple closed form, such as the maximum, minimum, mean, or quantile (see (2.43), (2.44), (2.42), and (3.15), respectively), considerably simpler optimization models can be introduced.

3.4 Quadratic ordered measures

In the study of statistical measures, linear ordered measures provide a flexible framework for capturing properties related to the magnitude and ordering of data. However, their linearity inherently limits their ability to model more complex relationships, specifically the interactions between ordered components of a vector. To overcome this limitation and to broaden the perspective developed in the previous section, a new family of measures that extends linear ordered measures is introduced by incorporating quadratic terms. This generalization not only enables the modeling of component interactions but also unifies many popular measures of dispersion. To this end, the family of *quadratic ordered measures* is formally defined, parameterized by a matrix that specifies the weights of the quadratic interactions as follows.

Definition 3.4.1 Let $\mathbf{x} \in \mathbb{R}^n_+$ and $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$ denote its ordered components. For any matrix $\mathbf{M} = (m_{ij})_{i,j=1}^n \in \mathbb{R}^{n\times n}$, a *quadratic ordered measure* is defined as:

$$\mathtt{Q}(\mathbf{x}, \mathbf{M}) = \sum_{i,j=1}^{n} m_{ij} x_{(i)} x_{(j)} \tag{3.28}$$

This operator is continuous and, by construction, symmetric with respect to its inputs. A key theoretical property of the $\mathtt{Q}$-measure is its convexity, which is determined entirely by the matrix $\mathbf{M}$. The measure $\mathtt{Q}(\mathbf{x}, \mathbf{M})$ is convex if and only if $\mathbf{M}$ is positive semidefinite, and it is strictly convex if and only if $\mathbf{M}$ is positive definite. This property is particularly important in optimization contexts, as it guarantees a unique minimum when $\mathbf{M}$ is positive definite.

The primary advantage of the quadratic structure is its capability to directly model interactions between the ordered elements of the vector. While linear measures are restricted to weighted sums of individual components, the $\mathtt{Q}$-measure can capture pairwise relationships. For instance, by selecting an appropriate matrix $\mathbf{M}$, one can measure the interaction between the minimum and maximum values, represented by the term $x_{(1)}x_{(n)}$, or quantify the aggregated relationship between consecutive pair of values, i.e., $x_{(1)}x_{(2)} + x_{(2)}x_{(3)} + \cdots + x_{(n-1)}x_{(n)}$. Hence, the choice of $\mathbf{M}$ provides a flexible mechanism for designing measures sensitive to specific distributional characteristics. It is worth noting that this framework can be naturally extended to capture higher-order interactions (e.g., triplets or quadruplets of ordered components) by replacing the matrix $\mathbf{M}$ with a higher-order tensor.

Before analyzing the proposed optimization-based approach to compute this type of operator, some well-known measures that can be expressed as a $\mathtt{Q}$-measure are presented.

■ **Example 3.2 — Squared deviation with respect to a linear ordered measure.** Consider the nested measure where ρ is the mean squared deviation function and $\ell(\mathbf{x}, \lambda) = \sum_{k=1}^n \lambda_k x_{(k)}$, this is:

$$\mathfrak{b}(\mathbf{x}, \rho) = \sum_{i=1}^{n} \big(x_i - \ell(\mathbf{x}, \lambda)\big)^2. \tag{3.29}$$

Note that:

$$\sum_{i=1}^{n} \big(x_i - \ell(\mathbf{x}, \lambda)\big)^2 = \sum_{i=1}^{n} \big(x_i - \sum_{j=1}^{n} \lambda_j x_{(j)}\big)^2 = \sum_{i=1}^{n} \big(x_{(i)} - \sum_{j=1}^{n} \lambda_j x_{(j)}\big)^2 \tag{3.30}$$

$$= \sum_{i=1}^{n} x_{(i)}^2 + \sum_{i=1}^{n} \Big(\sum_{j=1}^{n} \lambda_j x_{(j)}\Big)^2 - \sum_{i=1}^{n} 2x_{(i)} \sum_{j=1}^{n} \lambda_j x_{(j)} \tag{3.31}$$

$$= \sum_{i=1}^{n} x_{(i)}^2 + n\sum_{i=1}^{n} \lambda_i^2 x_{(i)}^2 + n \sum_{\substack{i,j=1 \\ i\neq j}}^{n} \lambda_i\lambda_j x_{(i)}x_{(j)} - 2\sum_{i=1}^{n} \lambda_i x_{(i)}^2 - 2 \sum_{\substack{i,j=1 \\ i\neq j}}^{n} \lambda_j x_{(i)}x_{(j)} \tag{3.32}$$

$$= \sum_{i=1}^{n} (1 + n\lambda_i^2 - 2\lambda_i) x_{(i)}^2 + \sum_{\substack{i,j=1 \\ i\neq j}}^{n} (n\lambda_i\lambda_j - 2\lambda_j) x_{(i)}x_{(j)}. \tag{3.33}$$

Thus, choosing $\mathbf{M} = (m_{ij})_{i,j=1}^n \in \mathbb{R}^{n\times n}$ with

$$m_{ij} = \begin{cases} 1 + n\lambda_i^2 - 2\lambda_i, & i = j, \\ n\lambda_i\lambda_j - 2\lambda_j, & i \neq j, \end{cases} \tag{3.34}$$

one obtains that $\mathfrak{b}(\mathbf{x}, \rho) = \mathtt{Q}(\mathbf{x}, \mathbf{M})$. ■

■ **Example 3.3 — Variance.** The variance of the vector $\mathbf{x}$ can be obtained as a quadratic ordered measure by choosing:

$$\mathbf{M} = \frac{1}{n} \cdot \begin{pmatrix} 1-\frac{1}{n} & -\frac{1}{n} & \dots & -\frac{1}{n} \\ -\frac{1}{n} & 1-\frac{1}{n} & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{1}{n} \\ -\frac{1}{n} & \dots & -\frac{1}{n} & 1-\frac{1}{n} \end{pmatrix} \tag{3.35}$$

then $\mathtt{Q}(\mathbf{x},\mathbf{M})$ represents the variance of the values, i.e., $\frac{1}{n}\sum_{i=1}^{n}\big(x_i - \frac{1}{n}\sum_{j=1}^{n} x_j\big)^2$. ■

■ **Example 3.4 — (r,s)-trimmed variance.** The proposed extension allows to express general measures based on deviations, as shown in Table 3.2. This is also the case of trimmed or winsorized variances (see, e.g., Welsh & Morrison, 1990; Wilcox, 2011) that can easily be tackled within this framework. For instance, the (r,s)-trimmed variance, whose expression is proportional to

$$\sum_{i=r+1}^{n-s}\big(x_{(i)} - \frac{1}{n-r-s}\sum_{i=r+1}^{n-s} x_{(i)}\big)^2, \tag{3.36}$$

can be computed by means of

$$m_{ij} = \begin{cases} 1-\frac{1}{n-r-s}, & r+1 \le i = j \le n-s, \\ -\frac{1}{n-r-s}, & r+1 \le i \ne j \le n-s, \\ 0, & \mathit{otherwise}. \end{cases} \tag{3.37}$$

■

■ **Example 3.5 — (r,s)-winsorized variance.** The (r,s)-winsorized variance is proportional to,

$$r\big(x_{(r+1)} - W_{(r,s)}(\mathbf{x})\big)^2 + \sum_{i=r+1}^{n-s}\big(x_{(i)} - W_{(r,s)}(\mathbf{x})\big)^2 + s\big(x_{(n-s)} - W_{(r,s)}(\mathbf{x})\big)^2 \tag{3.38}$$

where $W_{(r,s)}(\mathbf{x})$ is the (r,s)-winsorized mean in Table 3.1, can be computed by means of

$$m_{ij} = \begin{cases} 1+r-\frac{(r+1)^2}{n}, & i=j=r+1, \\ 1+s-\frac{(s+1)^2}{n}, & i=j=n-s, \\ 1-\frac{1}{n}, & r+1<i=j<n-s, \\ -\frac{(rs+r+s+1)}{n}, & i=r+1 \text{ and } j=n-1 \text{ (and viceversa)}, \\ -\frac{(r+1)}{n}, & i=r+1 \text{ and } r+1<j<n-s \text{ (and viceversa)}, \\ -\frac{(s+1)}{n}, & i=n-s \text{ and } r+1<j<n-s \text{ (and viceversa)}, \\ -\frac{1}{n}, & r+1<i\ne j<n-s, \\ 0, & \mathit{otherwise}. \end{cases} \tag{3.39}$$

■

Building on the analysis of linear and nested ordered measures in the preceding sections, suitable mathematical optimization models are introduced for computing the quadratic ordered measure $\mathtt{Q}(\mathbf{x},\mathbf{M})$, for any given matrix $\mathbf{M}$.

To develop an integer mathematical programming formulation for this family of quadratic ordered measures, the formulation (3.7) introduced in Section 3.2 can be adapted. Define the binary assignment variables $z_{ik} \in \{0,1\}$, $i,k \in N$. These variables are governed by the following set of constraints:

$$\mathcal{P}(\mathbf{x}) = \left\{ z \in \{0,1\}^{n\times n} \;\middle|\; \sum_{i\in N} z_{ik} = 1, k \in N; \sum_{k\in N} z_{ik} = 1, i \in N; \sum_{i\in N} z_{ik}x_i \le \sum_{i\in N} z_{i,k+1}x_i, k \in N : k < n \right\}$$

As previously discussed, this set $\mathcal{P}(\mathbf{x})$ defines all feasible assignments of the values in vector $\mathbf{x}$ to their sorted positions. Specifically, a solution $\bar{z} \in \mathcal{P}(\mathbf{x})$ where $\bar{z}_{ik} = 1$ indicates that the component x_i occupies the k-th position in the sorted arrangement of $\{x_1, \ldots, x_n\}$. If all components of $\mathbf{x}$ are distinct, $\mathcal{P}(\mathbf{x})$ contains a single vector corresponding to the unique sorted order. However, in the presence of ties (i.e., duplicate values in $\mathbf{x}$), the sorting is not unique, and $\mathcal{P}(\mathbf{x})$ will encompass multiple vectors representing all possible valid orderings.

Using this structure, the computation of the $\mathtt{Q}$-measure for a vector $\mathbf{x}$ can be formulated as the following integer quadratic optimization problem:

$$\min_{z\in\mathcal{P}(\mathbf{x})} \sum_{i,k\in N} \sum_{j,\ell\in N} m_{k\ell} x_i x_j z_{ik} z_{j\ell} \tag{3.40}$$

The formulation in (3.40) is a non-linear integer problem due to the product of binary variables $z_{ik}z_{j\ell}$ in the objective function. This non-linearity can be addressed using standard linearization techniques, such as McCormick envelopes (McCormick, 1976). To this end, a new set of non-negative variables is introduced, namely $y_{ijk\ell} = z_{ik}z_{j\ell}$, $i,j,k,\ell \in N$. This substitution allows to reformulate the problem into an equivalent integer linear program:

$$\begin{aligned}
\min_{z\in\mathcal{P}(\mathbf{x}),y} \quad & \sum_{i,k=1}^{n} \sum_{j,\ell=1}^{n} m_{k\ell} x_i x_j y_{ijk\ell} && && (3.41a)\\
\text{s.t.} \quad & y_{ijk\ell} \le z_{ik}, && i,k \in N, && (3.41b)\\
& y_{ijk\ell} \le z_{j\ell}, && j,\ell \in N, && (3.41c)\\
& y_{ijk\ell} \ge z_{ik} + z_{j\ell} - 1, && i,j,k,\ell \in N, && (3.41d)\\
& y_{ijk\ell} \ge 0, && i,j,k,\ell \in N. && (3.41e)
\end{aligned}$$

The nested ordered measures discussed in Section 3.3 can also give rise to quadratic models, as their objective functions frequently involve products of continuous variables, resulting in non-convex formulations. While such models may remain competitive when the number of these products is small, the framework proposed here provides greater flexibility in handling quadratic terms. This increased flexibility, however, comes at the cost of introducing variables with four indices, which substantially raises computational complexity and may limit scalability. Consequently, the choice between these two modeling approaches depends on the specific requirements and complexity of the application.

The quadratic measure framework can be readily extended to incorporate the linear ordered measures described earlier. This is achieved by considering a non-square matrix of parameters $(\lambda|\mathbf{M}) \in \mathbb{R}^{n\times(n+1)}$. The resulting generalized $\mathtt{Q}$ measure is then defined as:

$$\mathtt{Q}(\mathbf{x}, (\lambda|\mathbf{M})) = \sum_{k=1}^{n} \lambda_k x_{(k)} + \sum_{i,j=1}^{n} m_{ij} x_{(i)} x_{(j)} \quad \text{for } \mathbf{x} \in \mathbb{R}^n_+. \tag{3.42}$$

This generalized form unifies the linear and quadratic measures. If $\mathbf{M} = \mathbf{0}$, the measure simplifies to the linear ordered measure, $\mathtt{Q}(\mathbf{x}, (\lambda|\mathbf{0})) = \ell(\mathbf{x}, \lambda)$. Conversely, if $\lambda = 0$, it reduces to the purely quadratic ordered measure, $\mathtt{Q}(\mathbf{x}, (0|\mathbf{M})) = \mathtt{Q}(\mathbf{x}, \mathbf{M})$.

This unified measure can be formulated as a single optimization problem using the same z-variables:

$$\min_{z\in\mathcal{P}(\mathbf{x})}\left(\sum_{i,k=1}^{n}\lambda_k x_i z_{ik}+\sum_{i,k=1}^{n}\sum_{j,\ell=1}^{n} m_{k\ell}x_i x_j z_{ik}z_{j\ell}\right). \tag{3.43}$$

This generalized model provides a comprehensive and adaptable tool for a wide range of ordered weighting problems, which can be linearized as shown in formulation (3.41).

3.5 Computational study

This section presents the results obtained from our computational experiments, conducted to empirically compare the proposed models. Our formulations were implemented in Python (version 3.10) using the Gurobi Optimizer (version 9.5) as MILP solver. The experiments were conducted on a MacPro server equipped with a 2.7 GHz Intel Xeon W processor (24 cores) and 192 GB RAM, utilizing 8 threads per run. A time limit of 3600 seconds was set for computations. For evaluating the different families of measures, including the linear, nested and quadratic ordered measures, 20 data vectors of sizes 10, 20, 30, 40, 50, 100, 500 and 1000, were generated. Each element in the vectors $(x_i)_{i=1}^n$ followed a uniform distribution $U([0,100])$ and was rounded to three decimal digits.

Two primary metrics were used to assess model efficiency. First, `Time` represents the average time (in seconds) required to compute the solution for an instance over the 20 trials. Second, `Accuracy` shows the number of instances in which the value obtained by the models closely matches the reference values computed using Python's state-of-the-art functions from the `numpy`, `statictics`, and `scipy` packages. This is, given $\mathtt{f}_{\text{python}}$ as the value computed using the Python package and $\mathtt{f}_{\text{m}}$ as the solution obtained by the model "m", `Accuracy` shows the number of instances for which $(\mathtt{f}_{\text{python}}-\mathtt{f}_{\text{m}})\leq 10^{-5}$. In the worst case, our solution will be suboptimal with respect to the one computed by python, i.e, $\mathtt{f}_{\text{m}}\leq \mathtt{f}_{\text{python}}$.

Table 3.4 shows the results for some linear ordered measures, which are: the `mean`, $(n/10,n/10)$-trimmed mean (`trim-mean`), $(n/10,n/10)$-winsorized mean (`win-mean`), `range`, $(n/10,n/10)$-mid mean (`mid-mean`), `median`, first and third quartiles (`1quartile` and `3quartile`), the interquartile range (`iqr`), mean absolute deviations with respect to the median (`madev-median`) and the Gini differences (`gini-difs`), which expressions can be found in Tables 3.1 and 3.2. The tested models are ℓ_1, which is the one derived from formulation (3.7), ℓ_2 obtained from formulation (3.12), and ℓ_3 obtained from formulation (3.14). The results show that while ℓ_1 can handle small-scale problems effectively, it struggles with larger datasets, reaching the time limit of 3600 seconds. Conversely, ℓ_2 and ℓ_3 demonstrate significantly faster computational times, maintaining high accuracy across all instances.

Table 3.5 presents the results for several nested ordered measures, including the `variance` (mean squared deviations from the mean), mean squared deviations from the median, maximum, and minimum (`msdev-median`, `msdev-max`, and `msdev-min`), as well as their corresponding mean absolute deviations (`madev-median`, `madev-max`, and `madev-min`). These measures, previously introduced in Table 3.2, provide insight into the dispersion characteristics of the data. Additionally, the table includes the well-known `skewness` and `kurtosis` coefficients, where the mean serves as the tendency measure nested. The results compare two models: $\mathfrak{b}_1$, which is derived from formulation (3.20), and $\mathfrak{b}_2$, obtained from formulation (3.24). The computational time and accuracy for each measure are reported across varying sample sizes. Both models, $\mathfrak{b}_1$ and $\mathfrak{b}_2$, exhibit similar performance increasing computational time as the sample size grows, except for large instances sizes where $\mathfrak{b}_1$ is outperformed by $\mathfrak{b}_2$ in terms of time, especially for the `madev-max` and `skewness` criteria. Also, both $\mathfrak{b}_1$ and $\mathfrak{b}_2$ demonstrate stable accuracy across all measures, maintaining a consistent score of 20 in most cases. This suggests that both models provide reliable estimates across different dispersion and shape measures, except for the `msdev-min` measure where $\mathfrak{b}_1$ is clearly outperformed by $\mathfrak{b}_2$ for large instances.

Measure	Model	Time								Accuracy							
		10	20	30	40	50	100	500	1000	10	20	30	40	50	100	500	1000
`mean`	ℓ_1	0.0012	0.0055	0.0173	0.0437	0.1533	52.5193	3600.0000	3600.0000	20	20	20	20	20	20	20	20
	ℓ_2	0.0001	0.0001	0.0002	0.0002	0.0002	0.0007	0.0051	0.0081 20	20	20	20	20	20	20	20	20
	ℓ_3	0.0001	0.0001	0.0001	0.0001	0.0001	0.0002	0.0029	0.0005 20	20	20	20	20	20	20	20	20
`trim-mean`	ℓ_1	0.0011	0.0055	0.0173	0.0435	0.1236	83.1352	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0001	0.0001	0.0002	0.0002	0.0003	0.0007	0.0083	0.0118 20	20	20	20	20	20	20	20	20
	ℓ_3	0.0001	0.0001	0.0002	0.0002	0.0003	0.0009	0.0103	0.0106 20	20	20	20	20	20	20	20	20
`win-mean`	ℓ_1	0.0012	0.0055	0.0172	0.0437	0.1450	55.1646	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0002	0.0002	0.0003	0.0003	0.0004	0.0014	0.0097	0.0195	20	20	20	20	20	20	20	20
	ℓ_3	0.0002	0.0002	0.0003	0.0004	0.0005	0.0015	0.0155	0.0281	20	20	20	20	20	20	20	20
`range`	ℓ_1	0.0011	0.0055	0.0173	0.0436	0.1528	52.6488	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0002	0.0002	0.0002	0.0003	0.0003	0.0010	0.0064	0.0090	20	20	20	20	20	20	20	20
	ℓ_3	0.0002	0.0002	0.0003	0.0003	0.0004	0.0014	0.0094	0.0162	20	20	20	20	20	20	20	20
`mid-mean`	ℓ_1	0.0011	0.0055	0.0174	0.0437	0.1464	55.8178	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0002	0.0002	0.0002	0.0003	0.0003	0.0007	0.0064	0.0154	20	20	20	20	20	20	20	20
	ℓ_3	0.0002	0.0002	0.0002	0.0003	0.0004	0.0010	0.0088	0.0122	20	20	20	20	20	20	20	20
`median`	ℓ_1	0.0011	0.0054	0.0175	0.0434	0.1458	3.2773	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0002	0.0001	0.0002	0.0003	0.0003	0.0012	0.0088	0.0148	20	20	20	20	20	20	20	20
	ℓ_3	0.0001	0.0002	0.0002	0.0003	0.0003	0.0013	0.0142	0.0220	20	20	20	20	20	20	20	20
`1quartile`	ℓ_1	0.0011	0.0055	0.0172	0.0438	0.1665	107.7195	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0002	0.0002	0.0002	0.0002	0.0003	0.0009	0.0086	0.0141	20	20	20	20	20	20	20	20
	ℓ_3	0.0001	0.0001	0.0002	0.0002	0.0003	0.0010	0.0119	0.0157	20	20	20	20	20	20	20	20
`3quartile`	ℓ_1	0.0012	0.0055	0.0172	0.0436	0.1350	51.9655	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0002	0.0001	0.0002	0.0002	0.0003	0.0009	0.0070	0.0106	20	20	20	20	20	20	20	20
	ℓ_3	0.0001	0.0002	0.0002	0.0003	0.0003	0.0012	0.0122	0.0158	20	20	20	20	20	20	20	20
`iqr`	ℓ_1	0.0012	0.0055	0.0170	0.0433	0.1405	58.9756	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0002	0.0002	0.0003	0.0004	0.0005	0.0013	0.0129	0.0325	20	20	20	20	20	20	20	20
	ℓ_3	0.0002	0.0002	0.0003	0.0005	0.0006	0.0018	0.0178	0.0426	20	20	20	20	20	20	20	20
`madev-median`	ℓ_1	0.0011	0.0054	0.0172	0.0435	0.1446	52.9097	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0001	0.0002	0.0002	0.0003	0.0003	0.0008	0.0085	0.0143	20	20	20	20	20	20	20	20
	ℓ_3	0.0001	0.0002	0.0002	0.0003	0.0003	0.0009	0.0112	0.0151	20	20	20	20	20	20	20	20
`gini-difs`	ℓ_1	0.0011	0.0054	0.0168	0.0428	0.1403	52.7795	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0004	0.0012	0.0028	0.0060	0.0121	0.0249	1.7334	16.9960	20	20	20	20	20	20	20	20
	ℓ_3	0.0004	0.0015	0.0039	0.0086	0.0176	0.0404	2.1569	20.1885	20	20	20	20	20	20	20	20

Table 3.4: Results obtained evaluating linear ordered measures.

Measure	Model	Time								Accuracy							
		10	20	30	40	50	100	500	1000	10	20	30	40	50	100	500	1000
`variance`	$\mathfrak{b}_1$	0.0003	0.0005	0.0009	0.0014	0.0021	0.0098	0.3784	1.8353	20	20	20	20	20	20	20	20
	$\mathfrak{b}_2$	0.0002	0.0004	0.0009	0.0016	0.0024	0.0107	0.3871	2.1006	20	20	20	20	20	20	20	20
`msdev-median`	$\mathfrak{b}_1$	0.0004	0.0008	0.0013	0.0023	0.0034	0.0131	0.3337	1.7240	20	20	20	20	20	20	20	20
	$\mathfrak{b}_2$	0.0004	0.0009	0.0016	0.0029	0.0045	0.0166	0.3927	2.0073	20	20	20	20	20	20	20	20
`msdev-max`	$\mathfrak{b}_1$	0.0003	0.0006	0.0010	0.0017	0.0025	0.0107	0.3611	1.6875	20	20	20	20	20	20	20	20
	$\mathfrak{b}_2$	0.0003	0.0007	0.0013	0.0023	0.0034	0.0132	0.4167	1.9957	20	20	20	20	20	20	20	20
`msdev-min`	$\mathfrak{b}_1$	0.0004	0.0007	0.0013	0.0023	0.0034	0.0136	0.3471	1.8349	20	20	20	20	20	19	14	10
	$\mathfrak{b}_2$	0.0003	0.0007	0.0012	0.0021	0.0034	0.0130	0.4140	1.9671	20	20	20	20	20	20	20	20
`madev-mean`	$\mathfrak{b}_1$	0.0006	0.0014	0.0029	0.0054	0.0129	0.1243	0.5407	2.6555	20	20	20	20	20	20	20	20
	$\mathfrak{b}_2$	0.0003	0.0006	0.0011	0.0020	0.0032	0.0152	0.5710	2.7334	20	20	20	20	20	20	20	20
`madev-median`	$\mathfrak{b}_1$	0.0011	0.0019	0.0031	0.0047	0.0066	0.0270	0.6400	3.0635	20	20	20	20	20	20	20	20
	$\mathfrak{b}_2$	0.0012	0.0021	0.0035	0.0054	0.0075	0.0279	0.8018	3.5806	20	20	20	20	20	20	20	20
`madev-max`	$\mathfrak{b}_1$	0.0008	0.0012	0.0016	0.0025	0.0036	0.0167	6.4872	51.1322	20	20	20	20	20	20	20	20
	$\mathfrak{b}_2$	0.0008	0.0017	0.0031	0.0067	0.0104	0.0308	0.7752	3.9238	20	20	20	20	20	20	20	20
`madev-min`	$\mathfrak{b}_1$	0.0013	0.0034	0.0074	0.0093	0.0133	0.0439	0.7684	3.2349	20	20	20	20	20	20	20	20
	$\mathfrak{b}_2$	0.0005	0.0009	0.0015	0.0028	0.0041	0.0180	0.6795	3.2782	20	20	20	20	20	20	20	20
`skewness`	$\mathfrak{b}_1$	0.0018	0.0030	0.0054	0.0099	0.0204	0.1950	0.9368	52.8096	20	20	20	20	20	20	20	20
	$\mathfrak{b}_2$	0.0014	0.0019	0.0031	0.0050	0.0073	0.0286	10.4145	5.0563	20	20	20	20	20	20	20	20
`kurtosis`	$\mathfrak{b}_1$	0.0016	0.0029	0.0053	0.0095	0.0205	0.2081	0.9167	4.6692	20	20	20	20	20	20	20	20
	$\mathfrak{b}_2$	0.0012	0.0019	0.0031	0.0051	0.0073	0.0302	1.0423	5.0956	20	20	20	20	20	20	20	20

Table 3.5: Results obtained evaluating nested ordered measures.

Finally, Table 3.6 summarizes the results for several quadratic ordered measures, including mean squared deviations from the mean, median, maximum, and minimum (`variance`, `msdev-median`, `msdev-max`, and `msdev-min`, respectively). Additionally, the table includes the $(n/10, n/10)$-trimmed variance (`trim-variance`) and the $(n/10, n/10)$-winsorized variance (`win-variance`), which were introduced in Section 4. The only model tested for this set of measures is derived from formulation (3.41), denoted as `Q`. For the quadratic ordered measures the computational time increases significantly as the sample size grows. The model requires considerably more time as the sample size grows, indicating a higher computational cost compared to the linear and nested ordered measure models. In contrast, the accuracy of the model also remains stable at 20 across all tested measures, indicating that despite higher computational costs, the model maintains strong estimation consistency.

Measure	Model	Time					Accuracy				
		10	20	30	40	50	10	20	30	40	50
`variance`	`Q`	9.1381	13.6615	128.1267	902.3868	3292.905	20	20	20	20	20
`msdev-median`	`Q`	2.8683	13.9078	125.0910	793.5474	2187.096	20	20	20	20	20
`msdev-max`	`Q`	2.1225	12.4740	118.8592	746.7027	2225.038	20	20	20	20	20
`msdev-min`	`Q`	1.7471	12.2372	118.0600	750.9281	2199.251	20	20	20	20	20
`trim-variance`	`Q`	11.0785	13.0243	119.2869	844.5739	2952.002	20	20	20	20	20
`win-variance`	`Q`	10.5823	12.8667	120.0339	855.3233	3224.855	20	20	20	20	20

Table 3.6: Results obtained evaluating quadratic ordered measures.

Note that, although it is not our goal, methodology allows evaluating the measures using the *a priori* information about the sorted values of $\mathbf{x}$ by fixing the values of the variables in the optimization models, reducing to a simple calculation. A thorough analysis of the different systems of constraints results in a worst-case complexity of $\mathcal{O}(n^2)$ for the linear and nested measures, and $\mathcal{O}(n^4)$ for the quadratic measures.

3.6 Ordered measures in optimization problems

Mathematical optimization models for decision-making critically rely on the choice of the objective measure to be maximized or minimized. In most formulations found in the literature, as those that arise in facility location, network design, or even in machine learning, the quality of a solution is evaluated by aggregating into a single scalar the individual performance measures associated with the entities involved in the problem. For instance, in facility location, the allocation costs from each customer to a set of open facilities, the length of each of the arcs in a network design problems, the missclassification errors for each of the observations in a supervised classification problem, or the returns of the different items selected in a portfolio selection problem.

Classical models assume that this aggregation is performed either by the average (sum) or the worst case (max or min) behavior. These two approaches have given rise to a vast applied optimization literature in different fields. However, it has been largely recognized that many more point of views make sense in different frameworks. In this line, different aggregation measures have been proposed to derive solutions for decisions problems that allow the decision-maker to select alternatives not only focused on economically efficient returns, but also in fair allocations, robust criteria in uncertain situations, portfolio selection problems under risk, balanced criteria in supply chain management, or envy-free measures in social choice and distribution problems.

This is the case of computing the location of a facility with minimum variance of the distances, a portfolio with minimum Conditional Value at Risk (CVaR), or linear model estimators with minimum deviation residuals with respect their mean. Several of these measures have already been recognized in the literature on decision-making optimization problems, including, among others: minimax problems (Gross, 1959; Hansen, 1980; Schrijver, 1983; Puerto & Perea, 2018), combining minisum and minimax (Mazzola & Neebe, 1988; Hansen et al., 1991; Punnen et al., 1995; Averbakh et al., 1995; Tamir et al., 2002), k-centrum optimization (Slater, 1978; Tamir, 2001; Kalcsics et al., 2002; Ogryczak & Tamir, 2003; Puerto et al., 2017), k-th best solutions (Lawler, 1972; Martello et al., 1984; Hamacher & Queyranne, 1985; Pascoal et al., 2003; Yen, 1971), lexicographic optimization (Isermann, 1982; Ogryczak, 1997; Croce et al., 1999; Puerto & Perea, 2013), risk measures (Markowitz, 1952; Rockafellar et al., 2000, 2002; Benati, 2003, 2004; Benati & Rizzi, 2007; Fernández et al., 2019), dispersion measures (Maimon, 1986; Gupta &

Punnen, 1988; Berman, 1990; Punnen & Aneja, 1997; Martinez-Gavara et al., 2021), minimum-envy solutions (Espejo et al., 2009; Blanco et al., 2024), regret solutions (Savage, 1951; Averbakh, 2001; Puerto & Rodríguez-Chía, 2003; Puerto et al., 2011b; Conde et al., 2018; Conde & Leal, 2017), or fairness/equity measures (Marsh & Schilling, 1994; Young, 1995; Ogryczak, 2000; Mesa et al., 2003; López-de-los Mozos et al., 2008; Perea et al., 2012; Matl et al., 2018).

The order-based optimization approach developed in this dissertation offers three main advantages: 1) it provides simpler representations of these measures, making the modeling process more straightforward compared to other complex formulations in the literature, 2) it flexibly accommodates different measures within a unified framework, enabling a common algebraic structure that facilitates both analysis and model strengthening, and 3) it supports common solution strategies, thereby simplifying their implementation and making them more accessible to practitioners.

The shape of the linear ordered measures introduced above has already been applied in many contexts. Expressions of the form (3.1) are widely known in the literature as *Ordered Weighted Average* (OWA) operators. First introduced by Yager (1988), OWA operators provide a parameterized class of mean-type aggregation functions. While L-statistics are typically applied to known data points, OWA operators are often used to aggregate unknown values within a decision problem. Their flexibility has led to successful applications across numerous domains in Statistics, Data Science, and Operations Research.

In the fields of statistics and machine learning, OWA operators have become instrumental for tasks requiring robust and flexible aggregation. A prominent application is the development of robust estimators in linear regression, where the aggregation of generalized residuals using OWA operators helps to mitigate the influence of outliers (Yager & Beliakov, 2009; Blanco et al., 2018; Flores-Sosa et al., 2020; Blanco et al., 2021; Flores-Sosa et al., 2022; D'Urso & Chachi, 2022; Puerto & Torrejon, 2025). Further applications in this domain include modeling the behavior of neural networks with fuzzy quantifiers (Yager, 1987, 1992), clustering problems (Cheng et al., 2009), or enhancing support vector machines by aggregating misclassification errors for classification tasks (Maldonado et al., 2018; Marín et al., 2022) and for regression purposes (Martínez-Merino et al., 2025).

Beyond statistical modeling, the OWA framework has been applied to a diverse range of general optimization and decision-making problems. Its ability to model complex preferences has made it valuable in areas such as: combinatorial optimization (Fernández et al., 2013; Fernández et al., 2014), with several works addressing the formulation of fairness measures (Mesa et al., 2003; Blanco & Gázquez, 2023; Tsang & Shehadeh, 2025), the aggregation of expert information under uncertainty (Ahn & Yager, 2014; Blanco & Pérez-Sánchez, 2018; Xu, 2008), image segmentation based on the grouping of pixel intensities (Calvino et al., 2022; Muñoz-Ocaña et al., 2024), voting problems and social choice theory (Amanatidis et al., 2015; Ponce et al., 2018), and portfolio selection in finance (Laengle et al., 2017; Cesarone & Puerto, 2024; Benati et al., 2025).

One of the most significant and well-developed applications of OWA operators in the field of Operations Research, particularly in Facility Location Theory. In this context, they are commonly referred to as *ordered median functions* and provide a unified methodology to model a wide spectrum of distance-based or cost-based objective functions (Nickel & Puerto, 2005; Puerto & Rodríguez-Chía, 2019). The study of ordered median functions in location problems is extensive and spans several problem classes, e.g., the *Discrete Ordered Median Problem* (DOMP), which is typically set on networks and has been a subject of prolific research (Nickel & Puerto, 1999; Kalcsics et al., 2003; Puerto & Tamir, 2005; Domínguez-Marín et al., 2005; Boland et al., 2006; Puerto et al., 2009; Marín et al., 2009; Kalcsics et al., 2010; Tang et al., 2011; López-Redondo et al., 2016; Labbé et al., 2017; Deleplanque et al., 2020; Schnepper et al., 2019; Marín et al., 2020; Pozo et al., 2024; Ljubić et al., 2024; Cherkesly et al., 2025), the *Continuous Ordered*

Median Problem (COMP), which addresses the location problem in a continuous space (Drezner & Nickel, 2009; Krebs & Nickel, 2010; Blanco et al., 2013, 2014, 2016, 2023), or to generalize hub location problems, where the objective is to locate hubs where to allocate demand nodes (Puerto et al., 2011a, 2016, 2013; Pozo et al., 2021).

The simplicity of the ordered measure representation presented in this chapter facilitates its implementation in any scenario requiring an order-based approach. Its main advantage is that it only requires the encoding of the relevant decision variables, such as, e.g., assignment costs, routing costs, classification errors, or residuals. The last sections of this chapter explores the potential of this representation in problems like scenario analysis in Linear Programming, the Traveling Salesman Problem and the Weighted Multicover Set Problem. However, a fundamental complication arises, particularly in the linear case. When decision variables replace the fixed values, many of the existing models introduce non-linearities that render them inadequate in their current formulation. This limitation calls for the development of tailored methods to allow for the efficient resolution of such problems. In response to this challenge, the following chapters of this dissertation present specific contributions in some cases. The necessary adaptations and developments will be addressed in several key contexts:

- In the field of location problems, tailored models will be developed for the *Discrete Ordered Median Problem* (DOMP).
- DOMP extension concerning connecting between facilities, the *Ordered Median Tree Location Problem* (OMT), will be analyzed in the context of network design.
- For logistics and transportation, the integration of ordered measures will be presented for the first time in the context of the *Vehicle Routing Problem* (VRP), focusing on the potential of the ordered representation to integrate and manage complex fairness measures.
- Finally, in the field of robust statistics, *Ordered Linear Regression* will be addressed through the computation of the *Least Squares of Quantiles* (LQS) estimator, illustrating the robustness that these methods offer in the presence of outliers.

3.6.1 Scenario analysis in Linear Programming

Several optimization problems involve analyzing several scenarios and optimize over those. For instance, this is the case of, the classical approach to optimization under uncertainty, particularly in stochastic linear programming, which often involves a finite set of scenarios, each with a given probability of occurrence. The standard objective is typically to minimize the expected cost or maximize the expected profit over all scenarios. This is a risk-neutral approach, as it aggregates outcomes into a single weighted average, potentially obscuring low-probability events.

Ordered measures provide a powerful framework for introducing risk aversion into scenario analysis without resorting to more complex utility theory. Instead of minimizing the expected value, a decision-maker can minimize an ordered weighted average of the outcomes across the different scenarios. Let c_s be the cost associated with a decision under scenario $s \in S$. The objective function can be modeled as an ordered measure applied to the vector of costs $(c_s)_{s\in S}$.

This approach allows for a rich spectrum of risk profiles. For instance, by assigning all weight to the highest cost, the model reduces to the classical minimax (or worst-case) criterion, representing extreme risk aversion. Conversely, assigning weights only to a specific quantile of the worst outcomes allows for modeling measures like Value at Risk (VaR), which targets the quantile directly, or the Conditional Value at Risk (CVaR), which averages the costs of the worst $\alpha\%$ of scenarios. This is particularly useful in finance and supply chain management, where mitigating the impact of severe tail-risk events is crucial. The formulations of these ordered objectives, as discussed in this work, allow these risk-averse models to be solved with standard linear programming techniques, making them computationally tractable and accessible to practitioners.

Consider a problem where the goal is to select a decision vector $z \in \mathbb{R}^d$ from a feasible region defined by a polyhedron $P = \{z \in \mathbb{R}^d_+ : Az \leq b\}$. The decision's outcome is subject to uncertainty, modeled using n distinct scenarios. For each scenario $s \in \{1, \dots, n\}$, the cost is a linear function of the decision, given by $x_s = c_s^T z$, where c_s is the cost vector for that scenario. One can assemble these cost vectors into a single cost matrix $C \in \mathbb{R}^{n \times d}$. The vector of costs across all scenarios is then $\mathbf{x} = Cz$. The objective is to find the decision $z \in P$ that minimizes a chosen risk measure applied to the cost vector $\mathbf{x}$.

To exemplify, a two-dimensional decision space ($d = 2$) is used. The feasible polyhedron P is defined by the vertices $V = \{(2,3), (5,1), (7,4), (4,8), (1,6)\}$, see Figure 3.1 (left). Consider $n = 5$ cost scenarios, defined by the matrix:

$$C = \begin{pmatrix} 3 & 4 \\ 2 & 0 \\ -3 & -2 \\ -2 & -6 \\ 4 & -10 \end{pmatrix} \tag{3.44}$$

This problem is solved using different criteria via the ℓ and $\mathfrak{b}$ models (the $\mathtt{Q}$ model is a special case of $\mathfrak{b}$ with ρ defined as the squared deviation with respect to the specified λ) to identify an optimal point in the polyhedron that minimizes the measure aggregating the analyzed cost scenarios. The λ vectors corresponding to the mean and the maximum are used. The resulting solutions are shown in Figure 3.1 (right), where it can be observed that the solutions obtained under different summarizing criteria differ from one another.

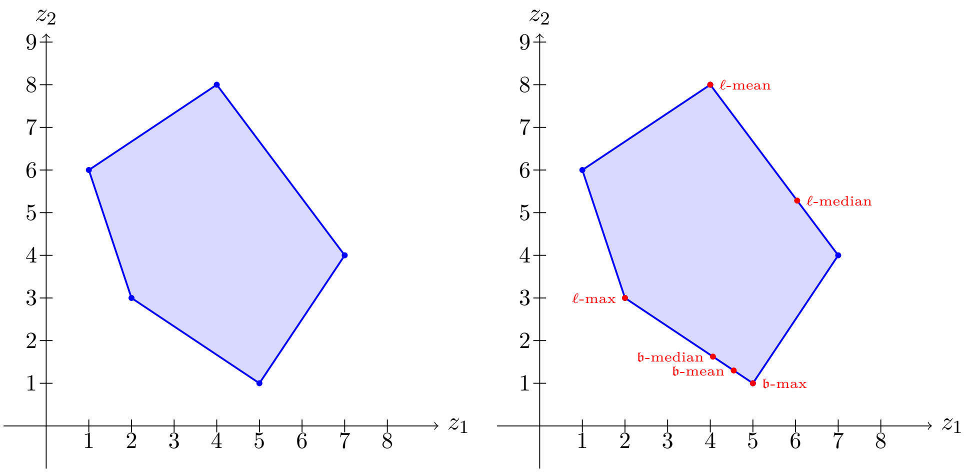


Figure 3.1: Polyhedron considered for the scenario analysis in linear programming (left) and solution obtained (right).

This result highlights a key insight: the "best" decision depends entirely on the decision-maker's attitude toward risk. A risk-neutral approach (minimizing the mean) may yield a different optimal decision than a strongly risk-averse one (minimizing the worst-case). Finding these optimal solutions requires dedicated optimization techniques. While one could easily evaluate the cost vector $\mathbf{x}$ for any *single* decision $z \in P$, it is impossible to find the optimum by simply checking all possibilities, as the polyhedron contains an infinite number of points. The strength of this framework lies in its ability to reformulate these problems into solvable LPs.

3.6.2 Traveling Salesman Problem

In the *Traveling Salesman Problem* (TSP), previously introduced in Section 2.6.3, ordered measures allow for the formulation of several TSP variants that capture these alternative objectives. Ordered measures can be used to find tours that are uniform or equitable. For instance, the minimax TSP, which seeks to minimize the length of the longest edge in the tour, or a generalization of this, the k-centrum TSP, which aims to minimize the length of the k longest edges, offering a flexible trade-off between average and worst-case performance. Other interesting objectives can be minimizing the range of edge costs (the difference between the longest and shortest edges) or their variance. Such goals are valuable in contexts like workload balancing for delivery drivers, ensuring no single route segment is disproportionately long, or when designing communication networks where consistent link latency is more important than the total latency.

Formally, given a directed graph $G = (V, A)$, with node set V and arc set A, and for each arc $a \in A$ a weight $w_a \geq 0$, the goal is to build a Hamiltonian tour visiting the nodes of G exactly once at minimum overall sum of the weights. To this end, the following family of binary variables is commonly introduced to identify the arcs that belong to the solution:

$$z_a = \begin{cases} 1 & \text{if a is in the tour,} \\ 0 & \text{otherwise} \end{cases}, \forall a \in A. \tag{3.45}$$

Note that since each node appears exactly once as an outgoing node in the tour, the cost of the outgoing arc from $v \in V$ can be expressed as:

$$x_v = \sum_{a \in \delta^+(v)} w_a z_a. \tag{3.46}$$

Thus, the overall sum of the weights can be written as $\sum_{v \in V} x_v$. Instead of minimizing the overall sum of the weights of the arcs involved in the solution, one can also minimize any of the measures proposed in the previous sections on the set of costs $\{x_v\}_{v \in V}$ described above.

To demonstrate the practical implications of using different ordered measures, next, several TSP variants are solved for a small, complete graph instance. The test instance is the complete graph shown in Figure 3.2, where nodes are represented by points in a plane and arc costs are their Euclidean distances.

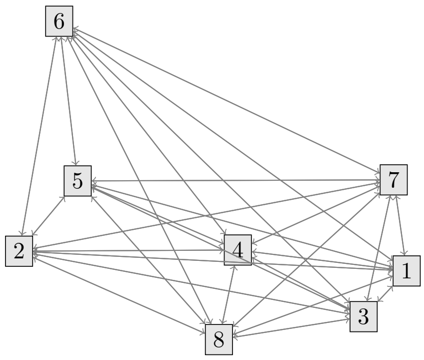
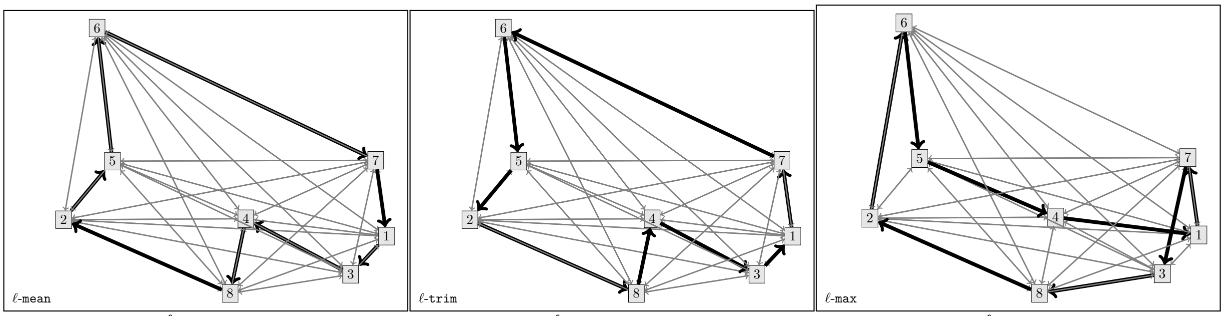


Figure 3.2: Graph used to illustrate the TSP under ordered measures.

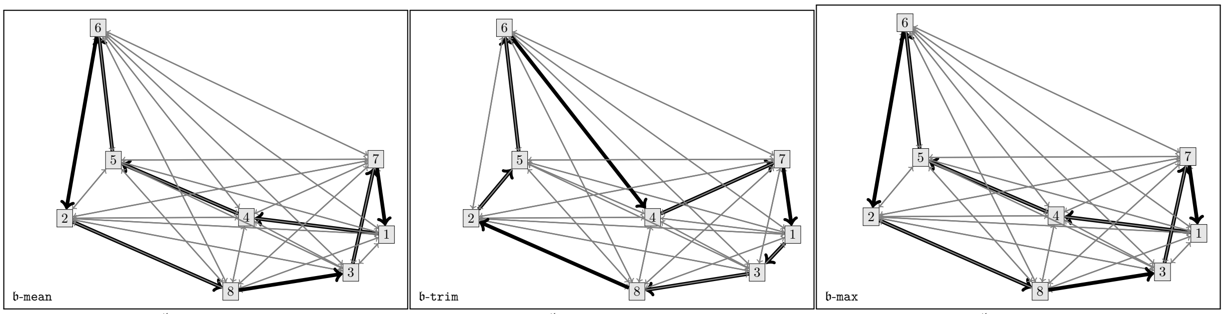
Our ordered framework is applied to solve the TSP using various objective functions based on the ℓ, $\mathfrak{b}$, and Q models. Three main families of ordered measures are considered in this work. The first are the linear ordered measures (ℓ-measures), which include the mean, the 25%-25% trimmed mean, and the maximum. The second are the nested ordered measures ($\mathfrak{b}$-measures), encompassing absolute deviations from the mean, the trimmed mean, and the maximum. Finally, the quadratic ordered measures (Q-measures) comprise squared deviations from the mean, the trimmed mean, and the maximum.

The resulting tours for the ℓ, $\mathfrak{b}$, and Q measures are displayed in Figures 3.3, 3.4, and 3.5, respectively. Even in this simple example, the choice of objective function produces different solutions, highlighting the flexibility of the framework. The solution to the classical TSP, which minimizes the total tour length, is equivalent to minimizing the mean edge cost (ℓ-mean). In contrast, the ℓ-max solution (minimax TSP) generates a different tour. To avoid the long arc $(6, 7)$ present in the classical solution, it reroutes the path, resulting in a tour with a longer total length but a shorter maximum leg. This demonstrates a trade-off between efficiency (minimum total cost) and robustness (no single very long edge). The solutions obtained by minimizing absolute ($\mathfrak{b}$) and squared (Q) deviations also differ significantly. These objectives penalize dispersion in edge lengths, forcing them to be more homogeneous. For example, the solution minimizing the squared deviation with respect to the maximum (Q-max) produces a tour where edge lengths are visibly more similar to one another, effectively seeking an equitable distribution of travel distances. In summary, these results confirm that ordered measures offer a powerful and unified methodology for addressing a broad spectrum of objectives in the TSP, enabling the design of solutions tailored to practical requirements that go beyond simple cost minimization. Nevertheless, the proposed models may suffer from limited computational efficiency and, in such cases, the development of tailored formulations would be particularly beneficial.



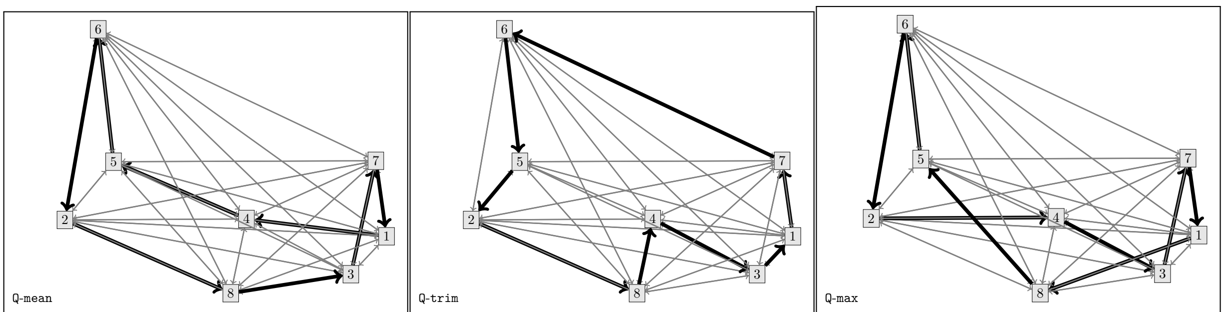
ℓ-mean ℓ-trimmed ℓ-max

Figure 3.3: TSP example solutions for different linear ordered measures.



$\mathfrak{b}$-mean $\mathfrak{b}$-trimmed $\mathfrak{b}$-max

Figure 3.4: TSP example solutions for different nested ordered measures.



Q-mean (variance) Q-trimmed Q-max

Figure 3.5: TSP example solutions for different quadratic ordered measures.

3.6.3 Weighted Multicover Set Problem

The *Set Covering Problem* (SCP) and its generalizations are fundamental combinatorial problems which provide models for resource allocation. In the *Weighted Multicover Set Problem* (WMCSP), a universe of elements is given, i.e., a collection of sets (each with an associated cost), and coverage requirements for each element. The goal is to select a minimum-cost subcollection of sets such that each element is covered at least a specified number of times. The standard objective is to minimize the sum of the costs of the selected sets. This problem has applications in various fields, such as healthcare, where items represent patients and sets correspond to different forms of medical care (e.g., nurses, physicians, medications, and medical assistance). Each patient is assigned a weight representing the severity of their illness or their level of urgency. A different application comes from telecommunication networks (as internet service providers or Wi-Fi routers to serve multiple users), were each service is allowed to *cover* a set of users, each of them with a different demand that can be supplied by different types of connections (activated sets).

Ordered measures introduce a new dimension to this problem, allowing for more sophisticated cost-management and fairness criteria. This is particularly relevant in capital budgeting or project selection, where the individual costs of selected projects (sets) are as important as their sum. For example, a decision-maker might want to minimize the cost of the most expensive set selected. This minimax WMCSP avoids solutions that, while potentially having a low total cost, rely on an individual component with an unacceptably high cost. This provides a mechanism to control budget spikes and ensure that the financial burden is not overly concentrated in a few selected items. In applications related to social good, where sets might represent social programs and costs represent resource allocation, such ordered objectives can be interpreted as measures of equity ensuring that no single program consumes a disproportionate share of the budget. The integration of these ordered measures into the WMCSP formulation maintains the problem's core structure while offering significantly more modeling flexibility.

Formally, in the WMCSP a set of m items and n subsets of these items are given, $S_1, \ldots, S_n \subset N = \{1, \ldots, n\}$ are given. Each item $j \in M = \{1, \ldots, m\}$ is endowed with a weight w_j that represents the preference of the item of its unit cost. The goal of the WMCSP is to decide which sets to accept to cover all the items by minimizing the weighted sum of the covered items.

$$y_i = \begin{cases} 1 & \text{if subset } S_i \text{ is selected,} \\ 0 & \text{otherwise,} \end{cases} \tag{3.47}$$

the number of items of type j can be calculated as $\sum_{\substack{i \in N: \\ j \in S_i}} y_i$, that is, the number of times an activated set contains the item j. Thus, the overall weight of item $j \in M$ for a solution induced by the y-variables is $x_j = w_j \sum_{\substack{i \in N: \\ j \in S_i}} y_i$. The usual aggregation of the weights is done using either the total sum or the average, which is then minimized to get a solution of the problem. Nevertheless, one can aggregate these costs using the measures described in this chapter.

The constraints of the WMCSP are the following:

$$\sum_{\substack{i \in N: \\ j \in S_i}} y_i \geq 1, \forall j \in M.$$

to assure that all the items are covered by at least one set.

To show the differences between some of the approaches, a simple instance of the problem is considered in Figure 3.6, with $m = 6$ items and $n = 10$ sets, where the sets are drawn at the top of the plot, items in the bottom (along with their weight), and the lines indicate the inclusion of the item in the set.

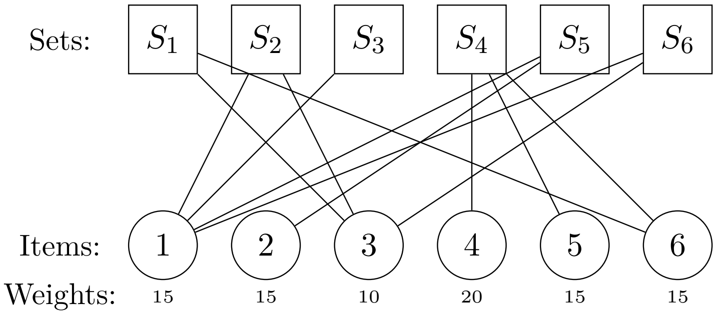
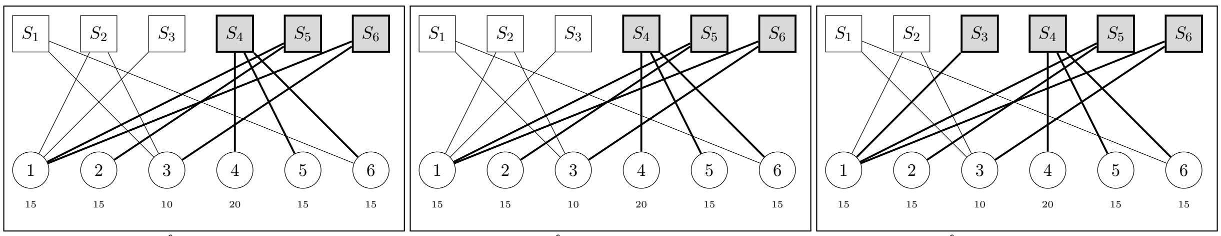


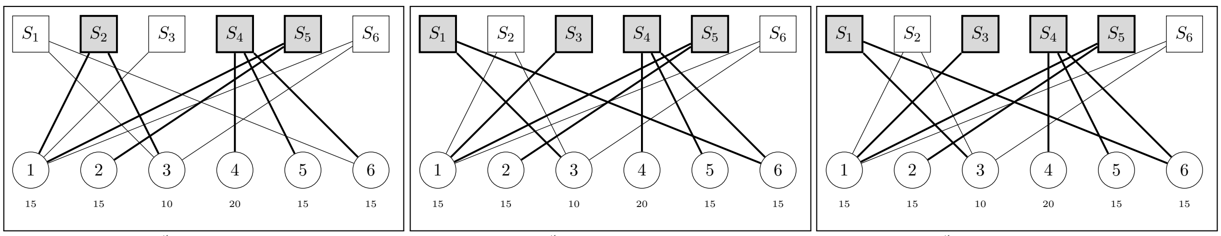
Figure 3.6: Instance for the WMCSP.

The solution of the WMCSP for this instance was computed using the ℓ, $\mathfrak{b}$, and Q measures, with λ induced by the mean (`mean`), 25%-25%-trimmed (`trimmed`), and the third quartile (`quartile3`). The matrix M is induced by the same λ for the squared loss (Example 3.2). Figures 3.7, 3.8, and 3.9 display the nine solutions obtained with these methodologies. In each case, the sets activated in the solution are highlighted in gray. Note that even in this toy example, the solutions obtained with different approaches differ from one another, reflecting the specific measure being minimized. This demonstrates that decision makers can leverage a unified framework to construct feasible solutions for the same problem while exploring different objectives.



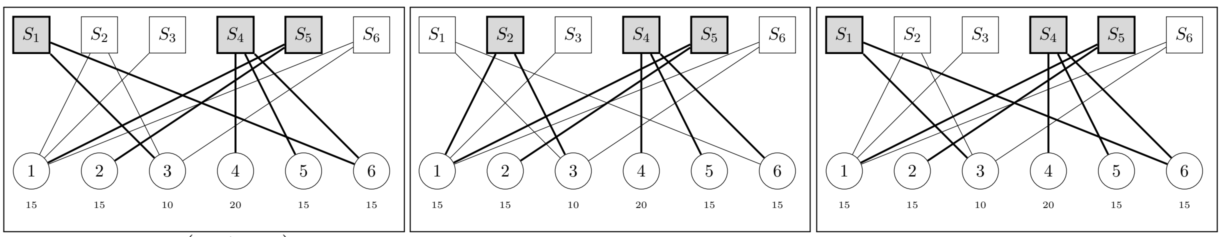
ℓ-`mean` ℓ-`trimmed` ℓ-`quartile3`

Figure 3.7: WMCSP example solutions for different linear ordered measures.



$\mathfrak{b}$-`mean` $\mathfrak{b}$-`trimmed` $\mathfrak{b}$-`quartile3`

Figure 3.8: WMCSP example solutions for different nested ordered measures.



`Q-mean` (variance) `Q-trimmed` `Q-quartile3`

Figure 3.9: WMCSP example solutions for different quadratic ordered measures.

"Per aspera ad astra"

Echo of the phrase *"non est ad astra mollis e terris via"*
from Seneca in his *Hercules furens* (1st century A.D.)

4

Application to Location Problems: Discrete Ordered Median Problem

This chapter presents the contributions of the dissertation related to the *Discrete Ordered Median Problem* (DOMP). New MILP formulations for DOMP are introduced, together with algorithmic enhancements that enable the problem to be solved to optimality using MILP-based techniques. The scope of the problem is also extended by allowing negative entries in λ, which makes it possible to address issues such as equity in facility location, obnoxious facility models, and other client preference settings. Starting from state-of-the-art formulations in the literature, several Benders decomposition reformulations are developed and applied. These approaches achieve substantial improvements, leading to new state-of-the-art results for a variety of ordered weighting vectors.

4.1 Introduction

The *Discrete Ordered Median Problem* (DOMP) is a well-known facility location problem where p facilities are to be located so that custumers are ranked according to their allocation cost to the nearest open facility, and then these costs are multiplied by a suitable weight vector λ. The goal is to decide which p facilities to open, so that the sum of λ-weighted allocation costs is minimized. The DOMP can be regarded as a flexible modeling framework in location analysis, as it generalizes several classical location models by incorporating a sorting feature into the underlying problem.

DOMP allows a straightforward and flexible generalization of many facility location problems since a large number of recognized problems in the literature can be modeled using an ordered median formulation. This is the case of the p-median, p-center or cent-dian problems, but also, of several models of undesirable or obnoxious location problems because the ordered median operator allows handling non-monotonic and negative vectors of weights. This type of objective function models preferences in location problems by means of the λ-vector of weights that multiplies the sorted vector of distances or costs from demand points to service facilities. This operator, namely the *ordered median objective function*, is described in full detail in Section 4.2.

An additional application of DOMP lies in its potential to provide a unified framework for incorporating and analyzing equity in location problems, specifically through the modeling of fairness in the allocation of customers to open facilities. The underlying rationale is that a solution that performs well at the system level may not be acceptable for individual parties if their allocation costs are disproportionately high compared to those of others. In this sense, DOMP enables the explicit consideration of compensation mechanisms for imbalances in allocation costs. The flexibility of the model allows for representing different forms of equity, thereby avoiding unfair outcomes. For instance, given the non-decreasing ordered sequence of allocation costs, by using a decreasing weight vector λ, clients with high allocation costs can be compensated (this is known as *reverse criterion*). In this regard, several criteria such as the *range* or the *sum of absolute differences*, among many others, can also be expressed within the ordered median framework (see, e.g., Mesa et al., 2003).

The ordered median objective function (Nickel & Puerto, 2005) has been successfully applied in the field of discrete location, typically set on networks (see, to cite some references, Nickel & Puerto, 1999; Kalcsics et al., 2003; Puerto & Tamir, 2005; Domínguez-Marín et al., 2005; Boland et al., 2006; Puerto et al., 2009; Marín et al., 2009; Kalcsics et al., 2010; Tang et al., 2011; López-Redondo et al., 2016; Labbé et al., 2017; Deleplanque et al., 2020; Marín et al., 2020; Pozo et al., 2024; Cherkesly et al., 2025), and also in continuous location (see, e.g., Drezner & Nickel, 2009; Krebs & Nickel, 2010; Blanco et al., 2013, 2014, 2016, 2023). Nevertheless, its applicability is not only limited to facility location analysis. The DOMP framework has also been incorporated into other well-known problems, giving rise to, e.g., ordered weighted average problems (Galand & Spanjaard, 2012; Fernández et al., 2014; Fernández et al., 2017) and ordered median hub location problems (Puerto et al., 2011a, 2013, 2016; Pozo et al., 2021). Moreover, it has found applications in disciplines a priori more distant to location analysis, including voting, portfolio optimization, outlier detection, and machine learning, among others, as discussed in Section 3.6 of Chapter 3.

Since the seminal works by Nickel & Puerto (1999) and Nickel (2001), several formulations and methods have been proposed to solve DOMP. For a short overview, in Boland et al. (2006) tailored branch-and-bound algorithms were presented which allow to solve problems of small size. More sophisticated methods, based on new formulations using radius variables, were presented in Marín et al. (2009, 2010); Labbé et al. (2017). In López-Redondo et al. (2016), a Lagrangian decomposition was applied to DOMP, in Deleplanque et al. (2020) a first branch-and-price-and-cut algorithm was specifically tailored for this problem and in Martínez-Merino et al. (2023) a row generation algorithm was also designed. More recently, a new solution approach that exploits partial monotonicity in efficient representations of the objective function has been presented in Marín et al. (2020). In spite of these many methods applied to solve DOMP, Benders decomposition has been never used to address this problem. Developing competitive Benders decompositions for DOMP and assessing their usefulness by comparing them with other methodologies is one of the goals of this chapter.

Benders decomposition is one of the most famous solution methods for difficult combinatorial optimization problems (Benders, 1962). Its hallmark is the capability of decomposing mixed-integer programming formulations into smaller subproblems, each of which can be solved individually to produce local information (notably, cutting planes) to be exploited by a centralized master problem. In general terms, the continuous variables of the original problem are projected out, resulting into an equivalent model with fewer variables, but many more constraints. To find an optimal solution, a large number of these constraints will not be required, suggesting then a strategy of separation that ignores all but a few of these constraints. The technique relies on projection and solving of smaller subproblems, combined with solution strategies of dualization, outer linearization and relaxation (Minoux, 1986; Lasdon, 2002). The reader is referred to Section 2.3.2 in Chapter 2 of this dissertation for further details on Benders decomposition.

The recent successful contributions of applying Benders decomposition on location problems (see, e.g., Fischetti et al., 2016, 2017; Cordeau et al., 2019; Coniglio et al., 2022; Duran-Mateluna et al., 2023; Gaar & Sinnl, 2022) and the lack of any attempt to solve DOMP using this technique motivate the developments in the current chapter. In this chapter, two state-of-the-art MILP formulations for the DOMP (Labbé et al., 2017; Marín et al., 2020) are extended, demonstrating how to further boost their computational performance by applying a clever but simple Benders decomposition.

After applying Benders decomposition techniques to the structure of the problem, state-of-the-art results are obtained for the most challenging instances of the DOMP, particularly in relation to the compensation vector λ. Most of the problems addressed in the existing DOMP literature focus on specific classes of λ-vectors, often restricted to non-negative values. In the few cases where negative entries are allowed, the structure of the λ-vectors typically ensures that the resulting objective function remains positive. Removing this assumption not only increases the difficulty of the problem but also introduces new features that are analyzed in this study. Also, although monotonicity of the λ-vector has been examined in several previous works on the DOMP, its *partial monotonicity* was explored in detail for the first time in Marín et al. (2020). However, no previous computational studies have developed a comprehensive methodology for solving problems with negative or partially negative λ-vectors that may yield negative non-monotonic objective function values. To the best of current knowledge, this section also represents the first use of the ordered median function to describe and solve obnoxious facility location problems through the selection of appropriate λ-vectors.

The remainder of the chapter is organized as follows. Section 4.2 formally defines the DOMP and introduces the notation used throughout. Section 4.3 reviews the radius formulation and proposes a first Benders decomposition approach. Section 4.4 presents the k-sums approach for DOMP along with three possible Benders decomposition reformulations. Section 4.5 describes the implementation details introduced to enhance the proposed models. Improvements for the bounds of the models and the strengthening of the Benders cuts are also discussed. The empirical performance of the resulting DOMP formulations is analyzed in Section 4.6, which compares these formulations across different scenarios.

4.2 Problem definition

In this section, potential facilities and customers are assumed to share the same set of locations indexed by $N = \{1, \dots, n\}$. This can be done without loss of generality, as otherwise, one could introduce dummy nodes and set the facility opening and allocation costs accordingly. Consider a cost matrix $C = (c_{ij})i, j \in N$, where $c_{ij} \geq 0$ represents the cost of allocating the client placed at position i to the facility opened at location j. It is consider the more general situation in which free self-service is not assumed, therefore in general c_{ii} is not necessarily equal to zero. DOMP assumes single-allocation, i.e., each client node has a unique facility where it is allocated.

To model the problem, the following binary variables are defined where $y_j = 1$ if and only if the facility at position j is open, and 0 otherwise, and $x_{ij} = 1$ if and only if the client located at position i is allocated to the facility at j, and 0 otherwise. With this notation, the p-median polytope (see Section 2.6.1 in Chapter 2) can be described as:

$$X = \{x \in [0,1]^{n \times n}, y \in [0,1]^n : \sum_{j \in N} y_j = p; \quad \sum_{j \in N} x_{ij} = 1,\ i \in N; \quad x_{ij} \leq y_j,\ i, j \in N; \tag{4.1a}$$

$$\sum_{\substack{j \in N: \\ c_{ij} > c_{im}}} x_{ij} + y_m \leq 1,\ i, m \in N\}. \tag{4.1b}$$

Since negative vector weights will be considered, in order to ensure that each client is allocated to a closest open facility (ties are randomly broken), the description of the p-median polytope includes the closest assignment constraints of clients to facilities (4.1b) reported in Espejo et al. (2012). These constraints guarantee that a client i cannot be allocated to a facility j if allocating i to j is is higher than the cost of allocating it to given facility m (that is, if $c_{ij} > c_{im}$), provided than facility m is opened. In the following, X_I denotes the integer lattice points within X. For a given feasible point $(x, y) \in X_I$, its assignment costs are given as:

$$c(x) := \left(\sum_{j \in N} c_{1j} x_{1j}, \dots, \sum_{j \in N} c_{nj} x_{nj} \right). \tag{4.2}$$

Next, the formal definition of DOMP is provided.

Definition 4.2.1 — DOMP. For a given solution $(x, y) \in X_I$, consider a reordering of $c(x)$ that satisfies $c_{(1)}(x) \leq c_{(2)}(x) \leq \cdots \leq c_{(n)}(x)$. Given a vector of (not necessarily non-negative) weights $(\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n$, DOMP seeks to find a solution $(x, y) \in X_I$ such that $\sum_{k \in N} \lambda_k c_{(k)}(x)$ is minimized.

Given a vector d of n real numbers, where $(d_{(\ell)})_{\ell=1,\dots,n}$ are the components of d sorted in non-decreasing order, let $S_k(d) = \sum_{\ell=k}^{n} d_{(\ell)}$ denote the sum of the largest $n - k + 1$ elements of d. Then the objective function of DOMP can be restated as

$$\sum_{k \in N} \lambda_k c_{(k)}(x) = \sum_{k \in N} \Delta_k S_k(c(x)), \tag{4.3}$$

where $\lambda_0 := 0$ and $\Delta_k := \lambda_k - \lambda_{k-1}$ for all $k \in N$. For further purposes, also let the sets $\Delta^- = \{k \in N | \Delta_k < 0\}$ and $\Delta^+ = \{k \in N | \Delta_k > 0\}$ be defined.

Using different definitions of the λ-vector of weights, or its Δ description given by (6.3), many prominent facility location problems can be defined. For instance, the p-median with $\lambda = (1, 1, ..., 1, 1)$, the p-center with $\lambda = (0, 0, ..., 0, 1)$, the α-centdian with $\lambda = (\alpha, \alpha, ..., \alpha, \alpha, 1)$, the Hurwitz criterion with $\lambda = (\alpha, 0, ..., 0, 1-\alpha)$, the k-max criterion with $\lambda = (0, \dots, 0, \underbrace{1}_{k-th}, 0, \dots, 0)$, the sum of absolute differences with $\lambda = (2(2i - n - 1))_{i \in N}$ or the range with $\lambda = (-1, 0, ..., 0, 1)$, among many others.

DOMP is NP-hard, as it contains the p-median problem as a special case ($\lambda_i = 1$, $\forall i \in N$). Many different MILP formulations for DOMP have been studied in the past two decades, see Nickel & Puerto (2005); Boland et al. (2006); Marín et al. (2009, 2010); Labbé et al. (2017); Puerto & Rodríguez-Chía (2019); Marín et al. (2020). The next section focuses on two state-of-the-art formulations and examines how Benders decomposition can be efficiently applied to them. The first is the *radius formulation*, for which a valid Benders reformulation can be obtained when λ takes non-increasing values. The second is the *partial monotonicity formulation*, which introduces three-index variables to enforce the ordering of allocation costs. This latter approach is more general, as it permits the derivation of a valid Benders reformulation for any $\lambda \in \mathbb{R}^n$.

4.3 DOMP radius formulation

4.3.1 Compact radius formulation

Consider the ordered sequence of unique allocation costs of $c_{ij} \geq 0$, $i, j \in N$:

$$c_{(0)} := 0 < c_{(1)} < c_{(2)} < \ldots < c_{(g)} = \max_{i,j \in N} c_{ij}, \tag{4.4}$$

where g is the number of different non-zero elements of the above allocation cost sequence and let $H := \{1, \dots, g\}$. This ordering can be used to perform the sorting process of the allocation costs. Let the following variables, namely *radius variables* in the following, be defined for $\ell \in N, h \in H$:

$$u_{\ell h} = \begin{cases} 1 & \text{if the } \ell\text{-th allocation cost is at least } c_{(h)}, \\ 0 & \text{otherwise,} \end{cases} \tag{4.5}$$

that is to say the ℓ-th smallest cost is equal to $c_{(m)}$ if and only if $u_{\ell m} = 1$ and $u_{\ell m+1} = 0$, $1 \leq m < g$.

Formulations using radius variables (see Elloumi et al., 2004; Puerto, 2008; Espejo et al., 2009; Marín et al., 2009, 2010; Puerto et al., 2011a; García et al., 2011; Pozo et al., 2024) can be used to reduce the number of variables and constraints in use. Indeed, note that radius formulation is defined over a set H whose cardinality depends on the number of different non-zero elements in the cost matrix. Thus, this formulation leverages of repeated entries in the cost matrix. This happens, for instance, when the location problem assumes free self-service and symmetrical allocation costs, what implies a significant number of repetitions in the cost matrix and a diagonal of zeros.

A general scheme which models DOMP using radius variables is shown below:

$$\begin{aligned}
\mathsf{DOMP}_{r0} : \min \sum_{\ell \in N} \sum_{h \in H} & \lambda_\ell u_{\ell h} (c_{(h)} - c_{(h-1)}), && && (4.6a)\\
\text{s.t.} \sum_{\ell \in N} u_{\ell h} = & \sum_{\substack{i,j \in N: \\ c_{ij} \geq c_{(h)}}} x_{ij}, && h \in H, && (4.6b)\\
& u_{\ell h} \leq u_{\ell+1 h}, && h \in H, \ell \in N : \ell < |N|, && (4.6c)\\
& u_{\ell h} \geq u_{\ell h+1}, && h \in H, \ell \in N : h < |H|, && (4.6d)\\
& (x, y) \in X_I, && && (4.6e)\\
& u_{\ell h} \in \{0, 1\}, && h \in H, \ell \in N. && (4.6f)
\end{aligned}$$

Constraints (4.6b) state that the number of allocations whose cost is at least $c_{(h)}$ must be equal to the number of sites that support allocation costs greater than or equal to $c_{(h)}$. Constraints (4.6c) and (4.6d) sort the values of u variables in non-decreasing order. As indicated in Marín et al. (2009) and Labbé et al. (2017), the extra sorting group of constraints (4.6d) are redundant, nevertheless, they have been shown to improve the computational performance, thus, they are included in the model.

4.3.2 Benders decomposition for the radius formulation

Radius formulation has been proven to be one of the most efficient formulations when λ is monotone increasing or decreasing (see, e.g., Labbé et al., 2017). Nevertheless, when turning to a non-monotone ordered weighting vector, the performance of the formulation has been recently improved by some alternative approaches (Marín et al., 2020). In this section, Benders decomposition approach based on projecting out u variables is proposed. The decomposition is obtained from the DOMP_{r0} model after relaxing the integrality conditions on the radius variables. However, for certain configurations of the λ-vector, the radius variables must remain binary, which is why the proposed Benders decomposition approach only provides valid lower bounds in such cases. In spite of that, it will proved that for monotone non-increasing values of λ, the decomposition approach is exact and it provides a new valid reformulation for DOMP.

The following result holds:

Proposition 4.3.1 The following compact model

$$\mathsf{DOMP}_{r0B} : \min \sum_{h \in H} \theta_h (c_{(h)} - c_{(h-1)}), \tag{4.7a}$$

$$\text{s.t. } \theta_h \geq \lambda_\ell \sum_{\substack{i,j \in N: \\ c_{ij} \geq c_{(h)}}} x_{ij} - \sum_{\ell < \ell'} (\lambda_\ell - \lambda_{\ell'}), \qquad h \in H, \ell \in N, \tag{4.7b}$$

$$(x, y) \in X_I. \tag{4.7c}$$

is a reformulation of the model DOMP_{r0} for any monotone non-increasing vector λ.

Proof. Start with the relaxation of the model DOMP_{r0} in which constraints (4.6f) are relaxed into $0 \leq u_{\ell h} \leq 1$, $h \in H, \ell \in N$. First, the focus is on monotone non-increasing values of λ and it is shown that the model (4.7) can be obtained by projecting out u variables from this relaxation. To this end, terms $\sum_{\ell \in N} \lambda_\ell u_{\ell h}$ in the objective function (4.6a) are replaced by new variables θ_h, $h \in H$.

Given $\bar{h} \in H$, and a point $(x^*, y^*) \in X$, denote by $x^*_{\bar{h}} = \sum_{\substack{i,j \in N: \\ c_{ij} \geq c_{(\bar{h})}}} x_{ij}$ and consider the following subproblem:

$$\min \sum_{\ell \in N} \lambda_\ell u_{\ell \bar{h}}, \tag{4.8a}$$

$$\sum_{\ell \in N} u_{\ell \bar{h}} = x^*_{\bar{h}}, \tag{4.8b}$$

$$u_{\ell \bar{h}} \leq u_{\ell+1 \bar{h}}, \qquad \ell \in N : \ell < |N|, \tag{4.8c}$$

$$0 \leq u_{\ell \bar{h}} \leq 1, \qquad \ell \in N. \tag{4.8d}$$

When λ is monotone non-increasing, the constraints (4.8c) are redundant, and the problem (4.8) can be restated as a continuous knapsack problem in which constraints (4.8c) are removed. Such obtained linear program is always feasible and bounded, so one can build its dual, in which a variable α is associated to constraint (4.8b) and variables β_ℓ, $\ell \in N$, are associated to constraints (4.8d):

$$\max \alpha x^*_{\bar{h}} - \sum_{\ell \in N} \beta_\ell, \tag{4.9a}$$

$$\alpha - \beta_\ell \leq \lambda_\ell, \qquad \ell \in N, \tag{4.9b}$$

$$(\alpha, \beta) \geq 0. \tag{4.9c}$$

Let $\mathcal{P} = \{(\alpha, \beta) \geq 0 \mid (\alpha, \beta) \text{ satisfy (4.9b)-(4.9c)}\}$ be the dual polyhedron and $\mathcal{P}^{ext}$ be the set of extreme points of $\mathcal{P}$, then the general form of Benders optimality cuts as a function of the dual and primal solution is given as:

$$\theta_h \geq \alpha^* \sum_{\substack{i,j \in N: \\ c_{ij} \geq c_{(\bar{h})}}} x_{ij} - \sum_{\ell \in N} \beta^*_\ell, \qquad (\alpha^*, \beta^*) \in \mathcal{P}^{ext}, \tag{4.10}$$

It is well known from knapsack theory that the following algorithm produces an optimal primal solution and optimal dual variables. Let k^* be the index of the *critical item* of the knapsack problem above, the optimal primal solution is

$$u^*_{\ell \bar{h}} = \begin{cases} 1 & \text{if } \ell > k^*, \\ x^*_{\bar{h}} - (|N| - k^*) & \text{if } \ell = k^*, \\ 0 & \text{otherwise,} \end{cases} \qquad \ell \in N. \tag{4.11}$$

Hence, the optimal value of the primal problem is

$$\sum_{\ell > k^*} \lambda_\ell + \lambda_{k^*}(x^*_{\bar{h}} - (|N| - k^*)) = \lambda_{k^*} x^*_{\bar{h}} - \sum_{\ell > k^*} (\lambda_{k^*} - \lambda_\ell), \tag{4.12}$$

and the dual variables can be computed as:

$$\begin{aligned} &\alpha^* = \lambda_{k^*}, \\ &\beta^*_\ell = \begin{cases} \lambda_{k^*} - \lambda_\ell & \text{if } \ell > k^*, \\ 0 & \text{otherwise}, \end{cases} \qquad \ell \in N. \end{aligned} \tag{4.13}$$

Generalizing for any possible critical item, one gets the following family of Benders optimality cuts:

$$\theta_h \geq \lambda_\ell \sum_{\substack{i,j \in N: \\ c_{ij} \geq c_{(h)}}} x_{ij} - \sum_{\ell' > \ell} (\lambda_\ell - \lambda_{\ell'}), \quad h \in H, \ell \in N. \tag{4.14}$$

Now, it remains to show that for monotone non-increasing values of λ, for the relaxation of the model DOMP_{r0} in which constraints (4.6f) are replaced by $0 \leq u_{\ell h} \leq 1$, $h \in H, \ell \in N$, there always exists an optimal solution in which the u variables take integer values. Indeed, this is immediate from formula (4.11): whenever a feasible point $(x^*, y^*) \in X_I$ is given, the optimal solution consists of setting the last $x^*_{\bar{h}}$ values of u to one, and the remaining ones to zero. ■

Corollary 4.3.2 For any general (not necessarily monotone non-increasing) vector λ, the optimal value of model (4.7) provides a valid lower bound for DOMP.

Proof. In case λ is not monotone non-increasing, constraints (4.8c) in Problem (4.8) cannot be avoided, and hence the reduction to the continuous knapsack problem does not work. Moreover, solutions to the LP-relaxation of DOMP_{r0} are not always integer so that (4.14) only provides valid inequalities which could not ensure optimality of the master problem. ■

4.4 DOMP k-sums formulation

As already announced, the objective is to provide improved reformulations for DOMP under general weights and general costs. The Benders reformulation developed in the previous section is only valid for monotone non-increasing λ-vectors. Therefore, this section addresses the problem in full generality. The exposition first elaborates on the state-of-the-art formulation for DOMP presented in Marín et al. (2020), and subsequently demonstrates how its variables can be projected out to obtain three different Benders reformulations.

4.4.1 Compact k-sums formulation

Marín et al. (2020) introduced a three-index formulation in which the variables θ^k_{ij} are set to 1 if the cost of allocating client i to facility j is sorted in position k, and they are set to 0 otherwise ($i, j, k \in N$). Even though the variables are defined as binary, without loss of generality, they can be relaxed to be continuous in the interval $[0, 1]$. This fact gives rise to a rather efficient problem formulation which is presented below:

$$\mathsf{DOMP}_\theta : \min \sum_{k \in \Delta^+} \Delta_k((n - k + 1)t_k + \sum_{i \in N} z_{ik}) \tag{4.15a}$$

$$+ \sum_{k\in\Delta^-} \Delta_k \sum_{i\in N}\sum_{j\in N} c_{ij}\theta_{ij}^k,$$

$$\text{s.t. } t_k + z_{ik} \geq \sum_{j\in N} c_{ij}x_{ij}, \qquad i\in N, k\in\Delta^+, \tag{4.15b}$$

$$\sum_{i\in N}\sum_{j\in N}\theta_{ij}^k = n-k+1, \qquad k\in\Delta^-, \tag{4.15c}$$

$$0 \leq \theta_{ij}^k \leq x_{ij}, \qquad i,j\in N, k\in\Delta^-, \tag{4.15d}$$

$$(x,y)\in X_I, \tag{4.15e}$$

$$z_{ik}, t_k \geq 0, \qquad i\in N, k\in\Delta^+. \tag{4.15f}$$

The objective function (4.15a) has two parts: the first one accounts for the sum of the terms $S_k(c(x))$ for $k\in\Delta^+$, and the second one does the same for $k\in\Delta^-$. Constraints (4.15b) allow for a valid representation of $S_k(c(x))$ for $k\in\Delta^+$. The remaining constraints, namely (4.15c) and (4.15d), are used to get a valid representation of $S_k(c(x))$ for $k\in\Delta^-$, where (4.15d) define the range of the variables θ_{ij}^k. Since x_{ij} are binary and θ_{ij}^k appears in the objective function with a negative coefficient ($\Delta_k < 0$), if $x_{ij}=1$ then $\theta_{ij}^k = 1$, thus in the optimal solution $\theta_{ij}^k \in \{0,1\}$. The remaining variables in (4.15f), namely z and t, are obtained from the dual of the k-sum problem as explained in Ogryczak & Tamir (2003); Kalcsics et al. (2002); Puerto et al. (2017); Marín et al. (2020), or as discussed in Section 3.2 of Chapter 3.

4.4.2 Benders decomposition for the k-sums formulation

Benders reformulations proposed in this section exploit the fact that the variables z, t and θ of the formulation DOMP_θ are continuous. Moreover, they belong to two well separated components in both the objective function and constraints, obtained by considering $k\in\Delta^+$ and $k\in\Delta^-$, respectively. Hence, Benders reformulations can be considered by projecting out (z,t) variables, θ variables, or both. For this purpose, it is necessary to introduce the following order relation between cost indexes, previously considered by Labbé et al. (2017):

$$ij \prec \hat{\imath}\hat{\jmath} := \begin{cases} c_{ij} < c_{\hat{\imath}\hat{\jmath}} \\ \text{or} \\ c_{ij} = c_{\hat{\imath}\hat{\jmath}} \quad \text{and } i < \hat{\imath} \\ \text{or} \\ c_{ij} = c_{\hat{\imath}\hat{\jmath}}, \quad i = \hat{\imath} \quad \text{and } j < \hat{\jmath}. \end{cases} \tag{4.16}$$

■ **Example 4.1** As an illustrative example, consider the following matrix:

$$\begin{pmatrix} 0 & 2 & 7 & 4 \\ 1 & 0 & 5 & 5 \\ 3 & 6 & 0 & 2 \\ 9 & 4 & 1 & 0 \end{pmatrix}, \tag{4.17}$$

then the order of couples ij by means of the above order preference is

$$11 \prec 22 \prec 33 \prec 44 \prec 21 \prec 43 \prec 12 \prec 34 \prec 31 \prec 14 \prec 42 \prec 23 \prec 24 \prec 32 \prec 13 \prec 41.$$

■

Let $H = \{1,\dots,n^2\}$, then there exists an bijection between costs in C and the elements of the ordered costs vector $(c_{ij(h)}|h\in H)^{\prec}_{i,j\in N}$, such that for every (i,j) pair there is a single $ij(h)$ associated to it for a certain $h\in H$. Note that while H in the previous section has a length of the number of unique costs of C, here the ordering of costs has the same number of elements as total costs.

Next, it is shown that for $k \in \Delta^-$ one can obtain a valid reformulation of DOMP_θ by projecting out θ variables. The advantage of this new model (that will be denoted by $\mathsf{DOMP}_{\theta 1}$) is in the reduction of the size of the underlying formulation. While the starting model DOMP_θ comprises $\mathcal{O}(|N|^3)$ variables and $\mathcal{O}(|N|^3)$ constraints, the new model requires only $\mathcal{O}(|N|^2)$ variables and $\mathcal{O}(|N|^3)$ constraints.

Proposition 4.4.1 The following model:

$$
\begin{aligned}
\mathsf{DOMP}_{\theta 1} : \min \ & \sum_{k \in \Delta^+} \Delta_k((n-k+1)t_k + \sum_{i \in N} z_{ik}) + \sum_{k \in \Delta^-} \Delta_k \varphi_k, && && (4.18a)\\
\text{s.t. } & t_k + z_{ik} \geq \sum_{j \in N} c_{ij} x_{ij}, && i \in N, k \in \Delta^+, && (4.18b)\\
& 0 \leq \varphi_k \leq (n-k+1) c_{ij(h)} + \sum_{ij: c_{ij} > c_{ij(h)}} (c_{ij} - c_{ij(h)}) x_{ij}, && h \in H, k \in \Delta^-, && (4.18c)\\
& (x, y) \in X_I, && && (4.18d)\\
& z_{ik}, t_k \geq 0, && i \in N, k \in \Delta^+. && (4.18e)
\end{aligned}
$$

is a reformulation of the model DOMP_θ.

Proof. Terms $\sum_{i \in N} \sum_{j \in N} c_{ij} \theta_{ij}^k$, are replaced in the objective function (4.15a) by new variables $\varphi_k \geq 0$, for all $k \in \Delta^-$. Then, the following relaxed master problem can be considered:

$$
\begin{aligned}
\min \ & \sum_{k \in \Delta^+} \Delta_k((n-k+1)t_k + \sum_{i \in N} z_{ik}) + \sum_{k \in \Delta^-} \Delta_k \varphi_k, && && (4.19a)\\
\text{s.t. } & (4.15b), (4.15e), (4.15f), && &&\\
& 0 \leq \varphi_k \leq \mathcal{U}_k, && k \in \Delta^-, &&
\end{aligned}
$$

where $\mathcal{U}_k$ are valid upper bounds for φ_k. Clearly, a simple, but valid upper bound for φ_k is the sum of the $(n-k+1)$ largest costs in C, while others will be derived below.

Let (x^*, y^*) be a solution of the above LP. To ensure that (x^*, y^*) corresponds to an optimal DOMP solution, one needs to solve the so-called Benders subproblem which is decomposable by k. For any $\bar{k} \in \Delta^-$, since $\Delta_{\bar{k}} < 0$, the $\bar{k}$-th Benders subproblem is a *maximization* problem given as:

$$
\begin{aligned}
\max \ & \sum_{i \in N} \sum_{j \in N} c_{ij} \theta_{ij}^{\bar{k}}, && && (4.20a)\\
\text{s.t. } & \sum_{i \in N} \sum_{j \in N} \theta_{ij}^{\bar{k}} \leq n - \bar{k} + 1, && && (4.20b)\\
& 0 \leq \theta_{ij}^{\bar{k}} \leq x_{ij}^*, && i, j \in N. && (4.20c)
\end{aligned}
$$

Note that constraints (4.20b) are stated as inequalities instead of equalities. This can be done without loss of generality, because the values $c_{ij} \geq 0$, for all $i, j \in N$. Moreover, observe that the LP-model (4.20) is always feasible and bounded, which follows from the fact that $\sum_{i,j \in N} x_{ij}^* = n$. Hence, one can consider its LP-dual in which a variable α is associated to constraint (4.20b) and variables β_{ij}, $i, j \in N$, are associated to constraints (4.20c):

$$
\min \ (n - \bar{k} + 1)\alpha + \sum_{i,j \in N} \beta_{ij} x_{ij}^*, \tag{4.21a}
$$

$$\text{s.t. } \beta_{ij} + \alpha \leq c_{ij}, \qquad i,j \in N \tag{4.21b}$$
$$(\alpha, \beta) \geq \mathbf{0}. \tag{4.21c}$$

Let $h^* \in H$ be the index of a critical item of the continuous knapsack problem defined by (4.20), namely the index h^* for which the following property holds:

$$\sum_{h>h^*} x^*_{ij(h)} < n - \bar{k} + 1 \leq \sum_{h \geq h^*} x^*_{ij(h)}. \tag{4.22}$$

Then, the optimal solution of the model (4.20) is given as

$$\theta^{\bar{k}}_{ij(h)} = \begin{cases} x^*_{ij(h)} & \text{if} \quad h > h^*, \\ (n - \bar{k} + 1) - \sum_{h>h^*} x^*_{ij(h)} & \text{if} \quad h = h^*, \\ 0 & \text{otherwise.} \end{cases} \tag{4.23}$$

It can be verified that the optimum value of the primal subproblem satisfies

$$\sum_{i \in N} \sum_{j \in N} c_{ij} \theta^{\bar{k}}_{ij} = \sum_{h>h^*} c_{ij(h)} x^*_{ij(h)} + c_{ij(h^*)} \Big[(n - \bar{k} + 1) - \sum_{h>h^*} x^*_{ij(h)} \Big] \tag{4.24}$$
$$= c_{ij(h^*)}(n - \bar{k} + 1) + \sum_{h>h^*} (c_{ij(h)} - c_{ij(h^*)}) x^*_{ij(h)} \tag{4.25}$$

and, therefore, the optimal dual solution is

$$\alpha = c_{ij(h^*)},$$
$$\beta_{ij} = \begin{cases} c_{ij(h)} - c_{ij(h^*)} & \text{if } h > h^* \quad (\iff c_{ij(h)} > c_{ij(h^*)}), \\ 0 & \text{otherwise.} \end{cases} \tag{4.26}$$

Generalizing for any possible critical item $h \in H$, one gets the following family of Benders optimality cuts for every $k \in \Delta^-$ and every $h \in H$:

$$\varphi_k \leq (n - k + 1) c_{ij(h)} + \sum_{ij \,:\, c_{ij} > c_{ij(h)}} (c_{ij} - c_{ij(h)}) x_{ij}. \tag{4.27}$$

■

For the values $k \in \Delta^+$, it is next shown how to project out (z, t) variables.

Proposition 4.4.2 The following model:

$$\mathsf{DOMP}_{\theta 2} : \min \sum_{k \in \Delta^+} \Delta_k \varphi_k + \sum_{k \in \Delta^-} \Delta_k \sum_{i \in N} \sum_{j \in N} c_{ij} \theta^k_{ij}, \tag{4.28a}$$
$$\text{s.t. } \varphi_k \geq \sum_{i \in N_L} \sum_{j \in N} c_{ij} x_{ij}, \qquad k \in \Delta^+, N_L \subseteq N : |N_L| = n - k + 1, \tag{4.28b}$$
$$\sum_{i \in N} \sum_{j \in N} \theta^k_{ij} = n - k + 1, \qquad k \in \Delta^-, \tag{4.28c}$$
$$0 \leq \theta^k_{ij} \leq x_{ij}, \qquad i, j \in N, k \in \Delta^-, \tag{4.28d}$$
$$(x, y) \in X_I, \tag{4.28e}$$

is a reformulation of the model DOMP_θ.

Proof. Terms $(n-k+1)t_k + \sum_{i\in N} z_{ik}$, for $k \in \Delta^+$, are replaced in the objective function (4.15a) by new variables $\varphi_k \geq 0$. Let (x^*, y^*) be a solution satisfying constraints (4.15c) and (4.15e). Again, Benders subproblem is separable by k, so the following LP associated to a given $\bar{k} \in \Delta^+$ is solved:

$$\min\ (n-\bar{k}+1)t_{\bar{k}} + \sum_{i\in N} z_{i\bar{k}}, \tag{4.29a}$$

$$\text{s.t. } t_{\bar{k}} + z_{i\bar{k}} \geq \sum_{j\in N} c_{ij}x^*_{ij}, \qquad i \in N, \tag{4.29b}$$

$$z_{i\bar{k}}, t_{\bar{k}} \geq 0, \qquad i \in N. \tag{4.29c}$$

This subproblem is always feasible and bounded, hence one can consider its LP-dual, in which variables $\alpha_i, i \in N$, are introduced for every constraint in (4.29b):

$$\max\ \sum_{i\in N} \alpha_i \sum_{j\in N} c_{ij}x^*_{ij}, \tag{4.30a}$$

$$\text{s.t. } \sum_{i\in N} \alpha_i \leq n-\bar{k}+1, \tag{4.30b}$$

$$0 \leq \alpha_i \leq 1, \qquad i \in N. \tag{4.30c}$$

In order to compute the optimal solution of model (4.30), one needs to sort $i \in N$ in increasing order according to $v^*_i = \sum_{j\in N} c_{ij}x^*_{ij}$. It is now enough to consider the set $N^*_L \subseteq N$ with the $(n-\bar{k}+1)$ indexes associated with the largest v^*_i values, i.e., $\alpha_i = 1$ if $i \in N^*_L$, and $\alpha_i = 0$, otherwise. Since the choice of N^*_L depends on x^*, after generalizing over all possible values of x^*, the following family of Benders optimality cuts associated to subproblem $\bar{k} \in \Delta^+$ is obtained:

$$\varphi_k \geq \sum_{i\in N_L} \sum_{j\in N} c_{ij}x_{ij} \qquad N_L \subseteq N, |N_L| = n-k+1. \tag{4.31}$$

■

By merging the results of Proposition 4.4.1 and 4.4.2, a third problem reformulation is obtained.

Proposition 4.4.3 The following model:

$$\mathsf{DOMP}_{\theta 3} : \min \sum_{k\in N} \Delta_k \varphi_k, \tag{4.32a}$$

$$\text{s.t. } \varphi_k \geq \sum_{i\in N_L} \sum_{j\in N} c_{ij}x_{ij}, \qquad k \in \Delta^+, N_L \subseteq N : |N_L| = n-k+1, \tag{4.32b}$$

$$\varphi_k \leq (n-k+1)c_{ij(h)} + \sum_{ij:\ c_{ij}>c_{ij(h)}} (c_{ij} - c_{ij(h)})x_{ij}, \quad k \in \Delta^-, h \in H, \tag{4.32c}$$

$$(x, y) \in X_I, \tag{4.32d}$$

is a reformulation of the model DOMP_θ.

Finally, it is pointed out that one can consider aggregated formulations in which a single variable ϕ^- or ϕ^+ is introduced to replace the term in the objective function associated with negative, respectively, positive multipliers Δ_k. This can be done for each of the previously introduced models. In the following, this scenario is illustrated for the model $\mathsf{DOMP}_{\theta 1}$, introducing a new variable $\phi = \sum_{k\in\Delta^-} \varphi_k$.

Proposition 4.4.4 The following model

$$\mathsf{DOMP}_{\theta 1a}: \min \sum_{k\in\Delta^+} \Delta_k((n-k+1)t_k + \sum_{i\in N} z_{ik}) + \phi, \tag{4.33a}$$

$$t_k + z_{ik} \geq \sum_{j\in N} c_{ij}x_{ij}, \qquad i\in N, k\in\Delta^+, \tag{4.33b}$$

$$\phi \geq \sum_{k\in\Delta^-} \Delta_k\big[(n-k+1)c_{ij(h_k)} + \sum_{ij:\, c_{ij}>c_{ij(h_k)}} (c_{ij}-c_{ij(h_k)})x_{ij}\big],$$

$$(h_k)_{k\in\Delta^-} \in H\times H\times\ldots\times H, \tag{4.33c}$$

$$(x,y)\in X_I, \tag{4.33d}$$

$$z_{ik}, t_k \geq 0, \qquad i\in N, k\in\Delta^+. \tag{4.33e}$$

is a reformulation of the model DOMP_θ.

While the reformulation $\mathsf{DOMP}_{\theta 1}$ contains polynomial number of constraints and variables, its aggregated counterpart $\mathsf{DOMP}_{\theta 1a}$ consists of an exponential number of Benders optimality cuts (4.33c).

4.4.3 Bounds and reformulation strengthening

To strengthen the linear relaxations of the previous formulations, several bounds can be proposed. These bounds are only discussed for the models that will be tested in the computational study (Section 4.6). To this end, let $c_{max} = \max\{c_{ij} \mid i,j\in N\}$ and let $c_{min} = \min\{c_{ij} \mid i,j\in N\}$. For the disaggregated model $\mathsf{DOMP}_{\theta 1}$, the following bounds on the value of φ_k, $k\in\Delta^-$, are valid:

$$(n-k+1)c_{min} \leq \varphi_k \leq (n-k+1)c_{max}. \tag{4.34}$$

The reader might observe that in the case of free self-service ($c_{ii}=0, c_{ij}>0, i\neq j$) one can obtain more accurate lower bounds. Indeed, let $c^+_{min} = \min\{c_{ij} \mid i,j\in N, c_{ij}\neq 0\}$, recall that φ_k represents the sum of the $n-k+1$ largest costs, i.e, $S_k = \sum_{l=k}^n c_{(l)}$. Therefore, under the free self-service assumptions, one can have:

- If $n-k+1\leq p$, all the addends could assume value 0 in the lower bound and therefore the lower bound in this case will be $0\leq\varphi_k$.
- If $n-k+1>p$, the number of addends in S_k is greater than p and, therefore, at most p of them could assume value zero in the lower bound and the remainder c^+_{min}, i.e, the lower bound in this case is $0\cdot p + (n-k+1-p)c^+_{min}$.

Hence the following bound are valid in this scenario:

$$\begin{cases} 0\leq\varphi_k\leq(n-k+1)c_{max} & \text{if } n-k+1\leq p,\\ (n-k+1-p)c^+_{min}\leq\varphi_k\leq(n-k+1)c_{max} & \text{otherwise.}\end{cases} \tag{4.35}$$

The above upper bound can also be enhanced. To this end, consider the set of maximum allocation costs for every client, $\tilde{c} = (\max\{c_{ij}|\ j\in N_{max}\})_{i\in N}$, where $N_{max}\subset N$ represents a set of open facilities obtained by solving the auxiliary problem

$$\max\{\sum_{i,j\in N} c_{ij}x_{ij} | (x,y)\in X_I\}, \tag{4.36}$$

where a subset of p facilities that maximizes the allocation costs is sought (with additional closest facility allocation constraints). The set N_{max} corresponds to p columns in the cost matrix C, where the $(n-k+1)$ highest costs in $\tilde{c}$ for each $k \in N$ can be considered to propose a tighter bound:

$$\varphi_k \leq \sum_{h=n-k+1}^{n} \tilde{c}_{(h)}, \tag{4.37}$$

where $\tilde{c}(h)$ is the h-th element of the non-decreasing sorted vector $\tilde{c}$.

For the aggregated model $\mathsf{DOMP}_{\theta 1a}$, several bounds can also be proposed for ϕ, e.g:

$$\sum_{k \in \Delta^-} \Delta_k (n-k+1) c_{max} \leq \phi \leq 0. \tag{4.38}$$

Instead one can consider, similarly to the disaggregated case, the $n-k+1$ highest single allocation costs,

$$\sum_{k \in \Delta^-} \Delta_k \Big(\sum_{h=n-k+1}^{n} \tilde{c}_{(h)} \Big) \leq \phi \leq 0. \tag{4.39}$$

Indeed, assuming non-negative λ, or a distribution of λ that ensures the objective function is non-negative, and given that the costs are non-negative, the optimal objective function of the problem (see (4.33a)) must be non-negative. Therefore, the sum of the contributions associated with positive $k \in \Delta^+$ (those appearing on the left of the above expression) cannot be smaller than the corresponding contributions associated with negative $k \in \Delta^-$ (those appearing on the right). Consequently, if $\Delta^+ \neq \emptyset$, a lower bound for ϕ can also be obtained:

$$\max\Big(- \sum_{k \in \Delta^+} \Delta_k \Big(\sum_{h=n-k+1}^{n} \tilde{c}_{(h)} \Big), \sum_{k \in \Delta^-} \Delta_k \Big(\sum_{h=n-k+1}^{n} \tilde{c}_{(h)} \Big) \Big) \leq \phi \leq 0. \tag{4.40}$$

In the general case, where λ can be positive and negative and non-negativity of the objective function cannot be assured, the bound above does not apply since it is based on the non-negativity of the objective function which is not valid in this case.

Furthermore, in the attempt to make our models efficient, the following lifting procedures for the coefficients of the Benders optimality cuts can be considered. For strengthening the cuts (4.18c), let $\mathcal{U}_k$ be a valid upper bound for φ_k, $k \in \Delta^-$. Then, for a given $k \in \Delta^-$, such that

$$\delta_k := \mathcal{U}_k - (n-k+1) c_{ij(h)} > 0 \tag{4.41}$$

holds, constraints

$$\varphi_k \leq (n-k+1) c_{ij(h)} + \sum_{c_{ij} > c_{ij(h)}} (c_{ij} - c_{ij(h)}) x_{ij}, \; h \in H, \tag{4.42}$$

can be strengthened as follows:

$$\varphi_k \leq (n-k+1) c_{ij(h)} + \sum_{c_{ij} > c_{ij(h)}} (\min\{c_{ij} - c_{ij(h)}, \delta_k\}) x_{ij}, \; h \in H : \mathcal{U}_k > (n-k+1) c_{ij(h)}. \tag{4.43}$$

Indeed, this follows from the fact that x variables are binary. Hence, if all values of the x variables from the right-hand-side of (4.43) are zero, a tighter bound on φ_k is obtained. If, on the other hand, at least one of these x variables is equal to one, the bound of φ_k should not surpass $\mathcal{U}_k$, and hence the coefficient next to x_{ij} can be downlifted correspondingly. Finally, it is not difficult to see that in the compact formulation (4.18), the remaining constraints (4.18c) (i.e., those for which $\mathcal{U}_k \leq (n-k+1) c_{ij(h)}$ holds) are redundant as they are dominated by $\varphi_k \leq \mathcal{U}_k$.

In order to strengthen the cuts (4.33c), let now $\mathcal{L}$ be a valid lower bound for ϕ (recall $\phi \leq 0$). For a given $(h_k)_{k\in\Delta^-}$, such that

$$\delta := \sum_{k\in\Delta^-} \Delta_k(n-k+1)c_{ij(h_k)} - \mathcal{L} > 0 \tag{4.44}$$

holds, constraints (4.33c), which can be rewritten as

$$\phi \geq \sum_{k\in\Delta^-} \Delta_k(n-k+1)c_{ij(h_k)} + \sum_{k\in\Delta^-} \Delta_k\Big[\sum_{c_{ij}>c_{ij(h_k)}} (c_{ij} - c_{ij(h_k)})x_{ij}\Big], \tag{4.45}$$

can be strengthened as follows:

$$\phi \geq \sum_{k\in\Delta^-} \Delta_k(n-k+1)c_{ij(h_k)} + \sum_{k\in\Delta^-} \sum_{c_{ij}>c_{ij(h_k)}} \max\{\Delta_k(c_{ij} - c_{ij(h_k)}), -\delta\}\ x_{ij}. \tag{4.46}$$

Indeed, as in the case of the disaggregated reformulation, this result follows from the fact that x variables are binary. If all values of the x variables from the right-hand-side of (4.46) are zero, a tighter lower bound on ϕ is obtained. If, on the other hand, at least one of these x variables is equal to one, the value of ϕ should not be smaller than $\mathcal{L}$, and hence the coefficient next to x_{ij} can be strengthened correspondingly. Finally, it is not difficult to see that in the aggregated formulation (4.33), the remaining constraints (4.33c) (i.e., those for which $\delta \leq 0$) are redundant as they are dominated by $\phi \geq \mathcal{L}$.

4.5 Implementation details

This section provides some details that play an important role in the design of effective implementations of the models proposed in Sections 4.3 and 4.4. The major bottleneck for Benders reformulations derived from the model DOMP_θ are the three-index variables θ_{ij}^k. Correspondingly, a significant computational improvement is only achieved when considering Benders decomposition approach just for $k \in \Delta^-$, i.e., when just projecting out those variables from the model as in $\mathsf{DOMP}_{\theta 1}$, or its aggregated counterpart. Therefore, this section restricts attention to presenting implementation details for $\mathsf{DOMP}_{\theta 1}$ and $\mathsf{DOMP}_{\theta 1a}$ models, together with DOMP_{r0B}.

4.5.1 Branch-and-Benders-cut implementations

For a more efficient performance, for each of our reformulations, Benders cuts are embedded into a branch-and-Benders-cut.

For the formulation DOMP_{r0B}, the model is initialized without the family of Benders cuts (4.7b) and with lower and upper bounds for θ_h given as $0 \leq \theta_h \leq \sum_{\ell\in N} \lambda_\ell$, $h \in H$. Benders cuts (4.7b) are then separated (in polynomial time) in a branch-and-bound (B&B) scheme. The pseudocode of the separation algorithm is shown in Algorithm 4.5.1. In this algorithm, a solution $(\bar{\theta}, \bar{x})$ found in a current node of the B&B tree is given. For each $h \in H$, following the rationale given in the proof of Proposition 4.3.1, one must first compute the value of $\bar{x}_h^*$. Next, determine the critical item index k as in (4.11). Unit capacity items are summed starting from the last index until the knapsack capacity is full, that is, until $\bar{x}_h^*$ minus this sum is less than one. Once the index is identified, a cut in the form of (4.14) can be proposed by computing the values of the cut expression. If the $\bar{\theta}$ value obtained as the node solution is less than the expression plus an ε tolerance, the cut is inserted into the master LP.

For the disaggregated and aggregated Benders reformulations, $\mathsf{DOMP}_{\theta 1}$ and $\mathsf{DOMP}_{\theta 1a}$, the separation is performed in a similar way. The respective families of Benders optimality cuts, namely (4.18c) and (4.33c), can be also separated in polynomial time. As for the aggregated reformulation the number of Benders optimality cuts (4.33c) is exponential, the branch-and-Benders-cut is the only computationally viable option for tackling instances of realistic size. The separation algorithms are summarized in Algorithms 4.5.2 and 4.5.3, respectively. These algorithms differ from Algorithm 4.5.1 in two aspects. First, the critical item index h is now computed summing the respective capacity for these cases, which is $\bar{x}_{ij(h)}$. Actually, for $\mathsf{DOMP}_{\theta 1a}$ one needs to determine a vector of indexes $h = (h_k)_{k \in \Delta^-}$. It should be noted that when searching for the values of h_k, $k \in \Delta^-$, an alternate sequence of indexes k must be processed, not necessarily consecutive, since only indexes k with $\Delta_k < 0$ are considered. Second, the proposed Benders cuts are now in the form of (4.18c) and (4.33c).

Algorithm 4.5.1 Branch-and-Benders-Cut Separation for DOMP$_{r0B}$

Consider a solution $(\bar{\theta}_h, \bar{x}_{ij})$ in a node of the branching tree
for $h \in H$ **do**
  Let $\bar{x}^*_h = \sum_{\substack{i,j \in N:\\ c_{ij} \geq c_{(\bar{h})}}} \bar{x}_{ij}$
  Determine the largest $k \in N$ such that $\bar{x}^*_h - \sum_{k'=k}^{n} 1 < 1$
  Let $RHS(\bar{x}_{ij}, k) = \lambda_k \bar{x}^*_h - \sum_{\ell' > k} (\lambda_k - \lambda_{\ell'})$
  if $\bar{\theta}_h < RHS(\bar{x}_{ij}, k) - \varepsilon$ **then**
    Add cut: $\theta_h \geq \lambda_k \sum_{\substack{i,j \in N:\\ c_{ij} \geq c_{(h)}}} x_{ij} - \sum_{\ell' > k} (\lambda_k - \lambda_{\ell'})$
  end if
end for

Algorithm 4.5.2 Branch-and-Benders-Cut Separation for DOMP$_{\theta 1}$

Consider a solution $(\bar{\varphi}_k, \bar{x}_{ij})$ in a node of the branching tree
for $k \in \Delta^-$ **do**
  Determine the largest $h \in H$ such that $\sum_{\substack{h' \geq h\\ h' \in H}} \bar{x}_{ij(h')} \geq n - k + 1$
  Let $RHS_k(\bar{x}_{ij}, h) = (n-k+1) c_{ij(h)} + \sum_{c_{ij} > c_{ij(h)}} (c_{ij} - c_{ij(h)}) \bar{x}_{ij}$
  if $\bar{\varphi}_k > RHS_k(\bar{x}_{ij}, h) + \varepsilon$ **then**
    Add cut: $\varphi_k \leq (n-k+1) c_{ij(h)} + \sum_{c_{ij} > c_{ij(h)}} (c_{ij} - c_{ij(h)}) x_{ij}$
  end if
end for

Algorithm 4.5.3 Branch-and-Benders-Cut Separation for DOMP$_{\theta 1a}$

Consider a solution $(\bar{\phi}, \bar{x}_{ij})$ in a node of the branching tree
Determine $h = (h_k)_{k \in \Delta^-}$ with $h_k \in H$ the largest such that $\sum_{\substack{h' \geq h_k\\ h' \in H}} \bar{x}_{ij(h')} \geq n - k + 1$
Let $RHS(\bar{x}_{ij}, h) = \sum_{k \in \Delta^-} \Delta_k \Big[(n-k+1) c_{ij(h_k)} + \sum_{c_{ij} > c_{ij(h_k)}} (c_{ij} - c_{ij(h_k)}) \bar{x}_{ij} \Big]$
if $\bar{\phi} < RHS(\bar{x}_{ij}, h) - \varepsilon$ **then**
  Add cut: $\phi \geq \sum_{k \in \Delta^-} \Delta_k \Big[(n-k+1) c_{ij(h_k)} + \sum_{c_{ij} > c_{ij(h_k)}} (c_{ij} - c_{ij(h_k)}) x_{ij} \Big]$
end if

The separation algorithms given in Algorithms 4.5.1-4.5.3 have to be applied at integer points, and alternatively, they can also be called for fractional LP-solutions within the branching tree. The latter case can be proven to be computationally profitable; however, if conditions limiting the number of fractional cuts are not set, these cuts may overload the master problem, reducing the efficiency of the algorithm. In order to manage this trade-off, the separation of fractional points is performed only at the root node of the search tree.

4.5.2 Stabilization procedure at the root node

Following observations in Fischetti et al. (2016, 2017), an in-and-out stabilization procedure can be applied at the root node before starting the branch-and-cut procedure. It aims to quickly determine a small set of Benders cuts that brings the master LP-relaxation value as close as possible to the “real optimal value”.

Next, it is explained how this procedure is implemented for the formulation $\mathsf{DOMP}_{\theta 1}$ (similar ideas are applied to $\mathsf{DOMP}_{\theta 1a}$ as well), starting from a feasible LP-solution $\hat{v} = (\hat{x}, \hat{y}, \hat{\varphi}, \hat{z}, \hat{t})$ used as the core or stabilizing point. This core point is computed as the analytic center (see, e.g., Bonami et al., 2020) obtained from solving the LP relaxation of the model DOMP_{θ} without the objective and using the barrier algorithm with crossover. If the computation of the LP relaxation takes much effort, then the core point is computed as a convex combination of feasible solutions. These feasible solutions are the result of randomly selecting p facilities, assigning the clients to its closest open facility and properly valuing the rest of continuous variables. This approach, which is highly dependable of the randomness in the facility selection, has been applied to the largest instances from our dataset.

At each cut loop iteration, two points in the v space are considered: the optimal solution of the current master LP, $v^* = (x^*, y^*, \varphi^*, z^*, t^*)$ as in Kelley (1960) method, and the core point $\hat{v}$. After initializing the parameter $\eta \in [0, 1]$, the current master LP solution v^* is moved towards the core point, obtaining $v^* = \eta v^* + (1-\eta)\hat{v}$. The Benders separation is applied to the current v^*, and if the corresponding $Benders\_cut(v^*)$ is violated, it is inserted into the pool of cuts $\mathcal{T}$. This procedure is iterated a given number of times ($iter_2$) without reoptimizing the current master LP, simply adding the cuts to $\mathcal{T}$. Afterwards, the LP relaxation is computed including all cuts from $\mathcal{T}$ and the process is repeated with the new master LP-solution v^*. The pseudo-code of this procedure is given in Algorithm 4.5.4.

Algorithm 4.5.4 In-and-Out Stabilization at the Root Node for DOMP Benders Decomposition

1: **Input:**
2:   Scaling parameter η
3:   Core point $\hat{v} = (\hat{x}, \hat{y}, \hat{\varphi}, \hat{z}, \hat{t})$
4:   Initial pool of Benders cuts $\mathcal{T} = \emptyset$
5:   Iteration counters $iter_1, iter_2$
6: **Procedure:**
7: **while** $iter_1$ **do**
8:   Solve to optimality the current master LP including cuts from $\mathcal{T}$
9:   Let $v^* = (x^*, y^*, \varphi^*, z^*, t^*)$ be its optimal solution
10:   **while** $iter_2$ **and** violated cuts added to $\mathcal{T}$ **do**
11:     $v^* \leftarrow \eta v^* + (1-\eta)\hat{v}$
12:     Perform Benders separation on u^* generating $Benders\_cut(v^*)$
13:     **if** $Benders\_cut(v^*)$ is violated **then**
14:       Add $Benders\_cut(v^*)$ to $\mathcal{T}$
15:     **end if**
16:   **end while**
17: **end while**
18: Remove from the master LP the cuts in $\mathcal{T}$ with positive slack

The procedure ends after a given number of iterations ($iter_1$). Also, to speed up the computation, non-binding cuts from the current LP are also removed (cf. Step 18 of Algorithm 4.5.4). This procedure has been computationally proven to be very helpful when solving instances of larger size, but not for the small ones. In the computational study, this stabilization procedure has been used for instances with 100 nodes or more. For the model based on the reformulation DOMP_{r0B} a lighter and faster warm-start phase is implemented. In this case, the LP relaxation of DOMP_{r0} was computed once and then the $|H|$ cuts in the form of (4.14) were added by means of the separation algorithm given in Algorithm 4.5.1.

4.6 Computational study

This section summarizes the results obtained from the computational experiments performed in order to empirically compare the proposed DOMP formulations. The algorithms were programmed in Python language (version 3.10) using Gurobi Optimizer 9.5 as a MIP solver. The experiments were run in a MacPro server with a 2,7 GHz Intel Xeon W processor of 24 cores and 192 GB RAM using a single thread and with all presolve and heuristic preprocessing disabled to ensure consistent comparisons. The time limit for computations was set to 7200 seconds per instance. For the violation threshold when separating Benders cuts (see Algorithms 4.5.1-4.5.3), $\varepsilon = 0.01$ was set. For the stabilization procedure given in Algorithm 4.5.4, $\eta = 0.9$, $iter_1 = 10$ and $iter_2 = 5$ were set. In the best-case scenario, this procedure introduces approximately 50 cuts.

Following experiments in Labbé et al. (2017) and Marín et al. (2020), two sets of benchmark instances were considered. The first dataset, namely `Beasley` instances, consists in p-median instances from Beasley (2015). In order to obtain the cost matrix the procedure explained in Labbé et al. (2017) was used, which gives rise to instances of sizes $n \in \{50, 100, 150, 200\}$ without symmetry but with (possibly repeated) integer costs from $[1, 1000]$. For the second dataset, namely `Random` instances, instances proposed in Labbé et al. (2017) were used. For each complete graph of size $n \in \{20, 30, 40, 50, 100, 140, 180, 200\}$, a set of five instances is generated by drawing costs c_{ij} uniformly at random from $[100, 1000]$ (up to two decimal digits). Following similar setting as in the literature, the number of facilities for each group of instances was chosen as $p \in \{\lfloor \frac{n}{4} \rfloor, \lfloor \frac{n}{3} \rfloor, \lfloor \frac{n}{2} \rfloor\}$.

Concerning the chosen criteria, i.e, the λ-vectors to be tested, the purpose was to broaden as possible the criteria selection including non-monotone weight vectors and also negative weights. In fact, that was the reason to introduce vectors not yet used in the existing DOMP literature. The final selection of λ-vectors (see Table 4.1) is a representative sample that depicts the variety of problems that can be generalized within the ordered median optimization framework. Among them, find some fairness criteria $(\lambda_1, \lambda_3, \lambda_7, \lambda_{11}, \lambda_{12})$, other particular customer preferences (λ_5, λ_9), and the obnoxious versions of most of them, where the ordered median objective is used to model the undesired facility location problems $(\lambda_2, \lambda_4, \lambda_6, \lambda_8, \lambda_{10})$. Furthermore, these vectors are among the most complex ones within the ordered median literature, primarily due to their non-monotone nature and the presence of negative weights when the minimization of the largest costs is required. Since λ-vectors with negative weights are tested, sets $\Lambda^+ = \{\lambda_1, \lambda_3, \lambda_5, \lambda_7, \lambda_9, \lambda_{11}, \lambda_{12}\}$ and $\Lambda^- = \{\lambda_2, \lambda_4, \lambda_6, \lambda_8, \lambda_{10}\}$ are defined in order to compute the optimality gap as

$$\mathtt{gap} = \begin{cases} (\mathtt{UB} - \mathtt{LB})/\mathtt{UB} & \text{for } \Lambda^+, \\ (\mathtt{UB} - \mathtt{LB})/|\mathtt{LB}| & \text{for } \Lambda^-, \end{cases} \tag{4.47}$$

where `UB` and `LB` stand for the best upper and lower bound found at termination, respectively.

The analysis begins with a comparison between DOMP_{r0} and its radius-Benders reformulation, DOMP_{r0B}, in both versions, implemented as a compact model and embedded within a Branch-and-Benders-cut algorithm (`BBC`) considering Algorithm 4.5.1. As remarked in Proposition 4.3.1, this reformulation is only valid for monotone non-increasing λ-vector. In the experimental setup, the focus was placed on testing λ_{11}, as it is the only strictly monotone non-increasing λ-vector. Preliminary tests also indicated that DOMP_{r0} outperformed DOMP_{θ} for this vector. Table 4.2 and Table 4.3 report `cpu` times for the `Beasley` and `Random` instances, respectively. For the

Notation	λ-vector	Criterion name	Purpose
λ_1	$(0, ..., 0, 1)$	p-center	Fairness
λ_2	$(0, ..., 0, -1)$	Obnoxious p-center	Obnoxious
λ_3	$(0, \dots, 0, 0, \underbrace{1, \dots, 1}_{k})$	k-centrum, $k = n/2$	Fairness
λ_4	$(0, \dots, 0, 0, \underbrace{-1, \dots, -1}_{k})$	Obnoxiuos k-centrum, $k = n/2$	Obnoxious
λ_5	$(0, \dots, 0, \underbrace{1}_{k-th\ position}, 0, \dots, 0)$	k-max	General preferences
λ_6	$(0, \dots, 0, \underbrace{-1}_{k-th\ position}, 0, \dots, 0)$	Obnoxious k-max	Obnoxious
λ_7	$(-1, 0, ..., 0, 1)$	Range	Fairness
λ_8	$(1, 0, ..., 0, -1)$	Obnoxious range	Obnoxious
λ_9	$(0, -1, 0, ..., 0, 1, 0)$	Second range	General preferences
λ_{10}	$(0, 1, 0, ..., 0, -1, 0)$	Second obnoxious range	Obnoxious
λ_{11}	$(n, n-1, \dots, 2, 1)$	Reverse	Fairness
λ_{12}	$(2(2i - n - 1))_{i \in N}$	Sum of absolute differences	Fairness

Table 4.1: λ-vectors tested in the Benders decomposition of the DOMP

`Beasley` instances, each instance is indexed as (`Ins-n-p`), where `Ins` identifies the number of the instance tested, while `n` and `p` indicate the number of nodes and the number of open facilities, respectively. For the `Random` instances, each row (`n-p`) shows average results over five random instances with the same number of nodes (`n`) and open facilities (`p`).

From Tables 4.2-4.3 it can be observed that DOMP_{r0} outperforms DOMP_{r0B} in mostly all cases. The compact reformulation DOMP_{r0B} (whose runtimes are given in the last column) is up to two orders of magnitude slower than the compact formulation DOMP_{r0}. This can be explained by the large number of constraints ($\mathcal{O}(|H||N|)$) of type (4.7b). On the other hand, for non-increasing λ vectors, the radius ordering constraints (4.6c)-(4.6d) are redundant (they can be removed by the MIP solver) so that the resulting DOMP_{r0} model requires only $\mathcal{O}(|H|)$ constraints of type (4.6b). This difference in the number of constraints, along with the structure of the constraint matrix, explains the inferior performance of DOMP_{r0B} for λ_{11}. When comparing the `BBC` implementation of DOMP_{r0B} against its compact version, speed-ups of approximately one order of magnitude are observed. Nevertheless, DOMP_{r0} remains the dominant model in this setting. The significant differences in the `cpu` times for `Beasley` and `Random` instances are mainly explained by the fact that there are more repetitions in the cost matrix for the former, resulting in a better performance of the models in that case. For these reasons, the remainder of this section focuses on Benders approaches applied to the state-of-the-art formulation DOMP_{θ}.

As commented in Section 4.5, Benders decomposition approaches tend to be promising when they result in reformulations with a significantly reduced number of decision variables. For the model DOMP_{θ}, this means projecting out θ_{ij}^k variables for $k \in \Delta^-$, $i, j \in N$. For this reason, the experiments focus on comparing two state-of-the-art compact formulations (DOMP_{r0} and DOMP_{θ}) against the branch-and-Benders-cut algorithms based on the reformulations $\mathsf{DOMP}_{\theta 1}$ and $\mathsf{DOMP}_{\theta 1a}$. Given the large number of experiments, the results are summarized using empirical cumulative distribution functions (ECDFs) and bar charts. ECDFs are reported with respect to runtimes and percentage gaps at termination, respectively. The ECDFs with runtimes can be interpreted as the number of instances (shown in y-axis) that can be solved within a certain amount of time (depicted in the x-axis).

For the `Beasley` instances, the ECDF in Figure 4.1 shows the cumulative percentage of

	DOMP_{r0}	DOMP_{r0B} (BBC)	DOMP_{r0B}
1-50-5	0.33	2.25	12.69
2-50-10	0.53	3.03	8.55
3-50-10	0.30	3.02	11.22
4-50-20	0.75	2.20	9.06
5-50-33	0.44	1.89	7.86
6-100-5	1.46	6.41	55.29
7-100-10	14.73	18.60	127.91
8-100-20	0.86	8.94	48.98
9-100-40	0.74	6.97	47.06
10-100-67	0.76	4.48	36.65
11-150-5	4.00	13.84	127.56
12-150-10	2.69	14.07	191.44
13-150-30	2.71	14.85	121.74
14-150-60	1.77	17.52	126.61
15-150-100	2.41	6.98	117.30
16-200-5	6.71	23.87	264.49
17-200-10	7.31	20.65	267.84
18-200-40	4.80	29.58	278.88
19-200-80	4.09	22.47	258.27
20-200-133	3.97	13.65	27.45

Table 4.2: `Beasley` instances `cpu` times for Reverse criterion (λ_{11})

	DOMP_{r0}	DOMP_{r0B} (BBC)	DOMP_{r0B}
20-5	0.09	0.39	0.41
20-6	0.27	0.47	0.76
20-10	0.19	0.32	0.39
30-7	4.08	4.50	8.89
30-10	65.76	2.92	5.35
30-15	0.57	2.12	3.75
40-10	51.39	12.89	93.83
40-13	1.03	6.47	203.82
40-20	0.92	8.49	22.43
50-12	7.74	212.08	792.50
50-16	2.17	201.28	393.59
50-25	1.70	12.56	189.62
100-25	227.04	2984.71	7200.00
100-33	55.96	1543.90	7200.00
100-50	53.79	585.79	7200.00
140-35	959.51	4656.04	7200.00
140-46	273.07	5680.54	7200.00
140-70	136.73	3695.69	7200.00
180-45	1722.72	7200.00	7200.00
180-60	956.11	7133.07	7200.00

Table 4.3: `Random` instances `cpu` times for Reverse criterion (λ_{11})

instances solved within a time limit and Figure 4.2 shows the cumulative percentage of instances solved within a gap value. Both ECDFs are included, as they capture the overall nature of the results, covering instances solved within the time limit as well as those that were not. In both cases, $\mathsf{DOMP}_{\theta 1}$ and $\mathsf{DOMP}_{\theta 1a}$ report improvements with respect to DOMP_{r0} and DOMP_{θ}. That is, the reformulations were able to solve more instances in less time and were able to obtain lower gaps in those instances for which optimality could not be certified. The reader may observe that, compared to other formulations, DOMP_{r0} improves its performance with an increasing value of `n` or, in other words, with an increasing number of ties in the cost matrix. Recall that `Beasley` instances include integer c_{ij} costs drawn uniformly in [1,100] for large instances. In order to have a better comparison between the different formulations, Figure 4.3 shows results disaggregated by `n` and Figure 4.4 shows results disaggregated by λ-vector. Similarly to Figure 4.1, it can be concluded from Figure 4.3 that $\mathsf{DOMP}_{\theta 1}$ and $\mathsf{DOMP}_{\theta 1a}$ are consistently better than DOMP_{r0} and DOMP_{θ} for all sizes, solving more instances in less time. From Figure 4.4, it can be concluded that $\mathsf{DOMP}_{\theta 1}$ and $\mathsf{DOMP}_{\theta 1a}$ are able to compute the same or more instances to optimality for mostly all different criteria in less running time. In this case, it should be highlighted that, while DOMP_{θ} is not able to prove optimality for any of the instances with λ_{11}, the reformulations can, although, as already noted, DOMP_{r0} performs very well for this specific criterion.

Regarding `Random` instances, the cumulative percentage of instances solved within a time limit and the cumulative percentage of instances solved within a gap value are shown in Figure 4.5 and Figure 4.6, respectively. A first comment on these results is that, as already pointed in Section 4.3, DOMP_{r0} performance was expected to be much worse for the `Random` set of instances than for `Beasley` due to the fact that DOMP_{r0} leverages from having a considerable number of repetitions in the cost matrix. On the other hand, it can also be observed that for this set of instances where there are almost no repeated costs, the performance of the Benders approaches is very similar in general to the compact DOMP_{θ} formulation. However, from Figure 4.7 where the number of instances and the time to solve them are shown, it can be observed that the contribution of the Benders reformulations for this set of `Random` instances is that for large `n` values (180 and 200) $\mathsf{DOMP}_{\theta 1}$ and $\mathsf{DOMP}_{\theta 1a}$ are able to certify optimality for more instances and in less time than DOMP_{r0} and DOMP_{θ}. Furthermore, as depicted in Figure 4.8, for the reverse criterion λ_{11}, optimality can be certified for more instances than with DOMP_{r0} and DOMP_{θ} and in less time, improving the performance of the models on a criterion previously considered difficult to solve, as shown in Tables 4.2–4.3.

For the sake of completeness and in order to give some more insights on the performance of the models based on $\mathsf{DOMP}_{\theta 1}$ and $\mathsf{DOMP}_{\theta 1a}$, some detailed results are included in Table 4.4 and Table 4.5, focusing on the strength of our cuts and their efficiency. Here, columns `lazy` and `user` indicate the total/average number of integer (or lazy) cuts and fractional (or user) cuts added until termination, respectively. As the stabilization procedures were not applied uniformly, the tables report only the number of cuts added within the Branch-and-Benders-cut algorithm. A first comment on these results (a result which is also known in the standard facility location literature) is that, the greater the number p of facilities to locate, the better results can be obtained in terms of running time or final gaps, in general. The reader might notice that there are certain λ-vectors for which Benders cuts are not being added. These are the cases for which Δ^- is empty, i.e., for λ_1 (shown in the summary result tables) and λ_3. However, even in these cases where no cuts are added, the reformulations $\mathsf{DOMP}_{\theta 1}$ and $\mathsf{DOMP}_{\theta 1a}$ are still making a contribution in the sense that they are lighter models with a smaller number of both variables and constraints. Overall, there is a correlation between the number of cuts that can be separated and the size of the set Δ^-. When $|\Delta^-|$ is large, i.e., many negative "jumps" in our Δ vector arise, in every separation call of the model $\mathsf{DOMP}_{\theta 1}$, up to $|\Delta^-|$ cuts are potentially added. In an extreme case, when λ_{11}, then $|\Delta^-| = |\Delta|$. In that case, many more cuts are separated for $\mathsf{DOMP}_{\theta 1a}$ than for $\mathsf{DOMP}_{\theta 1}$, further emphasizing the difference between these two models. On the contrary, when the size of Δ^+ is large, then $|\Delta^-|$ is relatively small. In that case, in every node fewer Benders cuts are added since the master LP contains all the constraints coming from Δ^+ part of the problem.

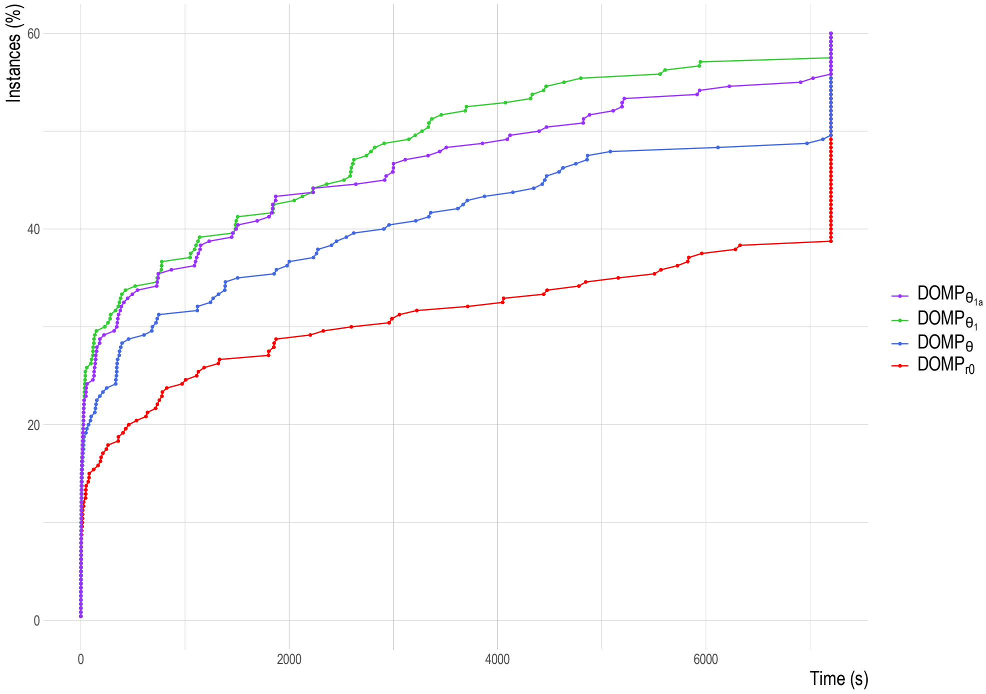


Figure 4.1: Cumulative % of `Beasley` instances solved to optimality to a given time

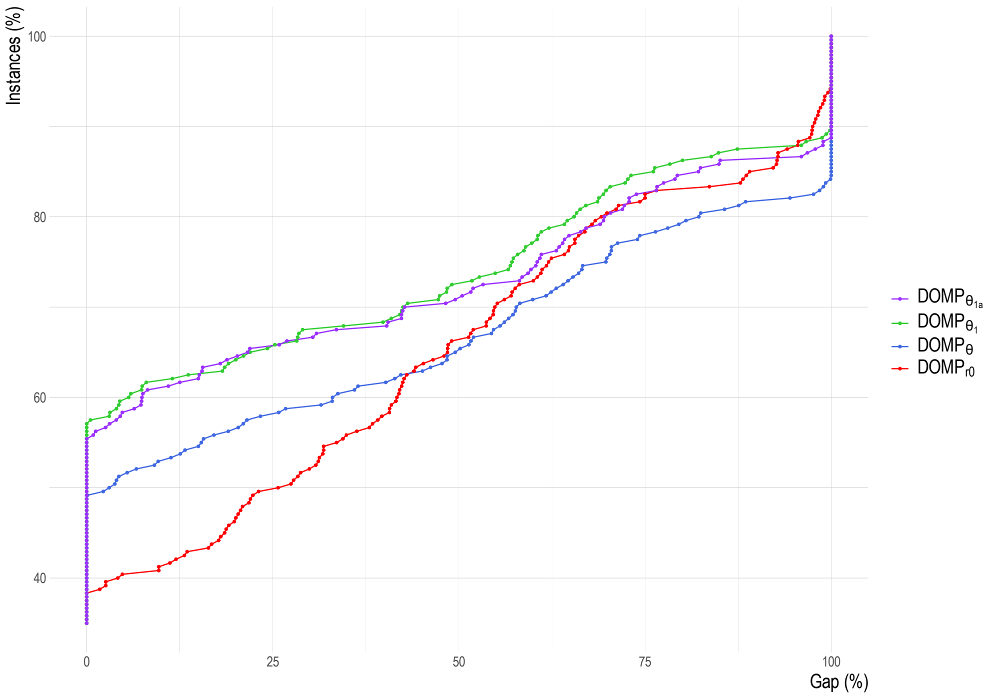


Figure 4.2: Cumulative % of `Beasley` instances solved within a gap value

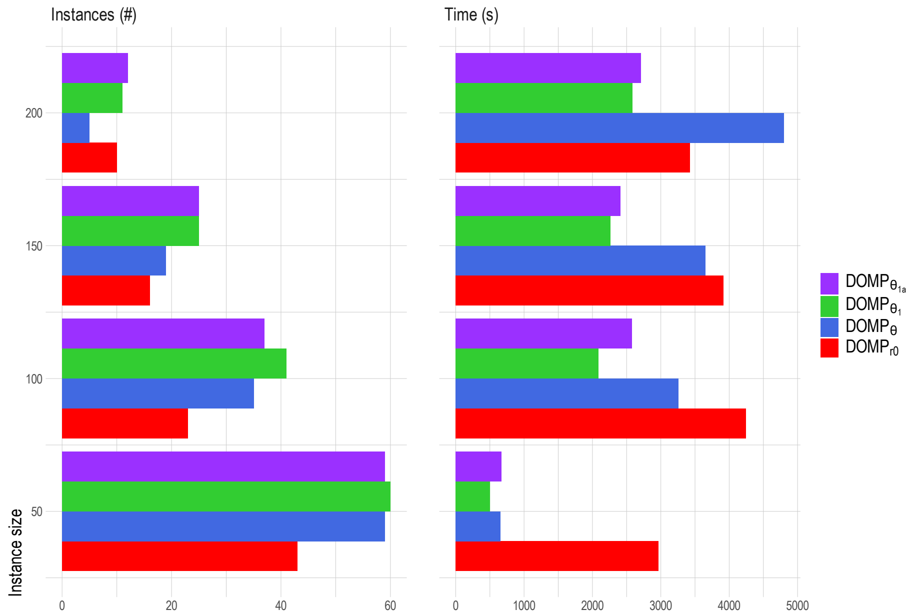


Figure 4.3: Number of `Beasley` instances solved to optimality and running time per instance size

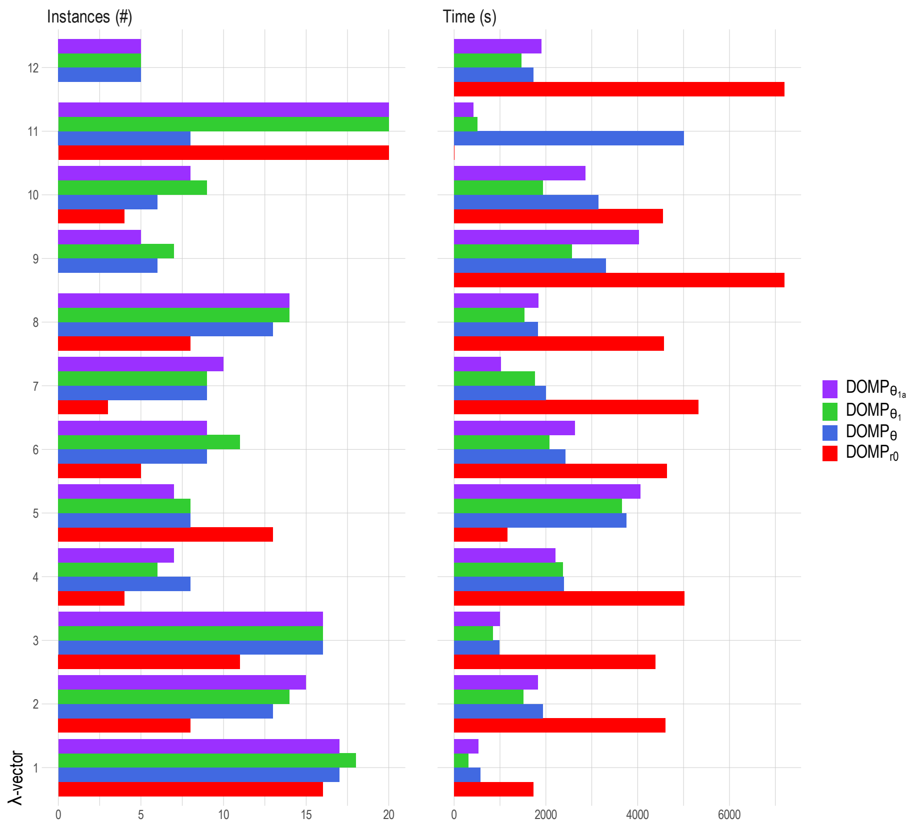


Figure 4.4: Number of `Beasley` instances solved to optimality and running time per criterion

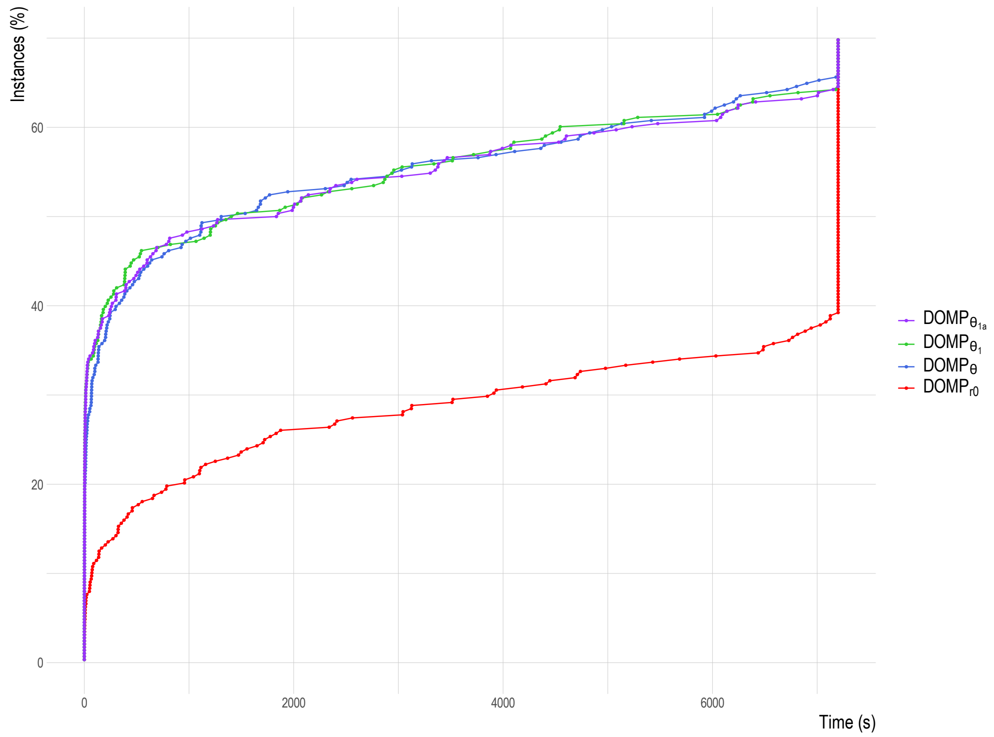


Figure 4.5: Cumulative % of `Random` instances solved to optimality to a given time

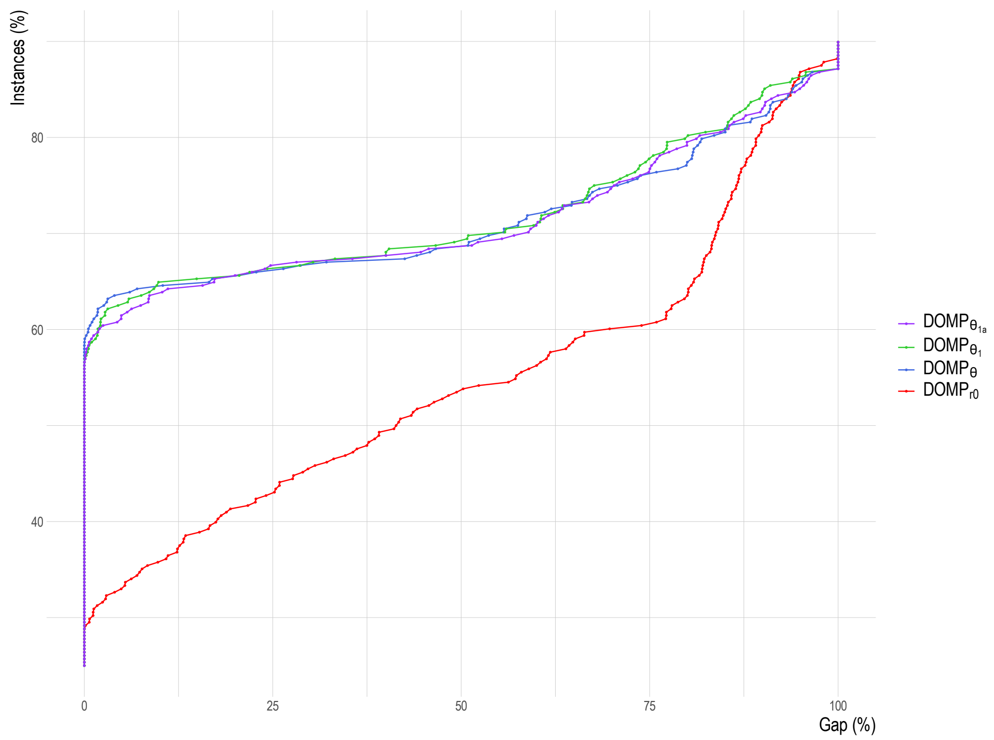


Figure 4.6: Cumulative % of `Random` instances solved within a gap value

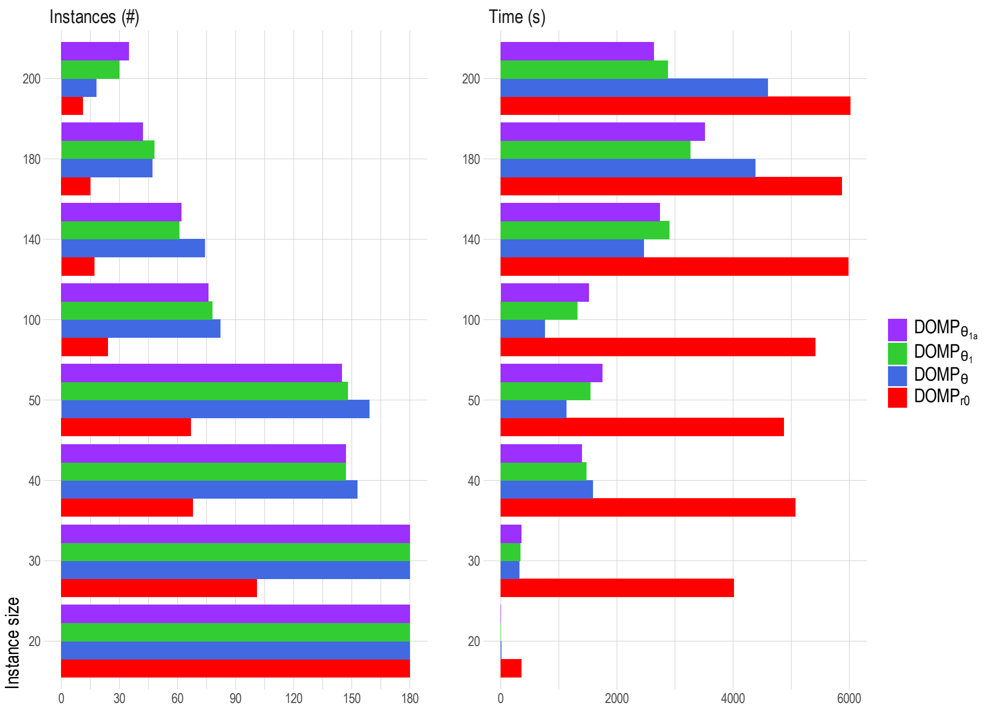


Figure 4.7: Number of `Random` instances solved to optimality and running time per instance size

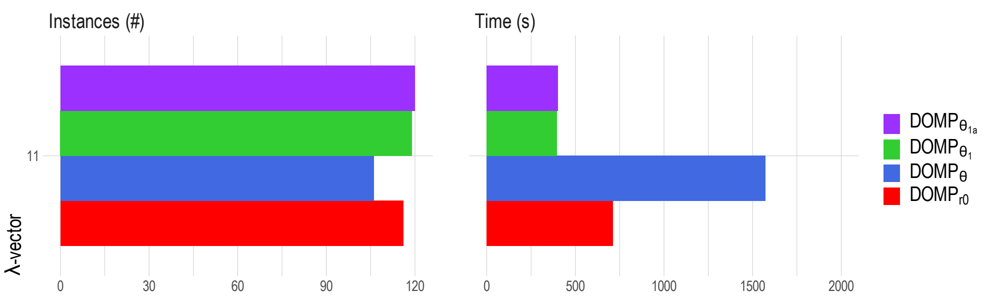


Figure 4.8: Number of `Random` instances solved to optimality and running time for λ_{11}

λ	DOMP$_{r0}$		DOMP$_{\theta}$		DOMP$_{\theta 1}$				DOMP$_{\theta 1a}$			
λ_3	cpu	gap	cpu	gap	cpu	gap	lazy	user	cpu	gap	lazy	user
1-50-5	624.74	0.00	1.49	0.00	1.40	0.00	0	0	1.43	0.00	0	0
2-50-10	2202.05	0.00	2.15	0.00	2.15	0.00	0	0	2.01	0.00	0	0
3-50-10	6282.73	0.00	3.17	0.00	2.64	0.00	0	0	6.37	0.00	0	0
4-50-20	778.56	0.00	0.08	0.00	0.07	0.00	0	0	0.07	0.00	0	0
5-50-33	26.20	0.00	0.03	0.00	0.03	0.00	0	0	0.02	0.00	0	0
6-100-5	7200.00	18.52	4858.98	0.00	2621.46	0.00	0	0	5216.46	0.00	0	0
7-100-10	7200.00	13.12	2233.24	0.00	2221.03	0.00	0	0	2230.18	0.00	0	0
8-100-20	4475.95	0.00	1387.51	0.00	1482.35	0.00	0	0	126.72	0.00	0	0
9-100-40	5568.64	0.00	0.49	0.00	0.44	0.00	0	0	0.51	0.00	0	0
10-100-67	2326.94	0.00	0.34	0.00	0.33	0.00	0	0	0.54	0.00	0	0
11-150-5	7200.00	20.66	4751.08	0.00	4799.92	0.00	0	0	4469.07	0.00	0	0
12-150-10	3057.69	0.00	2453.91	0.00	2360.36	0.00	0	0	3855.47	0.00	0	0
13-150-30	7200.00	20.03	7200.00	3.02	7200.00	0.51	0	0	7200.00	2.53	0	0
14-150-60	4780.37	0.00	1.10	0.00	1.08	0.00	0	0	1.12	0.00	0	0
15-150-100	4048.94	0.00	1.49	0.00	1.47	0.00	0	0	2.26	0.00	0	0
16-200-5	7200.00	51.92	7200.00	4.32	7200.00	4.32	0	0	7200.00	3.08	0	0
17-200-10	7200.00	42.30	7200.00	4.00	7200.00	3.99	0	0	7200.00	3.97	0	0
18-200-40	7200.00	19.09	7200.00	3.78	7200.00	7.37	0	0	7200.00	7.37	0	0
19-200-80	7200.00	11.18	4.02	0.00	2.46	0.00	0	0	3.83	0.00	0	0
20-200-133	7200.00	18.73	3.42	0.00	4.73	0.00	0	0	3.34	0.00	0	0
λ_7	cpu	gap	cpu	gap	cpu	gap	lazy	user	cpu	gap	lazy	user
1-50-5	7200.00	76.53	382.94	0.00	359.25	0.00	3	0	354.57	0.00	3	0
2-50-10	7200.00	89.05	25.14	0.00	15.77	0.00	3	1	17.82	0.00	3	1
3-50-10	7200.00	71.11	368.02	0.00	22.93	0.00	3	0	25.48	0.00	3	0
4-50-20	1803.42	0.00	10.19	0.00	6.38	0.00	3	0	6.39	0.00	3	0
5-50-33	187.68	0.00	4.78	0.00	4.02	0.00	3	0	3.93	0.00	3	0
6-100-5	7200.00	97.12	7200.00	63.07	7200.00	56.62	4	0	7200.00	18.84	4	0
7-100-10	7200.00	92.65	7200.00	59.91	7200.00	48.33	3	1	7200.00	63.93	4	0
8-100-20	7200.00	75.00	3214.78	0.00	4638.68	0.00	3	0	3444.02	0.00	3	0
9-100-40	7200.00	68.32	4453.26	0.00	1851.68	0.00	3	0	1870.44	0.00	3	0
10-100-67	825.64	0.00	145.38	0.00	125.62	0.00	3	0	126.26	0.00	3	0
11-150-5	7200.00	98.21	7200.00	100.00	7200.00	100.00	3	0	7200.00	100.00	3	0
12-150-10	7200.00	83.64	7200.00	13.21	7200.00	8.00	4	0	3507.46	0.00	4	0
13-150-30	7200.00	88.14	7200.00	85.67	7200.00	65.45	3	0	7200.00	82.17	3	0
14-150-60	7200.00	12.00	4146.97	0.00	3368.56	0.00	3	0	869.16	0.00	3	0
15-150-100	7200.00	53.66	7200.00	41.38	7200.00	28.21	3	0	7200.00	42.28	3	0
16-200-5	7200.00	100.00	7200.00	100.00	7200.00	99.34	4	0	7200.00	100.00	4	0
17-200-10	7200.00	100.00	7200.00	100.00	7200.00	100.00	4	0	7200.00	100.00	4	0
18-200-40	7200.00	100.00	7200.00	100.00	7200.00	100.00	3	0	7200.00	100.00	3	0
19-200-80	7200.00	74.29	7200.00	88.51	7200.00	84.86	3	0	7200.00	84.92	3	0
20-200-133	7200.00	46.51	7200.00	42.19	7200.00	42.49	3	0	7200.00	42.44	3	0
λ_8	cpu	gap	cpu	gap	cpu	gap	lazy	user	cpu	gap	lazy	user
1-50-5	5960.74	0.00	347.76	0.00	18.53	0.00	18	4	26.71	0.00	12	3
2-50-10	7200.00	9.68	10.92	0.00	21.26	0.00	12	6	12.36	0.00	9	3
3-50-10	4846.48	0.00	336.09	0.00	22.41	0.00	18	6	388.86	0.00	7	3
4-50-20	122.78	0.00	9.49	0.00	6.85	0.00	14	4	5.63	0.00	6	3
5-50-33	44.78	0.00	7.35	0.00	5.55	0.00	8	3	4.03	0.00	3	3
6-100-5	7200.00	16.77	3618.33	0.00	3340.95	0.00	14	6	1089.98	0.00	8	3
7-100-10	7200.00	37.97	2261.79	0.00	1846.71	0.00	13	6	1869.62	0.00	5	3
8-100-20	717.39	0.00	98.54	0.00	115.99	0.00	7	3	115.83	0.00	5	1
9-100-40	1331.89	0.00	248.32	0.00	260.38	0.00	8	4	222.06	0.00	4	3
10-100-67	639.79	0.00	182.25	0.00	134.55	0.00	8	3	142.86	0.00	9	3
11-150-5	7200.00	36.28	7200.00	9.09	7200.00	11.50	13	5	7200.00	8.18	8	3
12-150-10	7200.00	19.82	1269.31	0.00	1124.14	0.00	13	6	2994.33	0.00	7	3
13-150-30	7200.00	41.53	4429.17	0.00	5559.28	0.00	10	4	7200.00	0.88	6	3
14-150-60	4442.84	0.00	211.80	0.00	429.01	0.00	8	3	319.23	0.00	5	3
15-150-100	7200.00	62.04	7200.00	62.39	2911.70	0.00	7	3	6224.55	0.00	8	3
16-200-5	7200.00	34.41	7200.00	5.43	7200.00	7.45	6	6	7200.00	21.92	5	3
17-200-10	7200.00	43.96	7200.00	49.50	7200.00	42.05	8	6	7200.00	49.50	3	3
18-200-40	7200.00	57.01	7200.00	58.18	7200.00	29.00	9	4	6909.11	0.00	5	2
19-200-80	7200.00	21.79	7200.00	64.00	7200.00	60.60	7	3	7200.00	98.93	3	0
20-200-133	7200.00	20.93	7200.00	55.52	7200.00	57.88	6	3	7200.00	51.58	3	3
λ_{11}	cpu	gap	cpu	gap	cpu	gap	lazy	user	cpu	gap	lazy	user
1-50-5	0.33	0.00	29.55	0.00	4.47	0.00	147	0	3.48	0.00	3	0
2-50-10	0.53	0.00	344.28	0.00	4.30	0.00	178	0	4.49	0.00	4	0
3-50-10	0.30	0.00	26.97	0.00	2.19	0.00	166	0	2.50	0.00	4	0
4-50-20	0.75	0.00	17.99	0.00	2.14	0.00	147	0	2.08	0.00	3	0
5-50-33	0.44	0.00	344.76	0.00	4.10	0.00	147	0	3.96	0.00	3	0
6-100-5	1.46	0.00	7200.00	100.00	25.71	0.00	297	0	24.96	0.00	3	0
7-100-10	14.73	0.00	7200.00	100.00	229.08	0.00	566	0	137.14	0.00	6	3
8-100-20	0.86	0.00	3874.37	0.00	31.62	0.00	321	0	25.72	0.00	4	0
9-100-40	0.74	0.00	7200.00	100.00	40.53	0.00	302	0	51.23	0.00	4	0
10-100-67	0.76	0.00	7200.00	100.00	32.46	0.00	297	0	344.73	0.00	3	0
11-150-5	4.00	0.00	7200.00	100.00	1105.17	0.00	534	0	1127.73	0.00	4	0
12-150-10	2.69	0.00	4347.19	0.00	117.41	0.00	603	0	150.65	0.00	5	0
13-150-30	2.71	0.00	7200.00	100.00	285.98	0.00	503	0	182.58	0.00	4	0
14-150-60	1.77	0.00	4628.60	0.00	102.81	0.00	514	0	132.19	0.00	4	0
15-150-100	2.41	0.00	7200.00	100.00	777.31	0.00	447	0	1111.14	0.00	3	0
16-200-5	6.71	0.00	7200.00	100.00	772.47	0.00	1020	0	543.95	0.00	4	1
17-200-10	7.31	0.00	7200.00	100.00	332.55	0.00	796	0	413.16	0.00	4	0
18-200-40	4.80	0.00	7200.00	100.00	3210.45	0.00	712	0	493.90	0.00	6	3
19-200-80	4.09	0.00	7200.00	100.00	393.22	0.00	655	0	3113.80	0.00	6	0
20-200-133	3.97	0.00	7200.00	100.00	2591.39	0.00	597	0	448.69	0.00	3	0

Table 4.4: `Beasley` instances result tables for some λ-vectors

λ	DOMP_{r0}		DOMP_{θ}		$\mathsf{DOMP}_{\theta 1}$				$\mathsf{DOMP}_{\theta 1a}$			
λ_3	cpu	gap	cpu	gap	cpu	gap	lazy	user	cpu	gap	lazy	user
20-5	24.23	0.00	0.12	0.00	0.13	0.00	0.00	0.00	0.12	0.00	0.00	0.00
20-6	139.24	0.00	0.16	0.00	0.18	0.00	0.00	0.00	0.17	0.00	0.00	0.00
20-10	69.48	0.00	0.06	0.00	0.08	0.00	0.00	0.00	0.07	0.00	0.00	0.00
30-7	6582.21	8.38	1.81	0.00	1.83	0.00	0.00	0.00	3.12	0.00	0.00	0.00
30-10	3935.77	0.02	2.05	0.00	2.53	0.00	0.00	0.00	1.86	0.00	0.00	0.00
30-15	1042.16	0.00	0.61	0.00	0.62	0.00	0.00	0.00	0.56	0.00	0.00	0.00
40-10	7200.00	12.35	11.30	0.00	111.12	0.00	0.00	0.00	111.15	0.00	0.00	0.00
40-13	7200.00	6.23	3.80	0.00	8.94	0.00	0.00	0.00	73.46	0.00	0.00	0.00
40-20	2391.86	0.00	1.07	0.00	1.24	0.00	0.00	0.00	1.73	0.00	0.00	0.00
50-12	7200.00	13.20	386.72	0.00	241.66	0.00	0.00	0.00	172.03	0.00	0.00	0.00
50-16	6885.43	2.82	16.02	0.00	10.21	0.00	0.00	0.00	8.77	0.00	0.00	0.00
50-25	3044.59	0.69	1.63	0.00	1.87	0.00	0.00	0.00	1.65	0.00	0.00	0.00
100-25	7200.00	30.60	5923.57	0.76	5558.21	0.53	0.00	0.00	4780.92	0.19	0.00	0.00
100-33	7200.00	24.11	624.45	0.00	529.72	0.00	0.00	0.00	502.74	0.00	0.00	0.00
100-50	7200.00	17.44	40.16	0.00	35.01	0.00	0.00	0.00	41.35	0.00	0.00	0.00
140-35	7200.00	25.91	7200.00	1.80	7200.00	1.78	0.00	0.00	7200.00	1.79	0.00	0.00
140-46	7200.00	22.76	5991.38	0.26	6242.15	0.26	0.00	0.00	6083.40	0.24	0.00	0.00
140-70	7200.00	16.46	130.91	0.00	136.21	0.00	0.00	0.00	135.58	0.00	0.00	0.00
180-45	7200.00	38.53	7200.00	1.78	7200.00	1.81	0.00	0.00	7200.00	2.00	0.00	0.00
180-60	7200.00	18.16	7200.00	0.54	7200.00	0.55	0.00	0.00	7200.00	0.54	0.00	0.00
180-90	7200.00	13.12	567.99	0.00	472.68	0.00	0.00	0.00	511.89	0.00	0.00	0.00
200-50	7200.00	84.16	7200.00	2.58	7200.00	2.22	0.00	0.00	7200.00	2.51	0.00	0.00
200-66	7200.00	83.32	7200.00	1.02	7200.00	1.03	0.00	0.00	7200.00	1.23	0.00	0.00
200-100	7200.00	82.52	3760.83	0.06	2997.36	0.08	0.00	0.00	3187.14	0.18	0.00	0.00
λ_7	cpu	gap	cpu	gap	cpu	gap	lazy	user	cpu	gap	lazy	user
20-5	650.92	0.00	1.09	0.00	0.90	0.00	1.60	0.40	33.03	0.00	3.00	1.20
20-6	553.33	0.00	0.99	0.00	0.70	0.00	1.80	0.30	32.79	0.00	3.40	0.60
20-10	515.31	0.00	0.56	0.00	0.52	0.00	1.60	0.10	0.43	0.00	3.20	0.60
30-7	7200.00	57.18	23.48	0.00	142.40	0.00	1.60	0.30	76.24	0.00	3.20	0.40
30-10	6942.96	22.69	10.23	0.00	73.10	0.00	1.60	0.10	70.63	0.00	3.00	0.40
30-15	6436.51	4.02	7.13	0.00	5.12	0.00	1.50	0.10	36.03	0.00	3.00	0.20
40-10	7200.00	94.77	644.03	0.00	466.34	0.00	1.90	0.10	338.63	0.00	3.60	0.00
40-13	7200.00	85.43	356.56	0.00	224.65	0.00	1.50	0.00	179.34	0.00	3.00	0.00
40-20	7200.00	10.84	16.45	0.00	44.38	0.00	1.60	0.10	9.53	0.00	3.20	0.20
50-12	7200.00	78.75	132.87	0.00	113.39	0.00	1.50	0.10	100.48	0.00	3.20	0.00
50-16	7200.00	69.69	50.67	0.00	31.92	0.00	1.50	0.10	26.29	0.00	3.00	0.20
50-25	7200.00	28.97	25.78	0.00	17.04	0.00	1.60	0.00	14.48	0.00	3.20	0.00
100-25	7200.00	100.00	7200.00	10.41	7200.00	10.48	1.80	0.00	7200.00	9.50	2.70	0.00
100-33	7200.00	96.15	934.14	0.00	728.88	0.00	1.70	0.00	687.63	0.00	3.60	0.20
100-50	7200.00	81.92	477.54	0.00	432.68	0.00	1.70	0.00	397.38	0.00	3.40	0.00
140-35	7200.00	100.00	7017.64	42.50	7008.64	53.59	1.50	0.00	6806.11	61.98	1.50	0.00
140-46	7200.00	100.00	4827.15	0.00	6611.55	50.00	1.50	0.00	6140.11	53.27	1.50	0.00
140-70	7200.00	100.00	2891.72	0.00	3049.43	0.00	1.50	0.00	3108.95	0.01	1.50	0.00
180-45	7200.00	100.00	7200.00	100.00	7200.00	83.35	1.70	0.00	7200.00	100.00	1.70	0.00
180-60	7200.00	100.00	7200.00	100.00	6792.30	70.21	1.50	0.00	7106.24	80.08	1.60	0.00
180-90	7200.00	100.00	7200.00	100.00	6793.89	70.00	1.70	0.00	6720.80	70.00	1.70	0.00
200-50	7200.00	100.00	7200.00	100.00	7200.00	100.00	1.60	0.00	7200.00	100.00	1.60	0.00
200-66	7200.00	100.00	7200.00	100.00	7200.00	100.00	1.60	0.00	7200.00	100.00	1.60	0.00
200-100	7200.00	100.00	7200.00	100.00	7200.00	100.00	1.70	0.00	7200.00	100.00	1.70	0.00
λ_8	cpu	gap	cpu	gap	cpu	gap	lazy	user	cpu	gap	lazy	user
20-5	14.95	0.00	0.46	0.00	0.40	0.00	7.10	4.10	0.33	0.00	5.40	3.00
20-6	14.84	0.00	0.52	0.00	0.42	0.00	6.30	3.50	0.26	0.00	4.00	1.40
20-10	141.15	0.00	0.43	0.00	0.44	0.00	6.50	3.00	0.21	0.00	4.20	1.00
30-7	1649.23	0.00	3.03	0.00	34.53	0.00	7.70	3.60	1.76	0.00	6.20	3.00
30-10	1251.32	0.00	3.69	0.00	2.00	0.00	7.20	3.30	1.15	0.00	4.20	2.00
30-15	779.12	0.00	4.23	0.00	1.86	0.00	8.00	3.00	2.12	0.00	5.00	3.00
40-10	6728.33	43.37	71.07	0.00	40.56	0.00	8.60	3.90	7.30	0.00	6.00	2.60
40-13	5685.67	17.72	73.78	0.00	4.41	0.00	7.30	3.30	59.76	0.00	5.20	2.60
40-20	1498.25	0.00	7.97	0.00	36.33	0.00	7.10	3.10	34.96	0.00	5.80	3.00
50-12	4709.14	7.33	14.79	0.00	44.32	0.00	8.10	3.70	24.53	0.00	6.40	3.00
50-16	4978.01	25.22	12.34	0.00	7.07	0.00	7.90	3.30	4.24	0.00	4.40	1.80
50-25	3123.47	2.92	12.25	0.00	7.81	0.00	7.30	3.00	13.15	0.00	5.20	3.00
100-25	7200.00	82.24	294.98	0.00	2364.30	4.06	8.00	3.30	3032.64	4.91	6.80	3.00
100-33	7200.00	85.95	194.07	0.00	1035.15	2.29	6.70	3.40	128.23	0.00	4.80	3.00
100-50	7200.00	91.28	217.63	0.00	3518.56	3.58	6.60	3.00	658.22	0.00	5.60	3.00
140-35	7200.00	86.51	1536.35	0.00	2578.32	3.60	8.80	3.50	3378.45	7.47	5.20	3.00
140-46	7200.00	89.48	1683.21	0.00	5269.45	10.92	9.90	3.30	1125.79	0.00	5.80	3.00
140-70	7200.00	92.25	2510.94	0.00	5802.85	8.88	6.40	3.00	602.07	0.00	5.00	3.00
180-45	7200.00	88.69	6200.90	22.81	5128.99	3.88	7.60	3.90	6252.01	8.54	6.40	3.00
180-60	7200.00	89.95	6265.51	3.11	5137.93	7.17	7.70	3.30	5233.71	11.09	5.20	3.00
180-90	7200.00	94.02	7179.83	16.99	6732.91	18.06	7.50	3.10	6137.94	8.48	6.20	3.00
200-50	7200.00	89.78	7200.00	88.57	7200.00	51.47	7.20	4.30	7007.41	51.76	4.80	3.00
200-66	7200.00	91.78	7200.00	90.81	7200.00	76.74	7.40	3.60	7200.00	79.94	5.20	3.00
200-100	7200.00	93.87	7200.00	93.13	7200.00	76.03	6.50	3.00	7200.00	77.30	5.20	3.00
λ_{11}	cpu	gap	cpu	gap	cpu	gap	lazy	user	cpu	gap	lazy	user
20-5	0.09	0.00	0.13	0.00	0.17	0.00	63.90	0.00	0.19	0.00	4.00	0.00
20-6	0.27	0.00	0.30	0.00	0.36	0.00	67.10	6.00	0.46	0.00	4.60	0.60
20-10	0.19	0.00	0.23	0.00	0.35	0.00	59.70	0.00	0.39	0.00	3.80	0.00
30-7	4.08	0.00	68.02	0.00	1.47	0.00	125.20	20.70	2.97	0.00	6.20	1.80
30-10	65.76	0.00	1.25	0.00	1.09	0.00	103.60	0.00	1.33	0.00	4.40	0.20
30-15	0.57	0.00	1.25	0.00	1.12	0.00	98.40	0.00	1.42	0.00	4.80	0.00
40-10	51.39	0.00	70.69	0.00	5.22	0.00	160.80	31.20	70.01	0.00	5.60	2.40
40-13	1.03	0.00	2.78	0.00	1.39	0.00	137.10	3.70	1.63	0.00	4.20	0.40
40-20	0.92	0.00	2.63	0.00	1.19	0.00	133.40	0.00	1.40	0.00	4.20	0.00
50-12	7.74	0.00	14.12	0.00	5.04	0.00	211.70	34.60	8.35	0.00	10.60	1.20
50-16	2.17	0.00	7.37	0.00	2.67	0.00	183.10	0.00	2.75	0.00	4.40	0.00
50-25	1.70	0.00	6.77	0.00	2.50	0.00	172.40	0.00	2.86	0.00	4.20	0.00
100-25	227.04	0.00	377.72	0.00	229.34	0.00	550.00	196.20	218.29	0.00	23.10	2.50
100-33	55.96	0.00	213.63	0.00	156.31	0.00	402.40	78.40	161.86	0.00	7.10	2.10
100-50	53.79	0.00	204.16	0.00	138.82	0.00	327.50	0.00	139.80	0.00	4.20	1.50
140-35	959.51	0.00	1769.48	0.00	355.25	0.00	806.20	243.60	369.71	0.00	16.10	2.70
140-46	273.07	0.00	969.40	0.00	229.55	0.00	652.60	127.80	255.43	0.00	5.20	2.10
140-70	136.73	0.00	1013.76	0.00	173.47	0.00	468.80	0.00	198.92	0.00	3.70	0.60
180-45	1722.72	0.00	4948.15	0.00	534.47	0.00	796.20	391.70	600.39	0.00	4.50	2.40
180-60	956.11	0.00	3932.68	0.00	415.65	0.00	807.30	86.10	519.57	0.00	5.00	1.90
180-90	378.79	0.00	2939.66	0.00	360.76	0.00	694.10	0.00	392.59	0.00	4.50	1.20
200-50	5429.29	2.41	7200.00	80.86	4353.53	0.06	983.70	230.70	4150.96	0.45	4.90	2.20
200-66	4184.74	0.66	7200.00	80.57	3629.96	0.23	956.10	136.20	4519.16	0.33	5.30	1.80
200-100	2561.62	0.00	6803.08	80.00	1948.59	0.00	765.30	0.00	2582.17	0.00	4.30	0.60

Table 4.5: `Random` instances result tables for some λ-vectors

El herido

...

Retoñarán aladas de savia sin otoño
reliquias de mi cuerpo que pierdo en cada herida.
Porque soy como el árbol talado, que retoño:
porque aún tengo la vida.

Miguel Hernández - *El hombre acecha* (1939, but not published until 1981)

5 Application to Network Design Problems: Ordered Median Tree Location Problem

This chapter formally defines and addresses the *Ordered Median Tree Location Problem* (OMT), a single-allocation facility location model. In this problem, p facilities are to be located on a network represented by a undirected tree. The objective function is designed to minimize the sum of two key components: the ordered weighted average of allocation costs and the total cost of connecting the facilities within the tree. Given that ordered median location problems are notoriously difficult to solve, a significant portion of this chapter is dedicated to developing efficient solution methodologies. Several MILP formulations for the OMT are presented, together with enhancements which leverage properties from both minimum spanning tree problems and ordered optimization, and a comprehensive empirical analysis.

5.1 Introduction

As commented in previous chapters, facility location problems are concerned with attaining the optimal placement of facilities in order to minimize costs under certain considerations. In this context, facilities connection is an important feature to include in these problems. Gupta et al. (2001) introduced the Connected Facility Problem (ConnFL), which aims to find the optimal location of a set of facilities that require to be connected by a Steiner tree (see, e.g., Ljubić, 2021). Several papers and surveys cover different aspects of the ConnFL and new cases of study based on this problem (see, e.g, Fortz, 2015; Leitner et al., 2018; Ljubić, 2021). As detailed in Gollowitzer & Ljubić (2011), ConnFL is also related to other problems in the literature, for example, rent-or-buy problems (Kumar et al., 2002), Steiner tree-star problem (Lee et al., 1993; Khuller & Zhu, 2002), prize collecting capacitated connected facility location problem (Leitner & Raidl, 2011) or the tree of hubs location problem (Contreras et al., 2010; Pozo et al., 2021).

Although facilities connection can be modeled by using a Steiner tree problem, there are several other possibilities to guarantee such connectivity. Another common approach is exploiting the minimum spanning tree problem (MST) structural properties, (see, e.g., Anazawa, 2001). For further details on MST, the reader is referred to Section 2.6.4 in Chapter 2 of this dissertation. The most relevant properties of trees allow that the basic problem of finding a minimum cost spanning tree can be solved efficiently in polynomial time (Kruskal, 1956; Prim, 1957) and by means of linear programming tools (Miller et al., 1960; Edmonds et al., 1967; Gavish, 1983; Martin, 1991, among several others). Optimization problems related to spanning trees, or simply spanning tree problems, are among the core problems in combinatorial optimization, see, e.g., the former tree of hubs location problem (Contreras et al., 2010), the multi-objective spanning tree problem (Hamacher & Ruhe, 1994) or the Stackelberg minimum spanning tree problem (Cardinal et al., 2011; Labbé et al., 2021). Moreover, spanning trees are found in a wide range of applications (see, e.g., Clímaco & Pascoal, 2012).

This chapter extends the literature on connected facility location problems by introducing an ordered approach. To this end, the *Ordered Median Tree Location Problem* (OMT) is introduced and studied in detail, a single-allocation facility location problem where p facilities must be placed on a network connected by a undirected tree. The objective is to assign clients to its closest facility in order to minimize the ordered allocation average cost plus the facilities tree connection average cost. In this location problem, every client node can potentially become a facility and these facilities are uncapacitated, i.e., can serve as many clients as needed. Therefore, OMT has two main modeling aspects to model: the tree structure defined by the network and the rank dependent compensation factors applied to the costs of the system through the ordered function. Applications for the OMT arise in any discrete facility location problem requiring an interconnected network, such as in telecommunications, pipeline systems, or emergency response networks, where facilities must be linked to ensure minimal travel time and rapid response. The ordered median component not only enhances modeling flexibility but can also be used to address equity concerns, such as compensating for the worst-served clients (i.e., those located farthest from their assigned facility). A more detailed problem description will be provided in Section 5.2.

OMT is a new problem that generalizes both MST and DOMP. OMT includes as a particular case the MST when p is considered equal to the number of client nodes. If this is the case, clients would self-allocate to themselves giving rise to a MST among all nodes. As the number of facilities p decreases, clients not required to be self-allocated start to appear, as it happens in the ConnFL. Besides, if connectivity costs are set to zero, a “pure” DOMP arises. For these reasons, OMT can be seen as a bi-objective model in which the trade-off between connectivity and allocation costs requires to be balanced. Moreover, by considering the DOMP objective function, the possibilities in which client allocations are made are enriched, distinguishing OMT from the ConnFL. This chapter focuses on the median, k-center, and k-trimmed mean objectives, yet the range of possibilities is much broader. For instance, one may capture equity criteria through measures like the range, the sum of absolute differences, or weights inversely proportional to allocation costs, formulate obnoxious location problems, represent preferences using negative weight vectors, etc., exploring other approaches described in earlier chapters.

Furthermore, since solving the OMT implies solving a generalization of the DOMP and the MST, the problem is clearly separable. For this reason, a Benders decomposition arises naturally as a general OMT solving technique where the previous knowledge from DOMP and MST is jointly exploited. In general terms, the complicating variables of the original problem are projected out, resulting into an equivalent model with fewer variables, but many more constraints. To achieve optimality, a large number of these constraints will not be required, suggesting then a strategy of relaxation that ignores all but a few of these constraints. The interesed reader in referred to Martins de Sa et al. (2013) for a successful application of Benders decomposition in a hub location problem with inner tree structure.

The contributions of this chapter are the following. First, a new problem that was missing in the DOMP and ConnFL/MST literature is introduced. Specifically, OMT generalizes two well-studied, important problems in the sense that both facilities/clients connectivity and clients allocations are jointly included. Therefore, this new problem arises naturally as a bi-objective model that balances the trade-off between connectivity and allocation costs. Second, different

MILP formulations for the OMT that arise naturally from the properties of the DOMP and MST are presented comparing the number of variables and constraints as a function of the number of ties in the cost matrix. Regarding to the MST subproblem, ad hoc OMT connectivity cuts and flow formulations are provided. Third, theoretical results on the polytopes of the two main OMT formulations are derived, including the number of variables and constraints as a function of the number of ties in the cost matrix. This analysis can be extended to the DOMP and was not previously given in the literature. Fourth, also a different alternative formulation are studied in detail, one that treats the OMT as a single tree over the complete set of input nodes, and the Benders decomposition approach that arises naturally given the problem separability. This is complemented by several improvement techniques and extensive computational experiments that provide a detailed comparison of the different formulations, enhancements, and solution methods.

The remainder of the chapter is organized as follows. Section 5.2 formally defines the OMT and reviews a scheme of some well-known related problems in the literature. Section 5.3 introduces the catalogue of formulations and algorithms studied for the OMT. Section 5.5 describes the improvements developed to enhance the proposed models, the computation of initial solutions, and preprocessing phases for variable fixing. Finally, Section 5.6 analyzes the empirical performance of the resulting OMT formulations providing extensive numerical results.

5.2 Problem definition

This section formally introduces the OMT and establishes the notation used throughout the chapter. Given the necessity to model allocations that can be in either way, in or outside the facility, and since the MST is commonly modeled using arcs, particularly because several formulations describe arborescences in directed networks, the notation is slightly adapted from that of the previous chapter.

Let $G = (V, E)$ be a network where V is the set of nodes (assumed to be clients or potential facilities) and E the set of edges connecting nodes. Given the set of edges $E = \{(i, j) \mid i, j \in V, i < j\}$, the set of arcs $A = \{(i, j), (j, i) \mid (i, j) \in E\} \cup \{(i, i) \mid i \in V\}$, where loops connecting a node to itself are allowed, is defined. In the OMT, $p \leq |V|$ facilities must be placed on nodes of V and connected by a undirected tree. The model assumes single-allocation, i.e., each client node has a unique facility where it is allocated. A weight $c_{ij} \geq 0$ is defined for the cost of allocating client i to facility j or as the design cost of edge (i, j) whether both $i, j \in V$ are selected facilities. According to many OMT related problems described in the literature, these costs are assumed to be symmetrical between each pair of nodes, $c_{ij} = c_{ji}$, $i, j \in V$, and nodes can be allocated to themselves with no cost, $c_{ii} = 0$, $i \in V$, that is the so called free self-service assumption. In this context, free self-service is assumed, since it aligns more closely with the practical applications and potential uses of the OMT. In addition, the allocation costs of the $|V|$ clients to their corresponding facilities are compensated using scaling factor parameters $\lambda = (\lambda_1, \ldots, \lambda_{|V|})$. If client i is allocated to facility j at a cost c_{ij} and this cost is non-decreasingly ordered in the ℓ-th position among this type of costs, then this term would be scaled by λ_ℓ, i.e., the corresponding compensated cost would be $\lambda_\ell c_{ij}$. For the sake of understandability, the notation is summarized in Table 5.1.

Therefore, the OMT is to find the optimal location of p facilities in a network such that:

1. Each client is allocated to exactly one facility.
2. Facilities are connected together following a tree structure.
3. Edge costs connecting facilities plus compensated ranked allocation costs are minimized. Since the number of tree edges might be quite different from the number of allocation arcs, the objective function will minimize the average cost of the compensated ranked allocation

$G = (V, E)$	undirected network where V is the set of nodes and E is the set of edges
A	set of arcs associated to G where loops are allowed
p	fixed number of facilities to locate
c_{ij}	design cost of edge $(i, j) \in E$ or allocation cost between client i and a facility placed at node j
$\ell \in V$	index for the ℓ-th position of the sorted allocation costs sequence
λ_ℓ	scaling factor for the ℓ-th allocation cost
$E(S)$	subset of edges connecting nodes in $S \subset V$
$A(S)$	subset of arcs connecting nodes in $S \subset V$
$\delta(S)$	cut-set of $S \subset V$ (edges with one node in S and other node outside S)
$\delta^-(S)$	cut-set directed into $S \subset V$
$\delta^+(S)$	cut-set directed out of $S \subset V$

Table 5.1: Notation introduced for the OMT.

costs plus the average cost of a tree edge.

Figure 5.1 depicts different OMT solutions for a network example of $|V| = 10$ clients (*green circles*) and cost matrix (5.1), where $p = 5$ facilities (*red squares*) are located. Considering rounded costs proportional to Euclidean distances, client allocations to facilities (*arrows*) and the tree design connecting facilities (*dashed lines*) are different according to different criteria, namely (a) median $\lambda = (1, \overset{10}{\ldots}, 1)$, (b) k-centrum $\lambda = (0, \overset{6}{\ldots}, 0, 1, \overset{4}{\ldots}, 1)$ and (c) k-trimmed mean criterion $\lambda = (0, 0, 0, 1, 1, 1, 1, 0, 0, 0)$. For the latter, the three most expensive allocations, which correspond to nodes 8, 9 and 10, are ignored for the objective value computation. Due to this reason, for this criterion, those clients could be allocated to any facility, despite in Figure 5.1 appear allocated to its closest facility. Recall that OMT objective function includes the minimization of $p - 1$ edges costs plus a variable number of allocation costs that will vary depending on the instance size, the criteria selected, etc. That is, the contribution of the objective function corresponding to the tree design part may change considerably compared to the weight of the compensated ranked allocation costs. For that reason, since the number of tree edges might be quite different from the number of allocation arcs, the objective function minimizes the average cost of the compensated ranked allocation costs plus the average cost of a tree edge. Averaging the two features in the objective function is needed in order to normalize the trade-off between these two elements. As a consequence of this rationale, a higher value in the k-centrum optimum is observed because of the average between the higher costs divided by the number of costs considered in the lambda vector (smaller denominator) instead of the total average as in the median criterion.

$$\begin{pmatrix} 0 & 25 & 10 & 18 & 28 & 21 & 32 & 35 & 40 & 47 \\ 25 & 0 & 27 & 14 & 12 & 28 & 20 & 43 & 45 & 40 \\ 10 & 27 & 0 & 15 & 25 & 11 & 27 & 25 & 30 & 39 \\ 18 & 14 & 15 & 0 & 10 & 14 & 14 & 29 & 32 & 32 \\ 28 & 12 & 25 & 10 & 0 & 21 & 7 & 34 & 34 & 28 \\ 21 & 28 & 11 & 14 & 21 & 0 & 20 & 16 & 20 & 28 \\ 32 & 20 & 27 & 14 & 7 & 20 & 0 & 29 & 28 & 20 \\ 35 & 43 & 25 & 29 & 34 & 16 & 29 & 0 & 7 & 25 \\ 40 & 45 & 30 & 32 & 34 & 20 & 28 & 7 & 0 & 20 \\ 47 & 40 & 39 & 32 & 28 & 28 & 20 & 25 & 20 & 0 \end{pmatrix} \tag{5.1}$$

A first consideration about the OMT is that it is an NP-complete problem, which comes from the hardness results of the different combinatorial problems that give rise to the OMT. Both the p-median location problem and the ordered median problem are known to be NP-complete problems on its decision version (Nickel & Puerto, 2005).

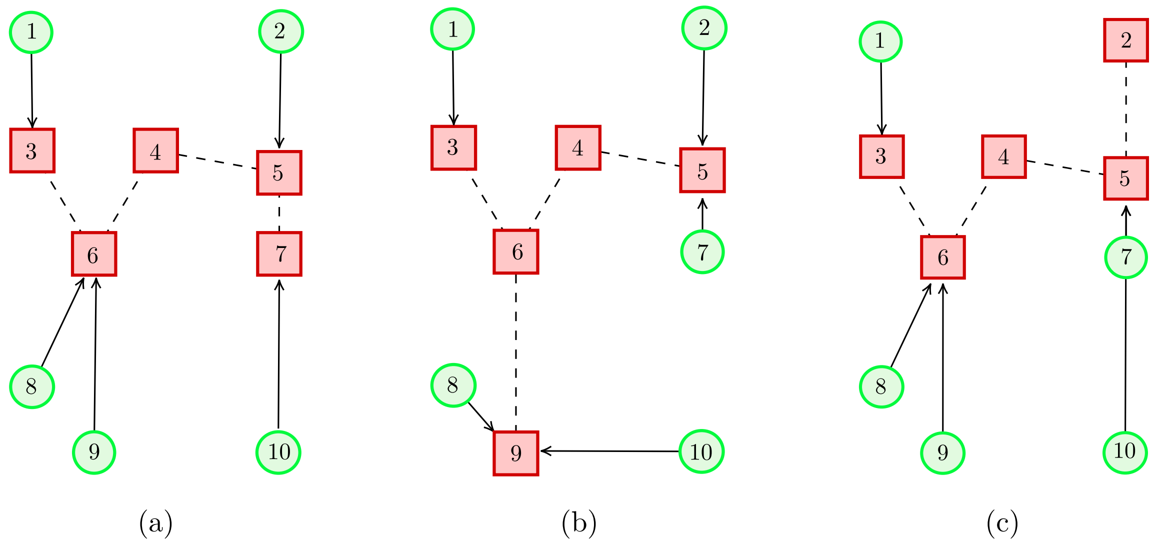


Figure 5.1: Three different OMT solutions according to (a) median [objective value: 18.3], (b) k-centrum [objective value: 26.0] and (c) k-trimmed mean criterion [objective value: 16.0], considering rounded costs proportional to Euclidean distances.

Second, as mentioned, the OMT has two main modeling aspects: the tree structure defined by the facilities network (tree connectivity) and the rank-dependent compensation factors applied to the operation cost of the system through the ordered median function (sorting). Therefore, the OMT can be seen as the classical p-Median Location Problem (pMED) adding two different features: *tree connectivity* and *sorting*. In this way, the OMT has as subproblems different well-known problems of the literature. The connectivity imposed to pMED gives rise to the pMED with inner Connected Structure (pMEDC). When the connected structure is a tree, one has the pMED with inner Tree Structure (pMEDT), that is closely related to several known problems in the literature as the tree-star network design problem (Nguyen & Knippel, 2007) or the already mentioned connected facility location problem (Gollowitzer & Ljubić, 2011). If the sorting feature is included in pMED, then it becomes the DOMP. The union of both features gives raises to the DOMP with inner Connected Structure (OMC) from which, when the connected structure considered is a tree, the problem under study is the OMT. The reader is referred to Figure 5.2 for an overview of the different considered problems and their relationships.

In addition, the OMT can be understood as a subproblem of the Ordered Median Tree of Hubs Location Problem (OMTHL), introduced in Pozo et al., 2021, where flow exchange between origin-destination pairs is considered for an OMT. The interested reader may also check the tree of hubs location problem (Contreras et al., 2010) and the ordered median hub location problem (Puerto et al., 2011a). Dealing with the OMT as a subproblem of the OMTHL is valuable in the sense that the model structure is inherited. Nevertheless, the inherent complexity of the OMTHL and the fact that the OMT holds its own interest, suggests that the OMT should be studied as a singular problem.

5.3 OMT formulations

This section details the core MILP formulations of this chapter. It begins by introducing the ordering polytope, which involves a comparison of two distinct approaches based on three-index sorting variables and radius variables, complemented by a discussion of key theoretical insights. Subsequently, the tree polytope is examined, with a focus on the adaptations necessary to embed its structure within the OMT formulations. Finally, the section concludes by presenting an alternative approach that models the entire problem as a tree.

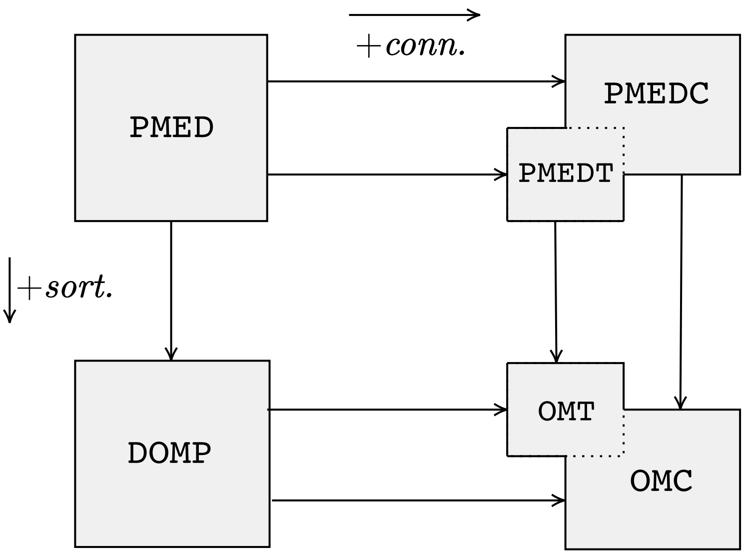


Figure 5.2: Diagram of relationships among the different OMT subproblems

5.3.1 Ordering polytope description

In the following, the general integer programming formulations for the OMT are introduced, focusing on describing the ordering feature first. Along the chapter, it is assumed working with complete networks. However, the proposed formulations are valid for general graphs with edge set E and arc set A, as long as the connectivity of the network is ensured. For modeling purposes the following variables are defined:

- $x_{ij} \in \{0,1\}$, $(i,j) \in A$, equal to 1 if client i is allocated to facility j, 0 otherwise.
- $z_{ij} \in \{0,1\}$, $(i,j) \in E$, equal to 1 if an edge (i,j) connects facilities $i,j \in V$, 0 otherwise.
- $x^{\ell}_{ij} \in \{0,1\}$, $(i,j) \in A, \ell \in V$, equal to 1 if client i is allocated to facility j and c_{ij} is ranked in the ℓ-th position, 0 otherwise.

The formulation presented below is based on the DOMP three-index variables formulation first introduced by Boland et al. (2006), including the tree connectivity feature of the OMT. Therefore, the formulation can be defined as follows:

$$F1^{\mathcal{T}}_{x^{\ell}} : \min \frac{1}{\sum_{\ell\in V}\lambda_\ell} \sum_{\ell\in V}\sum_{(i,j)\in A} \lambda_\ell c_{ij} x^{\ell}_{ij} + \frac{1}{p-1}\sum_{(i,j)\in E} c_{ij} z_{ij}, \tag{5.2a}$$

$$\text{s.t.} \sum_{i\in V} x_{ii} = p, \tag{5.2b}$$

$$\sum_{j\in V} x_{ij} = 1, \qquad i \in V, \tag{5.2c}$$

$$x_{ij} \le x_{jj}, \qquad (i,j) \in A : i \neq j, \tag{5.2d}$$

$$2z_{ij} \le x_{ii} + x_{jj}, \qquad (i,j) \in E, \tag{5.2e}$$

$$z_{ij} \in \mathcal{T}, \qquad (i,j) \in E, \tag{5.2f}$$

$$\sum_{\ell\in V} x^{\ell}_{ij} = x_{ij}, \qquad (i,j) \in A, \tag{5.2g}$$

$$\sum_{(i,j)\in A} x^{\ell}_{ij} = 1, \qquad \ell \in V, \tag{5.2h}$$

$$\sum_{(i,j)\in A} c_{ij} x^{\ell}_{ij} \le \sum_{(i,j)\in A} c_{ij} x^{\ell+1}_{ij}, \qquad \ell \in V : \ell < |V|, \tag{5.2i}$$

$$x_{ij} \in \{0,1\}, \qquad (i,j) \in A, \tag{5.2j}$$
$$x_{ij}^{\ell} \in \{0,1\}, \qquad (i,j) \in A, \ell \in V, \tag{5.2k}$$
$$z_{ij} \in \{0,1\}, \qquad (i,j) \in E. \tag{5.2l}$$

The objective function (5.2a) minimizes the compensated allocation costs weighted by the number of allocations considered (first component) plus the design cost of the tree connecting facilities weighted by the number of edges (second component). In other words, (5.2a) minimizes the average cost of an allocation plus the average cost of a tree edge. The group of constraints (5.2b), (5.2c) and (5.2d) models the allocation of clients to facilities. Specifically, constraint (5.2b) fixes the number of facilities in the network to p, constraints (5.2c) ensure that each client is allocated to exactly one facility and constraints (5.2d) ensure that no client is allocated to a non-facility node. Constraints (5.2e) relate the allocation variables to the design variables, imposing that each edge of the facilities spanning tree can only be selected if both edge nodes are facilities. Note that these constraints are obtained from the unification of two simpler group of constraints ($z_{ij} \leq x_{ii}$ & $z_{ij} \leq x_{jj}$ for $(i,j) \in E$). Constraints (5.2f) model the connectivity feature of the OMT describing the facilities spanning tree polytope $\mathcal{T}$. These constraints are different depending on the tree characterization considered (see Section 5.3.2). Finally, constraints (5.2g), (5.2h) and (5.2i) model the sorting feature of the problem. Specifically, constraints (5.2g) relate the sorting variables to the allocation variables guaranteeing that if the allocation of client i to facility j is selected, then only one sorting position can be assumed by this allocation, constraints (5.2h) ensures that every sorting position must be occupied and constraints (5.2i) sort allocation costs in non-decreasing order.

Note that, since $x_{ij} = \sum_{\ell \in V} x_{ij}^{\ell}$, by constraint (5.2g) it is possible to relax the integrality constraint (5.2j), defining $x_{ij} \in [0,1]$, $(i,j) \in A$. Furthermore, each x_{ij} can be replaced by a sum $\sum_{\ell \in V} x_{ij}^{\ell}$ unifying the allocation and sorting set of variables. For the last, despite the number of variables is reasonably reduced, the number of constraints increases and preliminary tests have shown that such unification is not efficient in computational terms. For this reason, disaggregated formulations are presented along the text.

Next, a second formulation based on the *radius variables*, already defined in (4.5) in Section 4.3 of the previous chapter, is introduced. To this end, recall the ordered sequence of unique allocation costs of $c_{ij} \geq 0$, $i,j \in V$, such that $c_{(0)} := 0 < c_{(1)} < c_{(2)} < \ldots < c_{(|H|)} = \max_{i,j \in V} c_{ij}$, where $|H|$ is the number of different non-zero elements of the above allocation cost sequence and $H = \{1, \ldots, |H|\}$. Then, the radius variables are defined as:

- $u_{\ell h} \in \{0,1\}$, $\ell \in V, h \in H$, if the ℓ-th allocation cost is at least $c_{(h)}$,

that is, the ℓ-th smallest cost is equal to $c_{(h)}$ if and only if $u_{\ell 1} = \ldots = u_{\ell h} = 1$ and $u_{\ell h+1} = \ldots = u_{\ell |H|} = 0$.

As reviewed in previous works (see Puerto et al., 2011a or Pozo et al., 2021 among others), sorting costs can be rather difficult in large instances when using the x^{ℓ} variables previously defined. Besides, formulations using radius variables (see Elloumi et al., 2004; Nickel & Puerto, 2005; Puerto, 2008; Espejo et al., 2009; Marín et al., 2009, 2010; Puerto et al., 2011a; García et al., 2011) can be used to downsize the number of variables in use, as shown in the next example.

■ **Example 5.1** Given $c = \begin{pmatrix} 0 & 2 & 4 \\ 2 & 0 & 0 \\ 4 & 0 & 0 \end{pmatrix}$, $p = 1$ and $\lambda = (1,1,1)$ an OMT solution implies $x_{12}^3 = x_{22}^1 = x_{32}^2 = 1$ (or equivalently $x_{12}^3 = x_{22}^2 = x_{32}^1 = 1$). Since $c_{(0)} := 0 < c_{(1)} = 2 < c_{(2)} = 4$, one has $u = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{pmatrix}$. ■

Recalling the radius formulation introduced in the previous chapter, one can rewrite $F1^{\mathcal{T}}_{x\ell}$ as follows:

$$
\begin{aligned}
F1^{\mathcal{T}}_{u} : \min & \frac{1}{\sum_{\ell \in V} \lambda_\ell} \sum_{\ell \in V} \sum_{h \in H} \lambda_\ell u_{\ell h}(c_{(h)} - c_{(h-1)}) + \frac{1}{p-1} \sum_{(i,j) \in E} c_{ij} z_{ij}, && \text{(5.3a)}\\
\text{s.t. } & \sum_{i \in V} x_{ii} = p, && \text{(5.3b)}\\
& \sum_{j \in V} x_{ij} = 1, \quad i \in V, && \text{(5.3c)}\\
& x_{ij} \leq x_{jj}, \quad (i,j) \in A : i \neq j, && \text{(5.3d)}\\
& 2z_{ij} \leq x_{ii} + x_{jj}, \quad (i,j) \in E, && \text{(5.3e)}\\
& z_{ij} \in \mathcal{T}, \quad (i,j) \in E, && \text{(5.3f)}\\
& \sum_{\ell \in V} u_{\ell h} = \sum_{i,j \in V : c_{ij} \geq c_{(h)}} x_{ij}, \quad h \in H, && \text{(5.3g)}\\
& u_{\ell h} \leq u_{\ell+1 h}, \quad h \in H, \ell \in V : \ell < |V|, && \text{(5.3h)}\\
& u_{\ell h} \geq u_{\ell h+1}, \quad h \in H, \ell \in V : h < |H|, && \text{(5.3i)}\\
& x_{ij} \in \{0,1\}, \quad (i,j) \in A, && \text{(5.3j)}\\
& u_{\ell h} \in \{0,1\}, \quad h \in H, \ell \in V, && \text{(5.3k)}\\
& z_{ij} \in \{0,1\}, \quad (i,j) \in E. && \text{(5.3l)}
\end{aligned}
$$

Constraints (5.3g) state that the number of allocations with a cost at least $c_{(h)}$ must be equal to the number of sites that support allocation costs greater than or equal to $c_{(h)}$ and constraints (5.3h) and (5.3i) sorts the values of the u variables in non-decreasing order. As noted by Marín et al. (2009) or Labbé et al. (2017), the additional sorting constraints (5.3i) are technically redundant. Nevertheless, they are included because they have been proven to strengthen the overall radius formulation. Indeed, these constraints become binding in some of the results discussed in this section, which underscores their practical value in tightening the model.

Note that $F1^{\mathcal{T}}_{u}$ is defined over a set H whose cardinality depends on the unique non-zero elements in the cost matrix. Recall that OMT assumes free self-service and symmetrical costs, what implies a significant number of repetitions in the cost matrix and a diagonal of zeros. To be more precise, let $\hat{\alpha}$ be the number of ties within the part of the cost matrix above the diagonal and by $\hat{\chi}_0$ a binary parameter equal to 1 if there is at least a 0 within the part of the cost matrix, zero otherwise. Thus,

- $0 \leq \hat{\alpha} \leq \frac{1}{2}(|V|^2 - |V|) - 1$.
- $\hat{\alpha} = \frac{1}{2}(|V|^2 - |V|) - \hat{\chi}_0 - |H|$.

Note that $\hat{\alpha}$ can also be easily computed as $\hat{\alpha} = \sum_{i=0}^{|H|}(m_i - 1)$, where m_i stands for the multiplicity of cost $c_{(h)}$ within the part of the cost matrix above the diagonal. Table 5.2 provides theoretical size values in terms of variables (*#variables*) and constraints (*#constraints*) of both, $F1^{\mathcal{T}}_{x\ell}$ and $F1^{\mathcal{T}}_{u}$, OMT formulations. Note that, the dimensions do not include the redundant set (5.3i) for $F1^{\mathcal{T}}_{u}$, and both formulations are specified independently of the $\mathcal{T}$ used but including (5.2f) and (5.3f) as constraints.

From Table 5.2, the following property is observed:

	#variables	#constrains
$F1^{\mathcal{T}}_{x^\ell}$:	$\|V\|^3 + \frac{3\|V\|^2}{2} - \frac{\|V\|}{2}$	$3\|V\|^2 + \|V\|$
$F1^{\mathcal{T}}_{u}$:	$\|V\|\|H\| + \frac{3\|V\|^2}{2} - \frac{\|V\|}{2} =$ $\frac{1}{2}\|V\|^3 + \|V\|^2 - (\hat{\chi_0} + \hat{\alpha} + \frac{1}{2})\|V\|$	$2\|V\|^2 + \|H\|\|V\| - \|V\| + 1 =$ $\frac{1}{2}\|V\|^3 + \frac{3}{2}\|V\|^2 - (\hat{\chi_0} + \hat{\alpha} + 1)\|V\| + 1$

Table 5.2: Theoretical model dimensions in the OMT formulations

Property 5.3.1

(a) $F1^{\mathcal{T}}_{u}$ has $\frac{1}{2}|V|^3 + \frac{1}{2}|V|^2 + (\hat{\chi_0} + \hat{\alpha})|V| > 0$ less variables than $F1^{\mathcal{T}}_{x^\ell}$.

(b) The difference of constraints between $F1^{\mathcal{T}}_{u}$ and $F1^{\mathcal{T}}_{x^\ell}$ is $\frac{1}{2}|V|^3 - \frac{3}{2}|V|^2 - (\hat{\chi_0} + \hat{\alpha} + 2)|V| + 1$. Therefore, $F1^{\mathcal{T}}_{u}$ has less constraints than $F1^{\mathcal{T}}_{x^\ell}$ if and only if

$$\hat{\alpha} \geq \frac{1}{2}|V|^2 - \frac{3}{2}|V| - \hat{\chi_0} - 1,$$

that is, $|H| \leq |V| + 1$.

Proof. (a) Is straightforward to prove.

(b) In terms of constraints, the difference between $F1^{\mathcal{T}}_{x^\ell}$ and $F1^{\mathcal{T}}_{u}$ is the sorting group of constraints, (5.2g)-(5.2i) and (5.3g)-(5.3h) respectively. For this group, there exists a total of $|V|^2 + 2|V| - 1$ constraints in $F1^{\mathcal{T}}_{x^\ell}$ and $|H||V| = \frac{1}{2}|V|^3 - \frac{1}{2}|V|^2 - (\hat{\chi_0} + \hat{\alpha})|V|$ constraints in $F1^{\mathcal{T}}_{u}$. Therefore, subtracting both expressions one gets that the difference in the number of constraints is $\frac{1}{2}|V|^3 - \frac{3}{2}|V|^2 - (\hat{\chi_0} + \hat{\alpha} + 2)|V| + 1$, that is negative only if $\hat{\alpha} > \frac{1}{2}|V|^2 - \frac{3}{2}|V| - \hat{\chi_0} - 2 + \frac{1}{|V|}$ (that is $\hat{\alpha} \geq \frac{1}{2}|V|^2 - \frac{3}{2}|V| - \hat{\chi_0} - 1$) and $|H| < |V| + 2 - \frac{1}{|V|}$ (that is $|H| \leq |V| + 1$). ∎

Formulation of the ordered median using x^ℓ variables is the general representation used to describe the problem, while formulation using radius variables performs best computationally. There exist several more formulations to represent the ordered median problem (see, e.g., Labbé et al. (2017), or the models presented in Chapter 4). However, the analysis of such other formulations embedded within the OMT structure is beyond the scope of this research.

Next, a theoretical comparison of the polytopes regarding the formulations $F1^{\mathcal{T}}_{x^\ell}$ and $F1^{\mathcal{T}}_{u}$ presented below is provided. Comparing OMT formulations is equivalent to comparing DOMP formulations because both problems differ only in the connection feature, which is modeled by the same group of constraints for every formulation. Therefore, some of the results adapted to the OMT polytopes comparison are those presented in Labbé et al. (2017) for DOMP. The goal is to state the formal relationships between the linear relaxations of the considered formulations and study the similarities between these polytopes, describing in detail the degenerate case of no ties in the cost matrix. Also, in this section, the OMT assumptions would be relaxed in an attempt to generalize these results.

First, the results presented next are valid for a general cost matrix independently of the number of ties. The reader may note that OMT does not really require free self-service and symmetric cost assumptions. Without these two assumptions our formulation is still valid although it slightly changes the meaning of costs c_{ii} that would turn to be the opening cost of facility i plus de allocation cost of client i to facility i (if a facility is opened i, client i has to be allocated to such facility).

Let $\Omega^{\mathcal{T}}_{x^\ell}$ be the set of points satisfying constraints (5.2b)-(5.2k) and $\Omega^{\mathcal{T}}_{u}$ be the set of points satisfying constraints (5.3b)-(5.3k). Let also $\phi_{x^\ell}(\cdot)$ and $\phi_u(\cdot)$ denote the value of the objective functions, (5.2a) and (5.3a) respectively, evaluated at a feasible point $(\cdot)$ and $P(\Omega^{\mathcal{T}}_{\mathcal{S}})$ for $\mathcal{S} \in \{x^\ell, u\}$ the polytope defined from the linear relaxation of $\Omega^{\mathcal{T}}_{\mathcal{S}}$. For the sake of understandability, the tree notation $\mathcal{T}$ is removed in this first analysis since is not relevant. A first property to recall is the following.

Property 5.3.2 There exist two mappings f and g,

$$f : \mathbb{R}^{|V|^2} \times \mathbb{R}^{|V|^2} \times \mathbb{R}^{|V|^3} \longmapsto \mathbb{R}^{|V|^2} \times \mathbb{R}^{|V|^2} \times \mathbb{R}^{|V||H|},$$
$$(x_{ij}, z_{ij}, x^\ell_{ij}) \longmapsto (x_{ij}, z_{ij}, u_{\ell h}).$$
$$g : \mathbb{R}^{|V|^2} \times \mathbb{R}^{|V|^2} \times \mathbb{R}^{|V||H|} \longmapsto \mathbb{R}^{|V|^2} \times \mathbb{R}^{|V|^2} \times \mathbb{R}^{|V|^3},$$
$$(x_{ij}, z_{ij}, u_{\ell h}) \longmapsto (x_{ij}, z_{ij}, x^\ell_{ij}).$$

such that,

- For any point $(x_{ij}, z_{ij}, x^\ell_{ij}) \in P(\Omega_{x^\ell})$, then $\phi_{x^\ell}(x_{ij}, z_{ij}, x^\ell_{ij}) = \phi_u(f(x_{ij}, z_{ij}, x^\ell_{ij}))$.
- For any point $(x_{ij}, z_{ij}, u_{\ell h}) \in P(\Omega_u)$, then $\phi_u(x_{ij}, z_{ij}, u_{\ell h}) = \phi_{x^\ell}(g(x_{ij}, z_{ij}, u_{\ell h}))$.

These mapping definitions are given in Labbé et al. (2017). Via these mappings, several polytope comparisons can be attained.

Property 5.3.3

(a) $g(P(\Omega_u)) \subset P(\Omega_{x^\ell})$.

(b) $f(P(\Omega_{x^\ell})) \not\subset P(\Omega_u)$.

Proof. (a) See *Theorem 1* in Labbé et al. (2017).

(b) Consider the network example depicted in Figure 5.3, where client nodes (*circles*) allocate to a single facility (*square*).

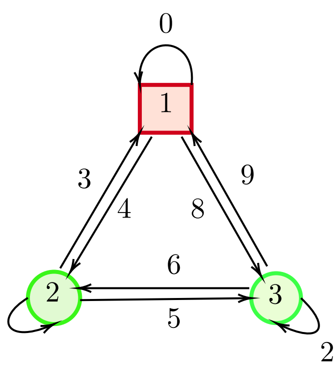


Figure 5.3: OMT complete network example.

Let the following feasible solution (non-optimal) defined by the values:

$$\begin{array}{lll} x_{11} = 1, & x_{12} = 0, & x_{13} = 0, \\ x_{21} = 1, & x_{22} = 0, & x_{23} = 0, \\ x_{31} = 1, & x_{32} = 0, & x_{33} = 0, \\ z_{12} = 0, & z_{13} = 0, & z_{23} = 0, \end{array}$$

$$\begin{array}{lll} x^1_{11} = 0.9, & x^2_{11} = 0, & x^3_{11} = 0.1, \\ x^1_{21} = 0, & x^2_{21} = 1, & x^3_{21} = 0, \\ x^1_{31} = 0.1, & x^2_{31} = 0, & x^3_{31} = 0.9, \end{array}$$

$$x^\ell_{ij} = 0 \text{ otherwise.}$$

One can verify that the previous solution satisfies the group of constraints defining $P(\Omega_{x^\ell})$ (shown only for sorting constraints):

- $\sum_{\ell \in V} x^\ell_{ij} = x_{ij}, \quad (i,j) \in A.$

$$\begin{aligned} x^1_{11} + x^2_{11} + x^3_{11} = 1 = x_{11}, \\ x^1_{21} + x^2_{21} + x^3_{21} = 1 = x_{21}, \\ x^1_{31} + x^2_{31} + x^3_{31} = 1 = x_{31}, \end{aligned}$$

Otherwise trivially equal to 0.

- $\sum_{(i,j) \in A} x^\ell_{ij} = 1, \quad \ell \in V.$

$$\begin{aligned} x^1_{11} + x^1_{21} + x^1_{31} + 0 \ldots = 1, \\ x^2_{11} + x^2_{21} + x^2_{31} + 0 \ldots = 1, \\ x^3_{11} + x^3_{21} + x^3_{31} + 0 \ldots = 1, \end{aligned}$$

- $\sum_{(i,j) \in A} c_{ij} x^\ell_{ij} \leq \sum_{(i,j) \in A} c_{ij} x^{\ell+1}_{ij}, \quad \ell \in V : \ell < |V.$

$$\begin{aligned} c_{11}x^1_{11} + c_{31}x^1_{31} + 0 \ldots = 0.9 \cdot 0 + 0.1 \cdot 9 = 0.9 \leq c_{21}x^2_{21} + 0 \ldots = 1 \cdot 3 = 3, \\ c_{21}x^2_{21} + 0 \ldots = 1 \cdot 3 = 3 \leq c_{11}x^3_{11} + c_{31}x^3_{31} + 0 \ldots = 0.9 \cdot 9 + 0.1 \cdot 0 = 8.1 \end{aligned}$$

An expression for u variables using f definition can be given by the system of equations:

$$\begin{aligned} f : (x_{ij}, z_{ij}, x^\ell_{ij}) \longmapsto (x_{ij}, z_{ij}, u_{\ell h}), \\ u_{\ell h} = \sum_{i,j \in V : c_{ij} \geq c_{(h)}} x^\ell_{ij} \quad h \in H, \ell \in V. \end{aligned} \tag{5.4}$$

Therefore,

$$\begin{array}{llllll} u_{11} = 0.1, & \cdots & \cdots & \cdots & \cdots & u_{18} = 0.1, \\ u_{21} = 1 & \cdots & u_{23} = 1, & u_{24} = 0, & \cdots & u_{28} = 0, \\ u_{31} = 0.9, & \cdots & \cdots & \cdots & \cdots & u_{38} = 0.9, \end{array}$$

which does not satisfy constraints $u_{\ell h} \leq u_{\ell+1 h}$ for $h \in H, \ell \in V : \ell < |V|$ in the relaxed polytope $P(\Omega_u)$. ■

As a consequence, there exists a feasible solution in $P(\Omega_{x^\ell})$ whose image by f is not feasible in $P(\Omega_u)$. Next, the objective function value of the points $(x_{ij}, z_{ij}, x^\ell_{ij}) \in P(\Omega_{x^\ell})$ such that $f(x_{ij}, z_{ij}, x^\ell_{ij}) \notin P(\Omega_u)$ is studied. To this end, consider first the following polytope namely $P(\Omega'_{x^\ell})$, where constraints (5.2i) are replaced by constraints

$$\sum_{i,j \in V : c_{ij} \geq c_{(h)}} x^\ell_{ij} + \sum_{i,j \in V : c_{ij} < c_{(h)}} x^{\ell+1}_{ij} \leq 1, \quad h \in H, \ell \in V : \ell < |V|, \tag{5.5}$$

which are a consequence of the *staircase inequalities* in Labbé et al. (2017). Observe that,

Property 5.3.4

(a) $f(P(\Omega'_{x^\ell})) = P(\Omega_u)$.

(b) $g(P(\Omega_u)) \subset P(\Omega'_{x^\ell})$.

Proof. (a) Following the results in Labbé et al. (2017), g satisfies the equality of f by construction, thus $f(g(u)) = u$. This implies $f(P(\Omega'_{x^\ell})) \supset P(\Omega_u)$ and it is only necessary to prove $f(P(\Omega'_{x^\ell})) \subset P(\Omega_u)$. Given $(x_{ij}, z_{ij}, x_{ij}^\ell) \in P(\Omega'_{x^\ell})$, it is proven next that $f(x_{ij}, z_{ij}, x_{ij}^\ell)$ satisfies constraints (5.3g)-(5.3i). Constraints (5.3i) are satisfied by means of f definition,

$$u_{\ell h} = \sum_{i,j \in V: c_{ij} \geq c_{(h)}} x_{ij}^\ell \geq \sum_{i,j \in V: c_{ij} \geq c_{(h+1)}} x_{ij}^\ell = u_{\ell h+1},$$

constraints (5.3g) are satisfied because of the following

$$\sum_{\ell \in V} u_{\ell h} = \sum_{\ell \in V} \sum_{i,j \in V: c_{ij} \geq c_{(h)}} x_{ij}^\ell = \sum_{i,j \in V: c_{ij} \geq c_{(h)}} \sum_{\ell \in V} x_{ij}^\ell \overset{(5.2g)}{=} \sum_{i,j \in V: c_{ij} \geq c_{(h)}} x_{ij},$$

and constraints (5.3h) because

$$\begin{aligned} u_{\ell h} &= \sum_{i,j \in V: c_{ij} \geq c_{(h)}} x_{ij}^\ell \overset{(5.5)}{\leq} 1 - \sum_{i,j \in V: c_{ij} < c_{(h)}} x_{ij}^{\ell+1} \overset{(5.2h)}{=} \sum_{(i,j) \in A} x_{ij}^{\ell+1} - \sum_{i,j \in V: c_{ij} < c_{(h)}} x_{ij}^{\ell+1} \\ &= \sum_{i,j \in V: c_{ij} < c_{(h)}} x_{ij}^{\ell+1} + \sum_{i,j \in V: c_{ij} \geq c_{(h)}} x_{ij}^{\ell+1} - \sum_{i,j \in V: c_{ij} < c_{(h)}} x_{ij}^{\ell+1} = \sum_{i,j \in V: c_{ij} \geq c_{(h)}} x_{ij}^{\ell+1} = u_{\ell+1,h}. \end{aligned}$$

(b) By *Property 5.3.3*, it is known that $g(P(\Omega_u)) \subset P(\Omega_{x^\ell})$. Hence, it is only necessary to prove that the constraints (5.5) are satisfied. By constraints (5.3h) one gets

$$u_{\ell h} \leq u_{\ell+1 h} \implies \sum_{i,j \in V: c_{ij} \geq c_{(h)}} x_{ij}^\ell \leq \sum_{i,j \in V: c_{ij} \geq c_{(h)}} x_{ij}^{\ell+1} \implies \sum_{i,j \in V: c_{ij} < c_{(h)}} x_{ij}^{\ell+1} + \sum_{i,j \in V: c_{ij} \geq c_{(h)}} x_{ij}^\ell \leq 1.$$

■

Consequently, the following result can be given as a conclusion when a general cost matrix is analyzed.

Property 5.3.5 $\phi_u = \phi'_{x^\ell} \geq \phi_{x^\ell}$.

Proof. This property is a consequence of *Property 5.3.2* and $P(\Omega_u) = f(P(\Omega'_{x^\ell})) \subset f(P(\Omega_{x^\ell}))$. ■

As an observation to Property 5.3.5, our computational experience shows the equality $\phi_u = \phi'_{x^\ell} = \phi_{x^\ell}$ holds in the optimum.

Next, the degenerate case with no ties in the cost matrix is analyzed in detail. As stated in Labbé et al. (2017), observe that in such case the previous definition of f holds while g transforms into the simpler expression:

$$\begin{aligned} g : \mathbb{R}^{|V|^2} \times \mathbb{R}^{|V|^2} \times \mathbb{R}^{|V|^3} &\longmapsto \mathbb{R}^{|V|^2} \times \mathbb{R}^{|V|^2} \times \mathbb{R}^{|V|^3}, \\ (x_{ij}, z_{ij}, u_{\ell h}) &\longmapsto (x_{ij}, z_{ij}, x_{ij}^\ell), \end{aligned}$$

$$x_{ij}^\ell = \begin{cases} 0 & \text{if } \ell < \ell(h) \text{ or } x_{ij} = 0, \\ \min\left\{ x_{ij} - \sum_{k<\ell} x_{ij}^k, u_{\ell h} - u_{\ell,h+1} \right\} & \text{if } \ell \geq \ell(h), \end{cases}$$

where $\ell(h) = \min\{\ell : u_{\ell h} - u_{\ell,h+1} > 0\}$ for $h = 1, \dots, H$, and $u_{\ell,H+1} = u_{\ell H}$. It is assumed that if $u_{\ell h} - u_{\ell,h+1} = 0$ for all $\ell \in V$ then $\ell(h) = +\infty$.

Furthermore, when there are no ties in the cost matrix, one can give an expression for f^{-1} using equations in (5.4). Since no ties exist in the cost matrix, each $c_{(h)}$ is uniquely associated with a c_{ij}. In other words, there exists a bijective mapping β between the sequence of costs defined as

$$\begin{aligned} \beta : H &\longmapsto \{ij \mid i,j \in V\}, \\ c_{(h)} &\longmapsto c_{\beta(h)} = c_{i_h j_h}. \end{aligned}$$

For every $\ell \in V$, system (5.4) can be expressed in matrix form as follows. Let $\underline{\mathbf{u}}_\ell = (u_{\ell 1}, \dots, u_{\ell |V|^2})'$ and $\underline{\mathbf{x}}^\ell = (x^\ell_{11}, \dots, x^\ell_{1|V|}, \dots, x^\ell_{|V|1}, \dots, x^\ell_{|V||V|})'$, then

$$\underline{\mathbf{u}}_\ell = \Sigma_\ell \underline{\mathbf{x}}^\ell \quad \forall \ell \in V, \tag{5.6}$$

where Σ_ℓ is a $|V|^2 \times |V|^2$ matrix. The first row in Σ_ℓ (corresponding to $h = 1$) has all coefficients equal to 1 since every cost in the sum is greater than $c_{(1)}$, the second row ($h = 2$) has all coefficients equal to 1 except for the coefficient corresponding to some position $i_1 j_1$ which is 0, and so until the last row ($h = |V|^2$), that has all coefficients equal to 0 except for the coefficient corresponding to some position $i_{|V|^2} j_{|V|^2}$ which is 1. Therefore, Σ_ℓ has the following structure

$$\Sigma_\ell = \begin{array}{cc} & \begin{array}{ccccccccccccccccc} \cdots & & i_1j_1 & & \cdots & & i_2j_2 & & \cdots & & i_3j_3 & & \cdots & & i_{|V|^2}j_{|V|^2} & & \cdots \end{array} \\ \begin{array}{c} h=1 \\ h=2 \\ \\ \vdots \\ \\ h=|V|^2 \end{array} & \begin{pmatrix} \cdots & 1 & 1 & 1 & \cdots & 1 & 1 & 1 & \cdots & 1 & 1 & 1 & \cdots & 1 & 1 & 1 & \cdots \\ \cdots & 1 & 0 & 1 & \cdots & 1 & 1 & 1 & \cdots & 1 & 1 & 1 & \cdots & 1 & 1 & 1 & \cdots \\ \cdots & 1 & 0 & 1 & \cdots & 1 & 0 & 1 & \cdots & 1 & 1 & 1 & \cdots & 1 & 1 & 1 & \cdots \\ \cdots & 1 & 0 & 1 & \cdots & 1 & 0 & 1 & \cdots & 1 & 0 & 1 & \cdots & 1 & 1 & 1 & \cdots \\ & & \vdots & & & & \vdots & & \ddots & & \vdots & & & & \vdots & & \ddots \\ \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 1 & 0 & \cdots \end{pmatrix} \end{array}. \tag{5.7}$$

Consequently, Σ_ℓ can be rearranged through elemental matrix transformations to an upper triangular matrix Σ^{upper}_ℓ, so one can rewrite system (5.6) as

$$\underline{\mathbf{u}}_\ell = \Sigma^{upper}_\ell \begin{pmatrix} x^\ell_{i_1 j_1} \\ x^\ell_{i_2 j_2} \\ \vdots \\ x^\ell_{i_{|V|^2} j_{|V|^2}} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 & 1 \\ 0 & 1 & 1 & \cdots & 1 & 1 \\ 0 & 0 & 1 & \cdots & 1 & 1 \\ & & & \vdots & & \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{pmatrix} \begin{pmatrix} x^\ell_{i_1 j_1} \\ x^\ell_{i_2 j_2} \\ \vdots \\ x^\ell_{i_{|V|^2} j_{|V|^2}} \end{pmatrix}. \tag{5.8}$$

Therefore, Σ_ℓ has maximum range which implies that f is an injection and (5.6) defines for every $\ell \in V$ a single unique solution for x^ℓ_{ij} variables for $i, j \in V$. A general expression for the inverse of Σ^{upper}_ℓ can be given to obtain the solution for x^ℓ_{ij} variables as follows

$$\underline{\mathbf{x}}^\ell = \begin{pmatrix} 1 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ & & & \vdots & & \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{pmatrix} \begin{pmatrix} u_{\ell 1} \\ u_{\ell 2} \\ \vdots \\ u_{\ell |V|^2} \end{pmatrix}, \tag{5.9}$$

being the mapping f^{-1} defined by the following relationship

$$x^{\ell}_{i_h j_h} = \begin{cases} u_{\ell h} - u_{\ell h+1} & h = 1, \dots, |V|^2 - 1, \\ u_{\ell |V|^2} & h = |V|^2. \end{cases} \tag{5.10}$$

Notice that constraints (5.3i) are needed in order for f^{-1} to be well defined. It can be verified that, in the case of no ties, f^{-1} and g coincide.

Property 5.3.6 If no ties exists in the cost matrix $(c_{ij})_{i,j\in V}$, then $g = f^{-1}$.

Proof. If $\ell \geq \ell(h)$, then $x_{ij} - \sum_{k<\ell} x^k_{ij} = \sum_{k\in V} x^k_{ij} - \sum_{k<\ell} x^k_{ij} = \sum_{k\geq \ell} x^k_{ij} \geq x^{\ell}_{i_h j_h} = u_{\ell h} - u_{\ell h+1} \implies \min\left\{x_{ij} - \sum_{k<\ell} x^k_{ij}, u_{\ell h} - u_{\ell,h+1}\right\} = u_{\ell h} - u_{\ell,h+1}$. Otherwise, if $\ell < \ell(h)$ or $x_{ij} = 0$, trivially equal to 0. ■

As a consequence, it holds that $g(P(\Omega'_u)) = P(\Omega'_{x^\ell})$ when no ties exists in the cost matrix.

5.3.2 Tree polytope description

Formulation $F1^{\mathcal{T}}_{x^\ell}$ and $F1^{\mathcal{T}}_u$ assume constraint $z \in \mathcal{T}$ to be included in the model. This constraint can be replaced by any representation of the MST polytope. This section extends the definition of the tree polytope $\mathcal{T}$ including some specific considerations that must be reviewed for the application of each tree characterization within the OMT. The reader is referred to Section 2.6.4 in Chapter 2 for further details on the MST formulations used.

In Table 2.1, the main properties of the MST formulations that would be considered are summarized. These are *subtour elimination*, *Miller-Tucker-Zemlim*, *flow*, and *Kipp Martin*. The criteria that have guided the selection of the formulations are either their good theoretical properties or some characteristic that seems useful as, for instance, a small number of variables or constraints. It is well known that, as seen in the last column, from the formulations described above $\mathcal{T}^{sub}$ and $\mathcal{T}^{km}$ hold the integrality porperty, conversely to $\mathcal{T}^{mtz}$ and $\mathcal{T}^{flow}$.

As a consequence of the formulations comparison in Fernández et al. (2017), a relationship between the different formulations derived from the combination of some sorting representations and tree polyhedra described above can be stated. To this end, let $P_{xz}(\Omega^{\mathcal{T}}_{\mathcal{S}})$ be the projected polytope onto the space of x and z of the linear relaxation of an OMT formulation. Then, the following property holds:

Property 5.3.7

$$P_{xz}(\Omega^{\mathcal{T}}_{x^\ell}) = P_{xz}(\Omega^{\mathcal{T}}_u). \tag{5.11}$$

$$P_{xz}(\Omega^{sub}_{\mathcal{S}}) = P_{xz}(\Omega^{km}_{\mathcal{S}}) \subseteq \begin{cases} P_{xz}(\Omega^{flow}_{\mathcal{S}}) \\ \neq \\ P_{xz}(\Omega^{mtz}_{\mathcal{S}}). \end{cases} \tag{5.12}$$

Proof. Is derived from the observations in Fernández et al. (2017). ■

Embedding the corresponding MST formulation into the OMT formulation requires addressing specific considerations. Regarding to $\mathcal{T}^{sub}$, note that the cardinality of the group of subtour elimination constraints is exponential in the number of nodes. Since many of these constraints are not needed to build the solution polytope, they can be separated in polynomial time by solving a series of minimum cut-problems. An effective branch-and-cut algorithm can be implemented to include dynamically a certain number of these constraints, preventing from using the entire set of subtour elimination constraints and reducing computational effort. Furthermore, a cut-set based group of constraints can be introduced replacing these constrains. Let $S \neq \emptyset, S \subset V$ be a connected component of a solution in the previous algorithm, the following connection cut can be defined:

$$\sum_{(i,j)\in\delta^+(S)} x_{ij} + \sum_{(i,j)\in\delta^-(S)} x_{ij} + \sum_{(i,j)\in\delta(S)} z_{ij} \geq 1, \qquad S \neq \emptyset, S \subset V. \tag{5.13}$$

This cut-set based connection cut implies that at least one allocation or one edge from the tree of facilities must connect S to a node of $V \setminus S$.

Regarding to $\mathcal{T}^{mtz}$, an arborescence is built rooted at an arbitrary root node in which arcs follow the direction from root to leaves. Note that if the given root node $r \in V$ is chosen as a facility, the arborescence will be rooted at r. Otherwise, r will be isolated and the root node will be arbitrarily chosen by the model. $\mathcal{T}^{flow}$ also relies on a source node $r \in V$ which distributes the flow, but for the OMT, contrary to $\mathcal{T}^{mtz}$, the choice of the root node is very influential as care must be taken when spreading the flow. This selection can be done in two ways: (1) adding a set of variables $r_i \in \{0,1\}$, $i \in V$, that are equal to 1 if the facility node $i \in V$ is selected as the source node for the tree of facilities, or otherwise (2) arbitrarily selecting the source node and ensuring flow units are only distributed between facilities, distinguishing whether the source node is a facility or not. If the source node is not a facility, then there must be no flow at the source node and the flow is distributed from the facility to which this source node is allocated.

Therefore, the flow formulation using additional variables includes the following group of constraints:

$$\mathcal{T}^{flow1}: \sum_{i\in V} r_i = 1, \tag{5.14a}$$

$$r_i \leq x_{ii}, \qquad i \in V, \tag{5.14b}$$

$$\sum_{(i,j)\in\delta^+(i)} f_{ij} - \sum_{(j,i)\in\delta^-(i)} f_{ji} = (p-1)r_i - (x_{ii} - r_i), \qquad i \in V, \tag{5.14c}$$

$$f_{ij} \leq (p-1)z_{ij}, \qquad (i,j) \in E, \tag{5.14d}$$

$$f_{ji} \leq (p-1)z_{ij}, \qquad (i,j) \in E. \tag{5.14e}$$

On the other hand, the flow formulation without using additional variables includes:

$$\mathcal{T}^{flow2}: \sum_{(i,j)\in\delta^+(i)} f_{ij} - \sum_{(j,i)\in\delta^-(i)} f_{ji} = px_{ri} - x_{ii}, \qquad i \in V, \tag{5.15a}$$

$$f_{ij} \leq (p-1)z_{ij}, \qquad (i,j) \in E, \tag{5.15b}$$

$$f_{ji} \leq (p-1)z_{ij}, \qquad (i,j) \in E. \tag{5.15c}$$

5.3.3 Alternative formulation

Previous $F1^{\mathcal{T}}_{x^\ell}$ and $F1^{\mathcal{T}}_{u}$ formulations use the set of design variables z to model a spanning tree connecting only facilities. However, since the OMT assumes single client-facility allocation, the tree structure of any OMT solution, uniquely determined by the allocation (x) and design (z) variables, is globally preserved. That is, by considering both the facilities tree and the allocations of clients to their corresponding facilities, a larger tree is obtained in which clients occupy the terminal nodes.

This rationale allows to provide an alternative formulation that models an OMT solution designing a spanning tree over the complete set of nodes V. Holding the definition of variables x and x^ℓ as before, let now $z_{ij} \in \{0,1\}$, $(i,j) \in E$, be equal to 1 if (i,j) is an edge connecting either facilities or client-facility pairs, and 0 otherwise. With this new definition of the z variables, one can reformulate $F1^{\mathcal{T}}_{x^\ell}$ as follows:

$$
\begin{aligned}
F2^{\mathcal{T}}_{x^\ell}: \ \min \quad & \frac{1}{\sum_{\ell\in V}\lambda_\ell}\sum_{\ell\in V}\sum_{(i,j)\in A}\lambda_\ell c_{ij}x^\ell_{ij} + \frac{1}{p-1}\sum_{(i,j)\in E} c_{ij}(z_{ij}-x_{ij}-x_{ji}), && && (5.16a)\\
\text{s.t.} \quad & \sum_{i\in V} x_{ii} = p, && && (5.16b)\\
& \sum_{j\in V} x_{ij} = 1, && i\in V, && (5.16c)\\
& 2x_{ij} \le 1 - x_{ii} + x_{jj}, && (i,j)\in A: i\neq j, && (5.16d)\\
& x_{ij} + x_{ji} \le z_{ij}, && (i,j)\in E, && (5.16e)\\
& 2z_{ij} \le x_{ii} + x_{jj} + x_{ij} + x_{ji}, && (i,j)\in E, && (5.16f)\\
& z_{ij} \in \mathcal{T}, && (i,j)\in E, && (5.16g)\\
& x_{ij} = \sum_{\ell\in V} x^\ell_{ij}, && (i,j)\in A, && (5.16h)\\
& \sum_{(i,j)\in A} x^\ell_{ij} = 1, && \ell\in V, && (5.16i)\\
& \sum_{(i,j)\in A} c_{ij}x^\ell_{ij} \le \sum_{(i,j)\in A} c_{ij}x^{\ell+1}_{ij}, && \ell\in V: \ell<|V|, && (5.16j)\\
& x_{ij}\in\{0,1\}, && (i,j)\in A, && (5.16k)\\
& x^\ell_{ij}\in\{0,1\}, && (i,j)\in A, \ell\in V, && (5.16l)\\
& z_{ij}\in\{0,1\}, && (i,j)\in E. && (5.16m)
\end{aligned}
$$

The objective function (5.16a) is similar to $F1^{\mathcal{T}}_{x^\ell}$, but needs to be modified in order to consider only the design cost of the part of the tree connecting facilities, this is, subtracting the associated cost to the edges allocating clients to facilities to the total tree design cost. Now, constraints (5.16d) guarantee that client-facility allocations are only possible if i is a client and j is a facility. Note that these constraints result from unification of two simpler group of constraints $x_{ij} \le 1 - x_{ii}$ and $x_{ij} \le x_{jj}$ for $(i,j) \in A : i \neq j$. Constraints (5.16e) ensure that only one allocation, client i to facility j or vicerversa, can be considered if there exists an edge connecting $i, j \in V$. In this sense, constraints (5.16f) impose that the only possibilities in which a tree edge can be selected are if both i or j are facilities or if i is a client and j its facility, or viceversa. The remaining constraints are unchanged with respect to the formulations presented above.

For this approach, similar $\mathcal{T}$ descriptions as in Section 5.3.2 can be considered, but the adaptation is straightforward since the main characteristic to model in this formulation is the tree. Now, the cut-set based group of constraints are simpler, $\sum_{(i,j)\in\delta(S)} z_{ij} \ge 1$, and for the flow formulation it is not necessary to differentiate whether the arbitrary source node selection is a facility or not, reducing formulations complexity. Formulation $F2^{\mathcal{T}}_{u}$ using radius variables for this alternative approach is left to the reader.

5.4 Benders decomposition for the OMT

Given that OMT is strongly separable into two subproblems (DOMP and MST), a Benders decomposition approach arises naturally. Further details on Benders decomposition can be found in Section 2.3.2 of Chapter 2 and in Chapter 4. In the context of OMT, it is natural to consider DOMP as the master problem (MP) and MST as the subproblem (SP). For example, the MP can be represented by $F1^{\mathcal{T}}_{x^\ell}$ as follows:

$$F^{MP}: \min \frac{1}{\sum_{\ell \in V} \lambda_\ell} \sum_{\ell \in V} \sum_{(i,j)\in A} \lambda_\ell c_{ij} x^\ell_{ij} + \mu, \tag{5.17a}$$

$$\text{s.t. } (5.2b) - (5.2d), (5.2g) - (5.2k), \tag{5.17b}$$

$$\mu \geq 0. \tag{5.17c}$$

Similar descriptions for F^{MP} can be introduced using other formulations in Section 5.3.

A classical Benders decomposition framework is illustrated in Algorithm 5.4.1. In this algorithm, the best upper and lower bounds obtained at each iteration, denoted by UB and LB, are maintained. The procedure is then as follows. The MP is solved to optimality and both the objective function obj^{MP}, which can be split into the sum of the DOMP objective (obj^{DOMP}) plus the μ term, and the variables representing the facilities selected $\bar{x} = \{x_{ii} \mid x_{ii} = 1, \forall i \in V\}$ are stored. At this moment, if possible, the lower bound is updated using the MP objective. Thereafter, the MST subproblem is solved setting as facilities the previously $\bar{x}$ identified in the MP solution and, if possible, the upper bound is updated using the (weighted) sum of the DOMP plus the SP objectives, $obj^{\mathsf{DOMP}} + obj^{SP}$. Finally, in any case, the optimality constraint *Opt.cut* is added to the MP. This procedure is repeated until the bounds are the same.

Algorithm 5.4.1 OMT Classical Benders Decomposition

$UB \leftarrow \infty$
$LB \leftarrow 0$
while $UB > LB$ **do**
  Solve F^{MP} to optimality and obtain $(obj^{MP}, \bar{x})$, where $obj^{MP} = obj^{\mathsf{DOMP}} + \mu$
  if $obj^{MP} > LB$ **then**
    Update LB
  end if
  Solve F^{SP} to optimality and obtain obj^{SP}
  if $obj^{\mathsf{DOMP}} + obj^{SP} < UB$ **then**
    Update UB
  end if
  Add optimality constraint *Opt.cut* to F^{MP}
end while

As aforementioned, the subproblem to deal with is the MST between the facilities computed in the MP solution, which can be solved in polynomial time using Kruskal algorithm. Once solved, the objective value of the subproblem obj^{SP} is used to build the following optimality constraint to add to the MP:

$$Opt.cut: \frac{obj^{SP}}{p-1}\left[\sum_{i\in V} \bar{x}_{ii} - (p-1)\right] \leq \mu. \tag{5.18}$$

For this *Opt.cut*, if the same p facilities that have been selected as inputs for the SP are selected in the MP, then the μ variable is increased in the amount of the cost of the tree obtained as output in the SP. Otherwise, if at least one different facility is selected, then μ does not change. In other words, μ is null until the same p facilities are selected again, where μ turns strictly positive and the algorithm ends.

Note that in every iteration, only one *Opt.cut* is added. For the general Benders framework, some authors (e.g. see Fischetti et al., 2010) force the addition of feasibility cuts even when the MP is feasible to boost the algorithm. In our case, since the SP is always feasible, it is not necessary to introduce feasibility cuts, but one can force a series of suboptimal cuts to be introduced in each iteration after the optimal cut is included.

Directly using *Opt.cut* avoids having to go through the dual formulation of our SP, as in most of the Benders decomposition frameworks. However, there exists an implementation of the Kipp Martin (km) MST dual formulation in Labbé et al. (2021) which can be used for this purpose. The km MST formulation and its dual form are adapted to the OMT structure as follows:

$$
\begin{aligned}
F^{km}: \min \ & \sum_{(i,j)\in E} c_{ij} z_{ij}, && && (5.19a)\\
\text{s.t. } & z_{ij} \le \bar{x}_{ii}, && (i,j)\in E && (5.19b)\\
& z_{ij} \le \bar{x}_{jj}, && (i,j)\in E && (5.19c)\\
& \sum_{(i,j)\in E} z_{ij} = p-1, && && (5.19d)\\
& \sum_{\substack{(i',j)\in E:\\ (i'=k\wedge j=i)\\ \vee\\ (i'=i\wedge j=k)}} z_{i'j} + \sum_{(i,j)\in A: j\neq k} q_{kij} \le 1, && k,i\in V: i\neq k, && (5.19e)\\
& q_{kij} + q_{kji} = z_{ij}, && k\in V, (i,j)\in E: i,j\neq k, && (5.19f)\\
& z_{ij} \ge 0, && (i,j)\in E, && (5.19g)\\
& q_{kij} \ge 0, && k\in V, (i,j)\in E. && (5.19h)
\end{aligned}
$$

$$
\begin{aligned}
F^{SP}: \max \ & \alpha(p-1) - \sum_{k,i\in V: i\neq k} \beta_{ki} - \sum_{(i,j)\in E} (\bar{x}_{ii}\tau_{ij} + \bar{x}_{jj}\eta_{ij}), && && (5.20a)\\
\text{s.t. } & \alpha - \beta_{ij} - \beta_{ji} - \sum_{k\in V: k\neq i,j} \gamma^k_{ij} - \tau_{ij} - \eta_{ij} \le c_{ij}, && (i,j)\in E, && (5.20b)\\
& -\beta_{ki} + \sum_{\substack{(i',j')\in E:\\ (i'=i\wedge j'=j)\\ \vee\\ (i'=j\wedge j'=i)}} \gamma^k_{i'j'} \le 0, && k\in V, (i,j)\in E: i,j\neq k, && (5.20c)\\
& \alpha \in \mathbb{R}, && && (5.20d)\\
& \beta_{ki} \ge 0, && k,i\in V: i\neq k, && (5.20e)\\
& \gamma^k_{ij} \in \mathbb{R}, && k\in V, (i,j)\in E: i,j\neq k, && (5.20f)\\
& \tau_{ij} \ge 0, && (i,j)\in E, && (5.20g)\\
& \eta_{ij} \ge 0, && (i,j)\in E. && (5.20h)
\end{aligned}
$$

Observe that, constraints (5.2e) are presented as disaggregated constraints (5.19b)-(5.19c) in the SP because, although it has been computationally proven to report worse results, dealing with continuous z variables force to use the disaggregated formulation. In this way, the *Opt.cut* introduced have the following expression:

$$
Opt.cut: \ \frac{1}{p-1}\Big[\bar{\alpha}(p-1) - \sum_{k,i\in V: i\neq k} \bar{\beta}_{ku} - \sum_{(i,j)\in E} (x_{ii}\bar{\tau}_{ij} + x_{jj}\bar{\eta}_{ij})\Big] \le \mu. \quad (5.21)
$$

The classical Benders decomposition can be introduced into a branch-and-cut framework for a more efficient performance as illustrated in Algorithm 5.4.2. This algorithm is initialized with the tree $\mathfrak{T}$, considering solving the MP at the root node, and an empty pool of cuts $\mathcal{P}$. Once solved the MP at the root node, the branching procedure is started. If at a particular node of the process the solution found is fractional, the algorithm keeps branching. Otherwise, when an integer solution is found, the lower bound is updated using the MP objective value of the current node $o' \in \mathfrak{T}$, the SP is solved fixing the facilities $\bar{x}$ obtained in the MP solution, the upper bound is updated if possible and the optimality constraint *Opt.cut* is added to a pool $\mathcal{P}$ of cuts. The cuts from pool $\mathcal{P}$ are then included for the MP solution in other nodes of $\mathfrak{T}$.

Algorithm 5.4.2 OMT Branch-and-Benders-Cut

Set tree $\mathfrak{T} = \{o\}$, where $o = F^{MP}$ has no branching constraints
Initialize a pool of cuts $\mathcal{P} = \emptyset$
while $\mathfrak{T}$ is nonempty **do**
    Select a node $o' \in \mathfrak{T}$
    $\mathfrak{T} \leftarrow \mathfrak{T} \setminus \{o'\}$
    Solve o' considering $\mathcal{P}$ and obtain solution $\bar{x}$
    if $\bar{x}$ is fractional **then**
        Branch, resulting in nodes o'' and o'''
        $\mathfrak{T} \leftarrow \mathfrak{T} \cup \{o'', o'''\}$
    else
        Solve F^{SD} and obtain $(\bar{\alpha}, \bar{\beta}, \bar{\tau}, \bar{\eta})$
        Add optimality constraint *Opt.cut* to $\mathcal{P}$
    end if
end while

This allows to use different strategies for producing the optimality cuts. For instance, these cuts can be generated in every feasible node, only in those nodes that have improved the lower bound beyond a predefined threshold or only when incumbent solutions are found along the search process. If caution is not taken, too many unnecessary cuts can be produced slowing down the attainment of a MP solution instead of speeding it up. Also, in the classical Benders decomposition, every *Opt.cut* added implies solving to optimality a MP, which is computationally costly. In the modern approach, the pool of cuts $\mathcal{P}$ considered in a specific node of the branch-and-cut procedure may contain cuts that have already been identified, slowing down the efficiency since it would reintroduce repeated cuts when branching at different levels of the search tree. For these reasons, it could be helpful to consider a strategy to initialize the pool of cuts $\mathcal{P}$ prior to starting the procedures using a so called *warm-start phase* (see, e.g., Martins de Sa et al., 2013).

This warm-start phase can be introduced in both classical and modern approaches. For the modern approach, the idea is to store the cuts generated in an independent warm-start phase in the pool of cuts $\mathcal{P}$ settled at the beginning of Algorithm 5.4.2. Given runtime^{MP}, elapsed time for a MP iteration, and runtime, total elapsed time for the completition of the algorithm, the parameters considered for tuning the warm-start phase are the following:

- max.time^{MP}. Time limit that every solution computation of the MP in the warm-start phase can take to obtain the best feasible solution possible. Once reached, the respective *Opt.cut* is added and a new MP is considered for solving.

- max.gap^{MP}. Gap percentage allowed to remain between the best lower and upper bound found in every MP of the warm-start phase, LB^{MP} and UB^{MP}, computed as $100 \cdot \frac{UB^{MP} - LB^{MP}}{LB^{MP}}$. Once reached, the respective *Opt.cut* is added and a new MP is considered.

- max.time. Total time limit that the entire warm-start phase can take. Once reached, the warm-start phase is finished and the algorithm starts searching for optimal solutions.

- max.gap. Gap percentage allowed to remain between best lower and upper bound found in the entire warm-start phase, computed as in max.gap^{MP} but using the overall bounds LB

and UB. Once reached, the warm-start phase is finished and the algorithm starts searching for optimal solutions.

The interest behind this warm phase is to be able to introduce a certain number of cuts at low cost before start branching. In order to introduce this cuts, the solution in the warm-start phase does not need to be computed to optimality, is enough to obtain a feasible solution. Furthermore, a procedure to build cuts from fractional values of the variables can be implemented.

As an observation to this Benders decomposition approaches, they deal with the DOMP as a subproblem which is more difficult to solve than MST. Therefore, this branch-and-Benders implies solving a DOMP adding optimality Benders cuts (costly to obtain), what seems to be less efficient than solving a DOMP with some MST additional constraints. Thus, this general technique for solving the OMT would require an alternative way for solving DOMP, which is out of the scope of this chapter.

5.5 Improvements

This section details the enhancements implemented to improve the computational performance of the proposed formulations. These techniques primarily consist of two components: solution initialization and variable fixing procedures.

For the first, note that standard branching frameworks, which form the basis for solving the formulations, can require significant computational time to identify an initial feasible solution, often referred to as an upper bound. To mitigate this issue and accelerate the solution process, two specialized heuristic algorithms were developed to generate a strong initial solution.

- **DOMP + MST heuristic**. This approach performs first the DOMP and, once the facilities have been assigned and compensated, they are fixed for searching a MST connecting those facilities in polynomial time using Kruskal MST algorithm.
- **pMEDT + DOMP heuristic**. This approach finds p facilities and the tree structure computing the pMEDT by means of a MST competitive formulation, for example the MTZ formulation. Both, facilities and the tree structure, are left fixed and the reassignment of clients to facilities is then performed using the DOMP. This algorithm has been proven to be more efficient providing an initial solution.

Next, several preprocessing steps aimed at reducing the size of the radius formulation $F1_u^{\mathcal{T}}$, and thereby improving its efficiency, are described. These variable fixing procedures are based on adaptations of some arguments already used in Puerto et al. (2011a, 2013) and Pozo et al. (2021) where fixing a certain number of variables improved the efficiency of the formulations.

Given the ordered sequence of unique costs and the *sorting* factor of the orders ℓ, the $u_{\ell h}$-variables matrix is expected to have a specific step structure. It might be expected that for small values of h, the number of allocations at cost smaller than $c_{(h)}$ should be small and therefore $u_{\ell h} = 1$ in the left-bottom part of the u-matrix from a certain ℓ to the end. Similarly, it might also be expected that many $u_{\ell h}$-variables in the right-upper part of the u-matrix will take value 0 in the optimal solution. Indeed, $u_{\ell h} = 0$ means that the ℓ-th sorted allocation cost is less than to $c_{(h)}$, which is very likely to be true if h is sufficiently large and ℓ is small enough. For example, since $c_{jj} = 0$, $j \in V$, one has that $u_{\ell h} = 0$, $\ell \in \{1, \dots, p\}, h \in H$. These preprocessings aim to fix as many of this $u_{\ell h}$-variables as possible.

Given a $h \in H$, in order to fix $u_{\ell h}$-variables to 1 one deals with an auxiliary problem that maximizes the number of allocations satisfying $c_{ij} \le c_{(h-1)}$, which is equivalent to the number of variables that can assume a 0 value, or what is the same, this will provide the number of allocations satisfying the opposite ($c_{ij} > c_{(h-1)} \equiv c_{ij} \ge c_{(h)}$) and, therefore, the minimum number of $u_{\ell h}$-variables that can assume value of 1. Using the variables previously defined, the formulation of this problem is:

$$
\begin{aligned}
F_1^{pre}:\ \max H_h^1 := & \sum_{i\in V}\sum_{j\in V} x_{ij}, && && (5.22a)\\
\text{s.t.}\ & \sum_{i\in V} x_{ii} = p, && && (5.22b)\\
& \sum_{j\in V} x_{ij} \le 1, && i\in V, && (5.22c)\\
& x_{ij} \le x_{jj}, && i,j\in V: i\ne j, && (5.22d)\\
& c_{ij}x_{ij} \le c_{(h-1)}, && i,j\in V, && (5.22e)\\
& x_{ij}\in\{0,1\}, && i,j\in V. && (5.22f)
\end{aligned}
$$

F_1^{pre} gives the maximum number (in the worst case) of allocations realized at a cost less than or equal to $c_{(h-1)}$. If H_h^1 is the optimal value of problem above, since there are $|V|$ total possible allocations, the number of allocations satisfying $c_{ij} \ge c_{(h)}$ must be necessarily greater than or equal to $|V| - H_h^1$, or equivalently, in any feasible solution of a radius formulation:

$$u_h^\ell = 1, \quad \forall \ell \in \{H_h^1 + 1, \ldots, |V|\}. \tag{5.23}$$

Also, for a given $h \in H$, to fix the maximum number of $u_{\ell h}$-variables possible to 0 one deals with the following auxiliary problem: maximize the number of allocations satisfying $c_{ij} \ge c_{(h)}$, which will provide the minimum number of 0 that the h-th column of the u-matrix must have. Using the variables defined previously, the formulation of this problem is:

$$
\begin{aligned}
F_0^{pre}:\ \max H_h^0 := & \sum_{i\in V}\sum_{j\in V} x_{ij}, && && (5.24a)\\
\text{s.t.}\ & \sum_{i\in V} x_{ii} = p, && && (5.24b)\\
& \sum_{j\in V} x_{ij} \le 1, && i\in V, && (5.24c)\\
& x_{ij} \le x_{jj}, && i,j\in V: i\ne j, && (5.24d)\\
& c_{ij} \ge c_{(h)}x_{ij}, && i,j\in V: i\ne j, && (5.24e)\\
& x_{ij}\in\{0,1\}, && i,j\in V. && (5.24f)
\end{aligned}
$$

In this problem, for constraints (5.24e) one could not consider allocations x_{ii}, $i \in V$, as considered in Puerto et al. (2011a, 2013) and Pozo et al. (2021) among others, because in such case our problem is not feasible. Infeasibility is due to having to consider constraints (5.24b) together with constraints (5.24e). If allocations such x_{ii}, $i \in V$, are considered, then one has $c_{ii} = 0$ and since $c_{(h)} > 0$, $h \in H$, constraints (5.24e) set all x_{ii} to 0, which is not compatible with constraints (5.24b). For this reason one must impose $i, j \in V : i \neq j$ in (5.24e).

For a given $h \in H$, F_0^{pre} is the maximum number of allocations done at a cost greater than or equal to $c_{(h)}$. Therefore, if H_h^0 is the optimal value of problem above, the h-th column of the u-matrix must have at least $|V| - H_h^0$ zeros. As it is known that exactly the p first rows of the u-matrix can be fixed to 0, which corresponds to the allocations of the facilities to itselves at a cost 0 not considered in the constraints (5.24e), in any feasible solution satisfies:

$$u_h^\ell = 0, \quad \forall \ell \in \{1, \ldots, |V| - H_h^0 + p\}. \tag{5.25}$$

For the ease of understanding, an example is presented next.

■ **Example 5.2** Consider the complete instance of the OMT with $|V| = 4$ in Figure 5.4 (left) and the allocation costs specified over its edges. The goal is to locate $p = 2$ facilities considering a λ-vector of $\lambda = (1, \overset{|V|}{\ldots}, 1)$ (median criterion). The optimal allocations (arrows) and facilities (squares) for this instance are depicted in Figure 5.4 (right). ■

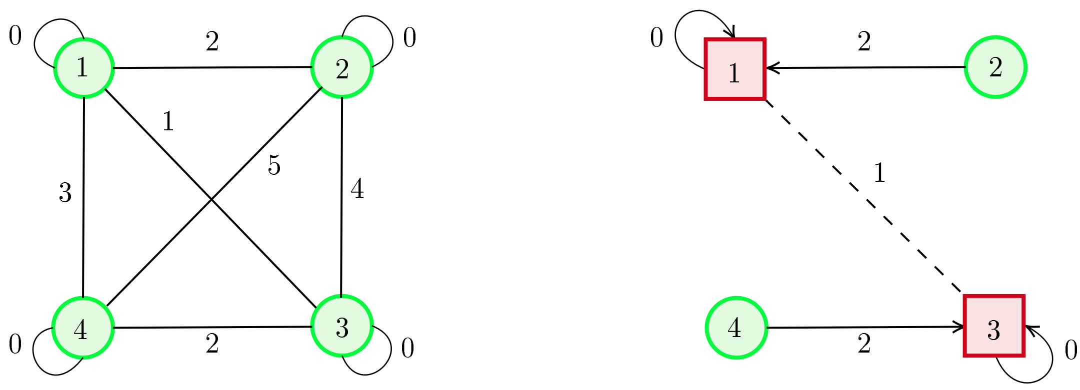


Figure 5.4: OMT instance example (left) and its optimal solution (right).

In the example, the optimal solution yields a total allocation cost of 4, given by c_{21} and c_{43} (together with c_{11} and c_{33}, which are zero). The facilities are connected through the edge $(1, 3)$, which preserves the tree structure of facilities expected in any solution of the OMT.

In order to illustrate the radius formulation preprocessings for fixing a certain number of $u_{\ell h}$-variables to 1 or 0 using the auxiliary problems F_1^{pre} and F_0^{pre}, one must first consider the ordered sequence of unique costs $c_{(0)} := 0 < c_{(1)} < c_{(2)} < \ldots < c_{(|H|)} = \max_{i,j \in V} c_{ij}$ of our problem as in Table 5.3.

$c_{(0)}$	$c_{(1)}$	$c_{(2)}$	$c_{(3)}$	$c_{(4)}$	$c_{(5)}$
0	1	2	3	4	5

Table 5.3: Sequence of unique ordered costs for the OMT example.

Note that cost 0 is repeated 4 times, since there are 4 possible allocations of facilities to themselves, and cost 2 is also repeated 2 times. For the radius formulation, the u-matrix of dimensions $|V| \times |H|$ of the optimal solution is given by the matrix

$$u_{\ell h} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{pmatrix},$$

which means that the $u_{\ell h}$ variables activated are $u_{31}, u_{32}, u_{41}, u_{42} = 1$. Note that the $c_{(0)}$ column is not being considered in the u-matrix, which in case of being considered must be $u_{\ell 0} = 1$ for $\ell \in V$. One can verify over this matrix that the allocation cost objective holds using radius variables:

$$\begin{aligned} \sum_{\ell \in V} \sum_{h \in H} \lambda_\ell u_{\ell h}(c_{(h)} - c_{(h-1)}) &= u_{31}(c_{(1)} - c_{(0)}) + u_{32}(c_{(2)} - c_{(1)}) + u_{41}(c_{(1)} - c_{(0)}) + u_{42}(c_{(2)} - c_{(1)}) \\ &= (1-0) + (2-1) + (1-0) + (2-1) = 4. \end{aligned}$$

Now, if one wants to previously fix a certain number of $u_{\ell h}$-variables to 1, then he deal with the auxiliary problem F_1^{pre}. In this problem, the objective is to maximize the number of allocations satisfying $c_{ij} \leq c_{(h-1)}$. Since there are $|V|$ possible allocations in total, this directly provides the minimum number of allocations satisfying $c_{ij} > c_{(h-1)} \equiv c_{ij} \geq c_{(h)}$, as specified in the radius formulation by constraints 5.3g. For example, if $h = 1$, the goal is to find allocations such that $c_{ij}x_{ij} \leq c_{(0)} = 0$. In this case, only the four allocations of potential facilities to themselves, at zero cost, can be considered. However, since constraints (5.22b) must also be satisfied, only two of these allocations can be selected, leading to $H_1^1 = 2$. Consequently, there are at least $|V| - H_1^1 = 2$ allocations that satisfy the opposite condition. Because these are placed in the lower part of the u-matrix, it follows that $u_{\ell 1} = 1$ for any $\ell \in \{|V| - H_1^1 + 1, \ldots, |V|\} = \{3, 4\}$. For $h = 2$, two of the four self-allocations are again considered, together with one additional allocation, namely x_{13}, which yields $H_2^1 = 3$. Hence, only $|V| - H_2^1 = 1$ variable can be fixed, corresponding to $u_{42} = 1$. By continuing this reasoning, one obtains $H_h^1 = 4$ for $h \geq 3$, implying that no further variables can be fixed.

On the other hand, for fixing $u_{\ell h}$-variables to 0 one deals with the auxiliary problem F_0^{pre}. In the example, for $h = 5$ there are only one possible allocation that satisfies $c_{ij} \geq c_{(5)}x_{ij} = 5x_{ij}$ plus the 2 allocations coming from constraints (5.24b). This means that as much $H_5^0 = 3$ of the $u_{\ell 5}$-variables could be fixed to 1 (there could be less), so the 5-th column of the u-matrix must have at least $|V| - H_5^0 = 4 - 3 = 1$ zeros. However, as it is known that the p first rows of the u-matrix can be fixed to 0, one ends up having $u_{\ell 5} = 0$ for any $\ell \in \{1, \ldots, |V| - H_5^0 + p\} = \{1, 2, 3\}$. For $h \leq 4$, the optimal solution of the auxiliary problem is $H_h^0 = 4$, and consequently only the first p variables of each column, already known in advance, can be fixed.

The following table summarizes this information:

h	$\overbrace{\#\ \text{alloc}\vert c_{ij} \leq c_{(h-1)}}^{H_h^1}$	$\vert V\vert - H_h^1$	$\overbrace{\#\ \text{alloc}\vert c_{ij} \geq c_{(h-1)}}^{H_h^0}$	$\vert V\vert - H_h^0 + p$
1	2	2	4	2
2	3	1	4	2
3	4	0	4	2
4	4	0	4	2
5	4	0	3	3

Table 5.4: Fixing solutions for the OMT example.

Thus, solving both auxiliary problems, one gets the following final preprocessing matrix of $u_{\ell h}$-variables fixed, where NF means not fixed by any of the auxiliary problems:

$$preproc(u_{\ell h}) = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & NF & NF & NF & 0 \\ 1 & 1 & NF & NF & NF \end{pmatrix}.$$

5.6 Computational study

This section reports on the results of some computational experiments performed in order to empirically compare the proposed OMT formulations. Table 5.5 catalogs all different formulations considering every combination for the tree polytope $\mathcal{T} \in \{sub, mtz, flow, km\}$, sorting type $\mathcal{S} \in \{x^\ell, u\}$ and whether $\mathcal{T}$ is built between facilities or in V. The interested reader can see the Appendix in this chapter for the whole catalogue of formulations described in detail.

$\mathcal{T}$	$\mathcal{S}$	(1): $\mathcal{T}$ in facilities	(2): $\mathcal{T}$ in V
sub	x^ℓ	$F1^{sub1}_{x^\ell}$, $F1^{sub2}_{x^\ell}$	$F2^{sub1}_{x^\ell}$, $F2^{sub2}_{x^\ell}$
	u	$F1^{sub1}_{u}$, $F1^{sub2}_{u}$	$F2^{sub1}_{u}$, $F2^{sub2}_{u}$
mtz	x^ℓ	$F1^{mtz}_{x^\ell}$	$F2^{mtz}_{x^\ell}$
	u	$F1^{mtz}_{u}$	$F2^{mtz}_{u}$
$flow$	x^ℓ	$F1^{flow1}_{x^\ell}$, $F1^{flow2}_{x^\ell}$	$F2^{flow1}_{x^\ell}$, $F2^{flow2}_{x^\ell}$
	u	$F1^{flow1}_{u}$, $F1^{flow2}_{u}$	$F2^{flow1}_{u}$, $F2^{flow2}_{u}$
km	x^ℓ	$F1^{km}_{x^\ell}$	$F2^{km}_{x^\ell}$
	u	$F1^{km}_{u}$	$F2^{km}_{u}$

Table 5.5: OMT formulations summary.

For the computational study, instances $G = (V, E)$ are generated as complete networks of sizes $|V| \in \{20, 30, 40, 50, 60, 70, 80, 90, 100\}$ with random integer costs c_{ij} following a uniform distribution in $[1, 1e5]$. In order to follow the structure of previous published works, where other ordered median problems were considered, the number of facilities for each group of instances is chosen as $p \in \{\lfloor\frac{|V|}{4}\rfloor, \lfloor\frac{|V|}{3}\rfloor, \lfloor\frac{|V|}{2}\rfloor\}$. Also, three commonly studied criteria for the λ-vector factor values are considered:

- **Median criterion**: $\lambda = (\underbrace{1, 1, ..., 1}_{|V|})$.
- **k-centrum criterion**: $\lambda = (\underbrace{0, ..., 0}_{\lfloor\frac{2}{3}|V|\rfloor}, \underbrace{1, ..., 1}_{|V|-\lfloor\frac{2}{3}|V|\rfloor})$.
- **k-trimmed mean criterion**: $\lambda = (\underbrace{0, ..., 0}_{\lfloor\frac{1}{3}|V|\rfloor}, \underbrace{1, ..., 1}_{|V|-\lfloor\frac{2}{3}|V|\rfloor}, \underbrace{0, ..., 0}_{\lfloor\frac{1}{3}|V|\rfloor})$.

In all tables, results correspond to groups of 5 instances with same $(|V|, p)$ pair. This gives a benchmark of 135 instances in total. Average results for each group of instances are presented. All instances were solved with Gurobi 9.1.1 optimizer, under a Windows 10 environment in an Intel(R) Core(TM) i7-2600 CPU 3.40 GHz processor and 16 GB RAM. The CPU time limit fixed for solving each instance is 3600 seconds. Tables are grouped in blocks. The first block contains the values of the instances parameters, namely $|V|$ and p. Then, a block of 6 columns is given for each formulation with the following information:

- Columns $|\#|$ indicate the number of instances in the group that could be solved to optimality within the CPU time limit.
- Columns *cpu* report the average computing time in seconds over the groups of instances.
- Columns $g\overline{U}R$ give the percentage relative root gap, computed as $100\frac{obj_{\overline{U}} - obj_R}{obj_{\overline{U}}}$, where $obj_{\overline{U}}$ denotes the best known upper bound obtained in all our experiments and obj_R denotes the optimal value of the linear relaxation at the root node.
- Columns $g\overline{U}L$ give the percentage relative lower bound gap, computed as $100\frac{obj_{\overline{U}} - obj_L}{obj_{\overline{U}}}$, where obj_L denotes the lower bound at termination.
- Columns $gU\overline{L}$ give the percentage relative upper bound gap, computed as $100\frac{obj_U - obj_{\overline{L}}}{obj_U}$, where obj_U denotes the upper bound at termination and $obj_{\overline{L}}$ denotes the best known lower bound obtained in all our experiments.

- Columns gUL give the percentage relative gap at termination, computed as $100\frac{obj_U - obj_L}{obj_U}$.
- Columns nod indicate the average number of nodes explored in the branch-and-bound ($B\&B$) search tree.

Note that, while $\overline{gUR}$ and $\overline{gUL}$ provide quality measures of the lower bounds (at the root node and at termination, respectively), $\overline{gUL}$ provides a quality measure of the upper bounds. In addition, gUL provides a measure of both upper and lower bounds for the average performance. If the reported value of gUL is 0 then all instances were solved to optimality. Along all experiments initial solutions coming from the heuristics developed in Section 5.5 were given, so in all cases it is possible to find at least a feasible solution.

The caption just above each block gives the formulation the block refers to. Due to the large number of formulations (see Table 5.5), it was decided to report just four final formulations. This choice was based on preliminary computational result based on different set of instances. Results among $F1^{(\mathcal{T})}_{(\mathcal{S})}$ with $\mathcal{S} \in \{x^\ell, u\}$ and $\mathcal{T} \in \{mtz, flow1, flow2, km\}$ are reported, since other approaches (subtour elimination, Benders decomposition and $F2^{(\cdot)}_{(\cdot)}$) were clearly outperformed in our preliminary tests, especially for large instance sizes. Then, the criteria that have guided us to present as final formulations $F1^{mtz}_{x^\ell}$, $F1^{flow1}_{x^\ell}$, $F1^{mtz}_{u}$ and $F1^{km}_{u}$ are the following:

1. Among the four formulations, two types of $F1^{(\cdot)}_{x^\ell}$ and two types of $F1^{(\cdot)}_{u}$ formulations were selected. Note that $F1^{(\cdot)}_{u}$ takes advantage of ties in the cost matrix. Selection among $F1^{(\cdot)}_{x^\ell}$ requires analysis of the gap left (many instances could not be solved within the time limit), while selection among $F1^{(\cdot)}_{u}$ requires analysis of running time (almost all instances were solved within the time limit).
2. According to Table 5.6, $F1^{flow1}_{x^\ell}$ slightly outperforms its variant $F1^{flow2}_{x^\ell}$. Given the similarity of these formulations, $F1^{flow2}_{x^\ell}$ is discarded.
3. Table 5.6 shows that $F1^{mtz}_{x^\ell}$ slightly outperforms $F1^{km}_{x^\ell}$ in terms of gaps for k-trimmed mean and k-centrum criteria.
4. Table 5.6 also indicates that $F1^{flow1}_{x^\ell}$ slightly outperforms $F1^{km}_{x^\ell}$ for the same criteria.
5. Performance profiles in Figure 5.5 indicate that $F1^{km}_{u}$ exhibits slightly better performance among all $F1^{(\cdot)}_{u}$ formulations with respect to the k-trimmed mean and k-centrum criteria (for the most difficult instances).
6. Finally, $F1^{mtz}_{u}$ was selected to provide a comparison of the same tree polytope description for both sorting formulations, i.e., $F1^{mtz}_{x^\ell}$ and $F1^{mtz}_{u}$. Additionally, $F1^{mtz}_{(\cdot)}$ is reported because mtz has proven to be a promising tree polytope description when combined with sorting constraints (see Fernández et al., 2014).

$\mathcal{T}$	Median		k-centrum		k-trimmed mean	
	$\lvert\#\rvert$	$\overline{gUL}$	$\lvert\#\rvert$	$\overline{gUL}$	$\lvert\#\rvert$	$\overline{gUL}$
mtz	57	1.26	13	27.14	30	9.59
$flow1$	57	0.53	14	25.12	32	8.85
$flow2$	57	0.65	13	25.90	32	9.58
km	60	-	13	28.47	33	10.19

Table 5.6: $F1^{\mathcal{T}}_{x^\ell}$ formulations comparison for the three different criteria, where $\overline{gUL}$ represents the mean gUL value across all runs.

A general overview of the OMT computational experience brings some common observations for all criteria. The reader may note that instance sizes have been chosen in such a way that many

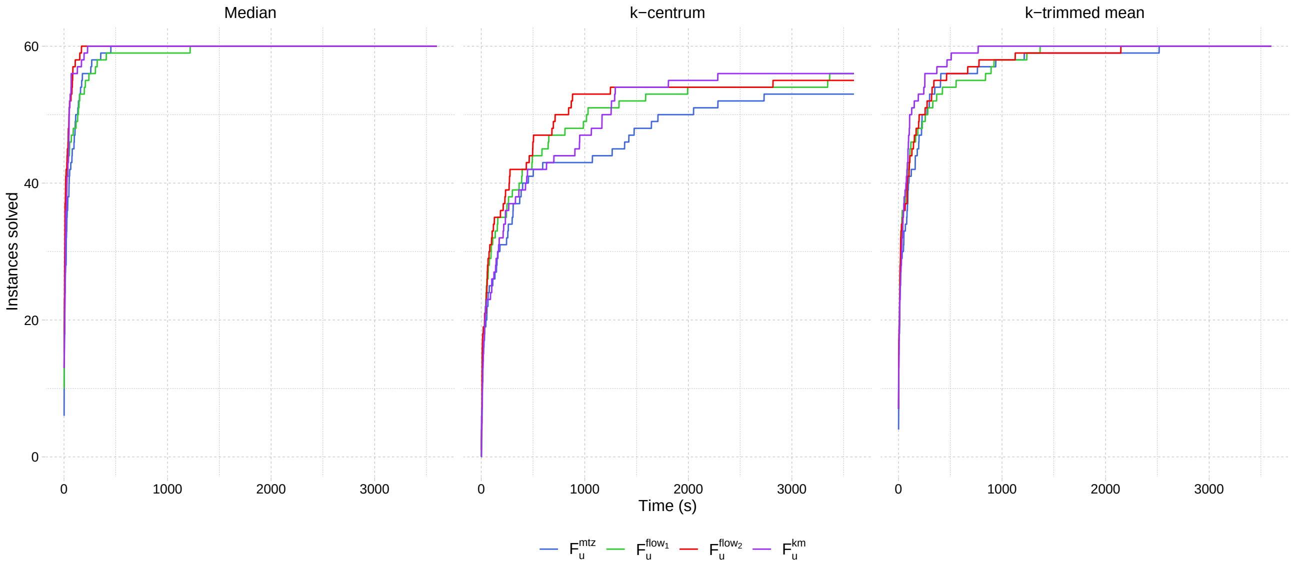


Figure 5.5: $F1_u^{(\cdot)}$ formulations comparison for the three different criteria

of the instances cannot be solved to optimality within the time limit established (3600 seconds). This is specially remarkable for the k-centrum and k-trimmed mean criteria that are those that reflect the computational complexity of the sorting problem. For that reason, CPU time is considered irrelevant in this particular study, while the reported gaps provide clearer information on the behavior of each formulation for each criterion and $(|V|, p)$. Nevertheless, CPU times are reported for the interested reader. A first evidence is that given a fixed tree polytope description, as a consequence of Property 5.3.2, values for $g\overline{U}R$ are equal for every $(|V|, p)$ pair among the two sorting descriptions. This can be observed in the tables of results, where for $F1_{x^\ell}^{mtz}$ and $F1_u^{mtz}$ the values of $g\overline{U}R$ coincide. However, this does not imply that for a fixed sorting description values for $g\overline{U}R$ should match, e.g., values for $F1_u^{km}$ do not coincide with $F1_u^{mtz}$ or other tree description. Reported values of $g\overline{U}L$, $gU\overline{L}$ and gUL are smaller for $F1_u^{(\cdot)}$ formulations, concluding that $F1_{x^\ell}^{(\cdot)}$ formulations are outperformed in these terms. Moreover, $F1_u^{(\cdot)}$ formulations are able to certify optimality for most of the instances (median and k-trimmed mean criteria) and more often than $F1_{x^\ell}^{(\cdot)}$ formulations. Concerning gUL values, instances with lower p are rather more difficult to solve in general terms than those with bigger p values. Besides, *nod* parameter is influenced by how well the problem is described (formulation used, number of fixed variables) and the number of variables and constraints introduced. Starting from the smaller size instances, formulations tend to increase *nod* values as long as they increase their size. However, at some point *nod* values start decreasing when variables and constraints considerably increase and computations at each node are more time-consuming. In general, the results differences between the $F1_{(\mathcal{S})}^{(\cdot)}$ formulations are small within the given sorting description and highly depend on the criterion.

Next some specific considerations regarding each criterion are provided. In Table 5.8, where results for median criterion are reported, all instances reach optimality for $F1_{x^\ell}^{(\cdot)}$ formulations when $|V| \leq 40$ and for $F1_u^{(\cdot)}$ formulations when $|V| \leq 60$. For the median criterion, values for $g\overline{U}R$ range between $28\% - 42\%$. Concerning *nod* values, $F1_{x^\ell}^{(\cdot)}$ formulations tend to explore more nodes of the $B\&B$ tree for small size instances than $F1_u^{(\cdot)}$ formulations. This exploration is gradually increased until $|V| = 70$, where computations at each node require more time. On the other hand, $F1_u^{(\cdot)}$ formulations need to explore less nodes in order to certify optimality. As shown in previous DOMP problems, median criterion does not reflect the computational complexity of the sorting problem since all lambdas are equal to 1. In Tables 5.9 and 5.10, where results for k-centrum criterion and k-trimmed mean criterion are respectively reported, problems to prove optimality start to appear for smaller size instances, being specially difficult for the k-centrum criterion. For this criteria, $g\overline{U}R$, $g\overline{U}L$, $gU\overline{L}$ and gUL values are greater than those reported for median criterion. Moreover, for k-centrum and k-trimmed mean, formulations tend to need to explore more $B\&B$ nodes for small size instances than for median criterion and are not able to explore many nodes for large size instances. Therefore, one can conclude that, as expected, the formulations perform better for median criterion than for k-trimmed mean, and specially k-center criterion reports the worst results.

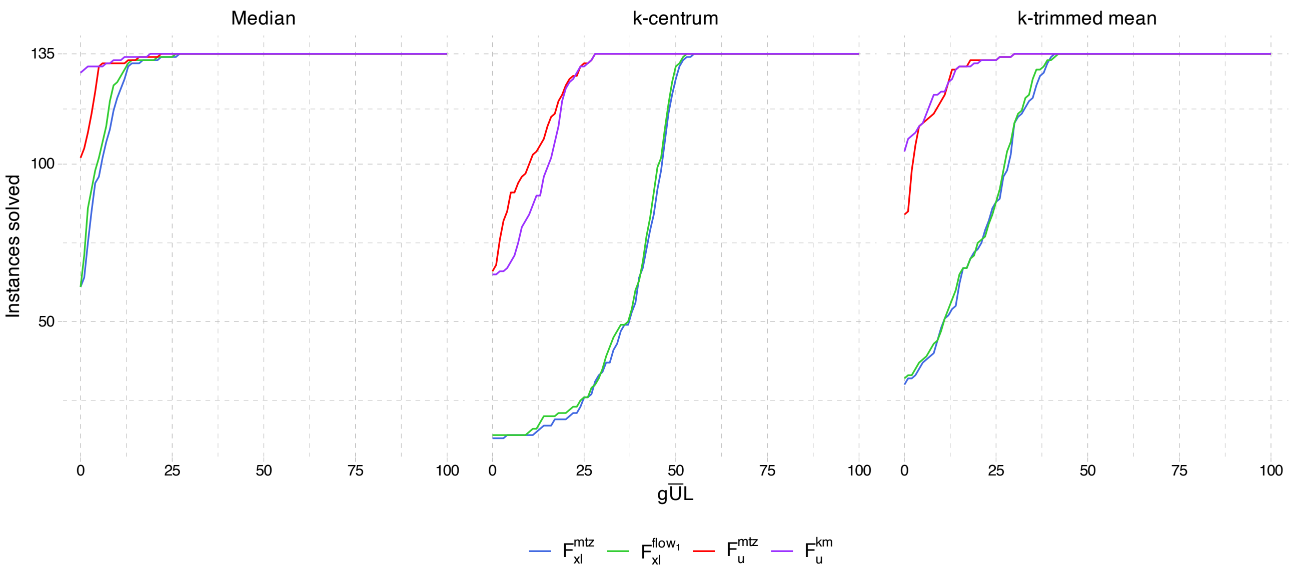


Figure 5.6: Cumulative number of instances solved within a $g\overline{U}L$ value.

Besides, the reader may note that particular behavior of gaps $g\overline{U}L$ and $gU\overline{L}$ in Tables 5.8, 5.9 and 5.10. Recall that $g\overline{U}L$ provide a quality measure of the lower bound at termination, $gU\overline{L}$ provides a quality measure of the upper bound at termination. For example, for median criterion $gU\overline{L}$ is significantly bigger than $g\overline{U}L$ for $F1_{x^\ell}^{mtz}$ and $F1_{x^\ell}^{flow1}$ (it seems difficult for the solver to find good feasible solutions). However, for median criterion $g\overline{U}L$ is bigger than $gU\overline{L}$ for $F1_u^{mtz}$ (it seems more difficult to increase the lower bound for certifying optimality). For the k-centrum criterion $g\overline{U}L$ is usually bigger than or equal to $gU\overline{L}$ for mostly all cases. However, this comparison depends on the formulation and $(|V|, p)$ for the k-trimmed mean criterion. These specific comments and previous general ones can also be seen in the cumulative number of instances solved within a $g\overline{U}L$ and $gU\overline{L}$ values (Figure 5.6 and Figure 5.7 respectively). In these graphics, $F1_u^{mtz}$ shows the best performance in many cases. However, nothing in general, can be concluded for all cases.

Table 5.7 resumes the final results for mean gUL ($\overline{gUL}$) values and number of instances solved to optimality across all final runs, and Figure 5.8 depicts the number of instances solved to optimality within a given time. On the one hand, these results show the similar performance of the $F1_{x^\ell}^{(.)}$ formulations. With respect to the differences between the $F1_u^{(.)}$ formulations, in terms of the final number of instances solved and the time required (Figure 5.8), $F1_u^{km}$ seems to perform better than $F1_u^{mtz}$. However, the poor performance of $F1_u^{km}$ regarding small p values makes similar average gaps for $F1_u^{mtz}$ and $F1_u^{km}$, what makes it hard to draw significant conclusions in general.

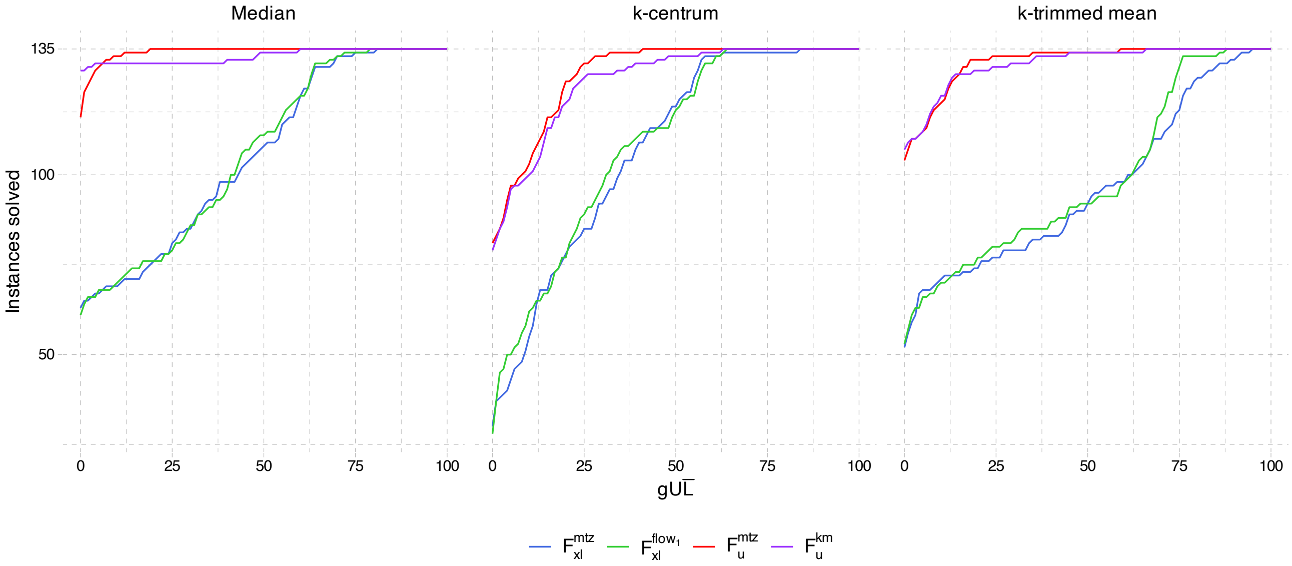


Figure 5.7: Cumulative number of instances solved within a $g\overline{UL}$ value.

$\mathcal{T}$	Median		k-centrum		k-trimmed mean	
	$\lvert\#\rvert$	$\overline{gUL}$	$\lvert\#\rvert$	$\overline{gUL}$	$\lvert\#\rvert$	$\overline{gUL}$
$F1_{x^\ell}^{mtz}$	60	23.13	13	45.10	30	37.29
$F1_{x^\ell}^{flow1}$	60	22.40	14	43.74	32	36.21
$F1_{u}^{mtz}$	102	1.25	67	6.23	90	2.10
$F1_{u}^{km}$	129	1.49	65	9.25	114	1.84

Table 5.7: Final results summary for mean gUL and total instances solved to optimality.

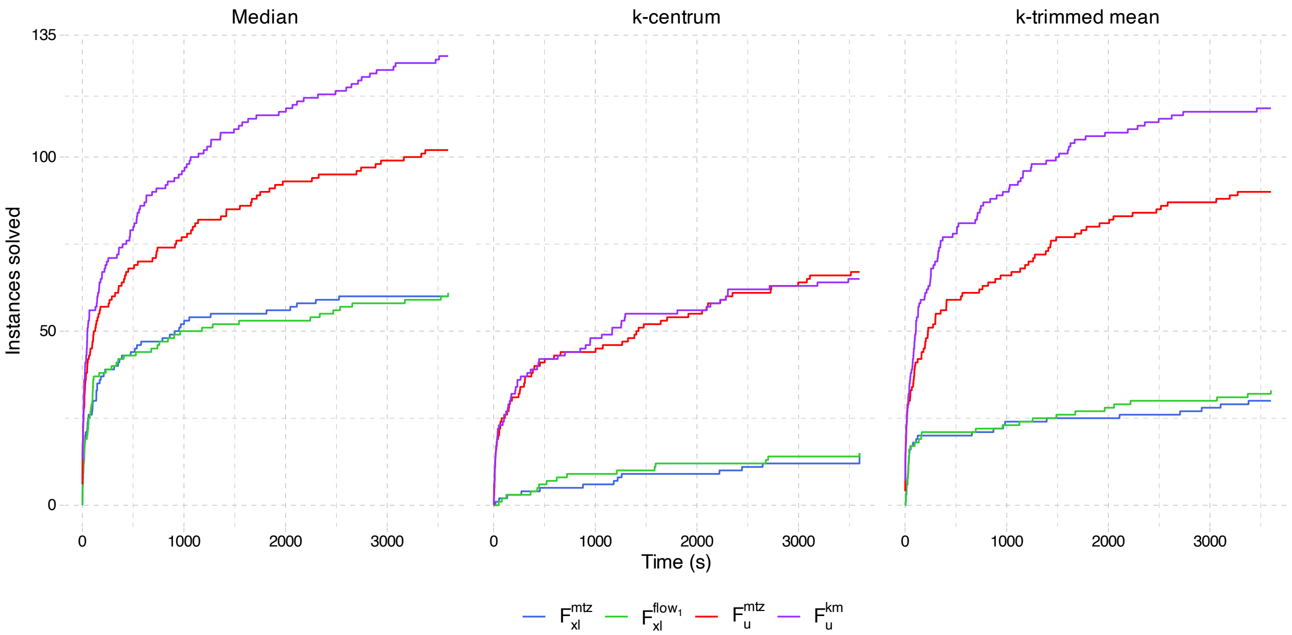


Figure 5.8: Instances solved to optimality within a given time comparison for the final results

		$F1_{x^\ell}^{mtz}$							$F1_{x^\ell}^{flow1}$							$F1_{u}^{mtz}$							$F1_{u}^{km}$						
$\|V\|$	p	$\|\#\|$	cpu	$g\overline{U}R$	$g\overline{U}L$	$g\overline{UL}$	gUL	nod	$\|\#\|$	cpu	$g\overline{U}R$	$g\overline{U}L$	$g\overline{UL}$	gUL	nod	$\|\#\|$	cpu	$g\overline{U}R$	$g\overline{U}L$	$g\overline{UL}$	gUL	nod	$\|\#\|$	cpu	$g\overline{U}R$	$g\overline{U}L$	$g\overline{UL}$	gUL	nod
20	5	5	11.8	37.0	0.0	0.0	0.0	951	5	6.3	37.0	0.0	0.0	0.0	284	5	1.5	37.0	0.0	0.0	0.0	1	5	0.9	37.0	0.0	0.0	0.0	1
20	6	5	12.2	35.0	0.0	0.0	0.0	762	5	10.2	35.0	0.0	0.0	0.0	741	5	1.3	35.0	0.0	0.0	0.0	1	5	0.7	35.0	0.0	0.0	0.0	0
20	10	5	6.4	28.9	0.0	0.0	0.0	516	5	5.7	28.9	0.0	0.0	0.0	404	5	1.2	28.9	0.0	0.0	0.0	34	5	0.4	28.5	0.0	0.0	0.0	0
30	7	5	127.5	39.6	0.0	0.0	0.0	1005	5	106.2	39.6	0.0	0.0	0.0	949	5	16.2	39.6	0.0	0.0	0.0	477	5	11.9	39.6	0.0	0.0	0.0	1
30	10	5	59.0	33.1	0.0	0.0	0.0	1110	5	62.4	33.1	0.0	0.0	0.0	1531	5	8.4	33.1	0.0	0.0	0.0	237	5	8.2	33.1	0.0	0.0	0.0	1
30	15	5	47.3	32.9	0.0	0.0	0.0	2090	5	44.2	33.0	0.0	0.0	0.0	710	5	16.4	32.9	0.0	0.0	0.0	2213	5	8.4	32.6	0.0	0.0	0.0	1
40	10	5	380.3	33.3	0.0	0.0	0.0	1669	5	502.2	33.3	0.0	0.0	0.0	2353	5	62.7	33.3	0.0	0.0	0.0	715	5	41.5	33.3	0.0	0.0	0.0	462
40	13	5	448.3	35.4	0.0	0.0	0.0	3080	5	701.1	35.4	0.0	0.0	0.0	6339	5	66.4	35.4	0.0	0.0	0.0	2400	5	30.1	35.4	0.0	0.0	0.0	1
40	20	5	141.2	28.6	0.0	0.0	0.0	1293	5	88.9	28.6	0.0	0.0	0.0	1611	5	40.9	28.6	0.0	0.0	0.0	2148	5	22.3	28.6	0.0	0.0	0.0	1
50	12	2	2944.8	35.8	1.4	14.0	15.1	2707	4	2913.2	35.8	0.5	4.5	4.9	2874	5	276.8	35.8	0.0	0.0	0.0	1165	5	153.7	35.8	0.0	0.0	0.0	116
50	16	5	1342.5	35.7	0.0	0.0	0.0	1908	4	1563.5	35.7	0.1	0.9	1.0	2769	5	145.1	35.7	0.0	0.0	0.0	1615	5	51.5	35.7	0.0	0.0	0.0	1
50	25	5	473.8	28.6	0.0	0.0	0.0	3616	4	927.2	28.6	0.5	0.0	0.5	12136	5	83.9	28.6	0.0	0.0	0.0	2028	5	44.6	28.5	0.0	0.0	0.0	1
60	15	0	3601.8	40.9	6.0	40.6	44.1	60	0	3601.5	40.9	4.3	31.4	34.4	39	5	827.7	40.9	0.0	0.0	0.0	1571	5	618.9	40.9	0.0	0.0	0.0	16
60	20	0	3600.9	35.5	4.5	24.9	28.2	592	0	3600.4	35.5	3.1	22.4	24.6	126	5	594.0	35.5	0.0	0.0	0.0	4050	5	375.4	35.5	0.0	0.0	0.0	20
60	30	1	3090.7	33.1	1.8	0.7	2.6	5686	1	3347.5	33.1	1.4	1.4	2.7	4322	5	820.2	33.1	0.0	0.0	0.0	6466	5	150.8	32.8	0.0	0.0	0.0	1
70	17	0	3603.1	37.5	8.7	58.1	61.8	14	0	3602.2	37.5	8.6	62.5	65.7	2	4	2571.9	37.5	0.6	0.0	0.6	2403	5	1468.3	37.5	0.0	0.0	0.0	25
70	23	0	3601.9	37.1	3.7	31.2	33.6	46	1	3375.2	37.1	3.7	24.7	27.2	125	4	2175.1	37.1	0.3	0.0	0.3	3503	5	371.2	37.1	0.0	0.0	0.0	1
70	35	2	3029.1	30.8	0.7	4.7	5.4	1224	0	3601.6	30.8	1.6	14.5	15.8	477	3	1933.5	30.8	1.2	0.0	1.2	9547	5	449.0	30.7	0.0	0.0	0.0	1
80	20	0	3600.7	39.3	11.6	61.1	65.6	3	0	3600.1	39.3	10.4	59.6	63.9	1	1	3365.4	39.3	2.6	2.2	4.7	3284	5	2076.7	39.3	0.0	0.0	0.0	607
80	26	0	3606.4	37.0	6.0	43.5	46.9	13	0	3602.0	37.0	4.6	39.3	42.0	4	3	2432.1	37.0	1.7	0.3	2.0	2327	5	1248.0	37.0	0.0	0.0	0.0	1
80	40	0	3601.9	31.2	3.6	19.0	21.8	740	1	3517.3	31.2	1.9	20.6	21.9	511	3	2016.6	31.2	1.0	0.0	1.0	12871	5	754.3	31.0	0.0	0.0	0.0	1
90	22	0	3600.3	41.7	14.5	59.9	64.3	1	0	3600.3	41.7	12.9	63.1	66.6	1	2	3097.7	41.7	5.9	5.4	7.5	1930	4	2873.8	41.7	3.7	9.4	9.4	456
90	30	0	3602.0	38.4	6.4	47.3	50.9	1	0	3600.8	38.4	4.7	47.6	50.1	1	2	3163.1	38.4	1.6	0.4	2.0	4826	5	1423.9	38.4	0.0	0.0	0.0	1
90	45	0	3601.2	29.3	2.6	32.4	34.2	2	0	3600.6	29.3	1.1	28.8	29.5	60	3	2433.2	29.3	0.4	0.2	0.6	18184	5	1140.6	29.3	0.0	0.0	0.0	1
100	25	0	3601.5	39.4	12.9	66.5	69.2	1	0	3600.3	39.4	11.6	57.7	60.7	1	0	3614.1	39.4	7.8	5.9	8.5	1241	1	3432.3	39.4	5.7	30.0	30.2	1
100	33	0	3601.2	35.4	6.6	58.5	61.2	1	0	3600.3	35.4	5.2	50.5	52.9	1	0	3606.4	35.4	2.7	0.7	3.2	861	4	2339.8	35.4	0.2	0.7	0.7	1
100	50	0	3600.3	29.7	1.9	17.9	19.5	1	0	3603.1	29.7	1.4	39.4	40.3	1	2	2959.8	29.7	2.2	0.0	2.2	10014	5	2095.2	29.7	0.0	0.0	0.0	1

Table 5.8: OMT summary results for median criterion

		$F1_{x^\ell}^{mtz}$							$F1_{x^\ell}^{flow1}$						
$\|V\|$	p	$\|\#\|$	cpu	$g\overline{UR}$	$g\overline{U}L$	$g\overline{U}\overline{L}$	gUL	nod	$\|\#\|$	cpu	$g\overline{UR}$	$g\overline{U}L$	$g\overline{U}\overline{L}$	gUL	nod
20	5	5	504.4	62.7	0.0	0.0	0.0	64813	5	307.3	62.7	0.0	0.0	0.0	47977
20	6	2	2739.8	65.0	6.0	0.0	6.0	319609	4	1981.0	65.0	2.5	0.0	2.5	204699
20	10	5	1472.8	56.8	0.0	0.0	0.0	260421	5	1056.2	56.8	0.0	0.0	0.0	185366
30	7	0	3600.4	65.4	30.5	3.2	32.7	31958	0	3600.2	65.4	28.9	1.0	29.6	33589
30	10	0	3600.3	61.2	25.5	0.3	25.7	36798	0	3600.4	61.2	24.0	0.5	24.4	39749
30	15	1	3600.2	59.9	13.4	0.0	13.4	235432	0	3600.2	59.9	11.6	0.0	11.6	165266
40	10	0	3600.3	63.8	37.0	3.8	39.3	16941	0	3600.2	63.8	37.1	5.3	40.2	12991
40	13	0	3600.2	61.0	31.9	7.0	36.8	15750	0	3600.3	61.0	31.4	3.2	33.6	15149
40	20	0	3600.3	59.0	27.0	0.3	27.2	41430	0	3600.6	59.0	26.0	0.6	26.4	40789
50	12	0	3600.7	63.6	44.0	22.8	55.4	2328	0	3601.0	63.6	42.8	22.3	54.4	2502
50	16	0	3600.5	63.5	40.5	19.3	52.1	2830	0	3600.7	63.5	39.8	8.8	45.1	1844
50	25	0	3600.2	56.7	32.3	7.2	37.2	5241	0	3600.2	56.7	32.0	2.4	33.6	3435
60	15	0	3601.0	67.4	47.9	31.9	62.2	114	0	3601.2	67.4	47.0	32.4	61.7	97
60	20	0	3600.6	61.7	43.4	17.9	52.9	2079	0	3600.6	61.7	42.0	18.9	52.5	1745
60	30	0	3600.5	60.9	37.1	6.3	40.8	5730	0	3600.5	60.9	37.1	5.3	40.2	4108
70	17	0	3603.9	63.9	46.6	49.0	69.2	24	0	3602.0	63.9	47.1	51.6	71.0	15
70	23	0	3602.5	62.7	42.7	22.7	55.2	221	0	3602.8	62.7	41.8	24.8	55.9	22
70	35	0	3600.9	59.9	42.3	8.8	47.3	1149	0	3600.8	59.9	41.7	7.9	46.3	285
80	20	0	3600.3	65.7	49.8	55.2	72.2	10	0	3600.8	65.7	49.2	55.3	71.9	2
80	26	0	3601.8	64.8	47.1	35.0	62.6	6	0	3601.8	64.8	46.3	32.9	60.7	3
80	40	0	3602.5	60.3	42.0	16.8	51.5	44	0	3601.0	60.3	42.1	18.7	52.9	16
90	22	0	3602.1	66.1	50.2	51.4	69.9	3	0	3600.6	66.1	49.0	52.4	69.9	1
90	30	0	3603.1	64.4	44.4	31.8	59.1	15	0	3601.9	64.4	44.2	28.7	57.0	5
90	45	0	3600.8	58.3	41.7	19.4	53.1	62	0	3602.0	58.3	39.9	17.5	50.4	6
100	25	0	3601.0	65.4	50.2	53.8	70.3	1	0	3600.2	65.4	49.2	54.4	70.1	1
100	33	0	3603.2	63.0	46.1	47.6	65.7	1	0	3601.4	63.0	45.6	44.5	63.2	1
100	50	0	3602.8	59.4	44.0	34.7	59.9	1	0	3600.8	59.4	43.2	29.3	55.8	1

		$F1_{u}^{mtz}$							$F1_{u}^{km}$						
$\|V\|$	p	$\|\#\|$	cpu	$g\overline{UR}$	$g\overline{U}L$	$g\overline{U}\overline{L}$	gUL	nod	$\|\#\|$	cpu	$g\overline{UR}$	$g\overline{U}L$	$g\overline{U}\overline{L}$	gUL	nod
20	5	5	8.7	62.7	0.0	0.0	0.0	2824	5	10.7	62.7	0.0	0.0	0.0	715
20	6	5	15.2	65.0	0.0	0.0	0.0	3192	5	13.0	65.0	0.0	0.0	0.0	983
20	10	5	7.9	56.8	0.0	0.0	0.0	2312	5	2.6	56.5	0.0	0.0	0.0	1
30	7	5	188.9	65.4	0.0	0.0	0.0	7802	5	136.4	65.4	0.0	0.0	0.0	2938
30	10	5	248.8	61.2	0.0	0.0	0.0	11008	5	95.4	61.2	0.0	0.0	0.0	2013
30	15	5	66.4	59.9	0.0	0.0	0.0	7673	5	28.6	59.7	0.0	0.0	0.0	623
40	10	5	1237.8	63.8	0.0	0.0	0.0	45176	5	621.2	63.8	0.0	0.0	0.0	13081
40	13	5	529.7	61.0	0.0	0.0	0.0	17580	5	285.2	61.0	0.0	0.0	0.0	4850
40	20	4	908.6	59.0	1.0	0.0	1.0	26517	5	334.3	59.0	0.0	0.0	0.0	3828
50	12	1	3426.7	63.6	4.7	2.9	4.8	28627	1	3337.1	63.6	3.5	3.1	3.9	14862
50	16	4	1955.3	63.5	0.4	0.0	0.4	23990	5	1311.3	63.5	0.0	0.0	0.0	5234
50	25	4	1034.1	56.7	0.2	0.0	0.2	32517	5	1000.7	56.6	0.0	0.0	0.0	1781
60	15	3	3080.3	67.4	2.1	6.3	2.2	24493	0	3601.4	67.4	13.3	6.2	13.4	5031
60	20	3	3012.2	61.7	1.2	1.1	1.2	28959	2	3026.0	61.7	6.4	1.1	6.4	4690
60	30	3	2053.1	60.9	0.7	0.3	0.7	36946	5	2061.6	60.9	0.0	0.3	0.0	5250
70	17	0	3602.9	63.9	12.3	12.2	12.6	7941	0	3601.8	63.9	18.2	11.8	18.1	1629
70	23	2	3246.9	62.7	4.8	1.5	5.2	7571	0	3602.5	62.7	11.6	1.7	12.1	1859
70	35	1	3145.3	59.9	2.9	0.0	2.9	24419	2	3496.1	59.9	4.5	0.0	4.5	3371
80	20	0	3603.7	65.7	22.3	27.2	29.9	4682	0	3604.0	65.7	21.2	19.7	21.8	44
80	26	0	3603.3	64.8	14.6	8.8	15.2	3180	0	3602.2	64.8	17.0	14.0	22.3	14
80	40	1	3426.6	60.3	2.7	0.0	2.7	13274	0	3600.8	60.3	6.5	0.0	6.5	2172
90	22	0	3600.3	66.1	19.7	20.0	20.4	3377	0	3605.0	66.1	21.2	22.0	23.7	445
90	30	0	3600.6	64.4	10.3	7.4	10.5	4525	0	3600.8	64.4	14.6	8.1	15.4	21
90	45	1	3263.6	58.3	1.9	0.0	1.9	14687	0	3600.3	58.3	9.4	0.3	9.7	502
100	25	0	3602.1	65.4	24.4	27.4	29.2	2126	0	3601.4	65.4	24.0	46.1	47.1	3
100	33	0	3601.4	63.0	18.1	17.9	18.3	2024	0	3602.5	63.0	17.9	22.7	22.9	1
100	50	0	3600.9	59.4	8.8	8.8	8.8	2973	0	3604.6	59.4	14.8	16.7	22.0	1

Table 5.9: OMT results summary for k-centrum criterion

		$F1^{mtz}_{x^\ell}$							$F1^{flow1}_{x^\ell}$							$F1^{mtz}_{u}$							$F1^{km}_{u}$						
$\|V\|$	p	$\|\#\|$	cpu	$g\overline{U}R$	$g\overline{U}L$	$gU\overline{L}$	gUL	nod	$\|\#\|$	cpu	$g\overline{U}R$	$g\overline{U}L$	$gU\overline{L}$	gUL	nod	$\|\#\|$	cpu	$g\overline{U}R$	$g\overline{U}L$	$gU\overline{L}$	gUL	nod	$\|\#\|$	cpu	$g\overline{U}R$	$g\overline{U}L$	$gU\overline{L}$	gUL	nod
20	5	5	36.6	50.7	0.0	0.0	0.0	2762	5	26.2	50.7	0.0	0.0	0.0	3047	5	3.0	50.7	0.0	0.0	0.0	125	5	3.8	50.7	0.0	0.0	0.0	56
20	6	5	25.9	55.2	0.0	0.0	0.0	2779	5	30.1	55.2	0.0	0.0	0.0	1641	5	1.3	55.2	0.0	0.0	0.0	1	5	0.9	55.2	0.0	0.0	0.0	0
20	10	5	13.2	30.2	0.0	0.0	0.0	2517	5	13.7	30.2	0.0	0.0	0.0	5279	5	2.1	30.2	0.0	0.0	0.0	1	5	1.2	29.9	0.0	0.0	0.0	0
30	7	3	3320.5	63.8	7.1	0.0	7.1	73408	2	3245.6	63.8	6.4	0.0	6.4	88561	5	36.1	63.8	0.0	0.0	0.0	1529	5	30.6	63.8	0.0	0.0	0.0	413
30	10	3	2334.0	52.7	4.1	0.0	4.1	87074	4	2021.7	52.7	2.1	0.0	2.1	89506	5	10.4	52.7	0.0	0.0	0.0	16	5	6.9	52.6	0.0	0.0	0.0	1
30	15	5	183.7	36.5	0.0	0.0	0.0	12498	5	95.0	36.5	0.0	0.0	0.0	4367	5	19.0	36.5	0.0	0.0	0.0	878	5	15.4	36.5	0.0	0.0	0.0	1
40	10	0	3600.2	62.2	22.8	0.1	22.9	28542	1	3323.9	62.2	18.4	0.0	18.4	22141	5	174.4	62.2	0.0	0.0	0.0	2423	5	94.6	62.2	0.0	0.0	0.0	975
40	13	0	3600.1	53.9	19.9	1.8	21.2	24033	0	3600.3	53.9	19.2	1.2	20.1	26373	5	102.4	53.9	0.0	0.0	0.0	919	5	107.7	53.9	0.0	0.0	0.0	14
40	20	3	2202.7	32.1	1.5	0.0	1.5	34646	5	1176.4	32.1	0.0	0.0	0.0	20527	5	89.3	32.1	0.0	0.0	0.0	833	5	89.9	32.1	0.0	0.0	0.0	1
50	12	0	3600.2	58.0	25.6	12.5	34.8	1488	0	3601.0	58.0	24.9	9.9	32.3	2793	5	925.2	58.0	0.0	0.0	0.0	5295	5	474.6	58.0	0.0	0.0	0.0	18481
50	16	0	3600.1	55.2	18.0	1.8	19.5	5109	0	3600.1	55.2	19.2	2.7	21.3	4825	5	426.9	55.2	0.0	0.0	0.0	696	5	84.7	55.0	0.0	0.0	0.0	1
50	25	1	3054.2	32.2	4.0	0.0	4.0	12362	0	3600.3	32.2	5.7	0.0	5.7	15313	5	265.5	32.2	0.0	0.0	0.0	1503	5	92.8	32.1	0.0	0.0	0.0	1
60	15	0	3600.8	65.5	33.8	48.2	65.8	3583	0	3600.7	65.5	33.0	39.5	59.6	1338	5	1921.8	65.5	0.0	0.0	0.0	5514	5	1105.8	65.5	0.0	0.0	0.0	3910
60	20	0	3600.5	53.2	22.0	15.9	34.4	1531	0	3600.5	53.2	24.6	18.6	38.7	2728	1	3057.5	53.2	1.6	0.4	2.0	486	5	273.6	53.2	0.0	0.0	0.0	1
60	30	0	3600.3	32.7	9.3	0.2	9.4	2178	0	3600.5	32.7	9.4	0.6	10.0	2325	5	1576.9	32.7	0.0	0.0	0.0	2626	5	155.8	32.7	0.0	0.0	0.0	1
70	17	0	3602.0	57.5	31.0	64.4	75.3	51	0	3602.9	57.5	30.2	59.9	71.8	52	3	2787.4	57.5	0.9	0.0	0.9	4735	2	2589.8	57.5	1.3	0.0	1.3	5319
70	23	0	3601.6	51.1	26.5	44.1	58.8	387	0	3601.1	51.1	25.5	31.6	49.3	136	2	2550.9	51.1	2.2	0.0	2.2	535	5	293.3	51.1	0.0	0.0	0.0	1
70	35	0	3600.6	30.9	9.2	6.8	15.4	568	0	3621.6	30.9	10.2	1.9	11.9	920	5	1333.7	30.9	0.0	0.0	0.0	2683	5	274.8	30.9	0.0	0.0	0.0	1
80	20	0	3603.8	60.4	28.9	71.0	79.4	27	0	3604.6	60.4	28.4	70.9	78.9	18	0	3602.2	60.4	6.2	11.8	7.6	1854	1	3406.0	60.4	7.3	9.1	2.4	2759
80	26	0	3603.0	50.3	22.5	67.3	75.7	32	0	3602.3	50.3	22.1	66.7	74.4	19	3	2141.3	50.3	12.9	11.9	1.9	556	4	1423.3	50.3	13.5	12.1	0.3	13
80	40	0	3601.1	32.8	13.3	34.1	33.1	94	0	3601.6	32.8	13.7	30.2	38.3	42	3	2554.8	32.8	13.8	15.6	0.3	2390	5	604.6	32.8	14.2	15.6	0.0	1
90	22	0	3601.7	64.4	38.2	80.0	86.4	1	0	3602.9	64.4	35.9	73.9	81.7	1	0	3605.1	64.4	11.0	8.4	11.4	1794	0	3613.4	64.4	8.3	18.6	18.7	11
90	30	0	3604.5	53.2	28.5	69.5	78.2	9	0	3602.5	53.2	28.6	64.7	74.6	6	2	2514.5	53.2	1.6	0.6	1.9	41	4	2725.8	53.2	0.3	0.3	0.3	1
90	45	0	3603.5	32.2	14.1	36.8	45.8	24	0	3601.4	32.2	14.2	51.6	58.4	2	0	3604.1	32.2	1.9	0.0	1.9	1981	5	1628.7	32.2	0.0	0.0	0.0	1
100	25	0	3602.0	62.8	37.3	75.8	83.2	1	0	3601.7	62.8	37.0	71.0	79.7	1	0	3602.2	62.8	12.2	20.6	22.5	38	0	3603.4	62.8	10.6	24.6	25.2	4
100	33	0	3601.4	53.8	28.7	78.5	84.5	1	0	3602.1	53.8	27.8	71.3	79.1	1	0	3675.0	53.8	3.2	1.4	3.2	77	4	2456.0	53.8	1.4	1.4	1.4	5
100	50	0	3601.6	30.1	14.7	60.3	66.2	2	0	3601.4	30.1	13.8	59.3	65.0	1	1	3246.0	30.1	1.1	0.0	1.1	658	4	2055.2	30.1	0.1	0.2	0.2	261

Table 5.10: OMT results summary for k-trimmed mean criterion

Appendix: OMT formulations description

OMT subtour elimination formulations

$$F1_x^{sub1} : \min \frac{1}{\sum_{\ell \in V} \lambda_\ell} \sum_{\ell \in V} \sum_{(i,j) \in A} \lambda_\ell c_{ij} x_{ij}^\ell + \frac{1}{p-1} \sum_{(i,j) \in E} c_{ij} z_{ij}, \tag{5.26a}$$

$$\text{s.t.} \sum_{i \in V} x_{ii} = p, \tag{5.26b}$$

$$\sum_{j \in V} x_{ij} = 1, \quad i \in V, \tag{5.26c}$$

$$x_{ij} \leq x_{jj}, \quad (i,j) \in A : i \neq j, \tag{5.26d}$$

$$2z_{ij} \leq x_{ii} + x_{jj}, \quad (i,j) \in E, \tag{5.26e}$$

$$\sum_{(i,j) \in E} z_{ij} = p - 1, \tag{5.26f}$$

$$\sum_{i,j \in S : i<j} z_{ij} \leq |S| - 1, \quad S \neq \emptyset, S \subset V, \tag{5.26g}$$

$$\sum_{\ell \in V} x_{ij}^\ell = x_{ij}, \quad (i,j) \in A, \tag{5.26h}$$

$$\sum_{(i,j) \in A} x_{ij}^\ell = 1, \quad \ell \in V, \tag{5.26i}$$

$$\sum_{(i,j) \in A} c_{ij} x_{ij}^\ell \leq \sum_{(i,j) \in A} c_{ij} x_{ij}^{\ell+1}, \quad \ell \in V : \ell < |V|, \tag{5.26j}$$

$$x_{ij} \in \{0,1\}, \quad (i,j) \in A, \tag{5.26k}$$

$$x_{ij}^\ell \in \{0,1\}, \quad (i,j) \in A, \ell \in V, \tag{5.26l}$$

$$z_{ij} \in \{0,1\}, \quad (i,j) \in E. \tag{5.26m}$$

(5.26a): Weighted minimization of the compensated allocation cost of the system plus the design cost of the tree connecting facilities (objective).
(5.26b): Exactly p facilities (allocation).
(5.26c): Each client is allocated to exactly one facility (allocation).
(5.26d): No client is allocated to a non-facility (allocation).
(5.26e): Edges of the tree of facilities can only be selected if both nodes are facilities (allocation).
(5.26f): Exactly $p-1$ edges connecting facilities (connectivity).
(5.26g): No subtours are allowed to be formed between facilities edges (connectivity).
(5.26h): Relationship between sorting and allocation variables (sorting).
(5.26i): Every sort position must be consider (sorting).
(5.26j): Correct sorting of the costs sequence (sorting).

$$F1_x^{sub2} : \min \frac{1}{\sum_{\ell \in V} \lambda_\ell} \sum_{\ell \in V} \sum_{(i,j) \in A} \lambda_\ell c_{ij} x_{ij}^\ell + \frac{1}{p-1} \sum_{(i,j) \in E} c_{ij} z_{ij}, \tag{5.27a}$$

$$\text{s.t.} \sum_{i \in V} x_{ii} = p, \tag{5.27b}$$

$$\sum_{j \in V} x_{ij} = 1, \quad i \in V, \tag{5.27c}$$

$$x_{ij} \leq x_{jj}, \quad (i,j) \in A : i \neq j, \tag{5.27d}$$

$$2z_{ij} \leq x_{ii} + x_{jj}, \quad (i,j) \in E, \tag{5.27e}$$

$$\sum_{(i,j) \in E} z_{ij} = p - 1, \tag{5.27f}$$

$$\sum_{i\in S, j\in V\setminus S} x_{ij} + \sum_{i\in S, j\in V\setminus S} x_{ji} + \sum_{i\in S, j\in V\setminus S: i<j} z_{ij} + \sum_{i\in S, j\in V\setminus S: i>j} z_{ji} \geq 1, \quad S\neq\emptyset, S\subset V \tag{5.27g}$$

$$\sum_{\ell\in V} x_{ij}^{\ell} = x_{ij}, \quad (i,j)\in A, \tag{5.27h}$$

$$\sum_{(i,j)\in A} x_{ij}^{\ell} = 1, \quad \ell\in V, \tag{5.27i}$$

$$\sum_{(i,j)\in A} c_{ij} x_{ij}^{\ell} \leq \sum_{(i,j)\in A} c_{ij} x_{ij}^{\ell+1}, \quad \ell\in V : \ell < |V|, \tag{5.27j}$$

$$x_{ij}\in\{0,1\}, \quad (i,j)\in A, \tag{5.27k}$$

$$x_{ij}^{\ell}\in\{0,1\}, \quad (i,j)\in A, \ell\in V, \tag{5.27l}$$

$$z_{ij}\in\{0,1\}, \quad (i,j)\in E. \tag{5.27m}$$

(5.27a): Weighted minimization of the compensated allocation cost of the system plus the design cost of the tree connecting facilities (objective).
(5.27b): Exactly p facilities (allocation).
(5.27c): Each client is allocated to exactly one facility (allocation).
(5.27d): No client is allocated to a non-facility (allocation).
(5.27e): Edges of the tree of facilities can only be selected if both nodes are facilities (allocation).
(5.27f): Exactly $p-1$ edges connecting facilities (connectivity).
(5.27g): At least one allocation or one edge from the tree of facilities must connect S to a node of $V\setminus S$ (connectivity).
(5.27h): Relationship between sorting and allocation variables (sorting).
(5.27i): Every sort position must be consider (sorting).
(5.27j): Correct sorting of the costs sequence (sorting).

$$F2_x^{sub1} : \min \frac{1}{\sum_{\ell\in V}\lambda_\ell} \sum_{\ell\in V}\sum_{(i,j)\in A} \lambda_\ell c_{ij} x_{ij}^{\ell} + \frac{1}{p-1}\sum_{(i,j)\in E} c_{ij}(z_{ij} - x_{ij} - x_{ji}), \tag{5.28a}$$

$$\text{s.t.} \sum_{i\in V} x_{ii} = p, \tag{5.28b}$$

$$\sum_{j\in V} x_{ij} = 1, \quad i\in V, \tag{5.28c}$$

$$2x_{ij} \leq 1 - x_{ii} + x_{jj}, \quad (i,j)\in A : i\neq j, \tag{5.28d}$$

$$x_{ij} + x_{ji} \leq z_{ij}, \quad (i,j)\in E, \tag{5.28e}$$

$$2z_{ij} \leq x_{ii} + x_{jj} + x_{ij} + x_{ji}, \quad (i,j)\in E, \tag{5.28f}$$

$$\sum_{(i,j)\in E} z_{ij} = |V| - 1, \tag{5.28g}$$

$$\sum_{i,j\in S: i<j} z_{ij} \leq |S| - 1, \quad S\neq\emptyset, S\subset V, \tag{5.28h}$$

$$\sum_{\ell\in V} x_{ij}^{\ell} = x_{ij}, \quad (i,j)\in A, \tag{5.28i}$$

$$\sum_{(i,j)\in A} x_{ij}^{\ell} = 1, \quad \ell\in V, \tag{5.28j}$$

$$\sum_{(i,j)\in A} c_{ij} x_{ij}^{\ell} \leq \sum_{(i,j)\in A} c_{ij} x_{ij}^{\ell+1}, \quad \ell\in V : \ell < |V|, \tag{5.28k}$$

$$x_{ij}\in\{0,1\}, \quad (i,j)\in A, \tag{5.28l}$$

$$x_{ij}^{\ell}\in\{0,1\}, \quad (i,j)\in A, \ell\in V, \tag{5.28m}$$

$$z_{ij}\in\{0,1\}, \quad (i,j)\in E. \tag{5.28n}$$

(5.28a): Weighted minimization of the compensated allocation cost of the system plus the design cost of the tree subtracting in the design cost part the cost associated to the edges that represent the allocation of client to facilities (objective).
(5.28b): Exactly p facilities (allocation).
(5.28c): Each client is allocated to exactly one facility (allocation).
(5.28d): Client-facility allocations are only possible if i is a client and j is a facility (allocation).
(5.28e): Only one allocation, whether client i to facility j or viceversa, can be considered if there exists an edge connecting $i, j \in V$ (allocation).
(5.28f): Only possibilities in which an edge can be selected is if both i or j are facilities or if either i, or j, is a client and j, or i respectively, its facility (allocation).
(5.28g): Exactly $p-1$ edges connecting facilities (connectivity).
(5.28h): No subtours are allowed to be formed between facilities edges (connectivity).
(5.28i): Relationship between sorting and allocation variables (sorting).
(5.28j): Every sort position must be consider (sorting).
(5.28k): Correct sorting of the costs sequence (sorting).

Formulations $F2_x^{sub2}$, $F1_u^{sub1}$, $F1_u^{sub2}$, $F2_u^{sub1}$ and $F2_u^{sub2}$ are left to the reader.

OMT MTZ formulations

Let $D = (V, A, r)$ be a rooted weighted directed network where A the set of arcs and r is a root node. An *arborescence* of D is a subgraph $D' = (V, A', r)$ where $A' \subseteq A$ such that each non-root node has exactly one incoming (or outcoming) edge (thus $|A'| = |V| - 1$) and D' has no cycles. The MTZ formulation builds an arborescence rooted at a arbitrarily selected root node $r \in V$, in which arcs follow the direction from root to leaves.

$$
\begin{aligned}
F1_x^{mtz} : \min\ & \frac{1}{\sum_{\ell \in V} \lambda_\ell} \sum_{\ell \in V} \sum_{(i,j) \in A} \lambda_\ell c_{ij} x_{ij}^\ell + \frac{1}{p-1} \sum_{(i,j) \in E} c_{ij} z_{ij} && && (5.29a)\\
\text{s.t. } & \sum_{i \in V} x_{ii} = p, && && (5.29b)\\
& \sum_{j \in V} x_{ij} = 1, && i \in V, && (5.29c)\\
& x_{ij} \leq x_{jj}, && (i,j) \in A : i \neq j, && (5.29d)\\
& 2z_{ij} \leq x_{ii} + x_{jj}, && (i,j) \in E, && (5.29e)\\
& \sum_{(i,j) \in E} z_{ij} = p - 1, && && (5.29f)\\
& \sum_{(j,i) \in A} y_{ji} = 1, && i \in V \setminus \{r\}, && (5.29g)\\
& y_{ij} + y_{ji} = z_{ij}, && (i,j) \in E, && (5.29h)\\
& l_j \geq l_i + 1 - |V|(1 - y_{ij}), && (i,j) \in A : i \neq j, && (5.29i)\\
& l_r = 1, && && (5.29j)\\
& 2 \leq l_i \leq p, && i \in V \setminus \{r\}, && (5.29k)\\
& \sum_{\ell \in V} x_{ij}^\ell = x_{ij}, && (i,j) \in A, && (5.29l)\\
& \sum_{(i,j) \in A} x_{ij}^\ell = 1, && \ell \in V, && (5.29m)\\
& \sum_{(i,j) \in A} c_{ij} x_{ij}^\ell \leq \sum_{(i,j) \in A} c_{ij} x_{ij}^{\ell+1}, && \ell \in V : \ell < |V|, && (5.29n)\\
& x_{ij} \in \{0,1\}, && (i,j) \in A, && (5.29o)\\
& x_{ij}^\ell \in \{0,1\}, && (i,j) \in A, \ell \in V, && (5.29p)
\end{aligned}
$$

$$z_{ij} \in \{0,1\}, \qquad (i,j) \in E, \tag{5.29q}$$
$$y_{ij} \in \{0,1\}, \qquad (i,j) \in A, \tag{5.29r}$$
$$l_i \geq 0, \qquad i \in V. \tag{5.29s}$$

(5.29a): Weighted minimization of the compensated allocation cost of the system plus the design cost of the tree of facilities (objective).
(5.29b): Exactly p facilities (allocation).
(5.29c): Each client is allocated to exactly one facility (allocation).
(5.29d): No client is allocated to a non-facility (allocation).
(5.29e): Edges of the tree of facilities can only be selected if both nodes are facilities (allocation).
(5.29f): Exactly $p-1$ edges connecting facilities (connectivity).
(5.29g): Exactly one arc goes into a non-root node of the arborescence (connectivity).
(5.29h): If considering an edge between facilities, the arborescence includes one of the corresponding arcs (connectivity).
(5.29i): If (i,j) arc is considered in the arborescence, then the position of j must be higher than i (connectivity).
(5.29j) - (5.29k): The root node is in 1st position of the arborescence and the non-root nodes must be distributed between the 2nd and p-th positions (connectivity).
(5.29l): Relationship between sorting and allocation variables (sorting).
(5.29m): Every sort position must be consider (sorting).
(5.29n): Correct sorting of the costs sequence (sorting).

$$F2_x^{mtz} : \min \frac{1}{\sum_{\ell \in V} \lambda_\ell} \sum_{\ell \in V} \sum_{(i,j)\in A} \lambda_\ell c_{ij} x_{ij}^\ell + \frac{1}{p-1} \sum_{(i,j)\in E} c_{ij}(z_{ij} - x_{ij} - x_{ji}), \tag{5.30a}$$
$$\text{s.t.} \sum_{i \in V} x_{ii} = p, \tag{5.30b}$$
$$\sum_{j \in V} x_{ij} = 1, \qquad i \in V, \tag{5.30c}$$
$$2x_{ij} \leq 1 - x_{ii} + x_{jj}, \qquad (i,j) \in A : i \neq j, \tag{5.30d}$$
$$x_{ij} + x_{ji} \leq z_{ij}, \qquad (i,j) \in E, \tag{5.30e}$$
$$2z_{ij} \leq x_{ii} + x_{jj} + x_{ij} + x_{ji}, \qquad (i,j) \in E, \tag{5.30f}$$
$$\sum_{(i,j)\in E} z_{ij} = |V| - 1, \tag{5.30g}$$
$$\sum_{(j,i)\in A} y_{ji} = 1, \qquad i \in V \setminus \{r\}, \tag{5.30h}$$
$$y_{ij} + y_{ji} = z_{ij}, \qquad (i,j) \in E, \tag{5.30i}$$
$$l_j \geq l_i + 1 - |V|(1 - y_{ij}), \qquad (i,j) \in A : i \neq j, \tag{5.30j}$$
$$l_r = 1, \tag{5.30k}$$
$$2 \leq l_i \leq |V|, \qquad i \in V \setminus \{r\}, \tag{5.30l}$$
$$\sum_{\ell \in V} x_{ij}^\ell = x_{ij}, \qquad (i,j) \in A, \tag{5.30m}$$
$$\sum_{(i,j)\in A} x_{ij}^\ell = 1, \qquad \ell \in V, \tag{5.30n}$$
$$\sum_{(i,j)\in A} c_{ij} x_{ij}^\ell \leq \sum_{(i,j)\in A} c_{ij} x_{ij}^{\ell+1}, \qquad \ell \in V : \ell < |V|, \tag{5.30o}$$
$$x_{ij} \in \{0,1\}, \qquad (i,j) \in A, \tag{5.30p}$$
$$x_{ij}^\ell \in \{0,1\}, \qquad (i,j) \in A, \ell \in V, \tag{5.30q}$$

$$z_{ij} \in \{0,1\}, \qquad (i,j) \in E, \tag{5.30r}$$
$$y_{ij} \in \{0,1\}, \qquad (i,j) \in A, \tag{5.30s}$$
$$l_i \geq 0, \qquad i \in V. \tag{5.30t}$$

(5.30a): Weighted minimization of the compensated allocation cost of the system plus the design cost of the tree subtracting in the design cost part the cost associated to the edges that represent the allocation of client to facilities (objective).
(5.30b): Exactly p facilities (allocation).
(5.30c): Each client is allocated to exactly one facility (allocation).
(5.30d): Client-facility allocations are only possible if i is a client and j is a facility (allocation).
(5.30e): Only one allocation, whether client i to facility j or viceversa, can be considered if there exists an edge connecting $i, j \in V$ (allocation).
(5.30f): Only possibilities in which an edge can be selected is if both i or j are facilities or if either i, or j, is a client and j, or i respectively, its facility (allocation).
(5.30g): Exactly $p-1$ edges connecting facilities (connectivity).
(5.30h): Exactly one arc goes into a non-root node of the arborescence (connectivity).
(5.30i): If considering an edge between facilities, the arborescence includes one of the corresponding arcs (connectivity).
(5.30j): If (i,j) arc is considered in the arborescence, then the position of j must be higher than i (connectivity).
(5.30k) - (5.30l): The root node is in 1st position of the arborescence and the non-root nodes must be distributed between the 2nd and p-th positions (connectivity).
(5.30m): Relationship between sorting and allocation variables (sorting).
(5.30n): Every sort position must be consider (sorting).
(5.30o): Correct sorting of the costs sequence (sorting).

Formulations $F1_u^{mtz}$ and $F2_u^{mtz}$ are left to the reader.

OMT flow based formulations

The flow formulation also relies on a source node $r \in V$ which distributes the flow but for the OMT, contrary to the MTZ formulation, the choice of the root node is very influential as care must be taken when spreading the flow. This selection can be done in two ways:

- ◇ Adding a set of variables $r_i \in \{0,1\}$ for $i \in V$, is 1 if the node $i \in V$ is selected as the source node for the tree of facilities.
- ◇ Arbitrarily selecting the source node and distributing the flow along the tree, distinguishing whether the node selected as the source is a facility or not.

Formulation using additional variables

$$F1_x^{flow1} : \min \frac{1}{\sum_{\ell \in V} \lambda_\ell} \sum_{\ell \in V} \sum_{(i,j) \in A} \lambda_\ell c_{ij} x_{ij}^\ell + \frac{1}{p-1} \sum_{(i,j) \in E} c_{ij} z_{ij}, \tag{5.31a}$$
$$\sum_{i \in V} x_{ii} = p, \tag{5.31b}$$
$$\sum_{j \in V} x_{ij} = 1, \qquad i \in V, \tag{5.31c}$$
$$x_{ij} \leq x_{jj}, \qquad (i,j) \in A : i \neq j, \tag{5.31d}$$
$$2z_{ij} \leq x_{ii} + x_{jj}, \qquad (i,j) \in E, \tag{5.31e}$$
$$\sum_{(i,j) \in E} z_{ij} = p-1, \tag{5.31f}$$
$$\sum_{i \in V} r_i = 1, \tag{5.31g}$$

$$r_i \leq x_{ii}, \qquad i \in V, \tag{5.31h}$$

$$\sum_{(i,j)\in\delta^+(i)} f_{ij} - \sum_{(j,i)\in\delta^-(i)} f_{ji} = (p-1)r_i - (x_{ii} - r_i), \quad i \in V \tag{5.31i}$$

$$f_{ij} \leq (p-1)z_{ij}, \qquad (i,j) \in E, \tag{5.31j}$$

$$f_{ji} \leq (p-1)z_{ij}, \qquad (i,j) \in E, \tag{5.31k}$$

$$\sum_{\ell\in V} x_{ij}^{\ell} = x_{ij}, \qquad (i,j) \in A, \tag{5.31l}$$

$$\sum_{(i,j)\in A} x_{ij}^{\ell} = 1, \qquad \ell \in V, \tag{5.31m}$$

$$\sum_{(i,j)\in A} c_{ij}x_{ij}^{\ell} \leq \sum_{(i,j)\in A} c_{ij}x_{ij}^{\ell+1}, \qquad \ell \in V : \ell < |V|, \tag{5.31n}$$

$$x_{ij} \in \{0,1\}, \qquad (i,j) \in A, \tag{5.31o}$$

$$x_{ij}^{\ell} \in \{0,1\}, \qquad (i,j) \in A, \ell \in V, \tag{5.31p}$$

$$r_i \in \{0,1\}, \qquad i \in V, \tag{5.31q}$$

$$z_{ij} \in \{0,1\}, \qquad (i,j) \in E, \tag{5.31r}$$

$$f_{ij} \geq 0, \qquad (i,j) \in A : i \neq j. \tag{5.31s}$$

(5.31a): Weighted minimization of the total compensated allocation cost of the system plus the design cost of the tree of facilities (objective).
(5.31b): Exactly p facilities (allocation).
(5.31c): Each client is allocated to exactly one facility (allocation).
(5.31d): No client is allocated to a non-facility (allocation).
(5.31e): Edges of the tree of facilities can only be selected if both nodes are facilities (allocation).
(5.31f): Exactly $p-1$ edges connecting facilities (connectivity).
(5.31g): Only one root node selected (connectivity).
(5.31h): Only facilities nodes can be activated as a root (connectivity).
(5.31i): Flow is properly distributed along the tree of facilities (connectivity).
(5.31j)-(5.31k): All variables sending flow must be activated (connectivity).
(5.31l): Relationship between sorting and allocation variables (sorting).
(5.31m): Every sort position must be consider (sorting).
(5.31n): Correct sorting of the costs sequence (sorting).

Formulation without using additional variables

$$F1_x^{flow2} : \min \frac{1}{\sum_{\ell\in V}\lambda_\ell} \sum_{\ell\in V}\sum_{(i,j)\in A} \lambda_\ell c_{ij}x_{ij}^{\ell} + \frac{1}{p-1}\sum_{(i,j)\in E} c_{ij}z_{ij}, \tag{5.32a}$$

$$\text{s.t.} \sum_{i\in V} x_{ii} = p, \tag{5.32b}$$

$$\sum_{j\in V} x_{ij} = 1, \qquad i \in V, \tag{5.32c}$$

$$x_{ij} \leq x_{jj}, \qquad (i,j) \in A : i \neq j, \tag{5.32d}$$

$$2z_{ij} \leq x_{ii} + x_{jj}, \qquad (i,j) \in E, \tag{5.32e}$$

$$\sum_{(i,j)\in E} z_{ij} = p-1, \tag{5.32f}$$

$$\sum_{(i,j)\in\delta^+(i)} f_{ij} - \sum_{(j,i)\in\delta^-(i)} f_{ji} = px_{ri} - x_{ii}, \qquad i \in V, \tag{5.32g}$$

$$f_{ij} \leq (p-1)z_{ij}, \qquad (i,j) \in E, \tag{5.32h}$$

$$f_{ji} \leq (p-1)z_{ij}, \qquad (i,j) \in E, \tag{5.32i}$$

$$\sum_{\ell \in V} x_{ij}^{\ell} = x_{ij}, \qquad (i,j) \in A, \tag{5.32j}$$

$$\sum_{(i,j) \in A} x_{ij}^{\ell} = 1, \qquad \ell \in V, \tag{5.32k}$$

$$\sum_{(i,j) \in A} c_{ij} x_{ij}^{\ell} \leq \sum_{(i,j) \in A} c_{ij} x_{ij}^{\ell+1}, \qquad \ell \in V : \ell < |V|, \tag{5.32l}$$

$$x_{ij} \in \{0,1\}, \qquad (i,j) \in A, \tag{5.32m}$$

$$x_{ij}^{\ell} \in \{0,1\}, \qquad (i,j) \in A, \ell \in V, \tag{5.32n}$$

$$z_{ij} \in \{0,1\}, \qquad (i,j) \in E, \tag{5.32o}$$

$$f_{ij} \geq 0, \qquad (i,j) \in A : i \neq j. \tag{5.32p}$$

(5.32a): Weighted minimization of the total compensated allocation cost plus the design cost of the tree of facilities (objective).
(5.32b): Exactly p facilities (allocation).
(5.32c): Each client is allocated to exactly one facility (allocation).
(5.32d): No client is allocated to a non-facility (allocation).
(5.32e): Edges of the tree of facilities can only be activated if both nodes are facilities (allocation).
(5.32f): Exactly $p-1$ edges connecting facilities (connectivity).
(5.32g): Flow is properly distributed along the tree of facilities (connectivity).
(5.32h)-(5.32i): All variables sending flow must be activated (connectivity).
(5.32j): Relationship between sorting and allocation variables (sorting).
(5.32k): Every sort position must be consider (sorting).
(5.32l): Correct sorting of the costs sequence (sorting).

$$F2_x^{flow} : \min \frac{1}{\sum_{\ell \in V} \lambda_\ell} \sum_{\ell \in V} \sum_{(i,j) \in A} \lambda_\ell c_{ij} x_{ij}^{\ell} + \frac{1}{p-1} \sum_{(i,j) \in E} c_{ij}(z_{ij} - x_{ij} - x_{ji}) \tag{5.33a}$$

$$\text{s.t.} \sum_{i \in V} x_{ii} = p, \tag{5.33b}$$

$$\sum_{j \in V} x_{ij} = 1, \qquad i \in V, \tag{5.33c}$$

$$2x_{ij} \leq 1 - x_{ii} + x_{jj}, \qquad (i,j) \in A : i \neq j, \tag{5.33d}$$

$$x_{ij} + x_{ji} \leq z_{ij}, \qquad (i,j) \in E, \tag{5.33e}$$

$$2z_{ij} \leq x_{ii} + x_{jj} + x_{ij} + x_{ji}, \qquad (i,j) \in E, \tag{5.33f}$$

$$\sum_{(i,j) \in E} z_{ij} = |V| - 1, \tag{5.33g}$$

$$\sum_{(r,j) \in \delta^+(r)} f_{rj} - \sum_{(j,r) \in \delta^-(r)} f_{jr} = |V| - 1, \tag{5.33h}$$

$$\sum_{(i,j) \in \delta^+(i)} f_{ij} - \sum_{(j,i) \in \delta^-(i)} f_{ji} = -1, \qquad i \in V \setminus r, \tag{5.33i}$$

$$f_{ij} \leq (p-1) z_{ij}, \qquad (i,j) \in E, \tag{5.33j}$$

$$f_{ji} \leq (p-1) z_{ij}, \qquad (i,j) \in E, \tag{5.33k}$$

$$\sum_{\ell \in V} x_{ij}^{\ell} = x_{ij}, \qquad (i,j) \in A, \tag{5.33l}$$

$$\sum_{(i,j) \in A} x_{ij}^{\ell} = 1, \qquad \ell \in V, \tag{5.33m}$$

$$\sum_{(i,j) \in A} c_{ij} x_{ij}^{\ell} \leq \sum_{(i,j) \in A} c_{ij} x_{ij}^{\ell+1}, \qquad \ell \in V : \ell < |V| \tag{5.33n}$$

$$x_{ij} \in \{0,1\}, \quad (i,j) \in A, \tag{5.33o}$$
$$x_{ij}^{\ell} \in \{0,1\}, \quad (i,j) \in A, \ell \in V, \tag{5.33p}$$
$$z_{ij} \in \{0,1\}, \quad (i,j) \in E, \tag{5.33q}$$
$$f_{ij} \geq 0, \quad (i,j) \in A : i \neq j. \tag{5.33r}$$

(5.33a): Weighted minimization of the total compensated allocation cost plus the design cost of the tree of facilities (objective).
(5.33b): Exactly p facilities (allocation).
(5.33c): Each client is allocated to exactly one facility (allocation).
(5.33d): Client-facility allocations are only possible if i is a client and j is a facility (allocation).
(5.33e): Only one allocation, whether client i to facility j or viceversa, can be considered if there exists an edge connecting $i, j \in V$ (allocation).
(5.33f): Only possibilities in which an edge can be selected is if both i or j are facilities or if either i, or j, is a client and j, or i respectively, its facility (allocation).
(5.33g): Exactly $p-1$ edges connecting facilities (connectivity).
(5.33h)-(5.33i): Flow is properly distributed along the tree of facilities (connectivity).
(5.33j)-(5.33k): All variables sending flow must be activated (connectivity).
(5.33l): Relationship between sorting and allocation variables (sorting).
(5.33m): Every sort position must be consider (sorting).
(5.33n): Correct sorting of the costs sequence (sorting).

Formulations $F1_u^{flow1}$, $F1_u^{flow2}$ and $F2_u^{flow}$ are left to the reader.

OMT KM formulation

$$F1_x^{km} : \min \frac{1}{\sum_{\ell \in V} \lambda_\ell} \sum_{\ell \in V} \sum_{(i,j) \in A} \lambda_\ell c_{ij} x_{ij}^{\ell} + \frac{1}{p-1} \sum_{(i,j) \in E} c_{ij} z_{ij} \tag{5.34a}$$
$$\text{s.t.} \sum_{i \in V} x_{ii} = p, \tag{5.34b}$$
$$\sum_{j \in V} x_{ij} = 1, \quad i \in V, \tag{5.34c}$$
$$x_{ij} \leq x_{jj}, \quad (i,j) \in A : i \neq j, \tag{5.34d}$$
$$2z_{ij} \leq x_{ii} + x_{jj}, \quad (i,j) \in E, \tag{5.34e}$$
$$\sum_{(i,j) \in E} z_{ij} = p - 1, \tag{5.34f}$$
$$q_{kij} + q_{kji} = z_{ij}, \quad k \in V, (i,j) \in E, \tag{5.34g}$$
$$\sum_{(k,j) \in \delta^+(k)} q_{kkj} \leq 0, \quad k \in V, \tag{5.34h}$$
$$\sum_{(i,j) \in \delta^+(i)} q_{kij} \leq 1, \quad k, i \in V : i \neq k, \tag{5.34i}$$
$$\sum_{\ell \in V} x_{ij}^{\ell} = x_{ij}, \quad (i,j) \in A, \tag{5.34j}$$
$$\sum_{(i,j) \in A} x_{ij}^{\ell} = 1, \quad \ell \in V, \tag{5.34k}$$
$$\sum_{(i,j) \in A} c_{ij} x_{ij}^{\ell} \leq \sum_{(i,j) \in A} c_{ij} x_{ij}^{\ell+1}, \quad \ell \in V : \ell < |V|, \tag{5.34l}$$
$$x_{ij} \in \{0,1\}, \quad (i,j) \in A, \tag{5.34m}$$
$$x_{ij}^{\ell} \in \{0,1\}, \quad (i,j) \in A, \ell \in V, \tag{5.34n}$$
$$z_{ij} \in \{0,1\}, \quad (i,j) \in E, \tag{5.34o}$$

$$q_{kij} \geq 0, \qquad (i,j) \in A, k \in V. \qquad (5.34p)$$

(5.34a): Weighted minimization of the compensated allocation cost of the system plus the design cost of the tree of facilities (objective).
(5.34b): Exactly p facilities (allocation).
(5.34c): Each client is allocated to exactly one facility (allocation).
(5.34d): No client is allocated to a non-facility (allocation).
(5.34e): Edges of the tree of facilities can only be selected if both nodes are facilities (allocation).
(5.34f): Exactly $p-1$ edges connecting facilities (connectivity).
(5.34g): If $(i,j) \in E$ is a tree edge, then there is only one variable related to the arcs underlying that can be selected (connectivity).
(5.34h): Forbids any arc leaving the root node k (connectivity).
(5.34i): Impose that no more that one arc leaves any node different from the root k (connectivity).
(5.34j): Relationship between sorting and allocation variables (sorting).
(5.34k): Every sort position must be consider (sorting).
(5.34l): Correct sorting of the costs sequence (sorting).

Formulations $F1_u^{km}$, $F2_x^{km}$ and $F2_u^{km}$ are left to the reader.

– *"C'est bien, jeune homme, c'est bien", continua Tréville, "je connais ces airs-là, je suis venu à Paris avec quatre écus dans ma poche, et je me serais battu avec quiconque m'aurait dit que je n'étais pas en état d'acheter le Louvre."*

Alexandre Dumas - *Les Trois Mousquetaires* (1844)

6

Application to Routing Problems: Fair Vehicle Routing

This chapter introduces a novel fairness-oriented approach to the Vehicle Routing Problem (VRP) based on ordering optimal route lengths. To avoid biased or inefficient route allocations, per-vehicle TSP-optimality is enforced, requiring each route to be the shortest possible for its assigned set of customers. This TSP-optimality constraint induces a combinatorial dependency between routing and assignment decisions, modeled via bilevel optimization. Computational experiments highlight the necessity of TSP-optimality, especially under non-monotonic objectives, and reveal how different fairness criteria shape route structures. Therefore, with the pretext of vehicle routing problems, this chapter elaborates on the contributions of the dissertation concerning the potential of ordered optimization in bringing fairness into optimization contexts.

6.1 Introduction

Traditional optimization models primarily emphasize *efficiency*, typically defined in terms of minimizing a global cost function. However, as noted by Savas (1978), focusing on *efficiency* may be insufficient when addressing, e.g., public administration decision-making, where all segments of the population must receive uniform or non-discriminatory treatment. It is therefore necessary to incorporate a second objective into the decision-making process: *fairness*. Rooted in Equity Theory (Adams, 1963), fairness draws attention to the perceived equity of outcomes. A solution that performs well at the system level but imposes disproportionate burdens on specific groups may be socially rejected or prove unsustainable. Despite its intuitive appeal, fairness remains an elusive and multifaceted concept, with no universally accepted definition. Consequently, the choice of a fairness metric can significantly influence the interpretation and quality of a given solution.

The literature has proposed a myriad of fairness approaches, each designed to capture a specific aspect of equity, such as proportionality, envy-freeness, or balance, but yet these metrics are often mutually incompatible and insufficient to simultaneously guarantee all desirable fairness properties. Furthermore, the notion of fairness entails intrinsic limitations that are, by nature, unavoidable. For example, one of the oldest and most accepted approaches to equity, introduced by Rawls (1971), consists in focusing on the worst-off entity. Some studies aim to maximize this worst outcome, while others impose a constraint to ensure it is above a predefined value. However, this concept has faced substantial criticism, as prioritizing only the least advantaged can lead to globally inefficient solutions, potentially resulting in significant aggregate losses for the remaining agents. Also, there exists an inherent trade-off between fairness and efficiency broadly discussed in the literature, commonly referred to as the *price of fairness*, see Mandell (1991) or Bertsimas et al. (2011). This notion quantifies the relative efficiency loss incurred when fairness considerations are imposed, highlighting that achieving a more balanced distribution often requires that some agents accept suboptimal outcomes to those they would receive under a purely efficiency-oriented allocation.

Mathematical optimization has traditionally served as the cornerstone of decision-making problems with its focus on maximizing efficiency. However, this perspective often neglects the critical dimension of distributive justice, wherein the allocation of costs and benefits among individual agents is a paramount. For this reason, a significant body of literature shifted the focus, advocating for the integration of fairness as a primary objective, compelling decision-makers to navigate the complex Pareto frontier between system efficiency and equitable distribution (see Hooker & Williams, 2012; Bertsimas et al., 2012, among others). One of the most extensively studied decision-making problems in the mathematical optimization literature serves as an ideal testbed for exploring this dichotomy: vehicle routing problems.

The *Vehicle Routing Problem* (VRP), first introduced by Dantzig & Ramser (1959), has long stood as a central topic in Operations Research and continues to grow in significance within the increasingly logistics-oriented landscape of contemporary courier companies, supply chains necessities, etc. (Archetti et al., 2025). Despite its simple formulation, the VRP is NP-hard, as it generalizes the TSP. Whereas the TSP seeks the shortest possible tour for a single vehicle that visits each customer exactly once and returns to the depot, the VRP extends this framework to multiple vehicles subject to capacity constraints, all serving the same set of customers. Current advances in VRP have been largely driven by mathematical programming techniques, most notably branch-and-price algorithms grounded in the set-partitioning formulation of Balinski & Quandt (1964). Among the leading exact solvers developed along these lines is the *VRPSolverEasy* (Pessoa et al., 2020; Errami et al., 2024). For comprehensive surveys covering both exact and heuristic methodologies, see Laporte (2009), Toth & Vigo (2014), or Archetti et al. (2025), among others.

While effective in optimizing global objectives like total travel distance or the demand attended, the resulting VRP solutions can exhibit severe imbalances in driver workload. These disparities can manifest in route duration, distance traveled, or service requirements, leading to decreased driver satisfaction, increased employee turnover, and long-term degradation of service quality, potentially fostering perceptions of unfairness, or even envy, among drivers. Consequently, embedding fairness within vehicle routing models is not merely an ethical consideration but a strategic imperative for achieving robust and sustainable logistics operations. To address this, alternative objective functions that are less sensitive to extreme values have been proposed. These include robust formulations and fairness measures aimed at promoting more equitable route distributions. For example, balance can be enforced by minimizing differences between route lengths or explicitly addressing *envy-freeness*. Envy-freeness, or no-envy, requires that no individual perceives another's allocation as more favorable. In the context of routing, this translates to assigning routes such that no driver would prefer another's route over their own. In practice, this concept is known to be partially captured by means of the well-known Gini coefficient (Gini, 1921), see, e.g., Blanco et al. (2024) for a successful application in the facility location context. Other common fairness metrics in the literature, include min-max, range (difference between maximum and minimum), the variance/standard deviation or the absolute deviations with respect to the mean (see Marsh & Schilling, 1994; Karsu & Morton, 2015; Lozano et al., 2016; Matl et al., 2018 or Chapter 3 for further details in this discussion).

A comprehensive and systematic study of fairness in routing, often termed as *workload equity or fairness*, was conducted by Matl et al. (2018). They provide a formal taxonomy, deconstructing fairness concepts into (1) the *fairness metrics*, which defines the aspect of workload to be equalized (e.g., route length, service time, or demand served); (2) the *fairness function or measure*, which quantifies inequality across agents (e.g., range, variance, maximum); and (3) the *fairness objective*, which results from the combination of a metric and a measure and is the actual function subject to optimization. A critical property they analyze is the *monotonicity* of the fairness objective. An objective is said to be monotonic if a unilateral increase in a single driver's workload cannot improve the overall equity value. Monotonicity is desirable, as it ensures that worsening an individual's condition does not falsely appear beneficial in terms of equity. Examples of monotonic measures include the min-max and lexicographic min-max criteria. However, many widely adopted and intuitive measures of fairness, such as the range, variance, and Gini coefficient, are unfortunately non-monotonic.

Much of the fairness-oriented literature in vehicle routing focuses on monotonic criteria, predominantly the min-max objective, which has been addressed with exact methods by authors such as Campbell et al. (2008), Huang et al. (2011), and Bertazzi et al. (2015). The trade-off between efficiency and equity has also been explicitly modeled as a bi-objective problem, notably in the *Balanced* VRP, first studied by Jozefowiez et al. (2002) and later advanced with exact methods by Reiter & Gutjahr (2012), Halvorsen-Weare & Savelsbergh (2016), and Artigues et al. (2018). Simultaneously, the challenge of enforcing TSP-optimal routing in multi-depot settings has been studied in the *multiple Traveling Salesman Problem* (mTSP), introduced by Svestka & Huckfeldt (1973). Fairness studies within the mTSP framework have also largely relied on monotonic criteria like the min-max objective (Applegate et al., 2002) or, more recently, ℓ_p norms (Bektaş & Letchford, 2020). Research on non-monotonic fairness objectives is comparatively scarce, with some contributions including an enumeration-based analysis by Matl et al. (2018) and post-hoc evaluations in both heuristic (Lozano et al., 2016) and exact (López-Sánchez et al., 2024) contexts. However, to the best of the knowledge, no existing work has proposed an optimization framework that guarantees globally optimal solutions for a VRP while simultaneously enforcing TSP-optimal routing under a non-monotonic fairness objective function.

Non-monotonicity induces a counterintuitive modeling dilemma, hereinafter termed as *route inconsistency*: an optimization model under a non-monotonic measure may favor a solution where a driver's route is made arbitrarily inefficient, i.e., not a TSP-optimal tour for the assigned customers, simply because the resulting increase in workload brings it closer to other routes, spuriously reducing the overall measure of dispersion. Clearly, this is not a desirable property, as non-TSP-optimal tours contradict the minimization of individual workloads. More generally, non-monotonic objectives can be improved even in the extreme case when all workloads are strictly worsened. This does not happen if the objective is monotonic, for which all the solutions are composed only by TSP-optimal tours. To further investigate these pathological behaviors, Matl et al. (2018) introduce the concept of *workload inconsistency*. This phenomenon occurs when a solution in which the individual fairness metrics are strictly worse, therefore incurring a higher total cost, yields a more favorable total value of the equity measure, i.e., a lower level of system inequality. Workload inconsistency exemplifies the paradoxical issue of the price of fairness, whereby improvements in equity require sacrifices in individual efficiency.

The discussion of these three concepts, monotonicity, route inconsistency, and workload inconsistency, leads to a central conclusion. When a fairness measure is monotonic, neither route nor workload inconsistency arises. Conversely, with non-monotonic measures, the problem of workload inconsistency persists even when route consistency is enforced (i.e., when all individual tours are TSP-optimal). This does not imply, however, that monotonic measures are without their own drawbacks. For instance, the common min-max objective, based on the rawlsian principle of equity, has its own well-documented issues (some of them already outlined at the beginning) and fails to capture as much information about the overall distribution of inequality as non-monotonic measures do. This creates a difficult trade-off, highlighting that it appears to be no universally ideal definition of fairness. Despite this dilemma, enforcing the optimality of individual routes is a crucial step towards efficient fairness. It is true that enforcing TSP-optimality eliminates route inconsistency but does not solve the workload inconsistency problem; a globally equitable solution may still require drivers to collectively travel further, meaning a price for fairness must often be paid. However, failing to ensure individual route optimality introduces a gratuitous form of inefficiency, as correcting a suboptimal tour would reduce a driver's effort without altering their customer assignments. As noted by Matl et al. (2018), imposing this optimality is a step toward improving workload consistency and, therefore, should be considered a fundamental requirement when addressing fairness in vehicle routing.

To address the multifaceted challenges associated with fairness in vehicle routing, this chapter introduces several novel contributions to the state-of-the-art. First, a general and flexible modeling framework based on the principles of ordered optimization is proposed. By ordering route costs, one can directly identify the most and least burdened agents, allowing the adoption of fairness measures that directly account for inequality. Furthermore, as commented along this dissertation, the ordering structure naturally encompasses a wide range of objectives in the literature, including many that have been specifically designed to capture fairness considerations. This paradigm is particularly well-suited for equity modeling, as it provides a unifying structure that accommodates both linear and non-linear fairness metrics, such as the variance. Within this framework, the use of fairness criteria based on minimizing absolute deviations from statistical anchors, such as the minimum or the median workload, are also studied.

Beyond this, the main technical contribution is a novel methodology to address the route inconsistency problem that arises when using non-monotonic fairness objectives. To the best of our knowledge, this is the first approach that enforces TSP-optimal routing for each individual driver in a provably optimal manner within a fair VRP context, accommodating both monotonic and non-monotonic objective functions. This is achieved through a sophisticated bilevel optimization model. Following the rationale in Cerulli et al. (2024), the upper-level problem sets the strategic partitioning of customers among drivers, imposing capacity and optimizing a given fairness objective, while the lower-level problem solves a collection of independent TSP problems to determine the exact cost of each route. This bilevel architecture allows for a clear conceptual and computational separation between the strategic design of fairness, i.e., the definition and selection of equitable criteria, and the tactical realization of routing efficiency. Consequently, the inherent trade-offs between fairness and efficiency are made explicit within the model, and the resulting operational plans remain unaffected by distortions commonly introduced by non-monotonic fairness objectives.

The remainder of the chapter is structured as follows. In Section 6.2, the general framework is introduced, based on the ordering of route lengths, to incorporate a wide range of fairness measures into vehicle routing, allowing the modeling of diverse loss functions and the analysis of their structural properties. Section 6.3 presents the bilevel optimization model that integrates the ordered objective framework while enforcing TSP-optimality on each driver's route, providing two formulations: one based on sorting the route lengths and another leveraging the k-sums operator. In Section 6.4, the bilevel model is reformulated into a single-level problem through the value function reformulation, highlighting the similarity of the lower-level problem to a profitable TSP. Section 6.5 explores alternative and more compact models in which routing variables are projected out, and additional bounding techniques are proposed to improve computational performance. Finally, Section 6.6 presents an empirical comparison of the different modeling and solution strategies developed throughout the paper.

Statistic	Expression	λ
Mean	$\frac{1}{n}\sum_{i=1}^{n} x_i$	$\frac{1}{n}(1,...,1)$
Maximum	$\max x_i$	$(0,...,0,1)$
Median	$\text{median}(x) = \begin{cases} x_{(m+1)}, & n=2m+1 \\ \frac{x_{(m)}+x_{(m+1)}}{2}, & n=2m \end{cases}$	$(0,...,0,\underbrace{1}_{(m+1)-th},0,...,0)$ $(0,...,0,\underbrace{\frac{1}{2}}_{m-th},\underbrace{\frac{1}{2}}_{(m+1)-th},0,...,0)$
q-th quantile (or τ-quantile)	$x_{(q)}$, $q = \lfloor \tau n \rfloor$	$(0,...,0,\underbrace{1}_{q-th},0,...,0)$
Range	$\max x_i - \min x_i$	$(-1,0,...,0,1)$
Gini differences or mean absolute differences	$\frac{1}{n^2}\sum_{i,j=1}^{n} \lvert x_i - x_j\rvert$	$\frac{2}{n^2}((2i-n-1))_{i\in N}$
Mean absolute deviation from the q-th quantile ($\mathsf{MeanAD}_{(q)}$)	$\frac{1}{n}\sum_{i=1}^{n} \lvert x_i - x_{(q)}\rvert$	$\frac{1}{n}(-1,...,-1,\underbrace{2q-n-1}_{q-th},1,...,1)$

Table 6.1: Different objective functions and associated λ-vectors fairness characteristics..

6.2 Ordered framework for vehicle routing

Formally, the VRP is defined on a complete directed graph $G = (V, A)$, where $V = \{0, 1, \ldots, n\}$ is a set of nodes and $A = \{(i, j) \in V \times V, i \neq j\}$ is a set of arcs. The node 0 represents the depot for a fleet of vehicles given by the set K with $|K| = m$. Each vehicle has identical capacity Q. The set $V' = V \setminus \{0\}$ represents the set of customers that need to be visited. Each customer holds a demand $q_i, i \in V'$. It is assumed that node 0 have no demand, i.e., $q_0 = 0$. Every arc arc $(i, j) \in A$ is associated with the positive travel cost c_{ij}. It is also assumed that the cost matrix is symmetric, i.e., $c_{ij} = c_{ji}$, $i, j \in V, i \neq j$, and it satisfies the triangular inequality, $c_{ij} + c_{jl} \geq c_{il}$, $i, j, l \in V$. The aim is to determine a minimum-cost set of m routes in order to satisfy the demands of all customers under the following constraints: each route starts and ends at the depot, each customer is visited exactly once, and the total demand of all customers on any route must not exceed the vehicle capacity.

This chapter introduces an ordered optimization framework to incorporate fairness into routing problems. To this end, let index k denote the k-th shortest route in a feasible solution, with $k \in K$, and $\Phi_k \geq 0$ a non-negative variable representing the length of the k-th shortest route. By arranging these route lengths in non-decreasing order and associating them with a weight vector $\lambda = (\lambda_\ell)_{\ell=1}^m$, where $\lambda_\ell \in \mathbb{R}$, a general modeling framework is established by means of ordered optimization. This framework allows the incorporation of fairness criteria by modeling objective functions through different choices of the weight vector λ. Table 6.1 illustrates several objective functions admitting a λ-vector representation and commonly associated with fairness criteria, equality or equity. Further discussion is provided in Chapter 3.

From a high-level perspective, the problem can be modeled as follows:

$$\min \sum_{k\in K} \lambda_k \Phi_k, \tag{6.1a}$$

$$\text{s.t.: } \Phi_k \leq \Phi_{k+1}, \qquad k \in K : k < m, \tag{6.1b}$$

$$\Phi_k \text{ is the length of route } k, \qquad k \in K, \tag{6.1c}$$

where in the objective function (6.1a) the variables ordered by constraints (6.1b) are weighted, and constraints (6.1c) ensure that Φ_k represents the length of the k-th route. Note that routing constraints that may depend on the problem under study (such as the number of visits of each customer, capacity with respect to demand, capacity with respect to the length of the routes, etc.) are omitted at this stage. This means that, although the intended application is within the VRP, the framework is sufficiently general to accommodate a wide range of vehicle routing problems.

In what follows, alternative models based on the sum of the largest-cost operator are introduced. Although the underlying rationale is the same as in previous chapters, the implementation is developed from scratch. Let Ψ_k denote the length of the route assigned to driver k (prior to sorting), and define the following operator as the sum of the largest $m - \ell + 1$ routes, namely:

$$S_\ell(\Psi) = \sum_{i=\ell}^{m} \Psi_{(i)}, \tag{6.2}$$

where $(\Psi_{(1)}, ..., \Psi_{(m)})$ is the sorted sequence of the route lengths in non-decreasing order, this is, $\Psi_{(1)} \leq ... \leq \Psi_{(m)}$. Using this operator, the expression of the objective function (6.1a) can be rewritten as:

$$\sum_{\ell=1}^{m} \lambda_\ell \Psi_{(\ell)} = \sum_{\ell=1}^{m} \Delta_\ell S_\ell(\Psi), \tag{6.3}$$

where $\lambda_0 := 0$ and $\Delta_\ell := \lambda_\ell - \lambda_{\ell-1}$, for all $\ell \in K$. Note that each route admits a distinct ordering, leading to as many possible orders as routes. For later use, define the sets $\Delta^- = \{\ell \in K \mid \Delta_\ell < 0\}$ and $\Delta^+ = \{\ell \in K \mid \Delta_\ell > 0\}$. As already discussed throughout this dissertation, computing (6.2) is equivalent to solving the following linear problem or its dual:

$$\begin{aligned} \max \ & \sum_{k\in K} \theta_k \Psi_k \\ \text{s.t.} \ & \sum_{k\in K} \theta_k = n - \ell + 1, \\ & 0 \leq \theta_k \leq 1, \ k \in K. \end{aligned} \tag{6.4a}$$

$$\begin{aligned} \min \ & (m - \ell + 1)t + \sum_{k\in K} v_k \\ \text{s.t.} \ & t + v_k \geq \Psi_k, \quad k \in K, \\ & t \in \mathbb{R}, \ v_k \geq 0, \ k \in K. \end{aligned} \tag{6.4b}$$

Observe that the variables θ_k correspond to the decision variables of a knapsack problem and, therefore, can be constrained to be binary, i.e., $\theta_k \in \{0, 1\}$, $k \in K$. Moreover, since $\Psi_k \in \mathbb{R}_+$ for all $k \in K$, the variable t can be assumed to be non-negative, i.e., $t \geq 0$.

Given the expression in (6.2) and the primal-dual relationships inherent to the operator, these elements can be integrated to formulate the following model.

$$\begin{aligned} \min \ & \sum_{\ell\in\Delta^+} \Delta_\ell((m - \ell + 1)t_\ell + \sum_{k\in K} v_{k\ell}) + \sum_{\ell\in\Delta^-} \Delta_\ell \sum_{k\in K} \theta_{k\ell}\Psi_k & & \quad (6.5a) \\ \text{s.t.} \ & t_\ell + v_{k\ell} \geq \Psi_k & k \in K, \ell \in \Delta^+, & \quad (6.5b) \\ & \sum_{k\in K} \theta_{k\ell} = m - \ell + 1 & \ell \in \Delta^-, & \quad (6.5c) \\ & t_\ell \in \mathbb{R}, v_{k\ell} \geq 0, & k \in K, \ell \in \Delta^+, & \quad (6.5d) \\ & \theta_{k\ell} \in \{0, 1\}, & k \in K, \ell \in \Delta^-, & \quad (6.5e) \\ & \Psi_k \text{ is the length of route } k, & k \in K. & \quad (6.5f) \end{aligned}$$

Note the objective function (6.5a) is non-linear because it includes the product of binary variables θ and non-negative Ψ variables. These products can be easily linearized via McCormick envelopes by defining a non-negative continuous variables $\eta_{k\ell} \geq 0$, $k \in K, \ell \in \Delta^-$, representing such products, and adding the constraints:

$$\eta_{k\ell} \leq \overline{\Psi}\theta_{k\ell} \qquad k \in K, \ell \in \Delta^-, \tag{6.6a}$$

$$\eta_{k\ell} \leq \Psi_k \qquad k \in K, \ell \in \Delta^-, \tag{6.6b}$$

$$\eta_{k\ell} \geq \Psi_k - (1 - \theta_{k\ell})\overline{\Psi} \qquad k \in K, \ell \in \Delta^-. \tag{6.6c}$$

It is important to observe that constraints (6.6c) are redundant due to the minimization structure of the problem, so they can be removed. Moreover, these constraints involve a big-M constant, $\overline{\Psi}$ which is an upper bound on the route length. This and other bounds for the route length variables will be discussed in Section 6.5.

Alternatively, the framework defined in (6.1) can be extended to incorporate non-linear fairness measures, for instance by minimizing overall deviations with respect to a reference ordered value, such as the mean or a selected quantile. To this end, the objective function (6.1a) can be replaced with the loss function:

$$\min \mathcal{L}(\Phi, l_{\text{ord}}(\Phi, \lambda)). \tag{6.7}$$

where $l_{\text{ord}}(\Phi, \lambda) = \sum_{k \in K} \lambda_k \Phi_k$ is the nested ordered value as defined in Section 3.3 of Chapter 3. For example, one can minimize the *least absolute deviation* with respect to the reference ordered value expressed as:

$$\mathcal{L}(\Phi, l_{\text{ord}}(\Phi, \lambda)) = \sum_{k \in K} |\Phi_k - l_{\text{ord}}(\Phi, \lambda)|. \tag{6.8}$$

Similarly, to minimize *the least squared deviation*, one obtains:

$$\mathcal{L}(\Phi, l_{\text{ord}}(\Phi, \lambda)) = \sum_{k \in K} (\Phi_k - l_{\text{ord}}(\Phi, \lambda))^2. \tag{6.9}$$

Moreover, if

$$\mathcal{L}(\Phi, l_{\text{ord}}(\Phi, \lambda)) = l_{\text{ord}}(\Phi, \lambda), \tag{6.10}$$

the formulation reduces to minimizing the ordered measure defined directly by the λ-vector.

For the sake of completeness, and since it is crucial in this research, the concept of monotonicity is formally reviewed in the next definition.

Definition 6.2.1 Let $\Phi, \Phi' \in \mathbb{R}_+^m$ be two vectors of route lengths for the m drivers, and $I(\Phi)$ denote the value of the inequality measure. $I(\cdot)$ is monotone if, provided that $\Phi'_k = \Phi_k + \delta_k$ for at least one $k \in K$. If $\delta_k \geq 0$ for all k with at least one strict inequality, then $I(\Phi') \geq I(\Phi)$.

Following definition 6.2.1, the mean, median (or any other quantile) or the maximum are monotonic measures. As stated in Matl et al. (2018), if monotonicity is ensured then the solution is composed of TSP-optimal tours. However, many interesting fairness objectives, such as the range or the variance involve non-monotonic measures, for which TSP-optimality cannot be ensured, and hence the model cannot ensure that every route is of minimum length, unless these constraints are explicitly imposed in the model. Furthermore, our framework allows to use negative λ entries, which is the case for the majority of these measures.

To illustrate the implications of imposing TSP-optimality in fair vehicle routing, consider a simple VRP instance with seven nodes, six customers, and a central depot, whose coordinates and demands are detailed in Table 6.2. Two identical vehicles are available, each with a capacity of five demand units. Table 6.3 reports the outcomes under a set of objective functions, including monotonic measures (mean, median, and maximum route length) and non-monotonic ones (range, Gini index, median absolute deviation, denoted as $\mathsf{MeanAD}_{(q)}$, and variance). Solutions are compared for each objective function under conditions where TSP-optimality is enforced for each route versus when it is not.

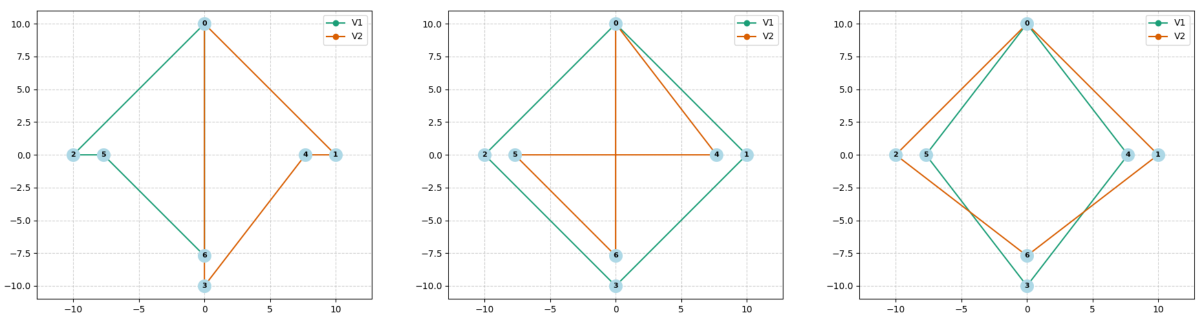
Three scenarios emerge from the analysis. In the case of monotonic objectives, the optimal routing remains unchanged regardless of whether TSP-optimality is enforced (Tour 1: 45.03, Tour 2: 49.06, Total cost: 94.09). This solution is illustrated in Figure 6.1 (a). In contrast, for non-monotonic objectives, solutions differ substantially depending on the enforcement of TSP-optimality. When this constraint is relaxed, the solution is more flexible to minimize inequality, producing nearly equal route lengths (Tour 1: 56.57, Tour 2: 56.61, Total cost: 113.18). As depicted in Figure 6.1 (b), this results in a route inconsistency for driver two, since its tour is not TSP-optimal. Conversely, when TSP-optimality is enforced, the resulting solution cannot fully equalize route lengths (Tour 1: 50.48, Tour 2: 53.52, Total cost: 104.00), but route inconsistency is eliminated as depicted in Figure 6.1 (c). This example illustrates two key trade-offs in equitable vehicle routing. First, while enforcing TSP-optimality solves the issue of route inconsistency, it does not guarantee workload consistency, since the total cost increases from the efficient baseline (94.09) to the fair solution (104.00). Second, the example shows that omitting TSP-optimality can paradoxically lead to even higher total costs (113.18), suggesting that incorporating this constraint may improve overall efficiency in certain fair-routing contexts.

Nodes	0	1	2	3	4	5	6
Coordinates	(0,10)	(10,0)	(-10,0)	(0,-10)	(7.7,0)	(-7.7,0)	(0,-7.7)
Demands	0	1	1	3	1	1	3

Table 6.2: Example parameters

		no TSP-optimal				TSP-optimal			
Measure	Monotonicity	Obj	Tour 1	Tour 2	Total	Obj	Tour 1	Tour 2	Total
Mean	yes	47.04	45.03	49.06	94.09	47.04	45.03	49.06	94.09
Median	yes	47.04	45.03	49.06	94.09	47.04	45.03	49.06	94.09
Max	yes	49.06	45.03	49.06	94.09	49.06	45.03	49.06	94.09
Range	no	0.04	56.57	56.61	113.18	3.04	50.48	53.52	104.00
Gini	no	0.02	56.57	56.61	113.18	1.52	50.48	53.52	104.00
$\mathsf{MeanAD}_{(q)}$	no	0.02	56.57	56.61	113.18	1.52	50.48	53.52	104.00
Variance	no	$4e^{-4}$	56.57	56.61	113.18	2.31	50.48	53.52	104.00

Table 6.3: Example results for different scenarios



(a) Monotonic measures (b) Non-monotonic measures (no TSP-optimal) (c) Non-monotonic measures (TSP-optimal)

Figure 6.1: Comparison of three routing solutions

This raises the question of bilevel optimization models, where, in the presence of non-monotonic measures, even more if negative λ values are present, it must be ensured that each Φ_k represents the shortest route (i.e., the optimal TSP solution) for the set of visited customers. Regarding the absolute and squared loss functions, (6.8) and (6.9) respectively, there also exist cases where one needs to impose bilevel cuts in order to have optimal routes. This can be seen in Remark 6.2.1 and 6.2.2, respectively.

Remark 6.2.1 Bilevel cuts are necessary in some scenarios to impose optimality of the TSP routes when the absolute deviation loss function (6.8) is minimized.

In order to check this assertion, let us consider the *mean absolute deviation from the q-th quantile*, defined as

$$\sum_{k\in K} |\Phi_k - \Phi_{(q)}|, \tag{6.11}$$

where $\Phi_{(q)}$ is the q-th value in the sorted vector Φ. This metric follows the same logic as the *mean absolute deviation* but substitutes the mean with a more robust central location, the q-th quantile. This measure admits a convenient λ-vector representation (see Harmati et al., 2024; Blanco et al., 2025) as follows:

$$\frac{1}{n}(-1, ..., -1, \underbrace{2q-n-1}_{q\text{-th}}, 1, ..., 1), \tag{6.12}$$

which have negative entries of the λ-vector, case in which we know bilevel cuts are needed to impose optimal routes.

Remark 6.2.2 Bilevel cuts are necessary in some scenarios to impose optimality of the TSP routes when the squared deviation loss function (6.9) is minimized.

Indeed, let us consider the squared deviation with respect to a general ordered measure $l_{\text{ord}}(\Phi, \lambda)$, given by (6.9). As shown in Section 3.4 in Chapter 3, this expression can be reformulated as a quadratic ordered measure:

$$\mathtt{Q}(\Phi, \mathbf{M}) = \sum_{i,j\in K} m_{ij}\Phi_{(i)}\Phi_{(j)}, \tag{6.13}$$

where $\mathbf{M} = (m_{ij})$ is a symmetric matrix determined by the weights λ. For instance, in the case of the variance objective (i.e., squared deviations from the mean),

$$\frac{1}{m}\sum_{i\in K}(\Phi_i - \frac{1}{m}\sum_{j\in K}\Phi_j)^2, \tag{6.14}$$

the matrix $\mathbf{M}$ takes the form:

$$\mathbf{M} = \frac{1}{m}\cdot\begin{pmatrix} (1-\frac{1}{m}) & -\frac{1}{m} & \cdots & -\frac{1}{m} \\ -\frac{1}{m} & (1-\frac{1}{m}) & \ddots & \vdots \\ \vdots & \ddots & \ddots & -\frac{1}{m} \\ -\frac{1}{m} & \cdots & -\frac{1}{m} & (1-\frac{1}{m}) \end{pmatrix}. \tag{6.15}$$

This matrix corresponds to the classical sample variance estimator and is known to define a non-monotonic quadratic form. This can easily be checked by taking two vectors Φ' and Φ'' such that $\Phi' \leq \Phi''$, e.g., $\Phi' = (1, 3, 5)$ and $\Phi'' = (2.5, 3, 5)$, for which $\mathtt{Q}(\Phi', \mathbf{M}) \geq \mathtt{Q}(\Phi'', \mathbf{M})$. That is, increasing some components of the ordered vector Φ does not necessarily increase the value of $\mathtt{Q}(\Phi, \mathbf{M})$.

Due to this non-monotonicity, the optimization process might favor artificially constructed route lengths Φ_k that are not true shortest routes, if that allows to reduce the total value of the quadratic loss. For example, increasing the length of already long routes and decreasing others might result in a lower objective. Hence, without explicit bilevel constraints ensuring that each Φ_k corresponds to an optimal TSP tour, the solution may contain suboptimal routes with respect to the underlying routing problem, what confirms our Remark 6.2.1.

6.3 Bilevel optimization model

This section introduces the bilevel approach to the VRP under the ordered framework. A single leader-multiple followers scheme is adopted, where the leader determines the allocation of customers to each of the k vehicles (followers), effectively dictating which vehicle visits which customers. That is, the leader sets a partition $\{V_1, ..., V_m\}$ of the set of customers into m groups. To model this, consider the upper-level binary decision variables $x_i^k \in \{0,1\}$, where x_i^k equal to 1 indicates that customer $i \in V$ is assigned to vehicle k, and 0 otherwise. Since each customer has an associated demand, additional constraints must be introduced to ensure feasibility with respect to vehicle capacities. Given the leader decisions in the upper-level, the lower-level enforces that the TSP routes are optimal. For this purpose, each one solves the TSP given the set of customers assigned to the driver from the upper-level.

Therefore, a bilevel model to solve this problem can be stated as follows:

$$
\begin{aligned}
& \min \mathcal{L}(\Phi, \nu), && && (6.16a)\\
& \text{s.t. } \nu = \sum_{k\in K} \lambda_k \Phi_k, && && (6.16b)\\
& \quad x_0^k = 1, && k \in K, && (6.16c)\\
& \quad \sum_{k\in K} x_i^k = 1, && i \in V', && (6.16d)\\
& \quad \sum_{i\in V} d_i x_i^k \le Q, && k \in K, && (6.16e)\\
& \quad \Phi_k \le \Phi_{k+1}, && k \in K : k < m, && (6.16f)\\
& \quad x_i^k \in \{0,1\}, && i \in V, k \in K, && (6.16g)\\
& \quad \text{where } \Phi_k = \phi_k(x^k) = \min \sum_{i\in V} \sum_{j\in V, j\neq i} c_{ij} z_{ij}^k, && && (6.16h)\\
& \qquad \sum_{i\in V} z_{ij}^k = x_j^k, && j \in V, && (6.16i)\\
& \qquad \sum_{j\in V} z_{ij}^k = x_i^k, && i \in V, && (6.16j)\\
& \qquad \texttt{Subtour\_elimination}(z^k), && && (6.16k)\\
& \qquad z_{ij}^k \in \{0,1\}. && && (6.16l)
\end{aligned}
$$

The objective function (6.16a) is designed to minimize a loss function $\mathcal{L}(\Phi, \nu)$, where the scalar quantity $\nu = \sum_{k\in K} \lambda_k \Phi_k$ represents a linearly weighted and ordered aggregation of the individual route lengths Φ_k. This formulation allows for considerable flexibility in modeling various fairness or efficiency objectives. In the particular case where the loss function $\mathcal{L}$ is the identity mapping, the objective reduces to the direct minimization of ν, and constraint (6.16b) becomes redundant.

The constraints defined in (6.16c) through (6.16g) correspond to the upper-level of the problem formulation. Specifically, constraints (6.16c) ensure that each driver visits the depot, while constraints (6.16d) enforce that every customer is served by exactly one driver. Capacity restrictions are imposed through constraints (6.16e), ensuring that the total demand on each route does not exceed vehicle capacity. Furthermore, constraint (6.16f) establishes a consistent ordering of the route lengths for the purpose of computing the ordered objective, and constraint (6.16g) imposes the integrality conditions on the upper-level assignment variables.

Constraints (6.16h) through (6.16l) constitute the lower-level structure of the model, which ensures the optimality of the route length Φ_k for each individual driver $k \in K$. This level introduces binary variables z_{ij}^k, which take the value 1 if arc $(i, j) \in A$ is included in the route of vehicle k, and 0 otherwise. The lower-level objective (6.16h) minimizes the total travel length associated with driver k, subject to several routing constraints. Constraints (6.16i) and (6.16j) enforce that the driver visits precisely the customers assigned to them by the upper-level. To prevent the formation of subtours, constraint (6.16k) imposes connectivity requirements, while constraint (6.16l) ensures integrality of the routing variables. The subtour elimination can be imposed by means of the MTZ constraints or similar compact formulations. Hereafter, generalized subtour elimination constraints are used, which are in (6.16k) given as follows:

$$\sum_{i \in S} \sum_{j \in S: j \neq i} z_{ij}^k \leq \sum_{i \in S \setminus \{h\}} x_i^k, \quad \forall S \subseteq V, |S| \geq 2, \forall h \in S, \tag{6.17}$$

which are restricted to the subset of nodes for which $x_i^k = 1$, and they remove subtours on the subgraph induced by these nodes.

If all components of λ are non-negative, there is no need to state the problem using bilevel optimization, i.e., one could simply solve the problem by inserting the constraints (6.16i)-(6.16l) into the upper-level and adding the connection between Φ_k and z variables:

$$\Phi_k = \sum_{i \in V} \sum_{j \in V, j \neq i} c_{ij} z_{ij}^k. \tag{6.18}$$

However, most of the interesting fairness and equity criteria, such as the range or Gini differences, involve negative λ components, which is why we will focus on solving these bilevel formulations.

Also, similarly, a second approach based on the ordering representation (6.5) can be introduced as follows:

$$\begin{aligned}
& \min \mathcal{L}(\Psi, \nu) && && (6.19a)\\
& \text{s.t. } \nu = \sum_{\ell \in \Delta^+} \Delta_\ell \big((m - \ell + 1)t_\ell + \sum_{k \in K} v_{k\ell}\big) + \sum_{\ell \in \Delta^-} \Delta_\ell \sum_{k \in K} \theta_{k\ell} \Psi_k && && (6.19b)\\
& \quad t_\ell + v_{k\ell} \geq \Psi_k && k \in K, \ell \in \Delta^+, && (6.19c)\\
& \quad \sum_{k \in K} \theta_{k\ell} = m - \ell + 1 && \ell \in \Delta^-, && (6.19d)\\
& \quad x_0^k = 1 && k \in K, && (6.19e)\\
& \quad \sum_{k \in K} x_i^k = 1 && i \in V', && (6.19f)\\
& \quad \sum_{i \in V} d_i x_i^k \leq Q && k \in K, && (6.19g)\\
& \quad \Psi_k = \phi_k(x^k) && i \in V, k \in K, && (6.19h)\\
& \quad x_i^k \in \{0, 1\}, && i \in V, k \in K, && (6.19i)\\
& \quad t_\ell \in \mathbb{R}, v_{k\ell} \geq 0, && k \in K, \ell \in \Delta^+, && (6.19j)\\
& \quad \theta_{k\ell} \in \{0, 1\}, && k \in K, \ell \in \Delta^-. && (6.19k)
\end{aligned}$$

This is an alternative bilevel model, where constraints (6.19h) enforce that Ψ_k is the length of the optimal TSP tour for the set of customers given by the vector x^k. Variable ν is equal to $l_{\text{ord}}(\Psi, \lambda)$ but considering the ℓ-sum representation in (6.19b).

6.4 Single-level reformulation

In what follows, it is discussed how to reformulate the bilevel problems into a single-level reformulation. In problem (6.16), the lower-level (6.16h)-(6.16l) is not convex. Therefore, one cannot solve this bilevel problem by reformulating it using the KKT conditions nor by exploiting the strong-duality, as commented in Section 2.4 of Chapter 2. Instead, here the value function reformulation is given as follows:

$$
\begin{aligned}
\min\ & \mathcal{L}(\Phi, \nu), && && \text{(6.20a)}\\
\text{s.t. } & \nu = \sum_{k \in K} \lambda_k \Phi_k && && \text{(6.20b)}\\
& x_0^k = 1 && k \in K, && \text{(6.20c)}\\
& \sum_{k \in K} x_i^k = 1 && i \in V', && \text{(6.20d)}\\
& \sum_{i \in V} d_i x_i^k \leq Q && k \in K, && \text{(6.20e)}\\
& \Phi_k \leq \Phi_{k+1} && k \in K : k < m, && \text{(6.20f)}\\
& \sum_{i \in V} z_{ij}^k = x_j^k && k \in K, j \in V, && \text{(6.20g)}\\
& \sum_{j \in V} z_{ij}^k = x_i^k && k \in K, i \in V, && \text{(6.20h)}\\
& \texttt{Subtour\_elimination}(z^k) && k \in K, && \text{(6.20i)}\\
& \sum_{i \in V} \sum_{j=0, j \neq i}^{n} c_{ij} z_{ij}^k = \Phi_k && k \in K, && \text{(6.20j)}\\
& \Phi_k \leq \phi_k(x^k) && k \in K, && \text{(6.20k)}\\
& x_i^k \in \{0, 1\} && i \in V, k \in K, && \text{(6.20l)}\\
& z_{ij}^k \in \{0, 1\} && i, j \in V : i \neq j, k \in K. && \text{(6.20m)}
\end{aligned}
$$

Constraints (6.20k) are non-convex, hence an attempt is made to convexify them. Note that the feasible set of the lower-level in problem (6.16) depends on the upper-level variables x^k. In lieu of solving the lower-level problem (6.16h)-(6.16l), one can solve an alternative problem whose feasible set does not depend on the upper-level variables. Instead, the problem contains additional penalty terms in the objective function, so that it is guaranteed that the same objective value is obtained.

The lower-level problem can be formulated as a Profitable TSP (PTP), see Feillet et al. (2005), in the same graph G, with node prizes $\tilde{\pi}_i$ and arc lengths $\tilde{c}_{ij}$ defined as follows:

$$
\tilde{\pi}_i = M \bar{x}_i^k \qquad \text{and} \qquad \tilde{c}_{ij} = \bar{x}_i^k c_{ij} + M(1 - \bar{x}_i^k) \tag{6.21}
$$

for sufficiently large values of M.

This is proven in the following proposition, where the penalization terms M^1 and M^2 are also specified.

Proposition 6.4.1 Given the upper-level decision variables fixed, $\overline{x}_i^k \in \{0,1\}$, $k \in K, i \in V$, define the binary decision variables $y_i^k \in \{0,1\}$, which are equal to 1 if the vehicle $k \in K$ decides to visit customer $i \in V'$. Then, for every $k \in K$, instead of solving problem (6.16g)-(6.16j) one can solve the following equivalent problem:

$$\phi_k(\bar{x}^k) = \min \sum_{i\in V} \overline{x}_i^k \sum_{j\in V, j\neq i} c_{ij} z_{ij}^k + \sum_{i\in V} M_i^1 \overline{x}_i^k (1-y_i^k) + \sum_{i\in V} M_i^2 (1-\overline{x}_i^k) y_i^k, \tag{6.22a}$$

$$\text{s.t.} \sum_{i\in V} z_{ij}^k = y_j^k, \quad j \in V, \tag{6.22b}$$

$$\sum_{j\in V} z_{ij}^k = y_i^k, \quad i \in V, \tag{6.22c}$$

$$\texttt{Subtour\_elimination}(z^k), \tag{6.22d}$$

$$z_{ij}^k \in \{0,1\}, \quad i,j \in V : i \neq j, \tag{6.22e}$$

where M_i^1 and M_i^2 are set as follows:

$$M_i^1 = \max\{c_{ri} + c_{it} - c_{rt} \ : \ r,t \in V, i \neq r,t\},$$
$$M_i^2 = \max\{c_{rt} - c_{ri} + c_{it} \ : \ r,t \in V, i \neq r,t\}.$$

Proof. To prove the desired result, it is sufficient to show that $\overline{x}_i^k = y_i^k$ holds for all $i \in V$ and $k \in K$. Given that the argument applies uniformly to all vehicles, it is omitted the explicit reference to the vehicle index $k \in K$.

Assume that $\overline{x}_i = 1$ for some $i \in V$. This means that customer i is visited in the TSP solution defined by the upper-level constraints. Let the path through i be given by $r \to i \to t$, with $i \neq r, t$, as depicted in Figure 6.2, where the blue path represents the one imposed by the upper-level, and the red path indicates the route the lower-level might take in the absence of any penalties. If $y_i = 0$ is chosen, a penalty cost of M_i^1 is incurred. The total cost of the solution includes this penalty plus the cost of a tour that does not visit i. If $y_i = 1$ is chosen, no penalty is incurred. The tour must now include node i. The additional cost of inserting node i into an existing tour between two nodes r and t is $c_{ri} + c_{it} - c_{rt}$. A rational decision-maker will choose $y_i = 1$ if the penalty for not doing so is higher than any possible cost of including i. That is, if M_i^1 is greater than the cost of inserting i. The definition $M_i^1 = \max_{r,t\in V, i\neq r,t} \{c_{ri} + c_{it} - c_{rt}\}$ correctly ensures that M_i^1 is an upper bound on the cost of inserting node i into any tour. Therefore, paying the penalty M_i^1 is always more expensive than including node i in the tour. Thus, any optimal solution must have $y_i = 1$.

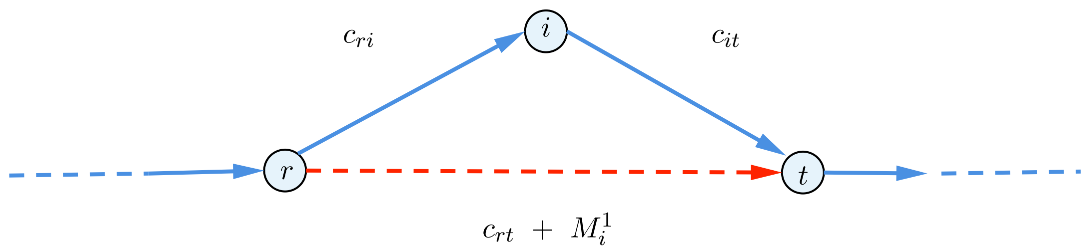


Figure 6.2: Subset penalization scenario

Now, consider the second case, where a node $s \in V$ is not prescribed to be visited by the upper-level, meaning $\overline{x}_s = 0$. To prove the proposition, it must shown that any optimal solution to the lower-level problem will have $y_s = 0$. Assume for contradiction that an optimal solution exists where $y_s = 1$. This implies that the tour created by the z_{ij}^k variables passes through node

s, as depicted in the red path in Figure 6.3. Let this path be $\dots \to i \to s \to j \to \dots$, where i must be a prescribed node (i.e., $\overline{x}_i = 1$) for the tour to have a non-zero cost contribution leading to s. The decision to visit s incurs a direct penalty of M_s^2 in the objective function. The critical aspect of this scenario lies in how the objective function (6.22a) calculates the total route cost. The cost of the arc (i, s), c_{is}, is included in the sum because its origin node i satisfies $\overline{x}_i = 1$. However, the cost of the subsequent arc (s, j), c_{sj}, is not included, because its origin node s has $\overline{x}_s = 0$. The optimization problem is thus presented with an artificial incentive to route through s, as it gains the use of arc (s, j) at zero cost to the objective function. To ensure that visiting s is never optimal, the penalty M_s^2 must be sufficiently large to avoid any potential savings from this *free* arc. A formally robust definition for the penalty, sufficient to guarantee correctness, is given by $M_s^2 > \max_{i,j\in V}\{c_{ij} - c_{is} + c_{sj}\}$. This choice for M_s^2 ensures that the penalized cost of the new path is always greater than the original cost. Therefore, any such deviation would result in a strictly more expensive solution, which contradicts optimality, concluding that it must be that $y_s = 0$ whenever $\overline{x}_s = 0$.

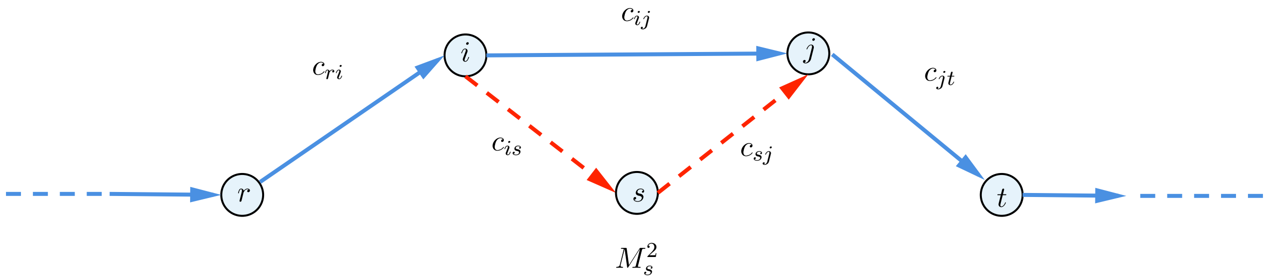


Figure 6.3: Superset penalization scenario

■

As a consequence of Proposition 6.4.1, since the feasible region of problem (6.22) does not depend on x variables the function ϕ can be represented as follows in Corollary 6.4.2 in terms of routes.

Corollary 6.4.2 Let $\mathcal{T}$ be the set of all the routes in graph G, and denote by $T \in \mathcal{T}$ an arbitrary route, with V_T and A_T the set of vertices and arcs visited and traversed by the route, respectively. Then, for every $k \in K$, the value function ϕ_k in problem (6.16) is equivalent to

$$\phi_k(x^k) = \min_{T=(V_T,A_T)} \sum_{(i,j)\in A_T} x_i^k c_{ij} + \sum_{i\notin V_T} M_i^1 x_i^k + \sum_{i\in V_T} M_i^2(1 - x_i^k) \tag{6.23}$$

In this way, the value function $\phi_k(x^k)$ is convexified. By replacing constraint (6.20k) in the model (6.20), with the following ones:

$$\Phi_k \leq \sum_{(i,j)\in A_T} x_i^k c_{ij} + \sum_{i\notin V_T} M_i^1 x_i^k + \sum_{i\in V_T} M_i^2(1 - x_i^k), \quad (V_T, A_T) \in \mathcal{T}. \tag{6.24}$$

one obtains a single-level reformulation for the first model. Constraints (6.20k) can be rearranged to

$$\Phi_k \leq \sum_{i\in V_T} \sum_{(i,j)\in A_T} \left(x_i^k c_{ij} + M_i^2(1 - x_i^k)\right) + \sum_{i\notin V_T} M_i^1 x_i^k, \quad (V_T, A_T) \in \mathcal{T}. \tag{6.25}$$

It is important to note that, one can introduce the aforementioned cuts for all vehicles across all criteria, including those associated with zero-valued components of λ. Also, note that the number of these constraints is exponential. In order to solve formulation (6.20) a separation procedure is next introduced separating them dynamically. To separate constraints (6.24), let the current solution of the master be $\bar{x}^k$ and the values $\bar{\Phi}_k$, one needs to solve the following TSP model:

$$\phi_k(\bar{x}^k) = \min \sum_{i\in V} \sum_{j\in V, j\neq i} c_{ij} z_{ij}^k, \tag{6.26a}$$

$$\text{s.t.} \quad \sum_{i\in V} z_{ij}^k = \bar{x}_j^k, \qquad j \in V, \tag{6.26b}$$

$$\sum_{j\in V} z_{ij}^k = \bar{x}_i^k, \qquad i \in V, \tag{6.26c}$$

$$\texttt{Subtour\_elimination}(z^k), \tag{6.26d}$$

$$z_{ij}^k \in \{0,1\}, \qquad i,j \in V : i \neq j. \tag{6.26e}$$

Let $(\bar{V}_T, \bar{A}_T)$ be the optimal tour obtained by solving the above model. If the length of this tour $\mathsf{TSP}(\bar{V}_T) > \bar{\Phi}_k$, the corresponding cut (6.24) is violated, so the following constraint is inserted:

$$\Phi_k \leq \sum_{(i,j)\in \overline{A}_T} x_i^k c_{ij} + \sum_{i\notin \overline{V}_T} M_i^1 x_i^k + \sum_{i\in \overline{V}_T} M_i^2 (1 - x_i^k), \tag{6.27}$$

to the master problem.

Furthermore, the penalization terms M^1 and M^2 depend on the tour, which means that they can be tighten computing them within the separation procedure. Therefore, cuts (6.24) can be represented by

$$\Phi_k \leq \sum_{(i,j)\in A_T} x_i^k c_{ij} + \sum_{i\notin V_T} M_i^{1,T} x_i^k + \sum_{i\in V_T} M_i^{2,T} (1 - x_i^k), \quad (V_T, A_T) \in \mathcal{T}, \tag{6.28}$$

where $M_i^{1,T}$ and $M_i^{2,T}$ are the penalization terms for each node $i \in V'$ and each tour $T \in \mathcal{T}$.

Regarding model (6.19), the value function reformulation is done similarly, and the bilevel cuts are

$$\Psi_k \leq \sum_{(i,j)\in A_T} x_i^k c_{ij} + \sum_{i\notin V_T} M_i^{1,T} x_i^k + \sum_{i\in V_T} M_i^{2,T} (1 - x_i^k), \quad (V_T, A_T) \in \mathcal{T}. \tag{6.29}$$

6.5 Improvements

This section presents alternative, sparser formulations for the ordered bilevel framework introduced in the previous chapters, with the aim of enhancing computational efficiency and scalability. In particular, these formulations reduce the number of variables and constraints required to model the allocation of customers to vehicles while preserving the structural properties of the original problem. Additionally, valid bounds for the route length variables are derived, providing tighter relaxations and improving the convergence of solution algorithms. Together, these enhancements will be later demonstrated to be crucial when solving large-scale instances of the problem.

6.5.1 Alternative sparser models

Note that in the value function reformulation, all lower-level constraints were included in the final model. They also include three-index binary variables z_{ij}^k that may negatively affect the tractability and models' performance. In this section, it is shown that in both of the proposed models, these z variables can be projected out and replaced by certain constraints, resulting in two much sparser formulations.

To this end, constraints

$$\begin{aligned}
&\sum_{i \in V} z_{ij}^k = x_j^k && k \in K, j \in V \\
&\sum_{j \in V} z_{ij}^k = x_i^k && k \in K, i \in V \\
&\texttt{Subtour_elimination}(z^k) && k \in K, \\
&\sum_{i \in V} \sum_{j \in V j \neq i} c_{ij} z_{ij}^k = \Phi_k && k \in K, \\
&z_{ij}^k \in \{0, 1\} && i, j \in V : i \neq j, k \in K,
\end{aligned}$$

are removed. In model (6.20), they are replaced by the following cuts:

$$\Phi_k \geq \underline{\Phi}_k + (\mathsf{TSP}(V_T) - \underline{\Phi}_k)(\sum_{i \in V_T} x_i^k - |V_T| + 1), \quad (V_T, A_T) \in \mathcal{T}, k \in K, \tag{6.31}$$

where $\underline{\Phi}_k$ represents a global lower bound on the k-th shortest route. These cuts resemble the integer $\mathcal{L}$-shaped cuts (see, e.g., Laporte & Louveaux, 1998). They ensure that if all nodes i from V_T are served by vehicle k, the lower bound on Φ_k must be $\mathsf{TSP}(V_T)$. However, if one of nodes from V_T is not included in the tour, the global lower bound on Φ_k is imposed, and if two or more nodes are excluded, this constraint is not active. In the second model, derived from (6.19), they are replaced by

$$\Psi_k \geq \underline{\Psi} + (\mathsf{TSP}(V_T) - \underline{\Psi})(\sum_{i \in V_T} x_i^k - |V_T| + 1), \quad (V_T, A_T) \in \mathcal{T}, k \in K, \tag{6.32}$$

where $\underline{\Psi}$ represents a global lower bound on any route in an optimal solution. How to obtain the values of $\underline{\Phi}_k$ and $\underline{\Psi}$ is discussed next in this section.

There exists another family of TSP optimality cuts that can be used instead, (see, e.g., in Fachini & Armentano, 2020). For model (6.20), these cuts are given as:

$$\Phi_k \geq \mathsf{TSP}(V_T) - \left(\sum_{i \in V_T} (1 - x_i^k)(\max_{ij \in \delta^+(i)} \{c_{ij}\} + \max_{ji \in \delta^-(i)} \{c_{ji}\}) \right), \quad (V_T, A_T) \in \mathcal{T}, k \in K. \tag{6.33}$$

These cuts ensure that if V_T are customers of the tour k (there could be more customers visited), then, because of the triangle inequality, Φ_k must be at least the length of the TSP tour connecting V_T. However, if a node $i \in V_T$ is not part of the tour, the lower bound on the right-hand side must be corrected. By excluding the node i, we need to subtract from $\mathsf{TSP}(V_T)$ the length of two arcs incident to i, and in the worst case, these are the most expensive incoming and the most expensive outgoing arc of i. Similarly, for the second model, the cuts enforce the lower bound on Ψ_k variables as follows:

$$\Psi_k \geq \mathsf{TSP}(V_T) - \left(\sum_{i \in V_T} (1 - x_i^k)(\max_{ij \in \delta^+(i)} \{c_{ij}\} + \max_{ji \in \delta^-(i)} \{c_{ji}\}) \right), \quad (V_T, A_T) \in \mathcal{T}, k \in K. \tag{6.34}$$

6.5.2 Bounds for route length variables

This section is devoted to providing valid lower and uppers bounds for the route length variables. First, lower bounds for Φ_k are derived. For this, the first step involves considering a relaxation model to estimate how many customers can be feasibly served by $m - 1$ vehicles, this is, all vehicles but one. For each node $i \in V$, binary variable α_i which is equal to 1 if the node is selected for service, and 0 otherwise, are introduced. Additionally, for each node i and vehicle

$j \in \{1, \ldots, m-1\}$, a binary variable β_{ij} is introduced, indicating whether node i is assigned to vehicle j. The model maximizes the total number of selected nodes as follows:

$$\max \sum_{i \in V} \alpha_i, \tag{6.35}$$

$$\text{s.t.} \sum_{j=1}^{K-1} \beta_{ij} = \alpha_i, \qquad i \in V, \tag{6.36}$$

$$\sum_{i \in V} d_i \beta_{ij} \leq Q, \qquad j \in \{1, \ldots, m-1\}. \tag{6.37}$$

Let obj^* be the optimal value of the objective function, then the number of unserved nodes is defined as $a = n - obj^*$. With a determined, a second model to compute a lower bound on the tour cost needed to visit exactly a nodes is presented. This is formulated as a Price Collecting TSP over the set of all customers, where the tour length is minimized and the constraints concerns that the capacity of served customers should not exceed Q and the number of served customers must be exactly a. If this model is solved successfully, the optimal tour length is denoted by A, serving as a conservative lower bound for the route lengths, i.e., Φ_k for all $k \in K$. In case the second model fails to solve due to time limits, a second bound is computed as $B = 2\min_{j \in V'} c_{0j}$, which corresponds to the shortest distance from the depot to any other node. The final output is the maximum of the two.

Second, to derive valid upper bounds for the ordered cost components Φ_k, it is proceeded as follows. A natural upper bound on individual route lengths is given by the optimal value of the classical TSP that ignores demand constraints. However, solving such a problem is itself NP-hard, and thus may not always be computationally feasible. When solvable within a prescribed time limit, this value is adopted as a valid upper bound. Otherwise, a greedy heuristic procedure is used to build a feasible TSP tour, which yields a looser but efficient approximation. Additionally, a more refined procedure to compute an upper bound can be propposed. To this end, initialize the bound UB to 0. Then, for each customer node $i \in V'$, the subset $V_i = V' \setminus \{i\}$ is defined and TSP over V_i is solved, denoted by $\mathsf{TSP}(V_i)$. The candidate upper bound is updated as: $UB = \max\{UB,\ c_{0i} + c_{i0},\ \mathsf{TSP}(V_i)\}$. The final upper bound on route lengths is then given by the minimum between this value and the TSP solution over the full customer set (if available).

6.6 Computational study

This section presents the results of the computational experiments conducted to empirically evaluate and compare the proposed models. All formulations were implemented in Python (version 3.10) and solved with the Gurobi Optimizer (version 9.5) as a Mixed Integer Programming (MIP) solver. The experiments were performed on a MacPro server equipped with a 2.7 GHz Intel Xeon W processor (24 cores) and 192 GB of RAM. To ensure comparability and practical runtime, a time limit of 3600 seconds (1 hour) was set on each instance run.

The benchmark instances were obtained from the well-established CVRP library (http://vrp.atd-lab.inf.puc-rio.br/), focusing on cases with 50 customers or less. This threshold preserves tractability while maintaining practical relevance. Due to the challenges in certifying optimality, instances were divided into two families according to their size. Set `A` represents instances with up to 25 customers, while set `B` comprises instances with more than 25 customers and equal or less to 50. This partition allows for a detailed assessment of model behavior under varying levels of difficulty while remaining within feasible computational bounds.

Four model formulations were evaluated:

- Φ: The ordered-constrained formulation (6.20) with the full variable set.

- Φ-`z`: The same ordered formulation with a sparse variable set (without z variables).
- Ψ: The ℓ-sum formulation with the complete variable set.
- Ψ-`z`: The ℓ-sum formulation with a sparse variable set.

Each formulation was tested under eight statistical measures as objective functions. Monotonic measures included MEAN, MEDIAN, and MAX. Non-monotonic, fairness-oriented measures included RANGE, GINI, $\text{MAD}_{(med)}$, $\text{MAD}_{(min)}$, and VARIANCE. The definitions of these measures are provided in Table 6.1, except for variance, which appears in (6.14). Two performance indicators were recorded to evaluate model effectiveness: the computational time in seconds (TIME), and the relative gap to the best known upper bound between all the models (GAP-BUB). The latter is defined as

$$\texttt{GAP-BUB} = (\texttt{f}_m - \texttt{f}_*)/\texttt{f}_* \times 100,$$

where $\texttt{f}_m$ denotes the objective value obtained by formulation m, and $\texttt{f}_*$ represents the best objective value identified for a given instance and metric, serving as upper bound on the optimal value if optimality is not certified. This implies that a zero value of GAP-BUB indicates that the formulation attained the best upper bound among all models.

The computational framework incorporated several cutting-plane strategies. Penalization was handled either through big-M constraints embedded within callback functions (`local`) or by precomputing them at the root node (`global`). In the dense formulations, i.e., those containing z variables, Dantzig–Fulkerson–Johnson (DFJ) cuts were used for generalized subtour elimination. In the sparse formulations, a combination of $\mathcal{L}$-shaped cuts, namely 6.31 and 6.33 for the Φ models, and 6.32 and 6.34 for the Ψ models, was applied. All models were initialized with a feasible heuristic solution that assigned each of the first $k-1$ drivers to a single-customer route, while the k-th driver was assigned a tour covering all remaining customers. The aggregated results, summarized in Tables 6.4 and 6.5, highlight several key insights.

Regarding computational times, results are reported only for instances in set A, as all runs for set B reached the 3600 second time limit. These times underscore notable differences in difficulty across objectives and model types. A clear performance gap emerges between monotonic and non-monotonic objectives. Problems optimizing for MEAN, MEDIAN, or MAX are solved relatively quickly, particularly with the dense formulations (Φ and Ψ), suggesting a more tractable solution space. In contrast, fairness-oriented non-monotonic objectives are substantially more demanding. For measures such as RANGE and GINI, nearly all models required close to 3000 seconds, reflecting persistent difficulties in converging within the one-hour limit. This underscores the intrinsic complexity of fairness optimization, which entails navigating a significantly more challenging combinatorial landscape.

Concerning the best upper bound gaps, the first observation also relates to the varying difficulty induced by different objective functions. For the A instances, models optimizing monotonic objectives such as MEAN or MAX consistently attained the best upper bounds at termination. As reported in Table 6.4, the Ψ formulation, in particular, achieved the strongest upper bounds within the time limit. By contrast, all formulations faced pronounced computational challenges with non-monotonic, fairness-oriented objectives. The longer solution times and larger final gaps observed for measures such as RANGE and GINI highlight the inherent complexity of balancing workload equity against aggregate cost minimization.

Another finding is the superior performance of the ℓ-sum formulations (Ψ and Ψ-`z`) over their ordering-based counterparts (Φ and Φ-`z`). This advantage is evident across both instance families but is especially pronounced on the larger B instances. As seen in Table 6.5, the ℓ-sum models, and specifically Ψ-`z`, yield solutions of substantially higher quality, consistently achieving the smallest gaps across all measures when the time limit is reached.

On the other hand, the trade-off between dense and sparse formulations reveals a size-dependent behavior. For the smaller `A` instances, the dense models (Φ and Ψ) were significantly better in terms of `GAP-BUB`, as the overhead of managing additional cutting planes outweighed the benefits. For the more challenging `B` instances, however, the advantage of a sparse structure becomes critical, but its effectiveness is highly dependent on the underlying model. The sparse ℓ-sum formulation (Ψ`-z`) markedly improves solution quality over its dense counterpart (Ψ). Conversely, the sparse ordering-based model (Φ`-z`) underperforms relative to its dense version (Φ). This suggests a synergistic effect: the strengthening cuts of the sparse structure are most potent when applied to the already stronger LP relaxation of the ℓ-sum formulation.

Finally, a comparison of penalty implementation schemes indicates a clear advantage for the `local` (callback-based) strategy in achieving high-quality solutions in general. For the difficult `B` instances, where minimizing the final gap is paramount, the `local` approach outperformed the `global` one in every test case. While the `global` scheme was occasionally marginally faster on simpler monotonic objectives in the `A` set, its static nature proved less effective than the dynamic generation of cuts within the branch-and-bound tree for complex problems. The ability of the `local` strategy to adaptively add cuts where needed is crucial for effectively pruning the search space.

In summary, the computational study validates the superiority of the ℓ-sum approach. For large-scale and complex instances, particularly when addressing fairness objectives, the most robust and effective configuration is the sparse Ψ`-z` formulation combined with dynamically generated `local` penalties. This combination consistently yields the highest-quality solutions, providing the tightest optimality gaps within practical time limits.

Metric	Measure	Φ		Φ-z		Ψ		Ψ-z	
		global	local	global	local	global	local	global	local
TIME	MEAN	**425.19**	430.43	2804.77	2804.36	430.23	440.35	3102.79	3041.43
	MEDIAN	231.83	**222.25**	2805.24	2804.72	714.16	701.17	3204.47	3204.25
	MAX	748.88	676.53	2806.15	2805.99	710.70	**572.51**	2827.55	2830.57
	RANGE	3201.52	3201.66	2857.44	**2812.58**	3275.59	3152.24	2973.21	2887.76
	GINI	3203.27	3187.84	2851.39	**2808.00**	3535.90	3292.31	2994.41	2916.13
	MAD$_{(med)}$	3204.02	3176.28	2864.75	**2811.47**	3531.37	3548.54	3182.40	3014.97
	MAD$_{(min)}$	3201.65	3201.77	2863.21	**2810.86**	3440.12	3424.94	3307.63	3209.67
	VARIANCE	3204.07	3203.18	2846.27	**2811.53**	3282.80	3106.88	2882.59	2857.97
GAP-BUB	MEAN	**0.00**	0.03	8.76	7.68	**0.00**	**0.00**	4.65	4.83
	MEDIAN	**0.00**	2.27	16.87	13.66	**0.00**	**0.00**	13.12	8.45
	MAX	11.11	11.11	10.32	6.85	**0.00**	**0.00**	6.57	5.12
	RANGE	75.41	49.08	52.01	51.94	53.01	37.27	42.26	**21.69**
	GINI	68.13	28.09	49.94	46.68	51.39	32.09	25.48	**16.60**
	MAD$_{(med)}$	75.37	25.32	48.02	50.19	57.76	31.19	24.22	**16.31**
	MAD$_{(min)}$	70.61	30.19	48.25	50.77	59.60	31.25	31.10	**14.06**
	VARIANCE	65.13	30.57	30.20	31.84	28.50	15.02	**10.53**	11.11

Table 6.4: Aggregated computational results for `A` instances.

Metric	Measure	Φ		Φ-z		Ψ		Ψ-z	
		global	local	global	local	global	local	global	local
GAP-BUB	MEAN	74.78	68.46	85.41	75.41	46.61	33.71	20.72	**20.34**
	MEDIAN	100.00	100.00	75.61	69.03	78.55	61.11	9.46	**8.57**
	MAX	100.00	100.00	84.73	70.82	83.55	73.91	**7.35**	9.80
	RANGE	100.00	99.57	93.97	83.04	85.89	42.23	46.07	**24.17**
	GINI	100.00	95.66	97.14	85.10	72.58	43.05	55.95	**30.88**
	MAD$_{(med)}$	100.00	96.45	86.94	85.94	87.87	83.50	40.95	**17.90**
	MAD$_{(min)}$	100.00	99.56	89.06	88.66	80.95	53.96	39.01	**38.17**
	VARIANCE	100.00	100.00	99.63	91.86	95.09	65.68	55.70	**42.13**

Table 6.5: Aggregated computational results for B instances (all runs reached the time limit of 3600 seconds).

En toda Andalucía, roca de Jaén y caracola de Cádiz, la gente habla constantemente del duende y lo descubre en cuanto sale con instinto eficaz. [...] Ángel y musa vienen de fuera; el ángel da luces y la musa da formas (Hesíodo aprendió de ellas). Pan de oro o pliegue de túnicas, el poeta recibe normas en su bosquecillo de laureles. En cambio, al duende hay que despertarlo en las últimas habitaciones de la sangre.

Federico García Lorca - *Conference "Teoría y juego del duende"* (Buenos Aires, October 20, 1933)

7

Application to Regression Problems: Least Quantile of Squares Regression

In Statistics, regression analysis is an important instrument to determine the effect of the explanatory variables on response variables. When outliers and bias errors are present, the standard weighted least squares estimator may perform poorly. For this reason, many alternative robust techniques have been studied in the literature. In these terms, the *Least Squares Quantile* (LQS), and in particular the *Least Squares Median*, are among the regression estimators that exhibit better robustness properties. However, the accurate computation of these estimators is computationally demanding, resulting in a difficult estimator to obtain. In this chapter, novel approaches to compute a global optimal solution for the LQS estimator based on single-level and bilevel optimization methods are proposed. An extensive computational study is provided to support the efficiency of the methods considered, and an ad-hoc procedure to address the scalability of the problem to larger instances is proposed. Therefore, with the pretext of linear regression, this chapter elaborates on the contributions of the dissertation concerning the potential of ordered optimization in fostering solution robustness.

7.1 Introduction

A regression analysis models the relationship between one or more independent variables, also known as covariates or explanatory variables, and a dependent variable, also known as the response or target variable. Consider the following multiple linear regression model with response $y \in \mathbb{R}^n$, model matrix $\mathbf{X}_{n\times p}$, regression coefficients $\beta \in \mathbb{R}^p$ and error $\varepsilon \in \mathbb{R}^n$, namely $y = \mathbf{X}\beta + \varepsilon$.

Classically, the error terms are assumed to be independent and identically distributed, $\varepsilon_i \sim N(0, \sigma^2), i \in N$, where N is the index set $\{1, \dots, n\}$. Hereafter, it is assumed that $\mathbf{X}$ contains a column of ones to account for the intercept in the model. Given data for the i-th sample $(y_i, \mathbf{x}_i), i \in N$, and regression coefficients β, the i-th residual is denoted by $r_i(\beta) = y_i - \mathbf{x}_i^\top \beta$ for $i \in N$. The traditional Least Squares estimator (LS) is given by

$$\hat{\beta}^{(\mathsf{LS})} := \arg\min_{\beta} \sum_{i=1}^{n} r_i^2, \tag{7.1}$$

which considers the minimization of the ℓ_2-norm on the residuals.

Standard types of regression, such as LS, have favorable properties if their underlying assumptions are true, but can give misleading results otherwise, i.e., are not robust to small deviations from the assumptions (such as non-constant variance, outliers or correlated data points). Robust regression methods provide an alternative to least squares regression by requiring less restrictive assumptions.

In this sense, M-estimators introduced by Huber (1973) are obtained by minimizing a loss function of the residuals of the form $\sum_{i=1}^{n} \rho(r_i)$, where $\rho(r)$ is a loss function with a unique minimum. Examples include the Huber function (Huber, 1964a, 1973) the Tukey's Biweight function (Rousseeuw & Leroy, 1987), the logistic function (Coleman et al., 1980), and the Talwar function (Hinich & Talwar, 1975), among others. In Koenker & Bassett Jr (1978), the Quantile Regression (QR) estimator was introduced as a form of robust (outlier resistant) regression, see Portnoy (1991). The τ-th conditional quantile of a random variable Y given X is the τ-th quantile of the conditional probability distribution of Y given X, i.e,

$$Q_{Y|X}(\tau) = \inf \left\{ y : F_{Y|X}(y) \geq \tau \right\}, \tag{7.2}$$

which is a random variable. In Quantile Regression, it is assumed that the τ-th conditional quantile is given as a linear function of the explanatory variables,

$$Q_{Y|X}(\tau) = X\beta. \tag{7.3}$$

Therefore, given the sample $(y_i, \mathbf{x}_i), i \in N$, the estimators of the regression coefficients can be obtained as

$$\hat{\beta}(\tau)^{(\mathsf{QR})} := \arg\min_{\beta} \sum_{i=1}^{n} \rho_\tau (r_i), \tag{7.4}$$

where, ρ_τ is the quantile loss function of the residuals defined in expression (2.46), which was introduced in Section 2.6.5 of Chapter 2 of this dissertation.

Different QR estimators $\hat{\beta}(\tau)$ can be obtained for different values of τ. The minimization of the quantile loss function of the residuals leads to the convex optimization problem (7.4), which can be transformed into a linear program by splitting the residual into its positive and negative parts, as it was discussed in Section 2.6.5. For this reason, QR has many expert and intelligence systems applications, e.g., Benoit & Van den Poel (2009); Zheng (2012); Reddy & Ravi (2013); Xu et al. (2015, 2017); Hesamian & Akbari (2019); Lyócsa & Stašek (2021); Wang et al. (2023); Sousa et al. (2024). Note that when $\tau = 0.5$, the loss function ρ_r is proportional to the absolute value function, and thus median regression is the same as linear regression by the Least Absolute Deviation (LAD) estimator, introduced by Edgeworth (1887), which is given by

$$\hat{\beta}^{(\mathsf{LAD})} := \arg\min_{\beta} \sum_{i=1}^{n} |r_i|, \tag{7.5}$$

considering the minimization of the ℓ_1-norm of the residuals.

Quantile Regression, and hence the LAD estimator, like other regression M-estimators, can correct robustness problems due to vertical deviations (that is, related to the response variable), but not those caused by horizontal deviations (that is, related to the explanatory variables), i.e., can be highly sensitive to outliers in the covariate space. The interested reader is referred, for instance, to He et al. (1990) for a detailed discussion on the tail behavior and breakdown points of many of these estimators. In order to control both factors, there exist many attempts in the literature that try to robustify the QR estimator, e.g., Adrover et al. (2004) or the Least Trimmed Quantile Regression estimator proposed by Neykov et al. (2012). Beyond the above, there is a whole range of robust estimators studied in the literature whose purpose is to resist up deviant data values without becoming extremely biased. One of the most studied robust estimators is the *Least Median of Squares* (LMS) regression estimator, studied in Hampel (1975) and Rousseeuw (1984), which minimizes the median of the squared residuals, although for further purposes it is similarly expressed it by means of the absolute residuals, i.e.,

$$\hat{\beta}^{(\mathsf{LMS})} := \arg\min_{\beta} \left(\operatorname*{median}_{i=1,\dots,n} |r_i| \right), \tag{7.6}$$

or its generalization by considering the q-th order statistic, the *Least Quantile of Squares* (LQS) regression estimator, which absolute value version was studied first in Tableman (1994),

$$\hat{\beta}^{(\mathsf{LQS})} := \arg\min_{\beta} \left| r_{(q)} \right|, \tag{7.7}$$

where $r_{(q)}$ denotes the residual, corresponding to the q-th ordered absolute residual,

$$\left| r_{(1)} \right| \leq \left| r_{(2)} \right| \leq \cdots \leq \left| r_{(n)} \right|. \tag{7.8}$$

Other estimators that achieve a high breakdown point and good statistical efficiency include the Least Trimmed Squares estimator (Rousseeuw, 1984; Rousseeuw & Leroy, 1987), which minimizes the sum of squares of the q smallest squared residuals, the Least Quartile Difference estimator, introduced by Croux et al. (1994), and the Least Trimmed Differences estimator, introduced by Stromberg et al. (2000).

The majority of the regression estimators listed above belong to larger families of estimators whose descriptive, asymptotic, and behavioral properties have been well studied in the literature. One of particular interest to us are the L-estimators introduced in Chapter 3. In order to formally define this structure for regression purposes, let us consider $r_{ord} = (|r_{(1)}|, |r_{(2)}|, \dots, |r_{(n)}|)^t$ the vector with the components sorted in non-decreasing sequence,

$$\hat{\beta}^{(L)} := \arg\min_{\beta} \ \lambda^t r_{ord}, \tag{7.9}$$

where $\lambda \in \mathbb{R}^n$ is a given rank dependant vector which does not necessarily have to be positive. Different definitions of the λ-vector also lead to different objective function based on the absolute residuals. Table 7.1 shows the ordered weighted formulation of different well-known robust regression estimators. In any case, the ordered structure in (7.9) allows great flexibility, not just to study a large number of robust estimators, but to extend the analysis to more general estimators. Furthermore, if quadratic terms are needed to meet the definition of the estimators, as in the case of the LS estimator (7.1), one can consider extending (7.9) by including an additional quadratic expression of the form $r_{ord}^t M r_{ord}$, where $M \in \mathbb{R}^{n \times n}$ is a rank dependant matrix which entries are the weighting factor of the interaction between residuals. For example, by considering $M = diag_n(1, ..., 1)$, where $diag_n(.)$ is the diagonal matrix with n entries, one meets the exact definition of (7.1).

Regression in the form of (7.9), known as *Ordered Regression*, was first studied by by Yager & Beliakov (2009). To name a few works in this direction, one can find Chachi & Chaji (2021), Flores-Sosa et al. (2020, 2022) or D'Urso & Chachi (2022). In Blanco et al. (2018, 2021), the authors proposed a framework to solve problems in the form of (7.9) based on ordered optimization models, where residuals are defined in a more general way. Moreover, this framework guarantees that any feasible solution provides a sub-optimal estimator (in those cases where optimality can not be proven) and it can also be used to compute estimators with extra side constraints. Specifically, this leads to,

$$\hat{\beta} := \arg\min_{\beta} \ \lambda^t r_{ord}, \quad \text{s.t. } \mathbf{A}\beta \leq \mathbf{b} \tag{7.10}$$

Estimator	Loss	Expression	λ-vector
Least Absolute Deviation (Edgeworth, 1887)	ℓ_1-norm	$\sum_{i=1}^{n} \|r_i\|$	$(1, ..., 1)$
Minimum Maximum Absolute Deviation (Sposito, 1976)	Chebyshev or ℓ_∞-norm	$\max \|r_i\|$	$(0, ..., 0, 1)$
Least Median of Squares (Hampel, 1975)	Median absolute residual	$\text{median}\|r_i\|$	$(0, ..., 0, \underbrace{1}_{\lfloor \frac{n}{2} \rfloor - th}, 0, ..., 0)$
Least Quantile of Squares (Rousseeuw, 1984)	q-th absolute residual	$\|r_{(q)}\|$	$(0, ..., 0, \underbrace{1}_{q-th}, 0, ..., 0)$
Least Trimmed of Squares (Rousseeuw, 1985)	α, β-trimmed mean $(\alpha, \beta \in [0, 1])$	$\sum_{i=\lfloor \alpha n \rfloor + 1}^{n - \lfloor \beta n \rfloor} \|r_{(i)}\|$	$(\underbrace{0, ..., 0,}_{\lfloor \alpha n \rfloor} \underbrace{1, ..., 1,}_{n - (\lfloor \alpha n \rfloor + \lfloor \beta n \rfloor)} \underbrace{0, ..., 0}_{\lfloor \beta n \rfloor})$
Gini Differences (Wainer & Thissen, 1976, Sievers, 1983)	Gini's mean or absolute differences	$\sum_{i,j=1}^{n} \|r_i - r_j\|$	$(2(2i - n - 1))_{i \in N}$
Least Trimmed Differences (Stromberg et al., 2000)	α-trimmed absolute differences $(\alpha \in [0, 1])$	$\sum_{i,j=1}^{\lfloor \alpha n \rfloor} \|r_i - r_j\|$	$(2(2i - \lfloor \alpha n \rfloor - 1)_{i \in \{1, ..., \lfloor \alpha n \rfloor\}}, 0, ..., 0)$

Table 7.1: Equivalent λ-vectors to express different robust estimators.

where $\mathbf{A}_{m \times p}, \mathbf{b}_{m \times 1}$ are given parameters in the problem representing side constraints useful to accommodate many forms of regularization on the variable β. Examples of these can be the well-known Ridge (Hoerl & Kennard, 1970), Lasso (Tibshirani, 1996) or Elastic-net (Zou & Hastie, 2005) regularizations.

The counterpart to the great generality of this model, rests on the non-linearity of the ordering operation that introduces a extra degree of complication and, therefore, solving these models can be computationally challenging. For this reason, tailored approaches for each particular case are needed which, for the LQS estimator, motivates the research in this chapter. The LQS, hence the LMS, estimator is known to be computationally demanding due to the combinatorial nature of the problem, proven to be NP-hard by Bernholt (2006). As a consequence, many different approaches in the literature to compute these estimators are not able to obtain a global minimum of the problem for large sample size instances, nor to provide certificates about the quality of the solution obtained. In this line, one can fin the works of Rousseeuw & Hubert (1997), Steele & Steiger (1986), Stromberg (1993), Watson (1998), Agulló (1997), Giloni & Padberg (2002), and Mount (2006), among others.

To the best of the knowledge, it is in the work of Bertsimas & Mazumder (2014) where, in addition to obtaining state-of-the-art results for medium/large instance sizes, a first-time resolution method is proposed that allows finding a provable global optimal solution for the LQS problem. In that work, LQS estimators (henceforth BM estimators), are obtained as solutions of a mixed integer mathematical programming model which is reviewed next for the sake of completeness. To this end, to model the q-th sorted residual, that is, $\left|r_{(q)}\right|$, the authors express the fact that $r_i \leq \left|r_{(q)}\right|$ for q many absolute residuals from $|r_i|, \ldots, |r_n|$ by introducing the binary variables

$$z_i = \begin{cases} 1, & \text{if } |r_i| \leq \left|r_{(q)}\right|. \\ 0, & \text{otherwise.} \end{cases} \tag{7.11}$$

In addition, two sets of auxiliary continuous variables, μ_i, $\bar{\mu}_i \geq 0$, are required such that they satisfy

$$|r_i| - \mu_i \leq \left|r_{(q)}\right| \leq |r_i| + \bar{\mu}_i, \quad i \in N, \tag{7.12}$$

together with the conditions that if $|r_i| \geq \left|r_{(q)}\right|$, then $\bar{\mu}_i = 0, \mu_i \geq 0$ and if $|r_i| \leq \left|r_{(q)}\right|$, then $\mu_i = 0, \bar{\mu}_i \geq 0$.

Finally, decision variables $r_i^+, r_i^- \geq 0$, $i \in N$, are introduced for the linearization of the absolute value of the residuals, $|r_i|$, such that

$$\begin{aligned} & r_i^+ + r_i^- = |r_i|, \\ & r_i^+ - r_i^- = r_i = y_i - \mathbf{x}_i^\top \beta, \\ & r_i^+, r_i^- \geq 0, r_i^+ r_i^- = 0, i \in N. \end{aligned} \tag{7.13}$$

Therefore, BM estimators are obtained by minimizing a variable $\gamma \in \mathbb{R}$, which represents q-th sorted absolute residual, by means of the following mixed integer model:

$$\begin{aligned}
\widehat{\beta}_{\mathsf{BM}}^{(\mathsf{LQS})} := \arg\min_{\beta} \ & \gamma, && && (7.14a)\\
\text{s.t. } & r_i^+ - r_i^- = y_i - \mathbf{x}_i^\top \beta, && i \in N, && (7.14b)\\
& r_i^+ + r_i^- - \gamma = \bar{\mu}_i - \mu_i, && i \in N, && (7.14c)\\
& \sum_{i=1}^{n} z_i = q, && && (7.14d)\\
& \gamma \geq \mu_i, && i \in N, && (7.14e)\\
& \mu_i, \bar{\mu}_i \geq 0, && i \in N, && (7.14f)\\
& r_i^+, r_i^- \geq 0, && i \in N, && (7.14g)\\
& (\bar{\mu}_i, \mu_i) \ : \mathsf{SOS1}, && i \in N, && (7.14h)\\
& (r_i^+, r_i^-) : \mathsf{SOS1}, && i \in N, && (7.14i)\\
& (z_i, \mu_i) \ \ : \mathsf{SOS1}, && i \in N, && (7.14j)\\
& z_i \in \{0, 1\}, && i \in N, && (7.14k)
\end{aligned}$$

where, $\mathsf{SOS1}$ state for the Special Ordered Sets of type 1.

This approach leads to an optimal solution for any dataset where the sample data points do not necessarily have to be in general position, that is, for any subset of $\mathcal{I} \subset N$ with $|\mathcal{I}| = p$, the $p \times p$ submatrix $X_{\mathcal{I}}$ has rank p. Furthermore, this framework enables us to provide a proof of the breakdown point of the LQS objective value that holds for any dataset. The authors also propose a regularized version of the mixed-integer optimization formulation, which they found to be effective in accelerating the convergence of the model without any loss of accuracy in the solution, that may also act as a regularizer to shrink coefficients.

The contributions of this chapter can be summarized as follows. Due to the complexity of the LQS problem, obtaining a global optimal solution of (7.7) is quite demanding, mainly since it is a non-linear and non-differentiable problem. In this chapter, a number of novel mathematical programming formulations are to be introduced that improve the efficiency of the computation of the global optimal solutions for the LQS estimator. These new formulations consider a smaller number of variables and constraints, allowing to provide better upper bounds of the objective value in less computing time. First, single-level formulations based on ordered optimization methods are proposed, yielding evidence of their equivalences and exploiting their properties to derive valid inequalities and bounds if needed. Second, exploiting the properties of the quantile loss function (2.46), a new result is derived that allows to propose a compact formulation with an additional constraint by resorting to the resolution of a bilevel optimization problem. Given the equivalences between all these formulations and (7.14), general and robustness properties regarding the LQS estimator, such as the high breakdown point, the validity of non-general position of the data samples or the sub-optimality of the solution when optimality cannot be certified, are the same than in Bertsimas & Mazumder (2014). These results are supported by an extensive and detailed computational study on instances of tractable size. In addition, to address the scalability of the problem, i.e., the resolution of larger size instances (up to 10000 points), a novel aggregation approach is proposed.

The remainder of the chapter is organized as follows. In Section 7.2, novel single-level approaches for the LQS problem are described. Section 7.3 develops the proposed bilevel optimization approaches. Section 7.4 reports the computational experiments supporting our results, and Section 7.5 shows the scalability of the proposed aggregation method to larger instances.

7.2 LQS single-level optimization approaches

In this section, three alternative single-level mixed-integer optimization formulations to solve problem (7.7) are presented. First, two formulations based on the ordered optimization approach introduced accross this dissertation are introduced in Section 7.2.1. Second, extending the work by Bertsimas & Mazumder (2014), a compact formulation based in the k-sum operator decomposition of the q-th ordered absolute residual is introduced in Section 7.2.2.

7.2.1 Sorting constraints formulation

In the current work, given the sequence of ordered absolute residuals in (7.8), our modeling purposes call for the minimization of the q-th element of this sequence to find LQS estimators for problem (7.7). This objective function can be formulated by means of the ordered median function via the λ-vector corresponding to the k-max criterion, i.e., $\lambda = (0, \ldots, 0, \underbrace{1}_{k-th}, 0, \ldots, 0)$, and stating the equivalence

$$\left|r_{(q)}\right| = \sum_{i=1}^{n} \lambda_i |r_{(i)}| = \sum_{i=1}^{n} z_{iq} |r_i|, \tag{7.15}$$

where, the following binary decision variables are defined

$$z_{ik} = \begin{cases} 1, & \text{if the } i\text{-th residual is in the } k\text{-th position of the ordered vector,} \\ 0, & \text{otherwise.} \end{cases} \tag{7.16}$$

Therefore, introducing sorting constraints for these assignment variables following the rationale in Boland et al. (2006), a solution to the LQS problem is obtained as follows:

$$\begin{aligned}
\hat{\beta}^{(\mathsf{LQS})} := \arg\min_{\beta} \quad & \sum_{i=1}^{n} z_{iq}|r_i|, && && (7.17a)\\
\text{s.t.} \quad & \sum_{i=1}^{n} z_{ik} = 1, && k \in N, && (7.17b)\\
& \sum_{k=1}^{n} z_{ik} = 1, && i \in N, && (7.17c)\\
& \sum_{i=1}^{n} z_{ik}|r_i| \leq \sum_{i=1}^{n} z_{i,k+1}|r_i|, && k \in N, k < n, && (7.17d)\\
& z_{ik} \in \{0, 1\}, && i, k \in N. && (7.17e)
\end{aligned}$$

Constraints (7.17b), (7.17c) and (7.17d) model the sorting of the absolute residuals. In particular, constraints (7.17b) guarantee that every order position is assigned to a residual, constraints (7.17c) ensure that every residual is assigned to a sorting position and constraints (7.17d) sort the absolute value of the residuals in non-decreasing order in the objective function. Note that formulation (7.17) is non-linear due to the product of a binary and a non-negative variables, in both the objective function (7.17a) and in the sorting constraints (7.17d), and also to the absolute value of the residuals. The first non-linearity can be easily linearized by introducing big-M notation via McCormick (1976) envelopes, and the latter by considering the equivalences in (7.13).

A first simplification of model (7.17) can be obtained stating that, constraints (7.17d) provide the sorting of the absolute residual values including them as coefficients in the right and left-hand side of the inequalities, thus weakening the formulation by introducing more non-linear terms. An a priori stronger formulation than (7.17) can be obtained by considering the equivalent constraints in Lemma 7.2.1.

Lema 7.2.1 The set of constraints

$$\sum_{k>\ell} z_{jk} + \sum_{k\leq\ell} z_{ik} \leq 1, \quad i, j, \ell \in N, \tag{7.18}$$

provide an alternative sorting of the absolute value of the residuals, thus being equivalent to constraints (7.17d).

Proof. It is necessary to show that constraints (7.18) sort the absolute value of the residuals in non-decreasing order.

To this end, if $|r_i| < |r_j|$ for two $i, j \in N$, and assuming $z_{j\ell} = 1$, this is, the absolute residual j is in the ℓ-th position, it is deduced that

$$\sum_{k\leq\ell} z_{ik} = 1 \text{ and } \sum_{k>\ell} z_{jk} = 0. \tag{7.19}$$

which is equivalent to

$$\sum_{k>\ell} z_{jk} + \sum_{k\leq\ell} z_{ik} \leq 1, \tag{7.20}$$

for every $i, j, \ell \in N$.

Conversely, the constraints in (7.18) determine the correct sorting of the absolute residuals, by pairwise comparison. ■

Constraints (7.18) are simpler than constraints (7.17d) in the sense that they do not include the absolute value of the residual terms as coefficients, thus removing many non-linear terms. However, in counterpart, the number of these constraints is greater than the number of constraints in (7.17d). In spite of that, these constraints can be dynamically added in a branch-and-cut procedure by means of a tailored separation algorithm, see Labbé et al. (2017).

The simplicity of the model (7.17) is clearly shadowed because it considers $\mathcal{O}(n^2)$ variables. However, for our purposes it is not necessary to sort all the residuals to obtain a valid formulation. Since it is known beforehand that one needs the absolute residual in the q-th position, it is sufficient to consider a set of binary variables which identifies the absolute residuals such that $|r_i| \leq |r_{(q)}|$, namely $z_i \in \{0, 1\}, i \in N$, and another set of binary variables, namely $\delta_i \in \{0, 1\}, i \in N$, which identifies the absolute residual value in the q-th position, $|r_i| = |r_{(q)}|$. Therefore, following a similar rationale that in (7.17), a much simpler model can be introduced as follows:

$$
\begin{aligned}
\hat{\beta}_1^{(\mathsf{LQS})} := \arg\min_{\beta} \; & \sum_{i=1}^{n} \delta_i s_i, && && \text{(7.21a)}\\
\text{s.t. } & r_i^+ - r_i^- = y_i - \mathbf{x}_i^\top \beta, && i \in N, && \text{(7.21b)}\\
& r_i^+ + r_i^- = s_i, && i \in N, && \text{(7.21c)}\\
& \sum_{i=1}^{n} \delta_i = 1, && && \text{(7.21d)}\\
& \sum_{i=1}^{n} z_i = q, && && \text{(7.21e)}\\
& z_i s_i \leq \sum_{i=1}^{n} \delta_i s_i, && i \in N, && \text{(7.21f)}\\
& r_i^+, r_i^-, s_i \geq 0, && i \in N, && \text{(7.21g)}\\
& (r_i^+, r_i^-) : \mathsf{SOS1}, && i \in N, && \text{(7.21h)}\\
& \delta_i, z_i \in \{0,1\}, && i \in N. && \text{(7.21i)}
\end{aligned}
$$

The objective function (7.22a) looks for the minimization of the absolute residual in the q-th position. Constraints (7.21b) and (7.21c) represent the residuals and the absolute value of the residuals, respectively. Constraint (7.21d) guarantees that only one absolute residual is selected in the q-th position and constraint (7.21e) indicates that there must be exactly q residuals which are less than or equal to the q-th residual. Finally, constraints (7.21f) ensure that there are at most q residuals less than or equal to the q-th one. Note that formulation (7.21) is non-linear due to the product of binary and non-negative variables in both the objective function (7.22a) and constraints (7.21d). Therefore, a first efficient model to calculate the LQS estimator can be obtained by linearizing these products via McCormick linearization, by introducing two non-negative variables $\eta_i = z_i s_i \geq 0$ and $\xi_i = \delta_i s_i \geq 0$, $i \in N$, giving rise to the following MILP formulation:

$$
\begin{aligned}
\hat{\beta}_1^{(\mathsf{LQS})} := \arg\min_{\beta} \; & \sum_{i=1}^{n} \xi_i, && && \text{(7.22a)}\\
\text{s.t. } & (7.21b) - (7.21e), (7.21g) - (7.21i), && && \text{(7.22b)}\\
& \eta_i \leq \sum_{i=1}^{n} \xi_i, && i \in N, && \text{(7.22c)}\\
& \eta_i \leq M_1 z_i, && i \in N, && \text{(7.22d)}\\
& \eta_i \leq s_i, && i \in N, && \text{(7.22e)}\\
& \eta_i \geq s_i - M_1(1 - z_i), && i \in N, && \text{(7.22f)}\\
& \xi_i \leq M_2 \delta_i, && i \in N, && \text{(7.22g)}\\
& \xi_i \leq s_i, && i \in N, && \text{(7.22h)}\\
& \xi_i \geq s_i - M_2(1 - \delta_i), && i \in N, && \text{(7.22i)}\\
& \eta_i, \xi_i \geq 0, && i \in N, && \text{(7.22j)}
\end{aligned}
$$

where M_1, M_2 are constants big enough.

Besides, one can further simplify formulation (7.21) by considering a continuous variable, namely $\omega \geq 0$, which will represent the q-th absolute residual value instead of representing it by means of binary variables. This rationale is already present in the work of Babat et al. (2018)

for the computation of the Value-at-Risk measure. To this end, consider the binary variables $z_i \in \{0,1\}$, which equal one if and only if the i-th absolute residual is less or equal than ω, i.e., $s_i > \omega$, and 0 if the i-th absolute residual is $s_i \leq \omega$. Then, an LQS estimator which does not include variable products is obtained as follows:

$$\hat{\beta}_2^{(\mathsf{LQS})} := \arg\min_{\beta} \omega, \tag{7.23a}$$
$$\text{s.t. } r_i^+ - r_i^- = y_i - \mathbf{x}_i^\top \beta, \quad i \in N, \tag{7.23b}$$
$$r_i^+ + r_i^- = s_i, \quad i \in N, \tag{7.23c}$$
$$\sum_{i=1}^n z_i \leq n - q, \tag{7.23d}$$
$$s_i \leq \omega + M z_i, \quad i \in N, \tag{7.23e}$$
$$r_i^+, r_i^-, s_i \geq 0, \quad i \in N, \tag{7.23f}$$
$$\omega \geq 0, \tag{7.23g}$$
$$z_i \in \{0,1\}, \quad i \in N, \tag{7.23h}$$

where M is a sufficiently large constant. It follows that $s_{(q)} \leq \omega$ if the number of residuals satisfying $s_i \geq s_{(q)}$ is at most $n-q$, a condition enforced by constraint (7.23d). Additionally, constraints (7.23e) ensure that one correctly counts the absolute residuals that are less than or equal to ω. Specifically, if $z_i = 0$, then $s_i \leq \omega$, whereas if $z_i = 1$, one obtains $s_i > \omega$ for a sufficiently large constant M. For instance, if it is imposed that $z_i = 0$, $i \in N$, constraint (16d) enforces $n - q = 0$, which implies $n = q$. This scenario corresponds to the problem of identifying estimators that minimize the maximum absolute residual value, denoted as $|r_{(max)}|$.

One can derive valid bounds for the big-M constant required in the formulation above. Assume that upper and lower bounds, β_j^+ and β_j^-, are given for the β coefficients in the model. Specifically, if β_0 denotes the solution obtained by a regression model, then for $\bar{M} = \eta\|\beta_0\|_\infty$, for $\eta \in [1,2]$ (say), the solution found within a box of diameter $2\bar{M}$ centered at β_0 is valid for the optimal solution of the LQS problem. Then, for a given β_0 one can take $\bar{M} = \eta\|\beta_0\|_\infty$ for $\eta \in [1,2]$. Hence, from the condition $\|\beta - \beta_0\|_\infty \leq 2\bar{M}$, one can obtain:

$$\beta_j^- := -2\bar{M} - |\beta_{0_j}| \leq \beta_j \leq \beta_j^+ := 2\bar{M} + |\beta_{0_j}|, \; j \in N. \tag{7.24}$$

Next, recall that s_i stands for the i-th absolute value residual. That is, $|y_i - \beta^t X_i|$. Let us denote by $P_i = \{j : x_{ij} > 0\}$ and $N_i = \{j : x_{ij} < 0\}$. Therefore,

$$\begin{aligned} s_i \leq & |y_i - \sum_{j \in N} \beta_j x_{ij}| \\ \leq & |y_i| + |\sum_{j \in P_i} \beta_j x_{ij}| + |\sum_{j \in N_i} \beta_j x_{ij}| \\ \leq & |y_i| + |\sum_{j \in P_i} \beta_j^+ x_{ij}| + |\sum_{j \in N_i} \beta_j^- x_{ij}|. \end{aligned}$$

Hence, a valid upper bound for M in the constraint (7.23e) is:

$$M_i = |y_i| + |\sum_{j \in P_i} \beta_j^+ x_{ij}| + |\sum_{j \in N_i} \beta_j^- x_{ij}|, \; i \in N. \tag{7.25}$$

7.2.2 k-sum formulation

In addition to the results reviewed in the introductory section, the authors in Bertsimas & Mazumder (2014) provide a decomposition of the q-th ordered absolute residual based on the

k-sum operator. To this end, let $x_{ord} = \left(x_{(1)}, \ldots, x_{(n)}\right)$ be the vector with the components of $x \in \mathbb{R}^n$ sorted in non-decreasing order, i.e., $x_{(1)} \leq \cdots \leq x_{(n)}$. Denote by

$$S_k(x) = \sum_{\ell=k}^{n} x_{(\ell)}, \tag{7.26}$$

the k-sum operator, then the q-th ordered absolute residual can be written as

$$|r_{(q)}| = |y_{(q)} - \mathbf{x}_{(q)}^\top \beta| = \sum_{i=q}^{n} |y_{(i)} - \mathbf{x}_{(i)}^\top \beta| - \sum_{i=q+1}^{n} |y_{(i)} - \mathbf{x}_{(i)}^\top \beta| = S_q(r) - S_{q+1}(r). \tag{7.27}$$

Computing (7.26) is equivalent to solve the following linear problem or its dual,

$$\left.\begin{array}{ll} \max & \sum_{i=1}^n d_i x_i, \\ \text{s.t.} & \sum_{i=1}^n d_i = n-k+1, \\ & 0 \leq d_i \leq 1, i \in N. \end{array}\right\} = \left\{\begin{array}{ll} \min & (n-k+1)t + \sum_{i=1}^n v_i, \\ \text{s.t.} & t + v_i \geq x_i, \quad i \in N, \\ & t \in \mathbb{R}, v_i \geq 0, \ i \in N. \end{array}\right. \tag{7.28}$$

Note that, the primal problem in the left side is a continuous knapsack problem for which variables $d_i \in [0,1]$ can be considered discrete, i.e., $d_i \in \{0,1\}$. In addition, if $x_i \geq 0$ for all $i \in N$, then the variable t can be considered non-negative, namely $t \geq 0$. Therefore, given (7.27) and these primal-dual relationships of the k-sum operator in (7.28), LQS estimators are obtained as a result of the following model

$$\hat{\beta}_3^{(\mathsf{LQS})} := \arg\min_{\beta} \ (n-q+1)t + \sum_{i=1}^{n} v_i - \sum_{i=1}^{n} d_i s_i, \tag{7.29a}$$

$$\text{s.t. } r_i^+ - r_i^- = y_i - \mathbf{x}_i^\top \beta, \qquad i \in N, \tag{7.29b}$$

$$r_i^+ + r_i^- = s_i, \qquad i \in N, \tag{7.29c}$$

$$t + v_i \geq r_i^+ + r_i^-, \qquad i \in N, \tag{7.29d}$$

$$\sum_{i=1}^{n} d_i = n - q, \qquad i \in N, \tag{7.29e}$$

$$t, v_i \geq 0, \qquad i \in N, \tag{7.29f}$$

$$r_i^+, r_i^-, s_i \geq 0, \qquad i \in N, \tag{7.29g}$$

$$(r_i^+, r_i^-) : \mathsf{SOS1}, \qquad i \in N, \tag{7.29h}$$

$$d_i \in \{0,1\}, \qquad i \in N. \tag{7.29i}$$

Note that this formulation is also non-linear because of the terms $d_i s_i$ for $i \in N$, which can be easily linearized by means of McCormick envelopes. Proposition 7.2.2 proves the validity of this formulation to compute the LQS estimator. Furthermore, Proposition 7.2.3 introduces a valid inequality to improve lower bounds for model (7.29).

Proposition 7.2.2 LQS problem (7.7) is equivalent to Problem (7.29) in the sense that the estimators $\hat{\beta}_3^{(\mathsf{LQS})}$ obtained by solving (7.29) also serve as solutions to the LQS problem (7.7).

Proof. Let $f(t,v)$ be the objective function of the problem $S_q^{dual}(r)$ and $g(d)$ the objective function of the problem $S_{q+1}^{primal}(r)$ where

$$
\begin{aligned}
S_q^{dual}(r) : \min\ & (n-q+1)t + \sum_{i=1}^{n} v_i \\
\text{s.t. } & t + v_i \geq |r_i|, && i \in N, \\
& t, v_i \geq 0, && i \in N,
\end{aligned}
\qquad
\begin{aligned}
S_{q+1}^{primal}(r) : \max\ & \sum_{i=1}^{n} d_i |r_i| \\
\text{s.t. } & \sum_{i=1}^{n} d_i = n-q, \\
& d_i \in [0,1], i \in N.
\end{aligned}
$$

As a consequence of equivalence (7.27), it follows that

$$\min_{t,v} f(t,v) - \max_{d} g(d) = \min_{t,v} f(t,v) + \min_{d}(-g(d)) = \min_{t,v,d}(f(t,v) - g(d)) = \min |r_{(q)}| \tag{7.30}$$

■

Proposition 7.2.3 The following inequality

$$(n-q)t + \sum_{i=1}^{n} v_i \geq \sum_{i=1}^{n} d_i s_i \tag{7.31}$$

is valid for the model (7.29).

Proof. In order to prove the validity of the inequality, the meaning of the optimal dual variables of problem $S_q^{primal}(r)$, namely t^*, v_i^*, $i \in N$, is first explained. To this end, let $|r_{(i)}^*|$, $i \in N$, denote the i-*th* optimal absolute residual values. It is well known from knapsack theory that the following algorithm produces optimal primal and dual solutions. Let q be the index of the *critical item* of the knapsack problem $S_q^{primal}(r)$, the optimal primal solution is

$$d_i^* = \begin{cases} 1 & \text{if } i > q, \\ (n-q+1) - \sum_{i>q} 1 & \text{if } i = q, \\ 0 & \text{otherwise.} \end{cases} \tag{7.32}$$

Therefore,

$$(n-q+1)t^* + \sum_{i=1}^{n} v_i^* = \sum_{i=1}^{n} |r_i^*| d_i^* = (n-q+1)|r^*(q)| + \sum_{i>q}(|r_i^*| - |r_{(q)}^*|) \tag{7.33}$$

and as a result, the optimal dual variables assume values

$$
\begin{aligned}
t^* &= |r_{(q)}^*|, \\
v_i^* &= \begin{cases} |r_i^*| - |r_{(q)}^*| & \text{if } i > q, \\ 0 & \text{otherwise.} \end{cases}
\end{aligned}
\tag{7.34}
$$

Given this, it follows that, in the optimum,

$$
\begin{aligned}
S_{q+1}(r) = S_q(r) - |r_{(q)}| &= (n-q+1)t + \sum_{i=1}^{n} v_i - |r_{(q)}| \\
&= (n-q)t + \sum_{i=1}^{n} v_i \geq S_q(r) = \sum_{i=1}^{n} d_i |r_i|
\end{aligned}
\tag{7.35}
$$

which proves the validity of the inequality. ■

Note that this formulation is also non-linear because of the terms $d_i s_i$, $i \in N$, which can be easily linearized by means of McCormick envelopes by considering a non-negative variable $\eta_i = d_i s_i \geq 0$, $i \in N$, giving rise to the following MILP formulation:

$$\begin{aligned}
\hat{\beta}_3^{(\mathsf{LQS})} := \arg\min_{\beta}\ & (n-q+1)t + \sum_{i=1}^{n} v_i - \sum_{i=1}^{n} \eta_i, && && (7.36a)\\
\text{s.t. }\ & (7.29b) - (7.29i), && && (7.36b)\\
& \eta_i \leq M d_i, && i \in N, && (7.36c)\\
& \eta_i \leq s_i, && i \in N, && (7.36d)\\
& \eta_i \geq s_i - M(1-d_i), && i \in N, && (7.36e)\\
& \eta_i \in \{0,1\}, && i \in N. && (7.36f)
\end{aligned}$$

where M is a constant big enough.

7.3 LQS bilevel optimization approaches

In this section, different novel bilevel optimization approaches for the LQS problem are introduced. To this end, recall first that the sample τ-quantile can be obtained by solving the minimization problem

$$\widehat{q}_\tau = \arg\min_{\gamma\in\mathbb{R}} \sum_{i=1}^{n} \rho_\tau(y_i - \gamma) = \arg\min_{\gamma\in\mathbb{R}} \mathcal{L}_\tau(y,\gamma), \tag{7.37}$$

where

$$\mathcal{L}_\tau(y,\gamma) = (\tau-1)\sum_{y_i<\gamma}(y_i-\gamma) + \tau\sum_{y_i\geq\gamma}(y_i-\gamma). \tag{7.38}$$

This $\mathcal{L}_\tau$ is the sample τ-quantile function for $y = (y_1, .., y_n)$, already reviewed in Section 2.6.5 of Chapter 2 and Section 3.2 of Chapter 3. As commented in those sections, quantiles given by (7.37) can be computed solving an optimization problem, considering two non-negative variables $u_i, v_i \geq 0$, $i \in N$, namely,

$$\begin{aligned}
\min_{\gamma\in\mathbb{R}}\ & \tau\sum_{i=1}^{n} u_i + (1-\tau)\sum_{i=1}^{n} v_i, && && (7.39a)\\
\text{s.t. }\ & \gamma - y_i \leq u_i, && i \in N, && (7.39b)\\
& y_i - \gamma \leq v_i, && i \in N, && (7.39c)\\
& u_i, v_i \geq 0, && i \in N. && (7.39d)
\end{aligned}$$

The bilevel approaches to the LQS problem in this section rely on the following observation.

Lema 7.3.1 Given $\tau \in [0,1]$, the equivalence

$$\min_{\beta} s_{(q)} = \min_{\beta}\left(\min_{\gamma} \mathcal{L}_\tau(s,\gamma)\right), \tag{7.40}$$

holds for $q = \lceil \tau n \rceil$, where $s_i = |r_i|$, $i \in N$.

Proof. In order to provide the equivalence above, one only has to show that given the values of the absolute residuals $\bar{s} = (s_i)_{i\in N}$ fixed, then the solution to the inner minimization problem in (7.40) is equivalent to the q-th sorted absolute residual value for $q = \lceil \tau n \rceil$. To this end, consider the ordered sequence of unique and different absolute residual values

$$0 \leq \bar{s}_{(1)} < \ldots < \bar{s}_{(i)} < \bar{s}_{(i+1)} < \ldots < \bar{s}_{(m)},$$

where $m \leq n$, such that for all $i \in \{1, ..., m\}$ there exists a $\ell \in N$ for which $s_\ell = \bar{s}_{(i)}$. Define

$$n_i := \sum_{\bar{s}_j \leq \bar{s}_{(i)}} 1 = \sum_{j=1}^{n} \mathbb{I}_{\{\bar{s}_j \leq \bar{s}_{(i)}\}}. \tag{7.41}$$

Then, for all $i \in \{1, ..., m-1\}$, it follows that

$$\begin{aligned}\mathcal{L}_\tau(\bar{s}, \gamma) &= (\tau - 1) \sum_{\bar{s}_j \leq \bar{s}_{(i)}} (\bar{s}_j - \gamma) + \tau \sum_{\bar{s}_j \geq \bar{s}_{(i+1)}} (\bar{s}_i - \gamma) \\ &= (\tau - 1) \sum_{\bar{s}_j \leq \bar{s}_{(i)}} \bar{s}_j + \tau \sum_{\bar{s}_j \geq \bar{s}_{(i+1)}} \bar{s}_j + \gamma(n_i - \tau n).\end{aligned}$$

For fixed $\bar{s}$, $\mathcal{L}_\tau(\bar{s}, \gamma)$ is a piecewise linear function of γ with a breakpoint at $\bar{s}_{(i)}$. Therefore, it can observed that, for all $i \in \{1, ..., m\}$, $\mathcal{L}_\tau(\bar{s}, \gamma)$ is decreasing if $n_i \leq \tau n$, and increasing if $n_i \geq \tau n$. Hence, $\mathcal{L}_\tau(\bar{s}, \gamma)$ has only one minimum in $[0, +\infty)$ which corresponds to the absolute residue at position $q = \lceil \tau n \rceil$. ■

Note that the right-hand side part of the equivalence in (7.40) has a bilevel problem structure. Based on this, a first bilevel approach is introduced where the upper-level problem sets the regression coefficients and the lower-level calls for the minimization of the sample quantile function (7.38), this is:

$$\begin{aligned}
\hat{\beta}_4^{(\mathsf{LQS})} := \arg\min_{\beta} \;& \gamma, && && (7.42a)\\
\text{s.t. } & r_i^+ - r_i^- = y_i - \mathbf{x}_i^\top \beta, && i \in N, && (7.42b)\\
& r_i^+ + r_i^- = s_i, && i \in N, && (7.42c)\\
& r_i^+, r_i^-, s_i \geq 0, && i \in N, && (7.42d)\\
& (r_i^+, r_i^-) : \mathsf{SOS1}, && i \in N, && (7.42e)\\
& \gamma = \arg\min_{\gamma} \; \mathcal{L}_\tau(s, \gamma). && && (7.42f)
\end{aligned}$$

A single-level reformulation of (7.42) is provided in Proposition 7.3.2.

Proposition 7.3.2 A solution to the LQS problem is given by

$$\begin{aligned}
\hat{\beta}_4^{(\mathsf{LQS})} := \arg\min_{\beta} \;& \gamma, && && (7.43a)\\
\text{s.t. } & r_i^+ - r_i^- = y_i - \mathbf{x}_i^\top \beta, && i \in N, && (7.43b)\\
& r_i^+ + r_i^- = s_i, && i \in N, && (7.43c)\\
& \sum_{j=1}^{n} \mathbb{I}_{\{s_j \leq \gamma\}} \geq \lceil \tau n \rceil, && && (7.43d)\\
& r_i^+, r_i^-, s_i \geq 0, && i \in N, && (7.43e)\\
& (r_i^+, r_i^-) : \mathsf{SOS1}, && i \in N, && (7.43f)\\
& \gamma \geq 0. && && (7.43g)
\end{aligned}$$

Proof. Constraint (7.43d) is obtained as a consequence of the result in Lemma 7.3.1. Given expression (7.41) and that the inner minimization problem in (7.40) is equivalent to the $\lceil \tau n \rceil$-th sorted absolute residual value, i.e., $s_{(\lceil \tau n \rceil)}$, the equivalence

$$\min \mathcal{L}_\tau(\bar{s}, \gamma) = \min \gamma, \text{ s.t.: } \sum_{j=1}^{n} \mathbb{I}_{\{s_j \leq \gamma\}} \geq \lceil \tau n \rceil, \tag{7.44}$$

holds, which proves that adding condition (7.43d) is equivalent to solving (7.42f). ∎

Note that the reformulation in Proposition 7.3.2 is non-linear because of constraints (7.43d), which involve the sum of the indicator functions $\mathbb{I}_{\{s_i \leq \gamma\}}$, $i \in N$. This set of constraints can be modeled by means of a set of binary variables $\kappa_i \in \{0, 1\}$, $i \in N$, which is 1 if $s_i \leq \gamma$, and 0 otherwise. Nevertheless, the indicator description of these constraint is kept since modern state-of-the-art general-purpose solvers allow to implement indicator functions directly.

A second bilevel approach is introduced where the lower-level calls for the minimization of the sample τ-quantile function expressed as the optimization problem in (7.39), as follows:

$$
\begin{aligned}
\hat{\beta}^{(\mathsf{LQS})} := \arg\min_{\beta} \ & \gamma, && && (7.45a)\\
\text{s.t. } & r_i^+ - r_i^- = y_i - \mathbf{x}_i^\top \beta, && i \in N, && (7.45b)\\
& r_i^+ + r_i^- = s_i, && i \in N, && (7.45c)\\
& s_i, r_i^+, r_i^- \geq 0, && i \in N, && (7.45d)\\
& (r_i^+, r_i^-) : \mathsf{SOS1}, && i \in N, && (7.45e)\\
& \gamma = \arg\min_{\gamma \geq 0} \ (1-\tau)\sum_{i=1}^n u_i + \tau \sum_{i=1}^n v_i, && && (7.45f)\\
& \qquad \text{s.t. } \gamma - s_i \leq u_i, && i \in N, && (7.45g)\\
& \qquad\qquad s_i - \gamma \leq v_i, && i \in N, && (7.45h)\\
& \qquad\qquad u_i, v_i \geq 0, && i \in N. && (7.45i)
\end{aligned}
$$

Since the lower-level problem (7.45f)-(7.45i) is a LP, one can apply the strong duality to provide a single-level reformulation, which is presented in Proposition 7.3.3.

Proposition 7.3.3 A solution to the LQS problem is given by

$$
\begin{aligned}
\hat{\beta}^{(\mathsf{LQS})} := \arg\min_{\beta} \ & \gamma, && && (7.46a)\\
\text{s.t. } & r_i^+ - r_i^- = y_i - \mathbf{x}_i^\top \beta, && i \in N, && (7.46b)\\
& r_i^+ + r_i^- = s_i, && i \in N, && (7.46c)\\
& (1-\tau)\sum_{i=1}^n u_i + \tau \sum_{i=1}^n v_i, = -\sum_{i=1}^n s_i\alpha_i + \sum_{i=1}^n s_i\beta_i, && && (7.46d)\\
& \gamma - s_i \leq u_i, && i \in N, && (7.46e)\\
& s_i - \gamma \leq v_i, && i \in N, && (7.46f)\\
& \alpha_i \leq 1 - \tau, && i \in N, && (7.46g)\\
& \beta_i \leq \tau, && i \in N, && (7.46h)\\
& \sum_{i=1}^n \alpha_i - \sum_{i=1}^n \beta_i \leq 0, && && (7.46i)\\
& s_i, r_i^+, r_i^- \geq 0, && i \in N, && (7.46j)\\
& (r_i^+, r_i^-) : \mathsf{SOS1}, && i \in N, && (7.46k)\\
& u_i, v_i \geq 0, && i \in N, && (7.46l)\\
& \alpha_i, \beta_i \geq 0, && i \in N, && (7.46m)\\
& \gamma \geq 0. && && (7.46n)
\end{aligned}
$$

Proof. Given the variables $\bar{s}_i \geq 0, i \in N$, fixed in the upper-level problem, the dual problem of the lower-level problem (7.45f)-(7.45i) is given by

$$\begin{aligned}
\min_{\beta} \quad & -\sum_{i=1}^{n} \bar{s}_i \alpha_i + \sum_{i=1}^{n} \bar{s}_i \beta_i, && && (7.47a)\\
\text{s.t.} \quad & \alpha_i \leq 1-\tau, && i \in N, && (7.47b)\\
& \beta_i \leq \tau, && i \in N, && (7.47c)\\
& \sum_{i=1}^{n} \alpha_i - \sum_{i=1}^{n} \beta_i \leq 0, && i \in N, && (7.47d)\\
& \alpha_i, \beta_i \geq 0, && i \in N. && (7.47e)
\end{aligned}$$

Therefore, problem (7.46) is obtained as single-level reformulation of (7.45) by applying strong-duality theorem, which is given by the constraints of the upper-level problem and primal and dual feasibility and the equality of the primal and dual objective functions of the lower-level problems. ■

Note that formulation (7.46) is also non-linear because of the products of the non-negative variables representing the absolute value of the residuals and the non-negative dual variables, which cannot be linearized via McCormick envelopes. For that reason, although the model has methodological interest, it is clearly outperformed by the previous one and is of limited applicability from a computational perspective. In spite of that, the reader may note that one can always initialize the formulation with the values obtained from different β estimators, as for instance the one obtained from standard LS regression. Once the initial β is known r^+, r^- and s are easily computed from the dataset and then, the remaining variables can be estimated solving an easy feasibility linear program.

7.4 Computational study

This section summarizes the results obtained from the computational experiments performed in order to empirically compare the proposed LQS formulations. The performance of the formulation to obtain $\widehat{\beta}_{\mathsf{BM}}$ estimators is compared with the formulations for $\widehat{\beta}_1$, $\widehat{\beta}_2$, $\widehat{\beta}_3$ and $\widehat{\beta}_4$ estimators. Notation will be overused by calling the formulation that computes the $\beta_{(.)}$ estimator by $\beta_{(.)}$. Our formulations were programmed in Python language (version 3.10) using Gurobi Optimizer 9.5 as a MIP solver. The experiments were run in a MacPro server with a 2,7 GHz Intel Xeon W processor of 24 cores and 192 GB RAM, using 8 of those threads per each run. The time limit for computations was set to 7200 seconds per instance. Following the computational experiments in Bertsimas & Mazumder (2014), the formulations are tested in two set of instances: real-world datasets and synthetic datasets.

Regarding real-world datasets, two set of instances available from the R-package `robustbase` have been considered (Todorov & Filzmoser, 2009; Maechler et al., 2024). First, the `Alcohol` dataset, which is aimed to study the solubility of alcohols in water to understand alcohol transport in living organisms. This dataset contains physicochemical characteristics of $n = 44$ aliphatic alcohols and measurements on seven numeric variables: SAG solvent accessible surface-bounded molecular volume ($\mathtt{x}_1$), logarithm of the octanol-water partitions coefficient ($\mathtt{x}_2$), polarizability ($\mathtt{x}_3$), molar refractivity ($\mathtt{x}_4$), mass ($\mathtt{x}_5$), volume ($\mathtt{x}_6$) and the response ($\mathtt{y}$) is taken to be the logarithm of the solubility. We consider two cases from the dataset, a first one with $n = 44$ and $p = 5$ where the five covariates were $\mathtt{x}_1, \mathtt{x}_2, \mathtt{x}_4, \mathtt{x}_5, \mathtt{x}_6$, and a second that has all the six covariates and an intercept term, which leads to $p = 7$. In this example, the LQS estimator was computed for $q = 31$. The `HBK` dataset created by Hawkins et al. (1984) is also considered in this experimentation set-up. The dataset consists of 75 observations with one response and three explanatory variables, that is, $n = 75$ and $p = 3$. In this example, the LQS estimator was computed $q \in \{60, 45\}$.

Table 7.2 shows results from real-world datasets. The performance of each formulation is evaluated for every instance using four metrics: `Time` (time to compute the best upper and lower bounds), `UB` (best upper bound), `LB` (best lower bound), and `TimeUB` (time to find the best upper bound). Generally, instances become harder to solve as the number of observations and covariates increases or as the quantile decreases. Across all instances, the formulations showed similar performance overall. The $\widehat{\beta}_4$ formulation performed best in terms of `Time` and `TimeUB`, followed by the $\widehat{\beta}_2$ formulation. In contrast, $\widehat{\beta}_3$ performed worse, especially with larger datasets. It could not certify optimality (i.e., close the gap between the best upper and lower bounds) in any instance. However, it performed well on `TimeUB` for the `Alcohol` dataset.

Estimator	Time	UB	LB	TimeUB	Time	UB	LB	TimeUB
Alcohol	(n = 44, p = 5, q = 31)				(n = 44, p = 7, q = 31)			
$\widehat{\beta}_{BM}$	27.79	0.196	0.196	9.86	729.54	0.156	0.156	51.96
$\widehat{\beta}_1$	15.41	0.196	0.196	2.96	60.29	0.156	0.156	6.57
$\widehat{\beta}_2$	3.20	0.196	0.196	0.25	22.87	0.156	0.156	4.39
$\widehat{\beta}_3$	7200.00	0.196	0.068	2.51	7200.00	0.156	0.046	0.08
$\widehat{\beta}_4$	1.78	0.196	0.196	0.21	13.39	0.156	0.156	0.55
HBK	(n = 75, p = 3, q = 45)				(n = 75, p = 3, q = 60)			
$\widehat{\beta}_{BM}$	55.93	0.585	0.585	48.59	1.01	0.819	0.819	0.91
$\widehat{\beta}_1$	1874.29	0.585	0.585	1.43	21.92	0.819	0.819	2.13
$\widehat{\beta}_2$	30.73	0.585	0.585	26.80	0.44	0.819	0.819	0.43
$\widehat{\beta}_3$	7200.00	0.612	0.054	7200.00	7200.00	0.819	0.107	5.94
$\widehat{\beta}_4$	4.79	0.585	0.585	0.37	0.33	0.819	0.819	0.13

Table 7.2: Real-world datasets results

Regarding synthetic datasets, following Rousseeuw & Van Driessen (2006), the model matrix $\mathbf{X}_{n\times p}$ was generated with i.i.d. Gaussian entries $N(0, 100)$ and $\beta \in \mathbb{R}^p$ was taken to be a vector of all ones. Subsequently, the response was generated as $\mathbf{y} = \mathbf{X}\beta + \varepsilon$, where $\varepsilon_i \sim N(0, 10)$, $i = N$. Once $(\mathbf{y}, \mathbf{X})$ have been generated, a certain proportion π of the samples is corrupted in two different ways:

A) $\lfloor \pi n \rfloor$ of the samples are chosen at random and the first coordinate of the data matrix $\mathbf{X}$, that is, x_{1j} is replaced by $x_{1j} \leftarrow x_{1j} + 1000$, hence outliers are only in the covariate space.
B) $\lfloor \pi n \rfloor$ of the samples are chosen at random out of which the covariates of half of the points are changed as in item (A). For the remaining half of the points the responses are corrupted as $y_j \leftarrow y_j + 1000$, so outliers are added in both the covariate and response spaces.

This provides two datasets in which outliers are artificially generated with outliers only in covariate space (type A) and with outliers in both covariate and response space (type B), which are more difficult to solve. Following Bertsimas & Mazumder (2014) experimentation set-up, to compare the different models in terms of the quality of solutions obtained it was proceeded as follows. For every instance, all formulations are run within the time limit and the best solution $\mathtt{f}_*$ among them is obtained. If $\mathtt{f}_m$ denotes the value of the LQS objective function for formulation "m", then the relative accuracy of the solution obtained by "m" is defined as

$$\texttt{Relative Accuracy} = (\mathtt{f}_m - \mathtt{f}_*)/\mathtt{f}_* \times 100.$$

Tables (7.3), (7.4) and (7.5), show the performance of the formulations in instances of 50, 75 and 101 data points, respectively. For every $(n, p, q, \pi, \text{perturbation type})$ combination, the column `Accuracy` accounts for the `Relative Accuracy` and the numbers within parenthesis denote the standard errors. Results are averaged over 20 different random instances of the problem. The tables have been arranged so that the left column shows the results for instances with type A perturbation and in the right column with type B. Similarly, Figures 7.1 and 7.2 depict average times and relative accuracy percentages for the different data sizes. Figures concerning averages times depict both, the time elapsed by the formulation for the computation of the best upper bound (green) and the time elapsed to finish solving the instance (yellow). Therefore, the yellow stripes identify the time that the formulations spend on average to close the gap by making the effort to raise the lower bound.

One can see that, for instances with 50 and 75 data points, the best-performing formulations are $\widehat{\beta}_2$ and $\widehat{\beta}_4$. Formulation $\widehat{\beta}_2$ performs better for type A perturbations, while $\widehat{\beta}_4$ excels for type B. For instances with 101 data points, $\widehat{\beta}_4$ is the top performer, particularly in large instances where it can certify optimality within the time limit, unlike the other formulations. It is followed by $\widehat{\beta}_2$. The only scenario where $\widehat{\beta}_{\mathsf{BM}}$ outperforms $\widehat{\beta}_4$ and $\widehat{\beta}_2$ in terms of `Time` is with the parameter combination $(75, 5, 40, 0.4, A)$. However, in this case, $\widehat{\beta}_4$ and $\widehat{\beta}_2$ compute the best upper bound much faster, outperforming $\widehat{\beta}_{\mathsf{BM}}$ in terms of `TimeUB`. Formulation $\widehat{\beta}_1$ occasionally outperforms $\widehat{\beta}_2$ and $\widehat{\beta}_4$ for `TimeUB`, such as in $(75, 10, 25, 0.5, B)$ instances. Meanwhile, $\widehat{\beta}_3$ consistently shows the worst performance across all metrics.

Estimator	Time	Accuracy	TimeUB	Time	Accuracy	TimeUB
	50 - 5 - 15 - 0.4 - A			50 - 5 - 15 - 0.4 - B		
$\widehat{\beta}_{\mathsf{BM}}$	112.74	0.00 (0.00)	65.89	121.52	0.00 (0.00)	73.8
$\widehat{\beta}_1$	285.01	0.00 (0.00)	81.16	274.37	0.00 (0.00)	109.61
$\widehat{\beta}_2$	79.62	0.00 (0.00)	23.04	146.63	0.00 (0.00)	85.05
$\widehat{\beta}_3$	498.58	0.00 (0.00)	139.35	1184.47	0.00 (0.00)	337.51
$\widehat{\beta}_4$	98.63	0.00 (0.00)	30.24	117.10	0.00 (0.00)	42.69
	50 - 5 - 30 - 0.4 - A			50 - 5 - 30 - 0.4 - B		
$\widehat{\beta}_{\mathsf{BM}}$	8.98	0.00 (0.00)	8.11	210.51	0.00 (0.00)	52.47
$\widehat{\beta}_1$	23.89	0.00 (0.00)	12.33	66.53	0.00 (0.00)	15.44
$\widehat{\beta}_2$	5.19	0.00 (0.00)	0.92	30.96	0.00 (0.00)	1.41
$\widehat{\beta}_3$	6810.73	0.00 (0.00)	107.92	7200.00	0.04 (0.04)	710.26
$\widehat{\beta}_4$	8.13	0.00 (0.00)	4.72	9.17	0.00 (0.00)	1.11
	50 - 10 - 15 - 0.5 - A			50 - 10 - 15 - 0.5 - B		
$\widehat{\beta}_{\mathsf{BM}}$	7200.00	55.94 (12.98)	6098.87	7200.00	68.47 (16.08)	6879.45
$\widehat{\beta}_1$	7200.00	59.68 (13.04)	5244.64	7200.00	96.32 (20.10)	6809.62
$\widehat{\beta}_2$	7200.00	32.57 (9.93)	5265.93	7200.00	44.14 (12.12)	5843.09
$\widehat{\beta}_3$	7200.00	63.08 (11.97)	7008.88	7200.00	56.30 (13.78)	6996.63
$\widehat{\beta}_4$	7200.00	39.74 (10.13)	5966.24	7200.00	34.91 (8.86)	6232.40
	50 - 10 - 30 - 0.5 - A			50 - 10 - 30 - 0.5 - B		
$\widehat{\beta}_{\mathsf{BM}}$	6309.04	4.07 (2.14)	4223.32	7139.45	0.49 (0.29)	2719.85
$\widehat{\beta}_1$	7148.67	2.14 (1.21)	3113.89	6813.67	0.00 (0.00)	1797.70
$\widehat{\beta}_2$	5315.74	0.64 (0.64)	1722.19	6504.35	0.35 (0.24)	1919.68
$\widehat{\beta}_3$	7200.00	7.22 (2.19)	5412.28	7200.00	5.47 (1.50)	5135.51
$\widehat{\beta}_4$	5052.41	0.17 (0.14)	1586.56	3055.81	0.00 (0.00)	606.24

Table 7.3: Synthetic data results for n = 50

Estimator	Time	Accuracy	TimeUB	Time	Accuracy	TimeUB
	75 - 5 - 25 - 0.4 - A			75 - 5 - 25 - 0.4 - B		
$\widehat{\beta}_{\mathsf{BM}}$	1633.62	0.00 (0.00)	724.56	4430.51	0.00 (0.00)	1247.38
$\widehat{\beta}_1$	2578.67	0.00 (0.00)	252.49	3913.49	0.00 (0.00)	1533.05
$\widehat{\beta}_2$	1314.71	0.00 (0.00)	451.75	2552.69	0.00 (0.00)	1628.06
$\widehat{\beta}_3$	7200.00	1.55 (0.71)	3587.03	7200.00	3.73 (0.90)	4946.81
$\widehat{\beta}_4$	959.56	0.00 (0.00)	407.40	958.24	0.00 (0.00)	350.15
	75 - 5 - 40 - 0.4 - A			75 - 5 - 40 - 0.4 - B		
$\widehat{\beta}_{\mathsf{BM}}$	104.06	0.00 (0.00)	39.29	7200.00	2.05 (1.14)	3080.55
$\widehat{\beta}_1$	391.33	0.00 (0.00)	25.20	3627.74	0.00 (0.00)	538.61
$\widehat{\beta}_2$	214.89	0.00 (0.00)	1.73	780.33	0.00 (0.00)	254.48
$\widehat{\beta}_3$	7200.00	0.00 (0.00)	1396.72	7200.00	4.40 (1.18)	5630.87
$\widehat{\beta}_4$	166.03	0.00 (0.00)	6.43	373.74	0.00 (0.00)	84.35
	75 - 10 - 25 - 0.5 - A			75 - 10 - 25 - 0.5 - B		
$\widehat{\beta}_{\mathsf{BM}}$	7200.00	60.78 (16.85)	5780.45	7200.00	52.43 (9.33)	6485.96
$\widehat{\beta}_1$	7200.00	95.97 (24.20)	5634.86	7200.00	31.94 (7.90)	5616.51
$\widehat{\beta}_2$	7200.00	41.26 (14.00)	4756.77	7200.00	20.96 (4.77)	6232.54
$\widehat{\beta}_3$	7200.00	117.97 (23.62)	6923.41	7200.00	31.73 (5.58)	6888.15
$\widehat{\beta}_4$	7200.00	54.13 (15.09)	4446.04	7200.00	31.59 (7.01)	6076.58
	75 - 10 - 40 - 0.5 - A			75 - 10 - 40 - 0.5 - B		
$\widehat{\beta}_{\mathsf{BM}}$	7200.00	10.72 (2.06)	6084.50	7200.00	20.17 (3.82)	6606.25
$\widehat{\beta}_1$	7200.00	5.43 (1.41)	4606.60	7200.00	19.45 (3.40)	6663.73
$\widehat{\beta}_2$	7200.00	2.57 (1.18)	3347.07	7200.00	4.92 (1.54)	4485.96
$\widehat{\beta}_3$	7200.00	16.33 (3.17)	6390.42	7200.00	16.13 (3.25)	6490.46
$\widehat{\beta}_4$	7200.00	2.63 (0.97)	3645.66	7200.00	12.85 (2.66)	5632.45

Table 7.4: Synthetic data results for n = 75

Estimator	Time	Accuracy	TimeUB	Time	Accuracy	TimeUB
	101 - 5 - 51 - 0.4 - A			101 - 5 - 51 - 0.4 - B		
$\widehat{\beta}_{\mathsf{BM}}$	2182.81	0.00 (0.00)	866.75	7200.00	1.98 (0.88)	3671.47
$\widehat{\beta}_1$	3443.78	0.00 (0.00)	546.49	7098.26	1.62 (0.66)	4538.83
$\widehat{\beta}_2$	2116.53	0.00 (0.00)	366.63	7200.00	0.16 (0.13)	1518.30
$\widehat{\beta}_3$	7200.00	0.74 (0.51)	3334.87	7200.00	7.84 (1.58)	6169.86
$\widehat{\beta}_4$	1289.68	0.00 (0.00)	21.03	5454.79	0.00 (0.00)	1303.71
	101 - 5 - 75 - 0.4 - A			101 - 5 - 75 - 0.4 - B		
$\widehat{\beta}_{\mathsf{BM}}$	227.67	0.00 (0.00)	118.18	6588.01	0.05 (0.05)	3428.24
$\widehat{\beta}_1$	1353.86	0.00 (0.00)	185.54	2295.74	0.00 (0.00)	665.38
$\widehat{\beta}_2$	477.77	0.00 (0.00)	29.47	4842.17	0.00 (0.00)	407.00
$\widehat{\beta}_3$	7200.00	0.69 (0.24)	3120.06	7200.00	1.44 (0.40)	4762.61
$\widehat{\beta}_4$	273.06	0.00 (0.00)	17.28	467.72	0.00 (0.00)	83.43

Table 7.5: Synthetic data results for n = 101

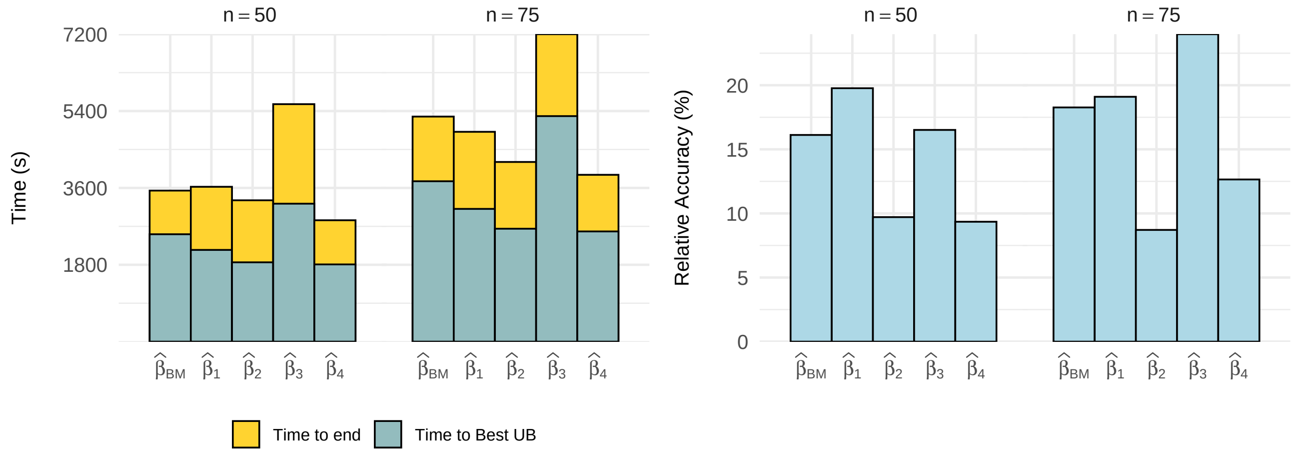


Figure 7.1: Average times and relative accuracy percentages results for n=50,75

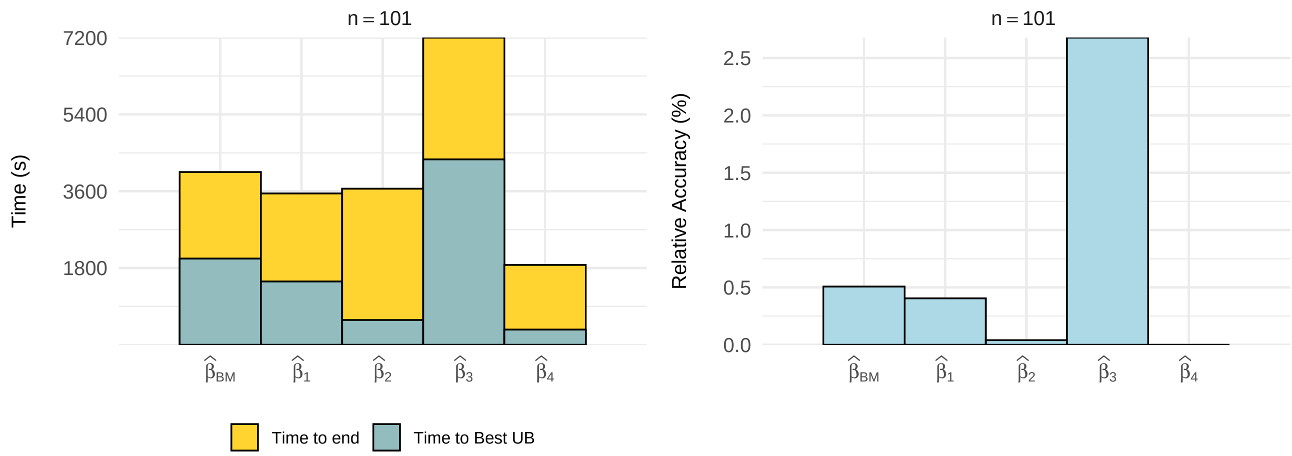


Figure 7.2: Average times and relative accuracy percentages results for n=101

7.5 Scalability to larger datasets

In this last section, the scalability of the LQS problem to larger size instances (up to 10000 points) is addressed by introducing a novel aggregation approach. To this end, denote by X the original input data and by $\hat{X}$ the multiset of the aggregate data. For each $x \in X$, an element $\hat{x} \in \hat{X}$ is associated. Possibly, several elements of X correspond to the same $\hat{x} \in \hat{X}$, so that the cardinality of distinct elements in $\hat{X}$ is much smaller than $|X|$. The goal is to provide some reduction in size result for which the error bound of the quantile regression is bounded and asymptotically converges to zero whenever $\hat{X} \to X$. For the sake of readability, denote $f_q(X, \beta) := |r_{(q)}(X, \beta)|$, the q-th quantile of the absolute residuals computed with respect to the data set X and the hyperplane defined by the vector β. The main result in this section is the following theorem.

Theorem 7.5.1 Let $D = \max_{x \in X} |x - \hat{x}|$ and $\beta^* = \arg\min_\beta f_q(X, \beta)$, $\hat{\beta} = \arg\min_\beta f_q(\hat{X}, \beta)$, the optimal q-quantile hyperplanes with respect to the data sets X and $\hat{X}$, respectively. Then,

$$|f_q(X, \beta^*) - f_q(X, \hat{\beta})| \leq 2D \tag{7.48}$$

Proof. Recall that $r_i(\beta) = y_i - \mathbf{x}_i^\top \beta$ and let $e_X(\beta) = (|r_i(\beta)|)_{i=1}^n$. First of all, it is observed that by the triangle inequality it always holds

$$e_X(\beta) \leq e_{\hat{X}}(\beta) + D\underline{1}^t,$$

being $D = \max_i\{|x_i - \hat{x}_i|\}$. Let $N = N$. Then, since the q-th residual is an isotonic operation

$$\begin{aligned} f_r(X,\beta) = r_{(q)}(e_X(\beta)) \leq & r_{(q)}(e_{\hat{X}}(\beta) + D\underline{1}^t) \\ \leq & \min_{\mathcal{I}\subset N:|\mathcal{I}|=q} \max_{j\in\mathcal{I}}\{|e_{\hat{x}_j}| + D\} \\ \leq & \max_{j\in\mathcal{I}}\{|e_{\hat{x}_j}| + D\},\ \forall \mathcal{I} \subset N : |\mathcal{I}| = q \end{aligned}$$

Thus, $f_q(\hat{X},\beta) \leq f_q(X,\beta) + D$ for all β. Analogously, $f_q(X,\beta) \leq f_q(\hat{X},\beta) + D$ for all β.

Next, the main inequality follows by the following

$$\begin{aligned} f_q(X,\hat{\beta}) \leq f_q(\hat{X},\hat{\beta}) + D \leq f_q(\hat{X},\beta^*) + D \leq f_q(X,\beta^*) + 2D \\ f_q(X,\hat{\beta}) \geq f_q(X,\beta^*) \geq f_q(\hat{X},\beta^*) - D \geq f_q(X,\beta^*) - 2D \end{aligned}$$

Joining both inequalities, one gets:

$$|f_q(X,\beta^*) - f_q(X,\hat{\beta})| \leq 2D.$$

■

Note that whenever $\hat{X} \to X$ the parameter $D \to 0$ and the approximation converges to the actual optimal solution. Therefore, the approach in this section works as follows. The main idea is to aggregate the dataset into a tractable number of points or clusters for the exact formulations previously presented, that is, to reduce the datasets to a size where the formulations can provide a solution in an admissible time limit. The clusters formed must be as homogeneous as possible, i.e., the number of original data points in each cluster must be similar.

Given the above conditions and the result in Theorem 7.5.1, an approximate LQS estimator can be obtained by solving the aggregated dataset by means of an adjusted version of the above formulations where the weight of each cluster is included. For example, let k be the number of clusters considered, N_i the number of data points in cluster i, and C_i represents the indexes belonging to cluster i, then formulation (7.49) can be rewritten modifying constraints (7.23d) and (7.23e) to take into account the weight of each cluster as follows,

$$\sum_{i=1}^{k} N_i z_i \leq n - q, \tag{7.49a}$$

$$s_j \leq \omega + M z_i, \qquad i \in \{1,\ldots,k\}, j \in C_i. \tag{7.49b}$$

Figure 7.3 depicts the performance of this approach, namely `SCALE`, comparing the solutions with those obtained by the `LQS` algorithm from the `R`-package `MASS` (Venables & Ripley, 2002). Following Bertsimas & Mazumder (2014) experimentation set-up for large instances, 20 instances are tested of the type $(2001, 10, 1201, 0.4, B)$, $(5001, 10, 3001, 0.4, B)$ and $(10001, 20, 6001, 0.4, B)$. Each of the original datasets are aggregated to 100 representative data points or clusters, letting this approach run for a time limit of 7200 seconds. It can be seen how the `SCALE` approach provides better solutions than those computed using the `MASS` package, especially, the difference is more remarkable the larger the instance size solved.

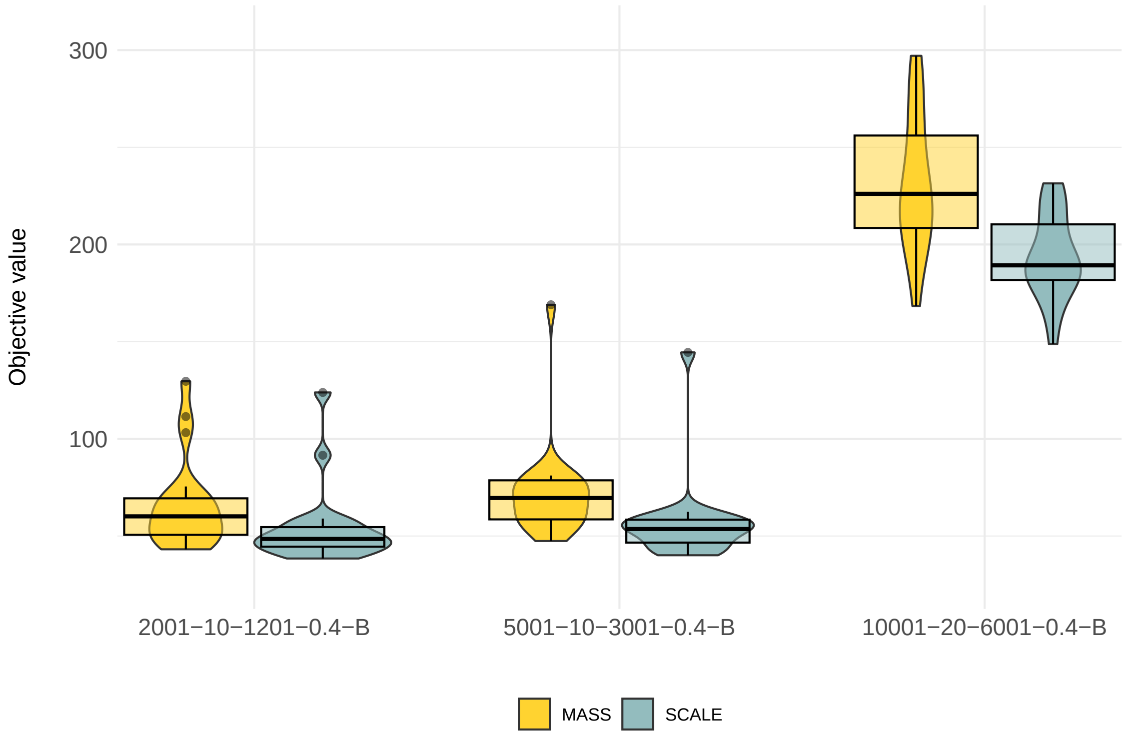


Figure 7.3: Comparison between `LQS` solutions obtained by the `R`-package `MASS` and the `SCALE` approach.

Como las cosas humanas no sean eternas, yendo siempre en declinación de sus principios hasta llegar a su último fin, especialmente las vidas de los hombres, y como la de don Quijote no tuviese privilegio del cielo para detener el curso de la suya, llegó su fin y acabamiento cuando él menos lo pensaba [...]

Miguel de Cervantes Saavedra - *El ingenioso hidalgo Don Quijote de la Mancha. Capítulo LXXIV: De cómo don Quijote cayó malo y del testamento que hizo y su muerte.*

8 Conclusions

In the realm of optimization, the sequence in which decisions are made is not merely a procedural detail, it is a fundamental factor that can dramatically influence both the efficiency of finding a solution and the quality of that solution itself. Ordering approaches are crucial because they provide a strategic framework for navigating vast and complex solution spaces. By moving beyond traditional aggregate measures such as sums or means, order based reasoning enables the explicit modeling of richer and more nuanced preferences, thereby unlocking new paths for both theoretical analysis and practical applications. This dissertation advances this perspective by developing a comprehensive and flexible framework, namely *Ordered Optimization*, in which ordering is embedded directly into mathematical models. The framework demonstrates both theoretical depth and practical utility across a diverse spectrum of optimization problems. The overall contributions of this work can be summarized in two main directions: *generalization* and *particularization*. First, it provides a generalization of multiple approaches that were previously considered independently, showing that diverse models can be expressed as particular cases of a single ordered formulation. Second, it enables particularization, in the sense that having such a powerful modeling tool, one can analyze specific concepts such as fairness or robustness. Together, these contributions demonstrate both the theoretical coherence and the practical versatility of Ordered Optimization as a modeling paradigm.

This dissertation has extended a mathematical optimization-based framework that formulates several different families of measures widely used in Statistics and Operations Research, among other contexts, formulating *linear*, *nested*, and *quadratic ordered measures*. The resulting linear and mixed-integer linear models have been shown to be competitive with respect to specialized software libraries through extensive computational testing. More importantly, this framework successfully integrates these sophisticated statistical measures into the core of the decision-making process, allowing the values being summarized to be variables themselves. This was effectively shown when applying ordered variants of classic problems, including scenario analysis in linear programming, the Traveling Salesman Problem, and the Set Cover Problem. However, given the wide variety of problems to which this framework can be applied, the development of tailored approaches may prove beneficial, as problem-specific adaptations can enhance both computational efficiency and the interpretability of the resulting solutions, giving rise to the subsequent chapters.

A significant focus has been placed on the *Discrete Ordered Median Problem* (DOMP). The analysis and solution techniques for this problem have been advanced through the application of

Benders decomposition approaches. Building from two different state-of-the-art formulations of DOMP, namely the one based on radius variables and the other based on the k-sums operator, new reformulations have been obtained. These methods have proven effective, especially for challenging instances with complex weight vectors, including those with both positive and negative entries. By strengthening bounds and leveraging various implementation enhancements, the Benders decomposition methods have shown performance comparable or superior to existing state-of-the-art approaches for DOMP.

This dissertation also introduces the *Ordered Median Tree Location Problem* (OMT), a complex network design challenge where p facilities must be placed on a network and connected by an undirected tree. Several mathematical formulations are proposed, combining properties of minimum spanning trees with ordered median objectives. These formulations are further strengthened by incorporating the use of radius variables and preprocessing techniques that significantly improve scalability, allowing the solution of larger instances. In addition, alternative modeling strategies and a tailored Benders decomposition algorithm were developed and thoroughly analyzed for this problem. Extensive computational experiments complement the theoretical contributions, providing a detailed comparison of the proposed formulations, enhancements, and solution methods.

The practical relevance of the ordered framework was also validated through its application to two significant, real-world domains: *fairness* and *robustness*.

Regarding fairness concerns, the ordered framework has been successfully applied to the *Capacitated Vehicle Routing Problem* (VRP), moving beyond the traditional objective of minimizing total transportation costs. By incorporating objectives related to the ordering of route lengths, fairness criteria have been addressed. The main contributions in this sense are two. First, the ordered framework is introduced for the first time in the context of vehicle routing, allowing one to model general objective functions. Beyond this, the main technical contribution is a novel methodology to address the route inconsistency problem that arises when using non-monotonic fairness objectives. To the best of our knowledge, this is the first approach that enforces TSP-optimal routing for each individual driver in a provably optimal manner within a fair VRP context, accommodating both monotonic and non-monotonic objective functions. This is achieved through a sophisticated bilevel optimization model. The resulting models generate more balanced solutions without a significant loss in efficiency, offering a flexible and integrative approach that aligns with the practical needs of complex logistics planning.

Finally, when it comes to robustness, algorithms for *Least Quantile of Squares* (LQS) regression problems have been proposed and validated as a particular case of ordered regression. The exact formulations developed can solve small to medium-sized instances to optimality and provide high-quality upper bounds for larger datasets, outperforming state-of-the-art methods in certain ranges. These estimators maintain desirable statistical properties, such as a high breakdown point, and the framework is flexible enough to be extended to more sophisticated forms of robust regression. Furthermore, the validity and scalability of the models have been shown by means of detailed computational experiments which have been conducted to check the performance, scope, and speed of the proposed algorithms and formulations.

In summary, this dissertation has established a powerful and unified paradigm for order-based optimization. The contributions are threefold: (1) the creation of a general and efficient framework for modeling a wide array of ordered measures, (2) the development of advanced, tailored algorithms, including Benders decomposition methods, that solve complex ordered optimization problems to optimality, and (3) the successful application of this framework to well-known optimization problems in the literature, demonstrating its practical value. The findings confirm that by explicitly modeling the order of outcomes, solutions can be created that better align with the multifaceted objectives of complex systems.

8.1 Future lines of research

The research presented in this dissertation opens up several promising lines for future consideration, building upon the theoretical, algorithmic, and practical foundations that have been established. The following sections outline key areas where the concepts of ordered optimization could be extended and refined.

One of the most significant directions for future work is the extension of the ordered framework to other problems in the literature. The current research has demonstrated the validity of ordered approaches for modeling concepts such as undesirability (obnoxious), fairness or robustness. A natural and compelling question arises from this: *What else can be effectively modeled within an ordered framework?* A prominent direction is the application of this paradigm to other classes of measures or indices, such as risk measures in finance or specialized loss functions in machine learning, where the relative ordering of outcomes is a critical component of the evaluation.

The successful integration of fairness and balance criteria into VRP suggests that similar benefits could be realized in other logistics and operational domains, such as scheduling, resource allocation, and network design. In particular, incorporating equity-oriented measures into these contexts has the potential to improve not only the efficiency of resource utilization but also the sustainability and acceptability of the resulting solutions. For instance, in scheduling, fairness can ensure a more balanced distribution of workloads among employees; in resource allocation, it can promote equitable access across competing demands; and in network design, it can help distribute service coverage more evenly across different regions. These extensions highlight the broader relevance of ordered optimization approaches and reinforce the idea that fairness and balance are not merely secondary objectives, but fundamental criteria that can enhance the long-term effectiveness of decision-making processes.

Similarly, the flexibility of the models developed for robust regression, based on ordering absolute residuals, could be leveraged to tackle other challenging problems in statistics and machine learning. This includes developing models for different types of data (e.g., time-series or non-Euclidean data), incorporating more complex structural constraints on regression coefficients (e.g., sparsity or group-level effects), and exploring novel applications in areas like finance, econometrics, and bioinformatics, where robust estimation is critical for drawing reliable conclusions from noisy data.

From an algorithmic perspective, the successful application of Benders decomposition techniques in the location context suggests a rich area for continued exploration. While the proposed methods for the DOMP projected out variables related to the "ordering" of values, there is untapped potential in reformulations that also project out allocation variables. Developing such "thinned-out" models could lead to new, more efficient Benders reformulations and is a compelling direction for future research. Likewise, while the Benders decomposition for the OMT did not uniformly outperform exact formulations in the conducted tests, further refinement of the algorithm could yield substantially improved results, perhaps through more sophisticated cut generation strategies or integration within a branch-and-cut framework. The application of Benders decomposition to other ordered optimization problems remains a promising and largely unexplored territory.

In the same direction, bilevel optimization provides another fertile path for extending the ordered framework. Many real-world decision-making processes are inherently hierarchical, where one set of decisions conditions or restricts another. Embedding ordered objectives within bilevel models could allow for the explicit incorporation of fairness, robustness, or other order-sensitive criteria at different decision levels. This line of research could be particularly impactful in areas such as pricing and tariff design, leader–follower network optimization, or bilevel facility location problems, where the interaction between the upper and lower-level decisions often gives rise to inequities or inefficiencies that ordered measures are well-suited to capture. Moreover, the successful application of bilevel optimization to develop valid and efficient models for the LQS estimator opens a complementary research avenue beyond ordered optimization, demonstrating the broader potential of mathematical programming to address complex estimation and decision-making problems in statistics and other related data science fields.

Furthermore, moving beyond a purely deterministic perspective is a natural next step. Investigating the incorporation of robustness criteria in the face of uncertainty, for instance, by integrating the ordered framework with stochastic programming or robust optimization techniques, would enhance the practical applicability of the models in dynamic and unpredictable environments. Finally, for large-scale instances of the problems studied in this dissertation where exact methods may become computationally prohibitive, there is significant potential in developing hybrid algorithms. Such approaches could combine the strengths of mathematical optimization with metaheuristics. This synergy could lead to high-quality solutions in a fraction of the time, bridging the gap between optimality and scalability.

8.2 Closing remarks

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The LaTeX template used for the writing of this dissertation can be found in Luque-Calvo (2017). The cover of this dissertation was designed by graphic artist Juan Jaramillo, from Los Barrios (Cádiz), and depicts an Alcornoque (cork oak), a native tree species characteristic of the region.

Abraham Lincoln, speaking of the Bible, once remarked: *"Take all that you can of this Book upon reason, and balance the rest on faith, and you will live and die a happier man."* In this manuscript, every effort has been made with the intent that each concept follows logically from the last, weaving a narrative so tightly that little or no refuge in faith is required. It is my hope that this dissertation, which ends with these lines, stands as a proof that the author has matured enough in the profession he once chose to pursue in life.

Caminante, son tus *decisiones*
el camino y nada más;
caminante, no hay camino,
se *optimiza* al andar.
Al *iterar* se hace el camino,
y al volver la vista atrás
se ve la senda que el *algoritmo*
ha de volver a *ajustar*.
Caminante no hay camino
sino *soluciones* a encontrar.

Abbreviations index

B&B	Branch-and-Bound
B&C	Branch-and-Cut
BM	Bertsimas-Mazumder
ConnFL	Connected Facility Location Problem
COMP	Continuous Ordered Median Problem
CVaR	Conditional Value at Risk
DFJ	Dantzig-Fulkerson-Johnson
DOMP	Discrete Ordered Median Problem
e.g.	For example
IP	Integer Programming
IQR	Interquartile range
i.e.	Id est
i.i.d	Independent and identically distributed
KnaP	Knapsack Problem
KKT	Karush-Kuhn-Tucker conditions
LAD	Least Absolute Deviation
LB	Lower bound
LMS	Least Median of Squares
LP	Linear Programming
LQS	Least Quantile of Squares
LS	Least Squares
MeanAD	Mean Absolute Deviation
$\mathsf{MeanAD}_{(q)}$	Mean Absolute Deviation with respect to the q-th quantile
MedianAD	Median Absolute Deviation
MILP	Mixed-Integer Linear Programming
MINLP	Mixed-Integer Non-Linear Programming
MIQP	Mixed-Integer Quadratic Programming
MP	Master Problem
MST	Minimum Spanning Tree Problem
mTSP	Multiple TSP
MTZ	Miller-Tucker-Zemlin
NLP	Non-Linear Programming
OMC	DOMP with inner Connected Structure
OMT	Ordered Median Tree Location Problem
OMTHL	Ordered Median Tree of Hubs Location Problem
pMED	p-Median Problem
pMEDC	pMED with inner Connected Structure
pMEDT	pMED with inner Tree Structure
PTP	Profitable Tour Problem

QP	Quadratic Programming
QR	Quantile Regression
SCP	Set covering Problem
SEC	Subtour Elimination Constraints
SOS1	Special Ordered Sets of Type 1
SP	Subproblem
s.t.	Subject to
TSP	Traveling Salesman Problem
UB	Upper bound
VaR	Value at Risk
VRP	Vehicle Routing Problem
WMCSP	Weighted Multicover Set Problem

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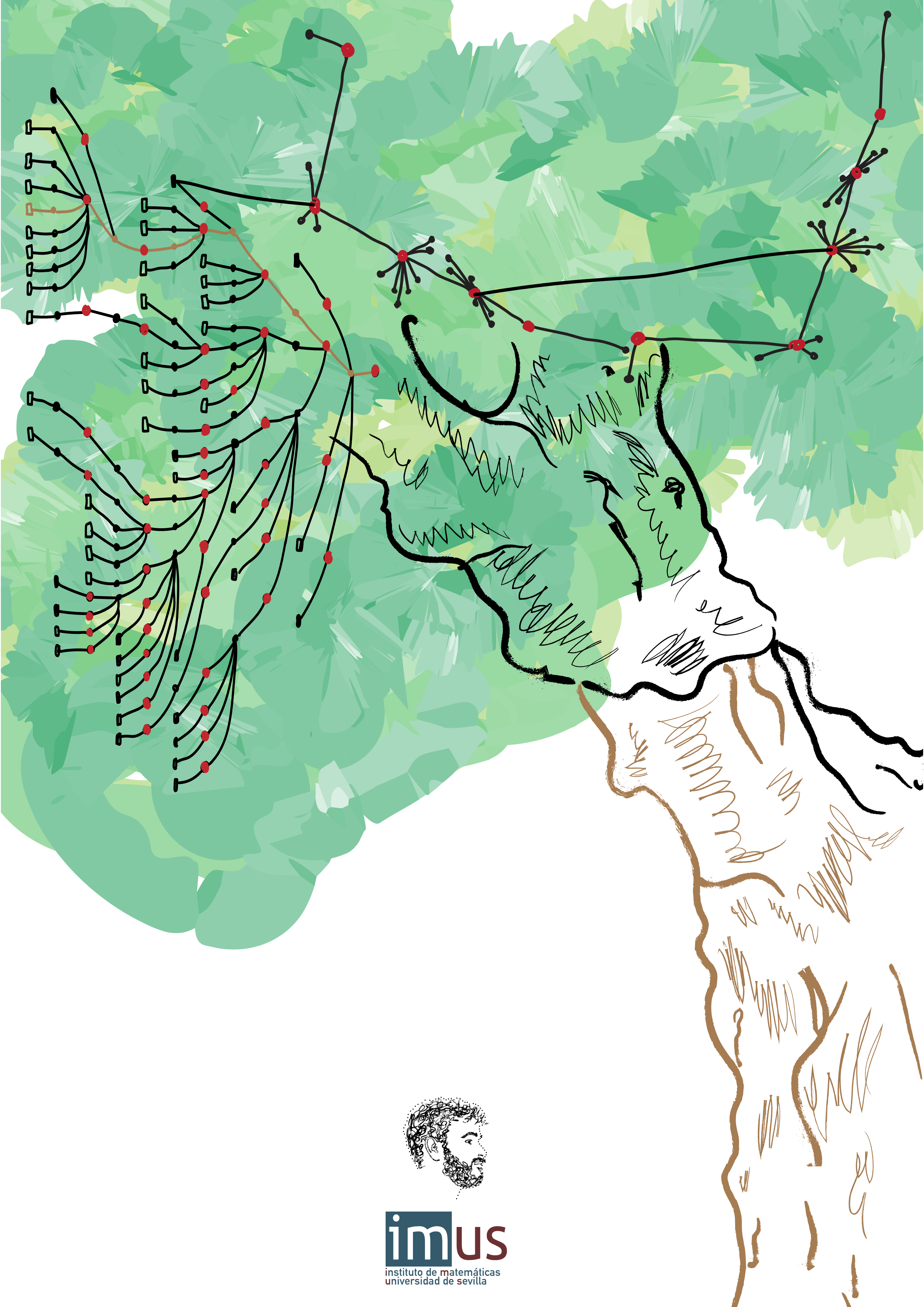

imus
instituto de matemáticas
universidad de sevilla